Story of
How POSTECH
Defeats KAIST

Mathematics for Science Quiz

Second Edition
September 2026

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POSTECH–KAIST Science War

Pohang University of Science and Technology
vs
Korea Advanced Institute of Science and Technology

Preface

This book is intended to provide complete preparation for the Mathematics section of Science Quiz in the POSTECH–KAIST Science War. Its structure begins with past Science Quiz problems: the mathematical concepts needed to solve them are developed in Mathematical Concepts, while the names, results, terminology, history, and other information worth remembering are organized in Facts to Memorize.

Science Quiz ranges across an unusually broad mathematical landscape, from middle- and high-school mathematics to undergraduate and more advanced university subjects. A strong competitor must therefore become fluent across many different areas of mathematics and be able to move between them quickly.

Accordingly, this book goes one step beyond the past problems themselves. It draws broadly from middle- and high-school mathematics competitions and olympiads; entrance examinations for gifted schools and science high schools; university entrance examinations; university midterm and final examinations; university mathematics competitions; and national and international olympiads.

Despite this breadth, the book retains its identity as a Science Quiz mathematics handbook. It emphasizes results, memorable formulas, elegant connections, interesting examples, and decisive ideas rather than long proofs or repetitive training. Its aim is to cultivate mathematical talent, intuition, and a reliable feel for unfamiliar problems.

Studying the mathematics of Science Quiz should also produce genuine mathematical growth beyond Science Quiz itself. The knowledge and habits of thought developed here are meant to strengthen mathematical understanding, judgment, and problem-solving ability in a lasting way.

I prepared this book as the 2026 Science Quiz Mathematics representative in the hope that POSTECH wins this year and continues to win in the years ahead. If circumstances permit, I will continue to revise and improve the book after 2026.

I sincerely hope that this book contributes to POSTECH's victory. Good luck, and let the game begin.

Acknowledgements

I am grateful to the following people for their comments, advice, and help in improving this book.

- Myeong Yoon (University of Cambridge)
- Youngjin Kim (Yonsei University)
- Woongju Lee (Massachusetts Institute of Technology)
- Junwoo June (Pohang University of Science and Technology)
- Jaegyeong Byun (Seoul National University)
- Taehyun Kong (Korea University)
- Jinu Kim (Pohang University of Science and Technology)

Contents

Notation and Conventions

The notation specified below is used throughout this book. When several typographic variants of the same letter exist, each variant is assigned a distinct role and the variants are not interchanged.

General Conventions

Scalars are written in ordinary italic letters, such as a, b, c, λ, t. Vectors are written in bold lowercase letters, such as $\mathbf{u}, \mathbf{v}, \mathbf{x}$. Matrices are written in bold uppercase letters, such as $\mathbf{A}, \mathbf{B}, \mathbf{P}$. In particular, the identity matrix, a diagonal matrix, and a projection matrix are written as $\mathbf{I}_n$, $\mathbf{D}$, and $\mathbf{P}$, respectively. Unless otherwise stated, vectors are column vectors.

Round parentheses are used for ordered tuples and matrix delimiters. Square brackets are used for closed intervals and for area notation such as $[ABC]$. Braces are used for sets. In set-builder notation, the condition is separated by $|$, as in $\{x \in A \mid P(x)\}$. Determinants may be enclosed by vertical bars because a determinant is a scalar.

In symbolic statements, logical conjunction, disjunction, and negation are written as $\wedge$, $\vee$, and $\neg$. The words "and" and "or" are used in ordinary prose. When several equivalent notations are listed in the same table entry, the first notation is the convention used in this book; the later forms are included only to identify common alternatives that a reader may encounter elsewhere.

Notation	Meaning
$\mathbb{N}$	the set of positive integers, $\{1, 2, 3, \dots\}$
$\mathbb{N}_0$	the set of nonnegative integers, $\{0, 1, 2, 3, \dots\}$
$\mathbb{Z}$	the set of integers
$\mathbb{Q}$	the set of rational numbers
$\mathbb{R}$	the set of real numbers
$\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$	the set of extended real numbers
$\mathbb{C}$	the set of complex numbers
$\mathbb{H}$	the quaternion division ring
$a := b$	a is defined to be b
s.t.	abbreviation used to introduce a condition or constraint
ε	a positive quantity in analysis, usually arbitrary
ϵ_{ijk}	the Levi–Civita symbol; the glyph ϵ is reserved for this use
$O(g(x))$	Big-O notation
$o(g(x))$	little-o notation
$\operatorname{sgn}(x)$	the sign of a real number or a permutation, according to context
$\square$	the end of a proof or solution

Theorem-like Environments

Label	Usage
Definition	a precise meaning of a term or object
Theorem	a major mathematical result
Proposition	a useful result of more limited scope
Lemma	a supporting result or technical tool
Corollary	a result that follows directly from another result
Formula	a computational identity or closed form
Algorithm	a step-by-step computational procedure
Fact	a verified mathematical or historical fact
Conjecture	a statement believed to be true but not proved in full generality
Hypothesis	a named assumption or proposed statement
Open Problem	a problem whose stated form remains unsolved
Idea	a concise account of a decisive mathematical idea
Strategy	a broadly useful problem-solving method
Tactic	a local, immediately applicable device
Remark	supplementary context or a caution

Set Theory and Logic

Uppercase letters such as A, B, X, Y usually denote sets, while lowercase letters such as a, b, x, y usually denote their elements unless the context indicates another meaning. The arrow $f : X \to Y$ specifies the domain and codomain of a function, while $x \mapsto f(x)$ specifies its action on an element.

Notation	Meaning
$x \in A$	x is an element of A
$x \notin A$	x is not an element of A
$A \subseteq B$	A is a subset of B
$A \subsetneq B$	A is a proper subset of B
$A \supseteq B$	A is a superset of B
$A \supsetneq B$	A is a proper superset of B
$A \cup B$	the union of A and B
$A \cap B$	the intersection of A and B
$A \setminus B$, $A - B$	the set difference of A and B
A^{c}	the complement of A in the understood ambient set
$\varnothing$	the empty set
$\mathcal{P}(A)$	the power set of A
$A \times B$	the Cartesian product of A and B
$A \sqcup B$	the disjoint union of A and B
$\lvert A \rvert$	the cardinality of a finite set A
$\{x \in A \mid P(x)\}$	the set of all $x \in A$ for which $P(x)$ holds
$x \sim y$	x and y are equivalent under an understood equivalence relation
$A/\sim$	the set of equivalence classes of A under $\sim$
$\forall x \in A$	for every $x \in A$
$\exists x \in A$	there exists $x \in A$
$\exists! x \in A$	there exists a unique $x \in A$
$\nexists x \in A$	there does not exist $x \in A$

Notation	Meaning
$P \Rightarrow Q$	P implies Q
$P \not\Rightarrow Q$	P does not imply Q
$P \Leftarrow Q$	Q implies P
$P \Leftrightarrow Q$	P if and only if Q
$P \wedge Q$	logical conjunction
$P \vee Q$	logical disjunction
$\neg P$	logical negation
$f : X \to Y$	a function from X to Y
$x \mapsto f(x)$	the elementwise rule of a function
$\operatorname{dom} f$	the domain of f
$\operatorname{codom} f$	the codomain of f
id_X	the identity map on X
$g \circ f$	composition of functions
$f\|_A$	restriction of f to A
$f : X \hookrightarrow Y$	an injective map
$f : X \twoheadrightarrow Y$	a surjective map
$f : X \xrightarrow{\sim} Y$	an isomorphism; for sets without additional structure, equivalently a bijection
$X \cong Y$	X and Y are isomorphic; in topology, this means homeomorphic
$f(A)$	the image of A under f
$f^{-1}(B)$	the inverse image, or preimage, of B
$\mathbb{1}_A(x)$	the indicator function of A
$\aleph_0$	the cardinality of a countably infinite set
$\mathfrak{c}$	the cardinality of the continuum

The symbol $\mathbb{1}_A$ is blackboard bold and does not denote a vector. The all-ones vector is $\mathbf{1}$. Boldface is otherwise reserved for vectors and matrices.

Calculus and Analysis

In polar, cylindrical, and spherical coordinates, θ is the azimuthal angle measured counterclockwise from the positive x-axis in the xy-plane. In spherical coordinates, ϕ is the polar angle measured from the positive z-axis. The glyph ϕ is reserved for this spherical polar angle. The glyph φ is used for functions, homomorphisms, Euler's totient function, and the golden ratio whenever one of these objects is introduced.

Notation	Meaning
$f'(a)$	the first derivative of f at a
$f^{(n)}(a)$	the nth derivative of f at a
$\dfrac{\mathrm{d}y}{\mathrm{d}x}$	the derivative of y with respect to x, with upright d
$\displaystyle\int_a^b f(x)\,\mathrm{d}x$	the definite integral of f over $[a, b]$
$\Gamma(z)$	the Gamma function
$B(x, y)$	the Beta function
$\zeta(s)$	the Riemann zeta function
$\ln x$	the logarithm to base e
$\lg x$	the logarithm to base 10

Notation	Meaning
$\operatorname{lb} x$	the logarithm to base 2
$\log_a x$	the logarithm to the explicitly specified base a
$\arcsin x$	the principal inverse sine, with values in $[-\pi/2, \pi/2]$
$\arccos x$	the principal inverse cosine, with values in $[0, \pi]$
$\arctan x$	the principal inverse tangent, with values in $(-\pi/2, \pi/2)$
$\operatorname{arcsec} x$	the principal inverse secant, with values in $[0, \pi] \setminus \{\pi/2\}$
$\operatorname{arccsc} x$	the principal inverse cosecant, with values in $[-\pi/2, \pi/2] \setminus \{0\}$
$\operatorname{arccot} x$	the principal inverse cotangent, with values in $(0, \pi)$
$\operatorname{sech} x$	the hyperbolic secant $1/\cosh x$
$\operatorname{csch} x$	the hyperbolic cosecant $1/\sinh x$
$\operatorname{arcsinh} x$	the inverse hyperbolic sine
$\operatorname{arccosh} x$	the principal inverse hyperbolic cosine, with values in $[0, \infty)$
$\operatorname{arctanh} x$	the inverse hyperbolic tangent on $(-1, 1)$
$C^k(I)$	the space of k times continuously differentiable functions on I
$C^\infty(I)$	the space of smooth functions on I
$\dfrac{\partial f}{\partial x_i}$	the partial derivative of f with respect to x_i
(x, y), (x, y, z)	Cartesian coordinates
(r, θ)	polar coordinates
(r, θ, z)	cylindrical coordinates
(ρ, θ, ϕ)	spherical coordinates
∇f	the gradient of f
$\nabla \cdot \mathbf{F}$	the divergence of $\mathbf{F}$
$\nabla \times \mathbf{F}$	the curl of $\mathbf{F}$
Δf	the Laplacian of f
$D\mathbf{F}(\mathbf{a})$	the total derivative, or Jacobian matrix, of $\mathbf{F}$ at $\mathbf{a}$
$\operatorname{Jac}(\mathbf{F})$	the Jacobian determinant of $\mathbf{F}$
$\operatorname{Hess} f$	the Hessian matrix of f
$\mathbf{r}(t)$	a parametrized curve
$\mathrm{d}s$	an element of arc length
$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r}$	a line integral along C
$\iint_S \mathbf{F} \cdot \mathbf{n}\, \mathrm{d}S$	a flux integral across S
$B_r(x)$	the open ball of radius r centered at x
$\overline{B}_r(x)$	the closed ball of radius r centered at x
$x_n \to x$	convergence of a sequence
$a_n \nearrow L$	monotone increasing convergence to L
$a_n \searrow L$	monotone decreasing convergence to L
$\limsup a_n$	the limit superior of (a_n)
$\liminf a_n$	the limit inferior of (a_n)
$f_n \to f$, $f_n \xrightarrow{\text{ptw}} f$	pointwise convergence
$f_n \rightrightarrows f$, $f_n \xrightarrow{\text{unif}} f$	uniform convergence
$f_n \xrightarrow{\text{a.e.}} f$	convergence almost everywhere
$\lVert f \rVert_\infty$	the supremum norm of f
$z = x + iy$	a complex number with real part x and imaginary part y
$\operatorname{Re} z$, $\Re z$	the real part of z
$\operatorname{Im} z$, $\Im z$	the imaginary part of z
$\overline{z}$	the complex conjugate of z

Notation	Meaning
$\|z\|$	the modulus of z
$\arg z$	an argument of z
$\oint_C f(z)\,\mathrm{d}z$	a contour integral around a closed curve C
$\operatorname{Res}(f;a)$	the residue of f at a

Linear Algebra

Notation	Meaning
$\mathbf{A} = (a_{ij})$	the matrix whose (i,j)-entry is a_{ij}
$\mathbf{A}^{\mathsf{T}}$	the transpose of $\mathbf{A}$
$\mathbf{A}^{\mathsf{H}}$	the conjugate transpose of $\mathbf{A}$
$\mathbf{A}^{-1}$	the inverse of $\mathbf{A}$
$\mathbf{I}_n$	the $n \times n$ identity matrix
$\mathbf{0}$	a zero vector or zero matrix, with dimension clear from context
$\mathbf{D}$	a diagonal matrix
$\operatorname{diag}(d_1, \ldots, d_n)$	the diagonal matrix with diagonal entries $d_1, \ldots, d_n$
$\mathbf{1}$	the all-ones vector
$\mathbf{J}_{m,n}$, $\mathbf{J}_n$	the $m \times n$ and $n \times n$ all-ones matrices
$\mathbf{P}_\sigma$	the permutation matrix associated with σ
$\det(\mathbf{A})$	the determinant of $\mathbf{A}$
$\operatorname{rk}(\mathbf{A})$	the rank of $\mathbf{A}$
$\operatorname{tr}(\mathbf{A})$	the trace of $\mathbf{A}$
$p_{\mathbf{A}}(\lambda)$	the characteristic polynomial of $\mathbf{A}$
$m_{\mathbf{A}}(\lambda)$	the minimal polynomial of $\mathbf{A}$
$\rho(\mathbf{A})$	the spectral radius of $\mathbf{A}$
$\sigma_i(\mathbf{A})$	the ith singular value of $\mathbf{A}$
M_{ij}	the minor of the entry a_{ij}
$C_{ij} = (-1)^{i+j} M_{ij}$	the cofactor of the entry a_{ij}
$\operatorname{adj}(\mathbf{A})$	the adjugate of $\mathbf{A}$
$\operatorname{Col}(\mathbf{A})$	the column space of $\mathbf{A}$
$\operatorname{Null}(\mathbf{A})$	the null space of $\mathbf{A}$
$\operatorname{Row}(\mathbf{A})$	the row space of $\mathbf{A}$
$\ker T$	the kernel of a linear map T
$\operatorname{im} T$	the image of a linear map T
$\mathbf{e}_i$	the ith standard basis vector
$\dim V$	the dimension of the vector space V
$\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k)$	the span of the listed vectors
$U \oplus V$	the direct sum of subspaces U and V
$\langle \mathbf{u}, \mathbf{v} \rangle$	the inner product of $\mathbf{u}$ and $\mathbf{v}$
$\\|\mathbf{v}\\|$	the norm of $\mathbf{v}$
$\operatorname{proj}_{\mathbf{v}} \mathbf{x}$	the orthogonal projection of $\mathbf{x}$ onto $\operatorname{span}(\mathbf{v})$
$\mathbf{u} \perp \mathbf{v}$	$\mathbf{u}$ and $\mathbf{v}$ are orthogonal
$\mathbf{u} \times \mathbf{v}$	the cross product in $\mathbb{R}^3$

Differential Equations

Notation	Meaning
y'	the first derivative of y with respect to the independent variable
y''	the second derivative of y
$\dot{x}$	the first derivative of x with respect to time t
$\ddot{x}$	the second derivative of x with respect to time t
y_h	the homogeneous part of a solution
y_p	a particular solution
$c_1, c_2, \dots$	arbitrary constants in a general solution
$W(y_1, y_2)(t)$	the Wronskian of y_1 and y_2
$\mathcal{L}\{f(t)\}(s)$	the Laplace transform of $f(t)$
$\widehat{f}(\omega)$	the Fourier transform under the convention stated where it is used

Combinatorics and Discrete Mathematics

Notation	Meaning
$[n]$	the set $\{1, 2, \dots, n\}$
$n!$	factorial
$\binom{n}{k}$, ${}_nC_k$	the number of k-element subsets of an n-element set
$\left(\!\binom{n}{k}\!\right)$, ${}_nH_k$	the number of size-k multisets chosen from an n-element set
$P(n, k)$	the number of k-permutations of an n-element set
$\left\{ {n \atop k} \right\}$	the Stirling number of the second kind
a_n	the nth term of a sequence
Δa_n	the forward difference $a_{n+1} - a_n$
$S_n = \displaystyle\sum_{k=1}^{n} a_k$	the nth partial sum
$A(x) = \displaystyle\sum_{n\geq 0} a_n x^n$	an ordinary generating function
$[x^n]A(x)$	the coefficient of x^n in $A(x)$
$\omega_n = e^{2\pi i/n}$	a primitive nth root of unity
$\sigma = (a_1\ a_2\ \cdots\ a_k)$	cycle notation for a permutation
$\mathrm{inv}(\sigma)$	the inversion number of σ
$\mathrm{Fix}(g)$	the set of objects fixed by g
$G = (V, E)$	a graph with vertex set V and edge set E
$\deg(v)$	the degree of the vertex v
K_n	the complete graph on n vertices
C_n	the cycle graph on n vertices
P_n	the path graph on n vertices

Probability and Statistics

Random variables are usually denoted by uppercase letters such as X, Y, Z; events are denoted by uppercase letters such as A, B, E.

Notation	Meaning
$\mathbb{P}(A)$	the probability of the event A
$\mathbb{P}(A \mid B)$	the conditional probability of A given B
$\mathbb{E}[X]$	the expectation of X
$\mathrm{Var}(X)$	the variance of X
$\mathrm{Cov}(X, Y)$	the covariance of X and Y
$X \sim \mathrm{Bin}(n, p)$	a binomial distribution
$X \sim \mathrm{Poi}(\lambda)$	a Poisson distribution
$X \sim \mathrm{N}(\mu, \sigma^2)$	a normal distribution with mean μ and variance σ^2
$X_n \xrightarrow{\text{a.s.}} X$	almost-sure convergence
$X_n \xrightarrow{\mathbb{P}} X$	convergence in probability
$X_n \xrightarrow{L^p} X$	convergence in L^p
$X_n \xrightarrow{\mathrm{d}} X$	convergence in distribution

Geometry

Uppercase letters such as A, B, C, D denote points. When the usual triangle notation is adopted, the corresponding lowercase letters a, b, c denote the side lengths opposite A, B, C, respectively.

Notation	Meaning
$\triangle ABC$	the triangle with vertices A, B, C
$\angle ABC$	the angle with vertex B
$\overline{AB}$	the segment joining A and B
AB	the length of $\overline{AB}$
$[ABC]$	the area of $\triangle ABC$
$\mathrm{length}(S)$	the length of a curve or one-dimensional geometric object S
$\mathrm{area}(S)$	the area of a planar region or surface S
$\mathrm{vol}(S)$	the volume of a three-dimensional region or solid S
$\mathrm{dist}(P, S)$	the distance from the point P to the set or geometric object S
$\mathrm{conv}(S)$	the convex hull of S
R	the circumradius of a triangle
r	the inradius of a triangle
s	the semiperimeter of a triangle
I	the number of interior lattice points in Pick's Theorem
B	the number of boundary lattice points in Pick's Theorem
$\parallel$	parallel
$\perp$	perpendicular

Number Theory

Unless otherwise stated, divisibility and congruence statements are made in $\mathbb{Z}$.

Notation	Meaning
$a \mid b$	a divides b
$a \nmid b$	a does not divide b

Notation	Meaning
$\gcd(a,b)$	the greatest common divisor of a and b
$\operatorname{lcm}(a,b)$	the least common multiple of a and b
$a \equiv b \pmod{n}$	a is congruent to b modulo n
$\pi(x)$	the number of primes not exceeding x
$[a]_n$	the residue class of a modulo n
$\mathbb{Z}/n\mathbb{Z}$	the ring of residue classes modulo n
$(\mathbb{Z}/n\mathbb{Z})^\times$	the multiplicative group of units modulo n
$a^{-1} \pmod{n}$	the multiplicative inverse of a modulo n
$\left(\frac{a}{p}\right)$	the Legendre symbol
$\varphi(n)$	Euler's totient function
$\operatorname{ord}_n(a)$	the multiplicative order of a modulo n
$\nu_p(n)$	the exponent of p in the prime factorization of n
$\tau(n)$	the number of positive divisors of n
$\sigma(n)$	the sum of the positive divisors of n
$\mu(n)$	the Möbius function
$\operatorname{rad}(n)$	the product of the distinct prime divisors of n
p	usually a prime number in number-theoretic contexts
$\mathbb{F}_q$	the finite field with q elements

Algebra

Capital letters such as G, H denote groups, R, S denote rings, and F, K, L denote fields; lowercase letters denote their elements unless the context requires otherwise.

Notation	Meaning
G	usually a group
S_n	the symmetric group on n symbols
A_n	the alternating group on n symbols
D_{2n}	the dihedral group of order $2n$
e	the identity element of a group
$\lvert G\rvert$	the order of a finite group G
$\langle S\rangle$	the subgroup generated by S
$H \leq G$	H is a subgroup of G
$H \trianglelefteq G$	H is a normal subgroup of G
G/H	the quotient group of G by H
$\ker\varphi$	the kernel of a homomorphism φ
$\operatorname{im}\varphi$	the image of a homomorphism φ
$\operatorname{Aut}(G)$	the automorphism group of G
$G \cdot x$	the orbit of x under a group action
G_x	the stabilizer of x under a group action
$Z(G)$	the center of G
$C_G(x)$	the centralizer of x in G
$\mathrm{GL}_n(F)$	the general linear group over F
$\mathrm{SL}_n(F)$	the special linear group over F
R	usually a ring
UFD	unique factorization domain

Notation	Meaning
PID	principal ideal domain
ED	Euclidean domain
$I \trianglelefteq R$	I is an ideal of R
R/I	the quotient ring of R by I
F	a base field in an algebraic context
K	a field extension of F, or a second field when required
$F[x]$	the polynomial ring over F
$\deg f$	the degree of a polynomial f
$\mathrm{Disc}(f)$	the discriminant of a polynomial f
$\mathrm{trdeg}_K L$	the transcendence degree of L over K
$\mathrm{char}\, F$	the characteristic of F
$\mathrm{Gal}(L/K)$	the Galois group of the extension L/K

Topology

Notation	Meaning
$(X, \mathcal{T})$	a topological space
$\mathrm{int}\, A$	the interior of A
$\overline{A}$	the closure of A
∂A	the boundary of A
$\pi_1(X, x_0)$	the fundamental group of X based at x_0
$\chi(X)$	the Euler characteristic of X

Miscellaneous Topics

Mathematical Constants

The table contains constants that occur frequently across several areas of mathematics. A defining expression is used whenever a single-letter symbol is not needed elsewhere in the book.

Symbol	Name	Meaning or definition	Exact value or approximation
i	Imaginary unit	The imaginary unit, defined by the relation $i^2 = -1$.	$i^2 = -1$
$\sqrt{2}$	Pythagoras constant	The positive length of the diagonal of a unit square.	$1.41421356237\ldots$
$\sqrt{3}$	Square root of 3	The positive square root of 3.	$1.73205080757\ldots$
$\sqrt{5}$	Square root of 5	The positive square root of 5.	$2.23606797750\ldots$
$\sqrt{7}$	Square root of 7	The positive square root of 7.	$2.64575131106\ldots$
e	Euler's number	The base of the natural logarithm: $e = \sum_{n=0}^{\infty} 1/n!$.	$2.71828182846\ldots$
π	Pi	The ratio of the circumference of a Euclidean circle to its diameter.	$3.14159265359\ldots$
φ	Golden ratio	The positive solution of $x^2 = x + 1$.	$\dfrac{1+\sqrt{5}}{2} \approx 1.61803398875\ldots$

Symbol	Name	Meaning or definition	Exact value or approximation
γ	Euler–Mascheroni constant	$\gamma = \lim_{n\to\infty}\left(\sum_{k=1}^{n}\frac{1}{k} - \ln n\right).$	$0.577215664902\ldots$
$\zeta(3)$	Apéry's constant	The value of the Riemann zeta function at 3: $\sum_{n=1}^{\infty} 1/n^3$.	$1.20205690316\ldots$
G	Catalan's constant	$G = \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2}.$	$0.915965594177\ldots$

Other Notation

Notation	Meaning
$\lfloor x \rfloor$	the floor of x
$\lceil x \rceil$	the ceiling of x
$\{x\}$	the fractional part $x - \lfloor x \rfloor$

Part I

Mathematical Concepts

Chapter 1

Set Theory and Logic

Related POSTECH Courses: (MATH202) Set Theory
Related Books: Introduction to Set Theory (Karel Hrbacek, Thomas Jech)

Definition 1.1. Set and Element
A set is a collection of objects, called its elements. We write $x \in A$ if x is an element of A, and $x \notin A$ otherwise. The relation $A \subseteq B$ means

$$\forall x, \qquad x \in A \Rightarrow x \in B.$$

Formula 1.2. Basic Set Operations
For sets A and B inside an ambient set U, we use

$$A \cup B = \{x \in U \mid (x \in A) \vee (x \in B)\},$$

$$A \cap B = \{x \in U \mid (x \in A) \wedge (x \in B)\},$$

and

$$A \setminus B = \{x \in A \mid x \notin B\}.$$

Formula 1.3. Set Algebra
For subsets $A, B, C \subseteq U$, the following identities are often useful:

$$A \cup \varnothing = A, \qquad A \cap U = A, \qquad A \cup A = A, \qquad A \cap A = A.$$

The distributive laws are

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

The absorption laws are

$$A \cup (A \cap B) = A, \qquad A \cap (A \cup B) = A.$$

Formula 1.4. Basic Logical Equivalences
The most useful logical equivalences are

$$(P \Rightarrow Q) \iff (\neg Q \Rightarrow \neg P),$$

$$\neg(P \Rightarrow Q) \iff P \wedge \neg Q,$$

$$\neg(P \wedge Q) \iff (\neg P) \vee (\neg Q), \qquad \neg(P \vee Q) \iff (\neg P) \wedge (\neg Q),$$

and

$$\neg(\forall x \in A,\ P(x)) \iff \exists x \in A \text{ s.t. } \neg P(x),$$
$$\neg(\exists x \in A \text{ s.t. } P(x)) \iff \forall x \in A,\ \neg P(x).$$

Strategy. To prove $P \Rightarrow Q$, it is often enough to prove the contrapositive $\neg Q \Rightarrow \neg P$. To disprove a universal statement, find a counterexample.

Formula 1.5. De Morgan's Laws
For subsets $A, B \subseteq U$, we have

$$(A \cup B)^{\mathsf{c}} = A^{\mathsf{c}} \cap B^{\mathsf{c}}, \qquad (A \cap B)^{\mathsf{c}} = A^{\mathsf{c}} \cup B^{\mathsf{c}}.$$

More generally,

$$\left(\bigcup_{i \in I} A_i\right)^{\mathsf{c}} = \bigcap_{i \in I} A_i^{\mathsf{c}}, \qquad \left(\bigcap_{i \in I} A_i\right)^{\mathsf{c}} = \bigcup_{i \in I} A_i^{\mathsf{c}}.$$

Definition 1.6. Power Set
The power set of a set A is

$$\mathcal{P}(A) = \{B \mid B \subseteq A\}.$$

If $|A| = n < \infty$, then

$$|\mathcal{P}(A)| = 2^n.$$

Definition 1.7. Function
A function $f : X \to Y$ assigns to each element $x \in X$ a unique element $f(x) \in Y$. The set X is the domain, and Y is the codomain. The notation $x \mapsto f(x)$ describes the rule on elements.

Definition 1.8. Injection, Surjection, and Bijection
A function $f : X \to Y$ is injective if distinct elements of X have distinct images. Equivalently,

$$\forall x_1, x_2 \in X, \qquad f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$

Once injectivity has been established, one may write $f : X \hookrightarrow Y$.

The function is surjective if every element of Y has a preimage:

$$\forall y \in Y, \qquad \exists x \in X \text{ s.t. } f(x) = y.$$

Once surjectivity has been established, one may write $f : X \twoheadrightarrow Y$. The function is bijective if it is both injective and surjective.

Proposition 1.9. Finite Injective-Surjective Criterion
If X and Y are finite sets with $|X| = |Y|$, then for a function $f : X \to Y$, the following are equivalent:

$$f \textit{ is injective} \iff f \textit{ is surjective} \iff f \textit{ is bijective.}$$

Proposition 1.10. Inverse Image Preserves Set Operations
If $f : X \to Y$ and $A, B \subseteq Y$, then

$$f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B),$$

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B),$$

and

$$f^{-1}(Y \setminus A) = X \setminus f^{-1}(A).$$

Definition 1.11. Relation and Equivalence Relation
A binary relation on a set A is a subset $R \subseteq A \times A$. An equivalence relation $\sim$ is reflexive, symmetric, and transitive; explicitly,

$$\forall x \in A, \qquad x \sim x,$$

$$\forall x, y \in A, \qquad x \sim y \Rightarrow y \sim x,$$

and
$$\forall x, y, z \in A, \qquad (x \sim y \wedge y \sim z) \Rightarrow x \sim z.$$
The equivalence class of $a \in A$ is
$$[a] = \{x \in A \mid x \sim a\}.$$

Proposition 1.12. Equivalence Relations and Partitions
Equivalence relations on a set A are the same structure as partitions of A: each equivalence relation partitions A into equivalence classes, and each partition of A determines an equivalence relation.

Definition 1.13. Cardinality and Equinumerosity
Two sets A and B have the same cardinality if there exists a bijection between them. We write
$$|A| = |B| \qquad \text{or} \qquad A \xrightarrow{\sim} B.$$
Thus cardinality compares the sizes of finite and infinite sets through one-to-one correspondences rather than through geometric containment.

Definition 1.14. Dedekind-Infinite Set
A set A is Dedekind-infinite if there exists a proper subset $B \subsetneq A$ s.t.
$$|A| = |B|.$$
In ZFC, a set is Dedekind-infinite if and only if it is infinite. Without the Axiom of Choice, this equivalence cannot be proved in full generality.

Definition 1.15. Countability
A set A is countable if it is finite or if there exists a bijection $f : A \to \mathbb{N}$. In the countably infinite case,
$$\exists f : A \xrightarrow{\sim} \mathbb{N}, \qquad |A| = |\mathbb{N}| = \aleph_0.$$
A set is uncountable if it is not countable.

Theorem 1.16. Basic Cardinalities
The standard countably infinite sets satisfy
$$|\mathbb{N}| = |2\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}| = \aleph_0,$$
whereas
$$|(0,1)| = |\mathbb{R}| = \mathfrak{c} = 2^{\aleph_0} > \aleph_0.$$
In particular, a proper subset of an infinite set may have the same cardinality as the whole set, while the real numbers cannot be listed by the natural numbers.

Theorem 1.17. Cantor's Theorem
For every set A, there is no surjection from A onto $\mathcal{P}(A)$. Equivalently,
$$|A| < |\mathcal{P}(A)|.$$

Definition 1.18. Aleph Numbers and the Continuum
The symbol $\aleph_0$ denotes the cardinality of a countably infinite set, and $\aleph_1$ denotes the least uncountable cardinal. The cardinality of the continuum is
$$\mathfrak{c} = |\mathbb{R}| = 2^{\aleph_0}.$$
The Continuum Hypothesis is the assertion
$$\mathfrak{c} = \aleph_1.$$

Theorem 1.19. Schröder–Bernstein Theorem; Cantor–Bernstein Theorem
If there exists an injection $f : A \hookrightarrow B$ and an injection $g : B \hookrightarrow A$, then there exists a bijection

$$h : A \xrightarrow{\sim} B.$$

Thus two sets have the same cardinality once each can be embedded into the other.

Algorithm 1.20. Mathematical Induction
To prove a statement $P(n)$ for all integers $n \geq n_0$:

1. Prove the base case $P(n_0)$.

2. Prove the induction step: for every $k \geq n_0$, show that $P(k)$ implies $P(k+1)$.

Then $P(n)$ holds for all integers $n \geq n_0$.

Theorem 1.21. Well-Ordering Principle
Every nonempty subset of $\mathbb{N}$ *has a least element.*

Fact 1.22. Russell's Paradox
The expression "the set of all sets that do not contain themselves" leads to a contradiction in naive set theory. This is one reason modern set theory is formulated axiomatically.

Fact 1.23. ZF and ZFC
ZF denotes Zermelo–Fraenkel set theory, and ZFC denotes ZF together with the Axiom of Choice. The axioms restrict set formation sufficiently to avoid Russell's paradox while supporting the standard foundations of modern mathematics.

Fact 1.24. Axiom of Choice and Zorn's Lemma
The Axiom of Choice, Zorn's Lemma, and the Well-Ordering Theorem are equivalent over the usual Zermelo–Fraenkel axioms. They are standard foundational tools in advanced algebra, analysis, and topology.

Chapter 2

Calculus and Analysis

Related POSTECH Courses: (MATH101) Calculus I, (MATH102) Calculus II, (MATH210) Applied Complex Variables, (MATH311) Analysis I, (MATH312) Analysis II
Related Books: Calculus With Applications (Peter D. Lax, Maria Shea Terrell), Multivariable Calculus with Applications (Peter D. Lax, Maria Shea Terrell), Principles of Mathematical Analysis (Walter Rudin), Complex Variables and Applications (James Ward Brown, Ruel V. Churchill)

One-Variable Calculus

Definition 2.1. Sequence, Series, and Partial Sum
A sequence is a function $a : \mathbb{N} \to \mathbb{R}$ or $a : \mathbb{N} \to \mathbb{C}$, usually written (a_n). A series is the formal sum

$$\sum_{n=1}^{\infty} a_n,$$

whose Nth partial sum is

$$S_N = \sum_{n=1}^{N} a_n.$$

The series converges to S exactly when $S_N \to S$ as $N \to \infty$; otherwise it diverges. Thus every convergence test for a series is a criterion for convergence of its sequence of partial sums.

Formula 2.2. Arithmetic and Geometric Progressions
For an arithmetic progression with first term a and common difference d, we have

$$a_n = a + (n-1)d, \qquad \sum_{k=1}^{n} a_k = \frac{n}{2}(a_1 + a_n).$$

For a geometric progression with first term a and common ratio r, we have

$$a_n = ar^{n-1}, \qquad \sum_{k=0}^{n-1} ar^k = a\frac{1-r^n}{1-r} \quad (r \neq 1).$$

If $|r| < 1$, then

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}.$$

Formula 2.3. Sums of Powers; Faulhaber's Formulas
The first power sums are

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2},$$

$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}, \qquad \sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2,$$

and

$$\sum_{k=1}^{n} k^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}.$$

More generally, for each fixed nonnegative integer m,

$$\sum_{k=1}^{n} k^m$$

is a polynomial in n of degree $m+1$ with leading coefficient $1/(m+1)$.

Formula 2.4. Finite Differences; Newton's Forward Difference Formula
For a sequence (a_n), define the forward difference by

$$\Delta a_n = a_{n+1} - a_n.$$

If $a_n = P(n)$ for a polynomial P of degree d with leading coefficient c, then

$$\Delta^d a_n = d!c, \qquad \Delta^{d+1} a_n = 0.$$

Conversely, a sequence with constant dth differences is polynomial in n of degree at most d. For every polynomial P of degree at most d,

$$P(n) = \sum_{k=0}^{d} \binom{n}{k} \Delta^k P(0).$$

Strategy. A table of successive differences is one of the most efficient methods to detect a hidden polynomial sequence. Constant first, second, and third differences indicate linear, quadratic, and cubic behavior, respectively.

Formula 2.5. Linear Recurrences with Constant Coefficients
Consider

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_m a_{n-m}.$$

Its characteristic polynomial is

$$r^m - c_1 r^{m-1} - c_2 r^{m-2} - \cdots - c_m.$$

If the roots $r_1, \ldots, r_m$ are distinct, then

$$a_n = A_1 r_1^n + \cdots + A_m r_m^n.$$

A root r of multiplicity q contributes terms

$$r^n,\ nr^n,\ \ldots,\ n^{q-1} r^n.$$

The constants are determined from the initial values.

Formula 2.6. First-Order Linear Recurrence
If

$$a_n = r a_{n-1} + b_n \qquad (n \geq 1),$$

then iteration gives

$$a_n = r^n a_0 + \sum_{j=1}^{n} r^{\,n-j} b_j.$$

In particular, if $b_n = b$ and $r \neq 1$, then

$$a_n = r^n a_0 + b\frac{r^n - 1}{r - 1}.$$

Formula 2.7. Nonhomogeneous Linear Recurrences
For a linear recurrence with constant coefficients, the general solution is

$$a_n = a_n^{(h)} + a_n^{(p)},$$

where $a_n^{(h)}$ solves the associated homogeneous recurrence and $a_n^{(p)}$ is one particular solution. If the forcing term has the form

$$P(n)\alpha^n,$$

try a particular solution of the form

$$n^s Q(n)\alpha^n,$$

where $\deg Q = \deg P$ and s is the multiplicity of α as a root of the characteristic polynomial.

Strategy. For recurrences, first identify whether the quickest language is characteristic roots, generating functions, a transfer matrix, or telescoping after a suitable normalization.

Definition 2.8. Limit of a Function; Epsilon–Delta Definition
Let $D \subseteq \mathbb{R}$, let a be an accumulation point of D, and let $f : D \to \mathbb{R}$. The statement

$$\lim_{x\to a} f(x) = L$$

means that $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to, but different from, a. Formally,

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in D, \qquad 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Definition 2.9. One-Sided Limits and Limits at Infinity
The right-hand limit $\lim_{x\to a^+} f(x) = L$ is defined by

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in D, \qquad 0 < x - a < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

The left-hand definition is analogous, with $0 < a - x < \delta$. Moreover,

$$\lim_{x\to\infty} f(x) = L$$

means

$$\forall \varepsilon > 0, \quad \exists M > 0, \quad \forall x \in D, \qquad x > M \Rightarrow |f(x) - L| < \varepsilon.$$

Finally, $\lim_{x\to a} f(x) = \infty$ means

$$\forall M > 0, \quad \exists \delta > 0, \quad \forall x \in D, \qquad 0 < |x - a| < \delta \Rightarrow f(x) > M.$$

Theorem 2.10. Squeeze Theorem; Sandwich Theorem
Suppose that $g(x) \le f(x) \le h(x)$ for all x sufficiently close to a, except possibly at a, and that

$$\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L.$$

Then

$$\lim_{x\to a} f(x) = L.$$

Definition 2.11. Continuity at a Point
Let $f : D \to \mathbb{R}$ and $a \in D$. The function f is continuous at a if

$$\lim_{x\to a} f(x) = f(a).$$

Equivalently,

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in D, \qquad |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon.$$

The function is continuous on D if it is continuous at every point of D.

Definition 2.12. Derivative
Let f be defined near a. The derivative of f at a is

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h},$$

provided this limit exists as a finite real number. Thus $f'(a) = L$ if and only if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall h \in \mathbb{R}, \qquad 0 < |h| < \delta \Rightarrow \left| \frac{f(a+h) - f(a)}{h} - L \right| < \varepsilon.$$

Formula 2.13. Standard Trigonometric and Exponential Limits
The following limits are fundamental:

$$\lim_{x \to 0} \frac{\sin x}{x} = 1, \qquad \lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2},$$

$$\lim_{x \to 0} \frac{e^x - 1}{x} = 1, \qquad \lim_{x \to 0} \frac{\ln(1+x)}{x} = 1,$$

and

$$\lim_{x \to 0} (1+x)^{1/x} = e.$$

Strategy. These limits are often faster than L'Hôpital's Rule. The expressions $\sin x$, $1 - \cos x$, $e^x - 1$, $\ln(1+x)$, and $(1+x)^{1/x}$ are strong signals for these standard patterns.

Formula 2.14. Elementary Function Reference Table
The following domains, ranges, symmetries, and periods are standard.

Function	Domain	Range	Symmetry	Period
e^x	$\mathbb{R}$	$(0, \infty)$	neither	none
$\ln x$	$(0, \infty)$	$\mathbb{R}$	neither	none
$\sin x$	$\mathbb{R}$	$[-1, 1]$	odd	2π
$\cos x$	$\mathbb{R}$	$[-1, 1]$	even	2π
$\tan x$	$\mathbb{R} \setminus \{\frac{\pi}{2} + k\pi \mid k \in \mathbb{Z}\}$	$\mathbb{R}$	odd	π
$\sec x$	$\mathbb{R} \setminus \{\frac{\pi}{2} + k\pi \mid k \in \mathbb{Z}\}$	$(-\infty, -1] \cup [1, \infty)$	even	2π
$\csc x$	$\mathbb{R} \setminus \{k\pi \mid k \in \mathbb{Z}\}$	$(-\infty, -1] \cup [1, \infty)$	odd	2π
$\cot x$	$\mathbb{R} \setminus \{k\pi \mid k \in \mathbb{Z}\}$	$\mathbb{R}$	odd	π
$\arcsin x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	odd	none
$\arccos x$	$[-1, 1]$	$[0, \pi]$	neither	none
$\arctan x$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$	odd	none
$\operatorname{arcsec} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] \setminus \{\frac{\pi}{2}\}$	neither	none
$\operatorname{arccsc} x$	$(-\infty, -1] \cup [1, \infty)$	$[-\frac{\pi}{2}, \frac{\pi}{2}] \setminus \{0\}$	odd	none
$\operatorname{arccot} x$	$\mathbb{R}$	$(0, \pi)$	neither	none
$\sinh x$	$\mathbb{R}$	$\mathbb{R}$	odd	none
$\cosh x$	$\mathbb{R}$	$[1, \infty)$	even	none
$\tanh x$	$\mathbb{R}$	$(-1, 1)$	odd	none
$\operatorname{sech} x$	$\mathbb{R}$	$(0, 1]$	even	none
$\operatorname{csch} x$	$\mathbb{R} \setminus \{0\}$	$\mathbb{R} \setminus \{0\}$	odd	none
$\coth x$	$\mathbb{R} \setminus \{0\}$	$(-\infty, -1) \cup (1, \infty)$	odd	none
$\operatorname{arcsinh} x$	$\mathbb{R}$	$\mathbb{R}$	odd	none
$\operatorname{arccosh} x$	$[1, \infty)$	$[0, \infty)$	neither	none
$\operatorname{arctanh} x$	$(-1, 1)$	$\mathbb{R}$	odd	none

Formula 2.15. Exact Trigonometric Values Table
For the standard first-quadrant angles,

x	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$
$\tan x$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$
$\sec x$	1	$\frac{2\sqrt{3}}{3}$	$\sqrt{2}$	2
$\csc x$	undefined	2	$\sqrt{2}$	$\frac{2\sqrt{3}}{3}$
$\cot x$	undefined	$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$

Moreover,

$$\sin\frac{\pi}{2} = 1, \qquad \cos\frac{\pi}{2} = 0,$$

and the remaining exact values follow from parity, periodicity, and reference-angle identities.

Formula 2.16. Trigonometric Identity and Synthesis Table
The principal trigonometric identities are

Type	Identities
Reciprocal	$\sec x = \dfrac{1}{\cos x}, \quad \csc x = \dfrac{1}{\sin x}, \quad \cot x = \dfrac{1}{\tan x}$
Quotient	$\tan x = \dfrac{\sin x}{\cos x}, \quad \cot x = \dfrac{\cos x}{\sin x}$
Pythagorean	$\sin^2 x + \cos^2 x = 1$, $1 + \tan^2 x = \sec^2 x$, $1 + \cot^2 x = \csc^2 x$
Angle addition	$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
Angle addition	$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
Angle addition	$\tan(x \pm y) = \dfrac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$
Double angle	$\sin 2x = 2 \sin x \cos x$
Double angle	$\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$
Double angle	$\tan 2x = \dfrac{2 \tan x}{1 - \tan^2 x}$
Half angle	$\sin^2 \dfrac{x}{2} = \dfrac{1 - \cos x}{2}, \quad \cos^2 \dfrac{x}{2} = \dfrac{1 + \cos x}{2}$
Product-to-sum	$2 \sin x \sin y = \cos(x - y) - \cos(x + y)$
Product-to-sum	$2 \cos x \cos y = \cos(x - y) + \cos(x + y)$
Product-to-sum	$2 \sin x \cos y = \sin(x + y) + \sin(x - y)$
Sum-to-product	$\sin x + \sin y = 2 \sin \dfrac{x + y}{2} \cos \dfrac{x - y}{2}$
Sum-to-product	$\cos x + \cos y = 2 \cos \dfrac{x + y}{2} \cos \dfrac{x - y}{2}$
Sum-to-product	$\sin x - \sin y = 2 \cos \dfrac{x + y}{2} \sin \dfrac{x - y}{2}$
Sum-to-product	$\cos x - \cos y = -2 \sin \dfrac{x + y}{2} \sin \dfrac{x - y}{2}$
Hyperbolic	$\cosh^2 x - \sinh^2 x = 1$, $1 - \tanh^2 x = \operatorname{sech}^2 x$
Hyperbolic	$\tanh x = \dfrac{\sinh x}{\cosh x}, \quad \coth x = \dfrac{\cosh x}{\sinh x}$
Inverse hyperbolic	$\operatorname{arcsinh} x = \ln\left(x + \sqrt{x^2 + 1}\right)$
Inverse hyperbolic	$\operatorname{arccosh} x = \ln\left(x + \sqrt{x^2 - 1}\right) \quad (x \geq 1)$
Inverse hyperbolic	$\operatorname{arctanh} x = \dfrac{1}{2} \ln\left(\dfrac{1 + x}{1 - x}\right) \quad (\lvert x \rvert < 1)$

Type	Identities
Synthesis	$a\cos x + b\sin x = R\cos(x-\alpha)$, $R = \sqrt{a^2+b^2}$, $\cos\alpha = \frac{a}{R}$, $\sin\alpha = \frac{b}{R}$ $(R > 0)$

Theorem 2.17. L'Hôpital's Rule
Let f and g be differentiable on a deleted neighborhood of a, and suppose that $g'(x) \neq 0$ there. Assume either

$$f(x) \to 0, \qquad g(x) \to 0,$$

or

$$|f(x)| \to \infty, \qquad |g(x)| \to \infty,$$

as $x \to a$. If

$$\lim_{x\to a} \frac{f'(x)}{g'(x)} = L$$

exists in the extended real numbers, then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = L.$$

Strategy. Apply L'Hôpital's Rule only after identifying a $0/0$ or ∞/∞ form. Rewrite $0 \cdot \infty$ and $\infty - \infty$ as a quotient first. For 0^0, 1^∞, or ∞^0, take logarithms before differentiating.

Formula 2.18. Derivative and Antiderivative Table
The following table records the basic one-variable forms. Affine changes of variable are handled by the chain rule and substitution. The constant of integration is omitted from the right column.

$f(x)$	$f'(x)$	$\int f(x)\,\mathrm{d}x$
c	0	cx
x	1	$\frac{x^2}{2}$
x^a	ax^{a-1}	$\frac{x^{a+1}}{a+1}$ $(a \neq -1)$
$\frac{1}{x}$	$-\frac{1}{x^2}$	$\ln\lvert x\rvert$
e^x	e^x	e^x
a^x	$a^x \ln a$	$\frac{a^x}{\ln a}$ $(a > 0,\ a \neq 1)$
$\ln x$	$\frac{1}{x}$	$x\ln x - x$
$\lg x$	$\frac{1}{x\ln 10}$	$\frac{x\ln x - x}{\ln 10}$
$\operatorname{lb} x$	$\frac{1}{x\ln 2}$	$\frac{x\ln x - x}{\ln 2}$
$\log_a x$	$\frac{1}{x\ln a}$	$\frac{x\ln x - x}{\ln a}$ $(a > 0,\ a \neq 1)$
$x\ln x$	$\ln x + 1$	$\frac{x^2}{2}\ln x - \frac{x^2}{4}$
$\frac{\ln x}{x}$	$\frac{1-\ln x}{x^2}$	$\frac{1}{2}(\ln x)^2$
$\frac{1}{x\ln x}$	$-\frac{\ln x + 1}{x^2(\ln x)^2}$	$\ln\lvert\ln x\rvert$
$\sin x$	$\cos x$	$-\cos x$
$\cos x$	$-\sin x$	$\sin x$
$\tan x$	$\sec^2 x$	$-\ln\lvert\cos x\rvert$

$f(x)$	$f'(x)$	$\int f(x)\,\mathrm{d}x$
$\cot x$	$-\csc^2 x$	$\ln\lvert\sin x\rvert$
$\sec x$	$\sec x\tan x$	$\ln\lvert\sec x+\tan x\rvert$
$\csc x$	$-\csc x\cot x$	$\ln\lvert\csc x-\cot x\rvert$
$\sec^2 x$	$2\sec^2 x\tan x$	$\tan x$
$\csc^2 x$	$-2\csc^2 x\cot x$	$-\cot x$
$\sec x\tan x$	$\sec x\tan^2 x+\sec^3 x$	$\sec x$
$\csc x\cot x$	$-\csc x\cot^2 x-\csc^3 x$	$-\csc x$
$\arcsin x$	$\dfrac{1}{\sqrt{1-x^2}}$	$x\arcsin x+\sqrt{1-x^2}$
$\arccos x$	$-\dfrac{1}{\sqrt{1-x^2}}$	$x\arccos x-\sqrt{1-x^2}$
$\arctan x$	$\dfrac{1}{1+x^2}$	$x\arctan x-\dfrac{1}{2}\ln(1+x^2)$
$\sinh x$	$\cosh x$	$\cosh x$
$\cosh x$	$\sinh x$	$\sinh x$
$\tanh x$	$\operatorname{sech}^2 x$	$\ln(\cosh x)$
$\coth x$	$-\operatorname{csch}^2 x$	$\ln\lvert\sinh x\rvert$
$\operatorname{sech} x$	$-\operatorname{sech} x\tanh x$	$\arctan(\sinh x)$
$\operatorname{csch} x$	$-\operatorname{csch} x\coth x$	$\ln\left\lvert\tanh\frac{x}{2}\right\rvert$
$\operatorname{arcsinh} x$	$\dfrac{1}{\sqrt{x^2+1}}$	$x\operatorname{arcsinh} x-\sqrt{x^2+1}$
$\operatorname{arccosh} x$	$\dfrac{1}{\sqrt{x-1}\sqrt{x+1}}$	$x\operatorname{arccosh} x-\sqrt{x^2-1}$
$\operatorname{arctanh} x$	$\dfrac{1}{1-x^2}$	$x\operatorname{arctanh} x+\dfrac{1}{2}\ln(1-x^2)$
$\dfrac{1}{x^2+a^2}$	$-\dfrac{2x}{(x^2+a^2)^2}$	$\dfrac{1}{a}\arctan\dfrac{x}{a}\quad(a>0)$
$\dfrac{1}{x^2-a^2}$	$-\dfrac{2x}{(x^2-a^2)^2}$	$\dfrac{1}{2a}\ln\left\lvert\dfrac{x-a}{x+a}\right\rvert\quad(a>0)$
$\dfrac{1}{a^2-x^2}$	$\dfrac{2x}{(a^2-x^2)^2}$	$\dfrac{1}{2a}\ln\left\lvert\dfrac{a+x}{a-x}\right\rvert\quad(a>0)$
$\dfrac{1}{\sqrt{a^2-x^2}}$	$\dfrac{x}{(a^2-x^2)^{3/2}}$	$\arcsin\dfrac{x}{a}\quad(a>0)$
$\dfrac{1}{\sqrt{x^2+a^2}}$	$-\dfrac{x}{(x^2+a^2)^{3/2}}$	$\ln\left\lvert x+\sqrt{x^2+a^2}\right\rvert$
$\dfrac{1}{\sqrt{x^2-a^2}}$	$-\dfrac{x}{(x^2-a^2)^{3/2}}$	$\ln\left\lvert x+\sqrt{x^2-a^2}\right\rvert\quad(a>0)$
$\dfrac{x}{\sqrt{x^2+a^2}}$	$\dfrac{a^2}{(x^2+a^2)^{3/2}}$	$\sqrt{x^2+a^2}$
$\dfrac{x}{\sqrt{a^2-x^2}}$	$\dfrac{a^2}{(a^2-x^2)^{3/2}}$	$-\sqrt{a^2-x^2}$
$\sqrt{a^2-x^2}$	$-\dfrac{x}{\sqrt{a^2-x^2}}$	$\dfrac{x}{2}\sqrt{a^2-x^2}+\dfrac{a^2}{2}\arcsin\dfrac{x}{a}$
$\sqrt{x^2+a^2}$	$\dfrac{x}{\sqrt{x^2+a^2}}$	$\dfrac{x}{2}\sqrt{x^2+a^2}+\dfrac{a^2}{2}\ln\left\lvert x+\sqrt{x^2+a^2}\right\rvert$
$\sqrt{x^2-a^2}$	$\dfrac{x}{\sqrt{x^2-a^2}}$	$\dfrac{x}{2}\sqrt{x^2-a^2}-\dfrac{a^2}{2}\ln\left\lvert x+\sqrt{x^2-a^2}\right\rvert$

Tactic. An antiderivative is often exposed by a simple substitution, symmetry, completing the square, or integration by parts. Try those before expanding into a long calculation.

Theorem 2.19. Intermediate Value Theorem
Let f be continuous on $[a,b]$. If N is any number between $f(a)$ and $f(b)$, then there exists

$c \in [a, b]$ *s.t.*

$$f(c) = N.$$

Strategy. The Intermediate Value Theorem is the theorem behind many existence arguments. To show that an equation has a solution, define a continuous function and show that its values cross the desired level.

Theorem 2.20. Extreme Value Theorem; Weierstrass Extreme Value Theorem
Let f be continuous on $[a, b]$. Then f attains both a maximum and a minimum on $[a, b]$.

Remark. The closed interval condition is important. A continuous function on an open interval need not attain a maximum or a minimum.

Theorem 2.21. Rolle's Theorem
Let f be continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$, then there exists $c \in (a, b)$ s.t.

$$f'(c) = 0.$$

Theorem 2.22. Mean Value Theorem; Lagrange Mean Value Theorem
Let f be continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Idea. The Mean Value Theorem says that somewhere on the interval, the instantaneous rate of change equals the average rate of change.

Theorem 2.23. Cauchy's Mean Value Theorem; Extended Mean Value Theorem
Let f and g be continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ s.t.

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

If $g'(x) \neq 0$ for every $x \in (a, b)$, then $g(b) \neq g(a)$ and the result may be written as

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Theorem 2.24. Derivative of an Inverse Function
Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be one-to-one, and suppose that f is differentiable at an interior point $x_0 \in I$ with $f'(x_0) \neq 0$. If the inverse $g = f^{-1}$ is continuous at $y_0 = f(x_0)$, then g is differentiable at y_0 and

$$g'(y_0) = \frac{1}{f'(x_0)}.$$

In particular, the continuity hypothesis is automatic when f is continuous and strictly monotone on an interval.

Definition 2.25. Riemann Integral
Let $f : [a, b] \to \mathbb{R}$ be bounded. For a partition

$$P : \quad a = x_0 < x_1 < \cdots < x_n = b,$$

choose $\xi_i \in [x_{i-1}, x_i]$ and form the Riemann sum

$$\sum_{i=1}^{n} f(\xi_i)(x_i - x_{i-1}).$$

If these sums approach the same limit I as the mesh $\max_i(x_i - x_{i-1}) \to 0$, independently of the sample points, then f is Riemann integrable and

$$I = \int_a^b f(x)\,\mathrm{d}x.$$

Every continuous function on a closed bounded interval is Riemann integrable.

Theorem 2.26. Fundamental Theorem of Calculus
If f is continuous on $[a, b]$ and

$$F(x) = \int_a^x f(t)\,\mathrm{d}t,$$

then $F'(x) = f(x)$. Conversely, if $F' = f$ on $[a, b]$, then

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Theorem 2.27. Taylor's Theorem
Let f be $(n+1)$ times differentiable on an open interval containing the closed interval with endpoints a and x. Then there exists a point ξ strictly between a and x s.t.

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}.$$

The final term is the Lagrange remainder.

Corollary 2.28. Taylor Error Bound
If

$$|f^{(n+1)}(t)| \leq M$$

for every t between a and x, then the Taylor remainder satisfies

$$|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}.$$

Strategy. Taylor expansion is one of the fastest tools for limits, approximations, and hidden series evaluations. Expressions involving e^x, $\sin x$, $\cos x$, $\ln(1+x)$, or $\arctan x$ often invite Taylor expansion.

Formula 2.29. Maclaurin Series
The most important Maclaurin series are

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \qquad \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!},$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \qquad (|x| < 1),$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} \qquad (-1 < x \leq 1),$$

and

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \qquad (|x| \leq 1).$$

Also,

$$-\ln(1-x) = \sum_{n=1}^{\infty} \frac{x^n}{n} \qquad (-1 \leq x < 1),$$

$$\sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}, \qquad \cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!},$$

and

$$\operatorname{arctanh} x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{2n+1} \qquad (|x| < 1).$$

Formula 2.30. Binomial Series
For any real or complex number β,

$$(1+x)^\beta = \sum_{n=0}^{\infty} \binom{\beta}{n} x^n \qquad (|x|<1),$$

where

$$\binom{\beta}{n} = \frac{\beta(\beta-1)\cdots(\beta-n+1)}{n!}.$$

Formula 2.31. Chebyshev Polynomials
The Chebyshev polynomials of the first kind are characterized by

$$T_n(\cos\theta) = \cos(n\theta).$$

They satisfy

$$T_0(x) = 1, \qquad T_1(x) = x, \qquad T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).$$

Hence multiple-angle trigonometric expressions can be converted into polynomial calculations almost immediately; for example,

$$\cos(3\theta) = 4\cos^3\theta - 3\cos\theta.$$

Formula 2.32. Classical Series Values
The following standard series values are useful in computation:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90},$$

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \ln 2, \qquad \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \frac{\pi}{4}.$$

Formula 2.33. Geometric-Series Differentiation
Starting from

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \qquad (|x|<1),$$

term-by-term differentiation gives

$$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} nx^{n-1} \qquad (|x|<1),$$

and hence

$$\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2} \qquad (|x|<1).$$

Formula 2.34. Telescoping Series
If

$$a_n = b_n - b_{n+1},$$

then the partial sums satisfy

$$\sum_{n=1}^{N} a_n = b_1 - b_{N+1}.$$

Hence, if $b_n \to L$, then

$$\sum_{n=1}^{\infty} a_n = b_1 - L.$$

Formula 2.35. Abel's Summation Formula; Summation by Parts
Let
$$A_k = \sum_{j=m}^{k} a_j.$$
Then
$$\sum_{k=m}^{n} a_k b_k = A_n b_n + \sum_{k=m}^{n-1} A_k (b_k - b_{k+1}).$$
This is the discrete analogue of integration by parts and is useful when one factor has simple partial sums while the other changes slowly.

Formula 2.36. Euler–Maclaurin Summation Formula
Let $p \geq 1$ and let $f \in C^{2p}([m, n])$, where $m < n$ are integers. Then
$$\sum_{k=m}^{n} f(k) = \int_m^n f(x)\,\mathrm{d}x + \frac{f(m) + f(n)}{2} + \sum_{r=1}^{p-1} \frac{B_{2r}}{(2r)!} \Big(f^{(2r-1)}(n) - f^{(2r-1)}(m) \Big) + R_p,$$
where B_{2r} are the Bernoulli numbers and
$$|R_p| \leq \frac{2\zeta(2p)}{(2\pi)^{2p}} \int_m^n |f^{(2p)}(x)|\,\mathrm{d}x.$$
The case $p = 1$ gives the trapezoidal approximation together with a rigorous error bound.

Formula 2.37. Asymptotic Growth Hierarchy
For fixed $p, q > 0$ and $c > 1$, as $n \to \infty$,
$$(\ln n)^p = o(n^q), \qquad n^q = o(c^n), \qquad c^n = o(n!), \qquad n! = o(n^n).$$
Strategy. For fast series calculations, first try to reduce the expression to a geometric series, a differentiated geometric series, a Maclaurin series, or a known special value such as $\pi^2/6$, $\pi^4/90$, $\ln 2$, or $\pi/4$.

Formula 2.38. Integration by Parts
If u and v are differentiable functions, then
$$\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u.$$
Equivalently,
$$\int_a^b u(x)v'(x)\,\mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x)\,\mathrm{d}x.$$

Theorem 2.39. Leibniz Integral Rule; Differentiation under the Integral Sign
Let I be an open interval, let $a, b : I \to \mathbb{R}$ be continuously differentiable, and suppose that f and $\partial f/\partial t$ are continuous on
$$\{(x, t) \mid t \in I,\ a(t) \leq x \leq b(t)\}.$$
Then
$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{a(t)}^{b(t)} f(x, t)\,\mathrm{d}x = f(b(t), t)b'(t) - f(a(t), t)a'(t) + \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x, t)\,\mathrm{d}x.$$
In particular, if the endpoints are constant, only the integral of the partial derivative remains.

Tactic. Introducing a parameter and differentiating under the integral sign is often called Feynman's trick. It is especially effective when direct integration is difficult but the parameter derivative is elementary.

Formula 2.40. Frullani's Integral
Let $f : [0, \infty) \to \mathbb{R}$ be continuous, suppose that the finite limits $f(0)$ and $f(\infty) := \lim_{x\to\infty} f(x)$ exist, and assume that the improper integral below converges. For $a, b > 0$,

$$\int_0^\infty \frac{f(ax) - f(bx)}{x}\,\mathrm{d}x = (f(0) - f(\infty)) \ln \frac{b}{a}.$$

A standard sufficient hypothesis is that f is continuously differentiable and $f' \in L^1(0, \infty)$.

Formula 2.41. Arc Length Formula
The length of the graph $y = f(x)$ on $[a, b]$ is

$$L = \int_a^b \sqrt{1 + (f'(x))^2}\,\mathrm{d}x.$$

More generally, if a curve is parametrized by $\mathbf{r}(t) = (x(t), y(t), z(t))$, then

$$L = \int_a^b \|\mathbf{r}'(t)\|\,\mathrm{d}t.$$

Formula 2.42. Viète's Infinite Product
For every real x,

$$\frac{\sin x}{x} = \prod_{k=1}^{\infty} \cos\left(\frac{x}{2^k}\right),$$

where the value at $x = 0$ is interpreted by continuity. Setting $x = \pi/2$ gives Viète's classical product for $2/\pi$.

Idea. Viète's formula is a product formula for π. The key identity is the double-angle formula

$$\sin(2x) = 2 \sin x \cos x.$$

Formula 2.43. Euler's Formula
For every real number x,

$$e^{ix} = \cos x + i \sin x.$$

In particular,

$$e^{i\pi} + 1 = 0.$$

Remark. The identity $e^{i\pi} + 1 = 0$ is called Euler's identity. It connects the constants e, i, π, 1, and 0 in a single formula.

Formula 2.44. Gaussian Integral
The Gaussian integral is

$$\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi}.$$

Equivalently,

$$\int_0^\infty e^{-x^2}\,\mathrm{d}x = \frac{\sqrt{\pi}}{2}.$$

Remark. The Gaussian integral is one of the most famous integrals in calculus. It is closely connected to the normal distribution in probability.

Formula 2.45. Fresnel Integrals
The Fresnel integrals satisfy

$$\int_0^\infty \cos(x^2)\,\mathrm{d}x = \int_0^\infty \sin(x^2)\,\mathrm{d}x = \sqrt{\frac{\pi}{8}}.$$

Remark. Fresnel integrals appear in wave optics, diffraction, and the Cornu spiral. They are memorable examples of non-elementary definite integrals with clean values.

Formula 2.46. Sine Integral; Dirichlet Integral
The sine integral is

$$\mathrm{Si}(x) = \int_0^x \frac{\sin t}{t}\,\mathrm{d}t.$$

The classical Dirichlet integral is

$$\int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}.$$

Formula 2.47. Arctangent Integral
For $a > 0$,

$$\int_{-\infty}^{\infty} \frac{1}{x^2 + a^2}\,\mathrm{d}x = \frac{\pi}{a}, \qquad \int_0^{\infty} \frac{1}{x^2 + a^2}\,\mathrm{d}x = \frac{\pi}{2a}.$$

More generally,

$$\int_{-\infty}^{\infty} \frac{c}{(x-b)^2 + a^2}\,\mathrm{d}x = \frac{c\pi}{a}.$$

Definition 2.48. Gamma Function
For $\mathrm{Re}(z) > 0$, the Gamma function is defined by

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\,\mathrm{d}t.$$

It extends meromorphically to $\mathbb{C}$, with simple poles at the nonpositive integers.

Proposition 2.49. Gamma Function and Factorials
The Gamma function satisfies

$$\Gamma(z+1) = z\Gamma(z)$$

wherever both sides are defined. In particular, for every positive integer n,

$$\Gamma(n) = (n-1)!.$$

Also,

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Definition 2.50. Beta Function; Euler Beta Integral
For $x, y > 0$, the Beta function is defined by

$$B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\,\mathrm{d}t.$$

Formula 2.51. Beta–Gamma Identity
The Beta and Gamma functions are related by

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}.$$

In particular, for nonnegative integers m, n,

$$\int_0^1 x^m (1-x)^n\,\mathrm{d}x = B(m+1, n+1) = \frac{m!n!}{(m+n+1)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}.$$

Formula 2.52. Gamma-Type Exponential Integrals
For $a > 0$ and every nonnegative integer n,

$$\int_0^\infty x^n e^{-ax}\,\mathrm{d}x = \frac{n!}{a^{n+1}}.$$

More generally, for $\mathrm{Re}(s) > 0$,

$$\int_0^\infty x^{s-1} e^{-ax}\,\mathrm{d}x = \frac{\Gamma(s)}{a^s}.$$

Formula 2.53. Damped Sine and Cosine Integrals
For $a > 0$,

$$\int_0^\infty e^{-ax}\cos bx\,\mathrm{d}x = \frac{a}{a^2+b^2}, \qquad \int_0^\infty e^{-ax}\sin bx\,\mathrm{d}x = \frac{b}{a^2+b^2}.$$

Strategy. Many fast integral problems reduce to this formula after completing a square or translating the variable. A shift such as $u = x - b$ does not change the infinite limits.

Formula 2.54. Symmetry Trick for Definite Integrals
If f is integrable on $[a, b]$, then

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^b f(a+b-x)\,\mathrm{d}x.$$

Hence, if

$$f(x) + f(a+b-x) = C$$

is constant, then

$$\int_a^b f(x)\,\mathrm{d}x = \frac{C(b-a)}{2}.$$

Strategy. This is one of the most useful Integration-Bee-style tricks. Typical signals are intervals such as $[0, \pi/2]$, expressions involving $\tan x$ and $\cot x$, or a denominator that becomes complementary after the substitution $x \mapsto a + b - x$.

Formula 2.55. Parity and Periodicity Shortcuts for Integrals
If f is integrable on $[-a, a]$, then

$$f(-x) = -f(x) \quad \Longrightarrow \quad \int_{-a}^a f(x)\,\mathrm{d}x = 0,$$

and

$$f(-x) = f(x) \quad \Longrightarrow \quad \int_{-a}^a f(x)\,\mathrm{d}x = 2\int_0^a f(x)\,\mathrm{d}x.$$

If f has period T, then for every c,

$$\int_c^{c+T} f(x)\,\mathrm{d}x = \int_0^T f(x)\,\mathrm{d}x.$$

Formula 2.56. Reciprocal-Substitution Identity
If the improper integrals converge, the substitution $x = 1/t$ gives

$$\int_0^\infty f(x)\,\mathrm{d}x = \int_0^\infty \frac{f(1/x)}{x^2}\,\mathrm{d}x.$$

Therefore,

$$\int_0^\infty f(x)\,\mathrm{d}x = \frac{1}{2}\int_0^\infty \left(f(x) + \frac{f(1/x)}{x^2}\right)\mathrm{d}x.$$

This often reveals a cancellation or symmetry hidden in an integral on $(0, \infty)$.

Formula 2.57. Weierstrass Substitution; Tangent Half-Angle Substitution
The substitution
$$t = \tan \frac{x}{2}$$
converts rational expressions in $\sin x$ and $\cos x$ into rational functions of t:
$$\sin x = \frac{2t}{1+t^2}, \qquad \cos x = \frac{1-t^2}{1+t^2}, \qquad \mathrm{d}x = \frac{2\,\mathrm{d}t}{1+t^2}.$$

Algorithm 2.58. Fast Integral Recognition Checklist
Before expanding or integrating directly, check in this order:

1. parity, periodicity, and interval symmetry;
2. the substitutions $x \mapsto a - x$, $x \mapsto 1/x$, or $x \mapsto a/x$;
3. a hidden derivative for substitution or integration by parts;
4. a parameter that permits differentiation under the integral sign;
5. Beta–Gamma, Gaussian, Wallis, or a standard trigonometric integral;
6. the tangent half-angle substitution for rational trigonometric expressions.

Formula 2.59. Wallis Product
The Wallis product is
$$\frac{\pi}{2} = \prod_{n=1}^{\infty} \frac{(2n)(2n)}{(2n-1)(2n+1)} = \frac{2}{1} \cdot \frac{2}{3} \cdot \frac{4}{3} \cdot \frac{4}{5} \cdot \frac{6}{5} \cdot \frac{6}{7} \cdots .$$

Formula 2.60. Euler's Reflection Formula
For $z \in \mathbb{C} \setminus \mathbb{Z}$,
$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)}.$$

Formula 2.61. Euler Beta Integral on the Positive Half-Line
If $p > 1$, then
$$\int_0^{\infty} \frac{\mathrm{d}x}{1+x^p} = \frac{1}{p} B\left(\frac{1}{p}, 1 - \frac{1}{p}\right) = \frac{\pi}{p} \csc\left(\frac{\pi}{p}\right).$$
The substitution
$$t = \frac{x^p}{1+x^p}$$
converts the improper integral into an Euler beta integral.

Formula 2.62. Stirling's Formula
As $n \to \infty$,
$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n .$$

Remark. Stirling's formula is the standard approximation for large factorials. It often appears in counting, probability, and asymptotic estimation.

Formula 2.63. Trapezoidal Rule
The trapezoidal approximation is
$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{b-a}{2}\big(f(a) + f(b)\big).$$

It has degree of precision 1.

Formula 2.64. Simpson's Rule
Simpson's rule is

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{b-a}{6}\left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right).$$

It has degree of precision 3.

Remark. Simpson's rule uses quadratic interpolation, but the resulting formula is exact for every polynomial of degree at most 3.

Theorem 2.65. p-Series Test
The series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

converges if and only if $p > 1$.

Theorem 2.66. Direct Comparison Test
Let $0 \leq a_n \leq b_n$ *for all sufficiently large* n. *If* $\sum b_n$ *converges, then* $\sum a_n$ *converges. If* $\sum a_n$ *diverges, then* $\sum b_n$ *diverges.*

Proposition 2.67. Absolute Convergence Implies Convergence
If

$$\sum_{n=1}^{\infty} |a_n|$$

converges, then $\sum_{n=1}^{\infty} a_n$ *converges.*

Theorem 2.68. Integral Test
Suppose f *is positive, continuous, and decreasing on* $[1, \infty)$, *and let* $a_n = f(n)$. *Then*

$$\sum_{n=1}^{\infty} a_n$$

and

$$\int_1^{\infty} f(x)\,\mathrm{d}x$$

either both converge or both diverge.

Theorem 2.69. Limit Comparison Test
Let $a_n, b_n > 0$. *If*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

for some finite number $c > 0$, *then*

$$\sum_{n=1}^{\infty} a_n \qquad \textit{and} \qquad \sum_{n=1}^{\infty} b_n$$

converge or diverge together.

Theorem 2.70. Ratio Test; d'Alembert's Ratio Test
For a series $\sum a_n$ *with* $a_n \neq 0$ *eventually, set*

$$L = \limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|.$$

If $L < 1$, *then* $\sum a_n$ *converges absolutely. If*

$$\liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1,$$

then $a_n \not\to 0$ and the series diverges. In the common case in which the ordinary limit

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L$$

exists, this reduces to the familiar rule: $L < 1$ gives absolute convergence, $L > 1$ gives divergence, and $L = 1$ is inconclusive.

Theorem 2.71. Root Test; Cauchy's Root Test
For a series $\sum a_n$, set

$$L = \limsup_{n\to\infty} \sqrt[n]{|a_n|}.$$

If $L < 1$, then $\sum a_n$ converges absolutely. If $L > 1$, then the series diverges. If $L = 1$, the test is inconclusive. When the ordinary limit $\lim\limits_{n\to\infty} \sqrt[n]{|a_n|}$ exists, it equals the displayed lim sup, *giving the usual limit form of the test.*

Theorem 2.72. Cauchy–Hadamard Theorem
For a power series

$$\sum_{n=0}^{\infty} a_n(x-a)^n,$$

the radius of convergence R is given by

$$\frac{1}{R} = \limsup_{n\to\infty} \sqrt[n]{|a_n|}.$$

Equivalently, the series converges absolutely for $|x-a| < R$ and diverges for $|x-a| > R$.

Remark. This is the standard theorem behind the phrase "radius of convergence" for a power series. It is also the source of the name Cauchy–Hadamard.

Theorem 2.73. Termwise Differentiation and Integration of Power Series
Suppose

$$f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n$$

has radius of convergence $R > 0$. For $|x-a| < R$,

$$f'(x) = \sum_{n=1}^{\infty} na_n(x-a)^{n-1},$$

and

$$\int_a^x f(t)\,\mathrm{d}t = \sum_{n=0}^{\infty} \frac{a_n}{n+1}(x-a)^{n+1}.$$

The differentiated and integrated series have the same radius of convergence R. Convergence at the endpoints must be checked separately.

Theorem 2.74. Alternating Series Test; Leibniz Test
If

$$a_1 \ge a_2 \ge a_3 \ge \cdots \ge 0$$

and $a_n \to 0$, then the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n$$

converges.

Formula 2.75. Alternating Series Error Estimate
Under the hypotheses of the Alternating Series Test, if

$$S = \sum_{n=1}^{\infty} (-1)^{n+1} a_n, \qquad S_N = \sum_{n=1}^{N} (-1)^{n+1} a_n,$$

then

$$|S - S_N| \le a_{N+1}.$$

Strategy. Factorials and powers often suggest the Ratio Test, nth powers suggest the Root Test, and alternating signs with decreasing terms suggest the Alternating Series Test.

Theorem 2.76. Divergence Test; Term Test
If the series

$$\sum_{n=1}^{\infty} a_n$$

converges, then

$$\lim_{n\to\infty} a_n = 0.$$

Equivalently, if $\lim_{n\to\infty} a_n$ *does not exist or is not* 0*, then* $\sum_{n=1}^{\infty} a_n$ *diverges.*

Strategy. Before trying a sophisticated convergence test, first check whether the terms go to 0. If the terms do not tend to 0, the series diverges immediately.

Algorithm 2.77. Fast Series Checklist
For a convergence or summation problem:

1. Check the term limit first.
2. Look for telescoping, a geometric series, or a known power series.
3. For positive terms, compare only the dominant asymptotic factors.
4. Use the Ratio Test for factorials and exponentials, and the Root Test for expressions raised to the nth power.
5. For alternating terms, check monotonicity and use the first omitted term to control the error.

Multivariable Calculus

Definition 2.78. Gradient, Divergence, Curl, and Laplacian
For a scalar-valued function $f(x, y, z)$ and a vector field $\mathbf{F} = (P, Q, R)^{\mathsf{T}}$, we write

$$\nabla f = \begin{pmatrix} \partial f/\partial x \\ \partial f/\partial y \\ \partial f/\partial z \end{pmatrix},$$

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z},$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix} = \begin{pmatrix} \partial R/\partial y - \partial Q/\partial z \\ \partial P/\partial z - \partial R/\partial x \\ \partial Q/\partial x - \partial P/\partial y \end{pmatrix},$$

and

$$\Delta f = \nabla^2 f = \nabla \cdot (\nabla f) = f_{xx} + f_{yy} + f_{zz}.$$

Idea. The gradient points in the direction of steepest increase. Divergence measures source or sink behavior. Curl measures local rotation. The Laplacian measures how a value differs from its surrounding average.

Theorem 2.79. Clairaut's Theorem; Mixed Partial Derivative Theorem
If f has continuous second partial derivatives near a point, then the mixed partial derivatives agree at that point:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

Definition 2.80. Differentiability; Total Derivative; Fréchet Derivative
Let $\mathbf{F} : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$, where U is open, and let $\mathbf{a} \in U$. The map $\mathbf{F}$ is differentiable at $\mathbf{a}$ if there exists a linear map $D\mathbf{F}(\mathbf{a}) : \mathbb{R}^n \to \mathbb{R}^m$ s.t.

$$\mathbf{F}(\mathbf{a}+\mathbf{h}) = \mathbf{F}(\mathbf{a}) + D\mathbf{F}(\mathbf{a})\mathbf{h} + \mathbf{r}(\mathbf{h}),$$

where

$$\lim_{\mathbf{h}\to\mathbf{0}} \frac{\|\mathbf{r}(\mathbf{h})\|}{\|\mathbf{h}\|} = 0.$$

Equivalently, in compact symbolic form,

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall \mathbf{h} \in \mathbb{R}^n : \quad \big(\mathbf{a}+\mathbf{h} \in U \wedge 0 < \|\mathbf{h}\| < \delta\big) \Rightarrow \|\mathbf{F}(\mathbf{a}+\mathbf{h}) - \mathbf{F}(\mathbf{a}) - D\mathbf{F}(\mathbf{a})\mathbf{h}\| < \varepsilon\|\mathbf{h}\|.$$

The linear map $D\mathbf{F}(\mathbf{a})$ is the total derivative; in standard coordinates, its matrix is the Jacobian matrix.

Definition 2.81. Directional Derivative
Let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ be differentiable at $\mathbf{a} \in U$, and let $\mathbf{u}$ be a unit vector. The directional derivative of f at $\mathbf{a}$ in the direction $\mathbf{u}$ is

$$D_{\mathbf{u}} f(\mathbf{a}) = \lim_{t\to 0} \frac{f(\mathbf{a}+t\mathbf{u}) - f(\mathbf{a})}{t} = \nabla f(\mathbf{a}) \cdot \mathbf{u}.$$

Consequently,

$$|D_{\mathbf{u}} f(\mathbf{a})| \le \|\nabla f(\mathbf{a})\|,$$

with maximum increase in the direction of $\nabla f(\mathbf{a})$ when the gradient is nonzero.

Theorem 2.82. Multivariable Chain Rule
Let $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$ and $\mathbf{G} : \mathbb{R}^m \to \mathbb{R}^k$ be differentiable. Then

$$D(\mathbf{G} \circ \mathbf{F})(\mathbf{x}) = D\mathbf{G}(\mathbf{F}(\mathbf{x}))D\mathbf{F}(\mathbf{x}).$$

Theorem 2.83. Fubini's Theorem
Let $R \subseteq \mathbb{R}^2$ be a bounded region of the form

$$R = \{(x, y) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x)\},$$

where g_1 and g_2 are continuous and $g_1 \le g_2$. If f is continuous on R, then

$$\iint_R f(x, y)\, \mathrm{d}A = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, \mathrm{d}y\, \mathrm{d}x.$$

If the same region is also described as

$$R = \{(x, y) \mid c \le y \le d,\ h_1(y) \le x \le h_2(y)\},$$

then the integral may equally be evaluated in the reverse order. More generally, if the underlying measure spaces are σ-finite and the function is integrable on their product, the two iterated integrals agree with the product-space integral.

Definition 2.84. Jacobian
For a map
$$\mathbf{F}(u_1,\ldots,u_n)=(x_1(u_1,\ldots,u_n),\ldots,x_n(u_1,\ldots,u_n)),$$
the Jacobian matrix is
$$D\mathbf{F}(\mathbf{u})=\left(\frac{\partial x_i}{\partial u_j}\right)_{i,j=1}^{n}.$$
Its determinant
$$\mathrm{Jac}(\mathbf{F})=\det(D\mathbf{F})$$
is called the Jacobian determinant.

Theorem 2.85. Change of Variables Formula
Let $U,V\subseteq\mathbb{R}^n$ be open, and let $\mathbf{F}:U\to V$ be a C^1 diffeomorphism. If $D\subseteq U$ is measurable and f is integrable on $\mathbf{F}(D)$, then
$$\int_{\mathbf{F}(D)} f(\mathbf{x})\,\mathrm{d}\mathbf{x}=\int_D f(\mathbf{F}(\mathbf{u}))\,|\det D\mathbf{F}(\mathbf{u})|\,\mathrm{d}\mathbf{u}.$$
The absolute value is essential because the integral measures unsigned volume, independently of orientation.

Remark. The Jacobian determinant is the local area or volume scaling factor of a change of variables.

Formula 2.86. Polar Coordinates
For polar coordinates (r,θ),
$$x=r\cos\theta,\qquad y=r\sin\theta,\qquad r\geq 0,\qquad 0\leq\theta<2\pi.$$
The area element is
$$\mathrm{d}A=r\,\mathrm{d}r\,\mathrm{d}\theta.$$
The area enclosed by a polar curve $r=r(\theta)$ is
$$A=\frac{1}{2}\int_\alpha^\beta r(\theta)^2\,\mathrm{d}\theta.$$
Its arc length is
$$L=\int_\alpha^\beta\sqrt{r(\theta)^2+(r'(\theta))^2}\,\mathrm{d}\theta.$$

Formula 2.87. Cylindrical and Spherical Coordinates
Cylindrical coordinates (r,θ,z) satisfy
$$x=r\cos\theta,\qquad y=r\sin\theta,\qquad z=z,$$
with volume element
$$\mathrm{d}V=r\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}z.$$
Spherical coordinates (ρ,θ,ϕ) satisfy
$$x=\rho\sin\phi\cos\theta,\qquad y=\rho\sin\phi\sin\theta,\qquad z=\rho\cos\phi,$$
where
$$\rho\geq 0,\qquad 0\leq\theta<2\pi,\qquad 0\leq\phi\leq\pi.$$
The volume element is
$$\mathrm{d}V=\rho^2\sin\phi\,\mathrm{d}\rho\,\mathrm{d}\theta\,\mathrm{d}\phi.$$
Here θ is the azimuthal angle measured from the positive x-axis in the xy-plane, and ϕ is the polar angle measured from the positive z-axis.

Formula 2.88. Differential Operators in Standard Coordinates
Let

$$\mathbf{F} = F_r\mathbf{e}_r + F_\theta\mathbf{e}_\theta + F_z\mathbf{e}_z$$

in cylindrical coordinates. Then

$$\nabla f = \frac{\partial f}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{\partial f}{\partial z}\mathbf{e}_z,$$

$$\nabla \cdot \mathbf{F} = \frac{1}{r}\frac{\partial}{\partial r}(rF_r) + \frac{1}{r}\frac{\partial F_\theta}{\partial \theta} + \frac{\partial F_z}{\partial z},$$

and

$$\Delta f = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial f}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2}.$$

In spherical coordinates (ρ, θ, ϕ), where θ is the azimuthal angle and ϕ is the polar angle, write

$$\mathbf{F} = F_\rho\mathbf{e}_\rho + F_\theta\mathbf{e}_\theta + F_\phi\mathbf{e}_\phi.$$

Then

$$\nabla f = \frac{\partial f}{\partial \rho}\mathbf{e}_\rho + \frac{1}{\rho\sin\phi}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{1}{\rho}\frac{\partial f}{\partial \phi}\mathbf{e}_\phi,$$

$$\nabla \cdot \mathbf{F} = \frac{1}{\rho^2}\frac{\partial}{\partial \rho}(\rho^2 F_\rho) + \frac{1}{\rho\sin\phi}\frac{\partial F_\theta}{\partial \theta} + \frac{1}{\rho\sin\phi}\frac{\partial}{\partial \phi}(\sin\phi\, F_\phi),$$

and

$$\Delta f = \frac{1}{\rho^2}\frac{\partial}{\partial \rho}\left(\rho^2\frac{\partial f}{\partial \rho}\right) + \frac{1}{\rho^2\sin^2\phi}\frac{\partial^2 f}{\partial \theta^2} + \frac{1}{\rho^2\sin\phi}\frac{\partial}{\partial \phi}\left(\sin\phi\frac{\partial f}{\partial \phi}\right).$$

Theorem 2.89. Inverse Function Theorem
Let $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable near $\mathbf{a}$. If

$$\det D\mathbf{F}(\mathbf{a}) \neq 0,$$

then $\mathbf{F}$ has a continuously differentiable local inverse near $\mathbf{a}$.

Theorem 2.90. Implicit Function Theorem
Let $F(x, y)$ be continuously differentiable near (a, b), with

$$F(a, b) = 0, \qquad \frac{\partial F}{\partial y}(a, b) \neq 0.$$

Then, near (a, b), the equation $F(x, y) = 0$ determines y as a differentiable function of x, and

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{F_x}{F_y}.$$

Formula 2.91. Second Derivative Test and Hessian
At a critical point (a, b) of a twice differentiable function $f(x, y)$, let

$$\Delta_H = f_{xx}(a, b)f_{yy}(a, b) - f_{xy}(a, b)^2.$$

If $\Delta_H > 0$ and $f_{xx}(a, b) > 0$, the point is a local minimum. If $\Delta_H > 0$ and $f_{xx}(a, b) < 0$, it is a local maximum. If $\Delta_H < 0$, it is a saddle point. If $\Delta_H = 0$, the test is inconclusive.

Theorem 2.92. Lagrange Multipliers
Suppose that f has a local maximum or minimum subject to the constraint

$$g(x_1, \ldots, x_n) = 0$$

at a point where $\nabla g \neq \mathbf{0}$. Then there exists a scalar λ s.t.

$$\nabla f = \lambda \nabla g.$$

Idea. At a constrained optimum, the level surface of f is tangent to the constraint surface. Thus their normal vectors are parallel.

Theorem 2.93. Euler's Homogeneous Function Theorem
Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable and homogeneous of degree k, meaning

$$f(t\mathbf{x}) = t^k f(\mathbf{x})$$

for $t > 0$. Then

$$\mathbf{x} \cdot \nabla f(\mathbf{x}) = k f(\mathbf{x}).$$

In coordinates,

$$\sum_{i=1}^{n} x_i \frac{\partial f}{\partial x_i} = kf.$$

Formula 2.94. Parametrized Curves
Let $\mathbf{r}(t)$ be a twice differentiable curve. Its velocity, speed, and acceleration are

$$\mathbf{v}(t) = \mathbf{r}'(t), \qquad \|\mathbf{v}(t)\| = \|\mathbf{r}'(t)\|, \qquad \mathbf{a}(t) = \mathbf{r}''(t).$$

If $\mathbf{r}'(t_0) \neq \mathbf{0}$, the tangent line at t_0 is

$$\mathbf{r}(t_0) + s\mathbf{r}'(t_0), \qquad s \in \mathbb{R}.$$

The speed is constant if and only if

$$\mathbf{r}'(t) \cdot \mathbf{r}''(t) = 0.$$

Definition 2.95. Scalar Line Integral
If a curve C is parametrized by $\mathbf{r}(t)$ for $a \leq t \leq b$, then

$$\int_C f \, \mathrm{d}s = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| \, \mathrm{d}t.$$

The value is unchanged by any regular reparametrization.

Definition 2.96. Line Integral
If a curve C is parametrized by $\mathbf{r}(t)$ for $a \leq t \leq b$, then the line integral of a vector field $\mathbf{F}$ along C is

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, \mathrm{d}t.$$

Theorem 2.97. Fundamental Theorem for Line Integrals; Gradient Theorem
If $\mathbf{F} = \nabla f$ and C is a smooth curve from $\mathbf{a}$ to $\mathbf{b}$, then

$$\int_C \mathbf{F} \cdot \mathrm{d}\mathbf{r} = f(\mathbf{b}) - f(\mathbf{a}).$$

Hence the integral is path independent.

Proposition 2.98. Conservative-Field Test
Let $U \subseteq \mathbb{R}^n$ be open and simply connected, and let $\mathbf{F} : U \to \mathbb{R}^n$ be continuously differentiable. Then $\mathbf{F}$ is conservative if and only if its Jacobian matrix is symmetric:

$$\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i} \qquad (1 \leq i, j \leq n).$$

In $\mathbb{R}^3$, this is equivalent to $\nabla \times \mathbf{F} = \mathbf{0}$. Simple connectedness cannot in general be omitted.

Theorem 2.99. Green's Theorem
Let $D \subseteq \mathbb{R}^2$ be a bounded region whose positively oriented boundary $C = \partial D$ is piecewise smooth, and let P, Q have continuous first partial derivatives on an open set containing D. Then

$$\oint_C P\,\mathrm{d}x + Q\,\mathrm{d}y = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}A.$$

Idea. Green's Theorem converts circulation along a boundary curve into curl over the region inside. It is the two-dimensional ancestor of Stokes' Theorem.

Definition 2.100. Surface Area, Scalar Surface Integral, and Flux
Let a smooth oriented surface be parametrized by

$$\mathbf{r}(u, v), \qquad (u, v) \in D.$$

Its surface element is

$$\mathrm{d}S = \|\mathbf{r}_u \times \mathbf{r}_v\|\,\mathrm{d}u\,\mathrm{d}v.$$

Hence

$$\iint_S f\,\mathrm{d}S = \iint_D f(\mathbf{r}(u, v))\|\mathbf{r}_u \times \mathbf{r}_v\|\,\mathrm{d}u\,\mathrm{d}v,$$

and the flux of a vector field $\mathbf{F}$ through the chosen orientation is

$$\iint_S \mathbf{F} \cdot \mathbf{n}\,\mathrm{d}S = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v)\,\mathrm{d}u\,\mathrm{d}v.$$

If S is the graph $z = g(x, y)$ with upward orientation, then

$$\mathrm{d}S = \sqrt{1 + g_x^2 + g_y^2}\,\mathrm{d}x\,\mathrm{d}y,$$

and

$$\mathbf{n}\,\mathrm{d}S = (-g_x, -g_y, 1)\,\mathrm{d}x\,\mathrm{d}y.$$

If instead S is a regular level surface $G(x, y, z) = 0$, then a unit normal is

$$\mathbf{n} = \pm\frac{\nabla G}{\|\nabla G\|}.$$

Theorem 2.101. Stokes' Theorem
Let S be an oriented piecewise smooth surface with positively oriented piecewise smooth boundary ∂S, and let $\mathbf{F}$ be continuously differentiable on an open set containing S. Then

$$\oint_{\partial S} \mathbf{F} \cdot \mathrm{d}\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\,\mathrm{d}S.$$

The orientation of ∂S is determined from that of S by the right-hand rule.

Idea. Stokes' Theorem says that circulation around the boundary equals total curl through the surface. It turns a boundary integral into an interior integral.

Theorem 2.102. Divergence Theorem; Gauss's Theorem
Let $V \subseteq \mathbb{R}^3$ be a bounded solid whose boundary $S = \partial V$ is piecewise smooth and oriented outward. If $\mathbf{F}$ is continuously differentiable on an open set containing V, then

$$\iint_S \mathbf{F} \cdot \mathbf{n}\,\mathrm{d}S = \iiint_V \nabla \cdot \mathbf{F}\,\mathrm{d}V.$$

Idea. The Divergence Theorem says that outward flux through the boundary equals total source strength inside the region.

Remark. Green's Theorem, Stokes' Theorem, and the Divergence Theorem share the same philosophy: a boundary integral is equal to an interior integral. This is the geometric heart of vector calculus.

Real Analysis

Theorem 2.103. Least Upper Bound Property; Completeness Axiom of $\mathbb{R}$
Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$. Equivalently, every nonempty subset bounded below has a greatest lower bound. This completeness property distinguishes $\mathbb{R}$ from $\mathbb{Q}$ and underlies the standard convergence theorems of analysis.

Definition 2.104. Metric Space and Open Ball
A metric space is a pair (X, d) in which $d : X \times X \to \mathbb{R}$ satisfies, for all $x, y, z \in X$,

$$d(x, y) \geq 0, \qquad d(x, y) = 0 \Leftrightarrow x = y,$$

$$d(x, y) = d(y, x), \qquad d(x, z) \leq d(x, y) + d(y, z).$$

The open ball of radius $r > 0$ centered at x is

$$B_r(x) = \{y \in X \mid d(x, y) < r\}.$$

Fact 2.105. Common Metrics
For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, each of

$$d_1(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^{n} |x_k - y_k|,$$

$$d_2(\mathbf{x}, \mathbf{y}) = \left(\sum_{k=1}^{n} |x_k - y_k|^2\right)^{1/2}, \qquad d_\infty(\mathbf{x}, \mathbf{y}) = \max_{1\leq k\leq n} |x_k - y_k|$$

is a metric. On any set X, the discrete metric is

$$d_{\text{disc}}(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Definition 2.106. Open and Closed Sets in a Metric Space
A subset $U \subseteq X$ is open if

$$\forall x \in U, \quad \exists r > 0, \qquad B_r(x) \subseteq U.$$

A subset $F \subseteq X$ is closed if $X \setminus F$ is open.

Definition 2.107. Point and Set Taxonomy in a Metric Space
Let $A \subseteq (X, d)$. The standard point types and associated sets are summarized below.

Term	Criterion
Interior point	$x \in A$ and $\exists r > 0$ s.t. $B_r(x) \subseteq A$
Exterior point	$\exists r > 0$ s.t. $B_r(x) \subseteq X \setminus A$
Boundary point	$\forall r > 0$, $B_r(x) \cap A \neq \varnothing$ and $B_r(x) \cap (X \setminus A) \neq \varnothing$
Accumulation point; limit point	$\forall r > 0$, $(B_r(x) \setminus \{x\}) \cap A \neq \varnothing$
Isolated point	$x \in A$ and $\exists r > 0$ s.t. $B_r(x) \cap A = \{x\}$
Interior A°	the set of all interior points of A
Closure $\overline{A}$	A together with all of its accumulation points; equivalently, the smallest closed set containing A
Boundary ∂A	$\overline{A} \setminus A^\circ$
Dense set	$\overline{A} = X$

Term	Criterion
Perfect set	A is closed and has no isolated points; equivalently, every point of A is an accumulation point of A
Bounded set	$A \subseteq B_R(x_0)$ for some $x_0 \in X$ and $R > 0$
Compact set	every open cover of A has a finite subcover

In a metric space, A is open iff every point of A is interior, and A is closed iff it contains all of its accumulation points. In $\mathbb{R}^n$, compactness is equivalent to being closed and bounded by the Heine–Borel Theorem; this equivalence does not hold in arbitrary metric spaces.

Fact 2.108. Standard Subsets of the Real Line
In the usual topology on $\mathbb{R}$:

- $\mathbb{Z}$ is closed but not open;
- $\mathbb{Q}$ is countable and is neither open nor closed;
- the Cantor set is nonempty, compact, perfect, uncountable, and contains no nondegenerate interval.

These examples are standard tests of the distinctions among countability, openness, closedness, and perfectness.

Definition 2.109. Limit of a Map Between Metric Spaces
Let (X, d_X) and (Y, d_Y) be metric spaces, let $D \subseteq X$, let a be an accumulation point of D, and let $f : D \to Y$. The statement

$$\lim_{x\to a} f(x) = L$$

means

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in D, \qquad 0 < d_X(x, a) < \delta \Rightarrow d_Y\big(f(x), L\big) < \varepsilon.$$

Definition 2.110. Convergence of a Sequence
A sequence (x_n) in a metric space (X, d) converges to $x \in X$, written $x_n \to x$, if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \qquad d(x_n, x) < \varepsilon.$$

Definition 2.111. Cauchy Sequence and Complete Metric Space
A sequence (x_n) in (X, d) is Cauchy if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m, n \geq N, \qquad d(x_m, x_n) < \varepsilon.$$

The metric space is complete if every Cauchy sequence in X converges to a point of X.

Definition 2.112. Continuity in Metric Spaces
Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f : X \to Y$ is continuous at $x_0 \in X$ if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x \in X, \qquad d_X(x, x_0) < \delta \Rightarrow d_Y\big(f(x), f(x_0)\big) < \varepsilon.$$

It is continuous on X if it is continuous at every $x_0 \in X$.

Theorem 2.113. Sequential Criterion for Continuity
A map $f : X \to Y$ between metric spaces is continuous at $x \in X$ if and only if

$$x_n \to x \Rightarrow f(x_n) \to f(x)$$

for every sequence (x_n) in X.

Definition 2.114. Uniform Continuity
A map $f : (X, d_X) \to (Y, d_Y)$ is uniformly continuous if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall x, y \in X, \qquad d_X(x, y) < \delta \Rightarrow d_Y\big(f(x), f(y)\big) < \varepsilon.$$

Unlike ordinary continuity, the same δ must work for every pair $x, y \in X$.

Definition 2.115. Pointwise and Uniform Convergence
Let $f_n : E \to Y$ and $f : E \to Y$, where (Y, d) is a metric space. The sequence (f_n) converges pointwise to f, written $f_n \to f$, if

$$\forall x \in E, \quad \forall \varepsilon > 0, \quad \exists N = N(x, \varepsilon) \in \mathbb{N}, \quad \forall n \geq N, \qquad d\big(f_n(x), f(x)\big) < \varepsilon.$$

It converges uniformly to f, written $f_n \rightrightarrows f$, if

$$\forall \varepsilon > 0, \quad \exists N = N(\varepsilon) \in \mathbb{N}, \quad \forall n \geq N, \quad \forall x \in E, \qquad d\big(f_n(x), f(x)\big) < \varepsilon.$$

For real- or complex-valued functions, this is equivalent to

$$\|f_n - f\|_\infty = \sup_{x \in E} |f_n(x) - f(x)| \longrightarrow 0.$$

Definition 2.116. Equicontinuity and Uniform Boundedness
A family $\mathcal{F}$ of maps from (X, d_X) to (Y, d_Y) is equicontinuous at $x_0 \in X$ if

$$\forall \varepsilon > 0, \quad \exists \delta > 0, \quad \forall f \in \mathcal{F}, \quad \forall x \in X, \qquad d_X(x, x_0) < \delta \Rightarrow d_Y\big(f(x), f(x_0)\big) < \varepsilon.$$

A sequence of real-valued functions (f_n) is uniformly bounded if

$$\exists M > 0, \quad \forall n \in \mathbb{N}, \quad \forall x \in X, \qquad |f_n(x)| \leq M.$$

Definition 2.117. Limit Superior and Limit Inferior
For a real sequence (a_n), define

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup_{k \geq n} a_k, \qquad \liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf_{k \geq n} a_k,$$

whenever these extended real limits are allowed.

Theorem 2.118. Monotone Convergence Theorem for Sequences
Every bounded monotone real sequence converges. More precisely, every increasing sequence bounded above converges to its supremum, and every decreasing sequence bounded below converges to its infimum.

Theorem 2.119. Bolzano–Weierstrass Theorem
Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Theorem 2.120. Heine–Borel Theorem
A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

Theorem 2.121. Heine–Cantor Theorem; Uniform Continuity on Compact Sets
If $K \subseteq \mathbb{R}^n$ is compact and $f : K \to \mathbb{R}^m$ is continuous, then f is uniformly continuous on K.

Strategy. In analysis questions, compactness usually means that one may extract a convergent subsequence or that a continuous function attains extrema. In $\mathbb{R}^n$, the fast recognition phrase is "closed and bounded."

Theorem 2.122. Cauchy Criterion for Convergence
In a complete metric space, a sequence converges if and only if it is Cauchy. In particular, for (a_n) in $\mathbb{R}$ or $\mathbb{C}$,

$$(a_n) \text{ converges} \Leftrightarrow \forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall m, n \geq N,\ |a_m - a_n| < \varepsilon.$$

Theorem 2.123. Uniform Cauchy Criterion
Let (Y,d) be complete and let $f_n : E \to Y$. Then (f_n) converges uniformly on E if and only if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m,n \geq N, \quad \forall x \in E, \qquad d\big(f_m(x), f_n(x)\big) < \varepsilon.$$

Theorem 2.124. Weierstrass M-Test
Let (f_n) be a sequence of functions on a set E. If

$$|f_n(x)| \leq M_n \quad \textit{for all } x \in E$$

and

$$\sum_{n=1}^{\infty} M_n$$

converges, then the series

$$\sum_{n=1}^{\infty} f_n(x)$$

converges uniformly on E.

Theorem 2.125. Uniform Limit Theorem
If (f_n) is a sequence of real- or complex-valued continuous functions on a metric space E and $f_n \rightrightarrows f$ on E, then f is continuous on E.

Theorem 2.126. Term-by-Term Differentiation
Suppose (f_n) is a sequence of continuously differentiable functions on a closed bounded interval $[a,b]$, and suppose $(f_n(x_0))$ converges for some $x_0 \in [a,b]$. If $f_n' \rightrightarrows g$ on $[a,b]$, then $f_n \rightrightarrows f$ for a differentiable function f, and

$$f' = g = \lim_{n\to\infty} f_n'.$$

Theorem 2.127. Arzelà–Ascoli Theorem
Every uniformly bounded and equicontinuous sequence of real-valued continuous functions on a compact metric space has a uniformly convergent subsequence.

Theorem 2.128. Stone–Weierstrass Theorem
A subalgebra of $C(K,\mathbb{R})$ that contains the constants and separates points is dense in $C(K,\mathbb{R})$ under the supremum norm, provided K is compact.

Theorem 2.129. Banach Fixed-Point Theorem; Contraction Mapping Theorem
Let (X,d) be a nonempty complete metric space, and suppose that $T : X \to X$ is a contraction: there exists $q \in [0,1)$ s.t.

$$\forall x,y \in X, \qquad d\big(T(x), T(y)\big) \leq q d(x,y).$$

Then T has a unique fixed point $x^* \in X$, and the iteration $x_{n+1} = T(x_n)$ converges to x^* for every initial point $x_0 \in X$.

Theorem 2.130. Baire Category Theorem
A nonempty complete metric space cannot be written as a countable union of nowhere dense closed sets. Equivalently, a countable intersection of open dense subsets of a complete metric space is dense.

Theorem 2.131. Tonelli's Theorem
Let (X,μ) and (Y,ν) be σ-finite measure spaces, and let $f : X \times Y \to [0,\infty]$ be measurable. Then the iterated integrals and the product-space integral agree, possibly with value $+\infty$:

$$\int_{X\times Y} f \, \mathrm{d}(\mu \times \nu) = \int_X \left(\int_Y f(x,y) \, \mathrm{d}\nu(y) \right) \mathrm{d}\mu(x) = \int_Y \left(\int_X f(x,y) \, \mathrm{d}\mu(x) \right) \mathrm{d}\nu(y).$$

In particular, a nonnegative series may be integrated term by term.

Theorem 2.132. Fatou's Lemma
If $f_n \geq 0$ are measurable, then

$$\int \liminf_{n\to\infty} f_n \, \mathrm{d}\mu \leq \liminf_{n\to\infty} \int f_n \, \mathrm{d}\mu.$$

Theorem 2.133. Lebesgue Monotone Convergence Theorem; Beppo Levi Theorem
If $0 \leq f_1 \leq f_2 \leq \cdots$ and $f_n \xrightarrow{\text{a.e.}} f$, then

$$\lim_{n\to\infty} \int f_n \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu.$$

Theorem 2.134. Lebesgue Dominated Convergence Theorem
If $f_n \xrightarrow{\text{a.e.}} f$ and there exists an integrable function g s.t. $|f_n| \leq g$ for all n, then

$$\lim_{n\to\infty} \int f_n \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu.$$

Theorem 2.135. Density of Irrational Rotations
If $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, then the sequence of fractional parts

$$\{n\alpha\}, \qquad n = 0, 1, 2, \ldots,$$

is dense in $[0, 1]$. Equivalently, the orbit of any point under rotation of the circle through an irrational multiple of 2π is dense in the circle.

Idea. Write $\|x\|_{\mathbb{R}/\mathbb{Z}}$ for the distance from x to the nearest integer. By the pigeonhole principle, for every $N \geq 2$ there is a positive integer q s.t.

$$0 < \|q\alpha\|_{\mathbb{R}/\mathbb{Z}} < \frac{1}{N}.$$

Put $\delta = \|q\alpha\|_{\mathbb{R}/\mathbb{Z}}$ and $M = \lfloor 1/\delta \rfloor$. Up to reversing orientation on the circle, the integer orbit contains

$$0, \delta, 2\delta, \ldots, M\delta,$$

whose successive circular gaps are at most $\delta < 1/N$. Thus the two-sided orbit $\{n\alpha \bmod 1 \mid n \in \mathbb{Z}\}$ is dense.

Moreover, irrationality gives an unbounded sequence of positive integers q_j for which

$$\|q_j\alpha\|_{\mathbb{R}/\mathbb{Z}} \longrightarrow 0.$$

For each fixed $m \geq 1$, the nonnegative multiples $(q_j - m)\alpha$ converge modulo 1 to $-m\alpha$, and $q_j - m \geq 0$ for all sufficiently large j. Hence the closure of the nonnegative orbit contains every negative multiple as well as every nonnegative multiple. It therefore contains the dense two-sided orbit, proving the claim.

Complex Analysis

Definition 2.136. Complex Differentiability; Holomorphic and Entire Functions
Let $U \subseteq \mathbb{C}$ be open and let $f : U \to \mathbb{C}$. The complex derivative of f at $z_0 \in U$ is

$$f'(z_0) = \lim_{z\to z_0} \frac{f(z) - f(z_0)}{z - z_0},$$

provided the limit exists independently of the direction of approach. The function is holomorphic on U if it is complex differentiable at every point of U, and entire if it is holomorphic on all of $\mathbb{C}$.

Theorem 2.137. Cauchy–Riemann Equations
Write

$$f(z) = u(x, y) + iv(x, y), \qquad z = x + iy.$$

If f is complex differentiable at $z_0 = (x_0, y_0)$ and the first partial derivatives exist there, then

$$\frac{\partial u}{\partial x}(x_0, y_0) = \frac{\partial v}{\partial y}(x_0, y_0), \qquad \frac{\partial u}{\partial y}(x_0, y_0) = -\frac{\partial v}{\partial x}(x_0, y_0).$$

Conversely, if these partial derivatives are continuous near z_0 and satisfy the equations there, then f is complex differentiable at z_0.

Formula 2.138. Cauchy–Riemann Equations in Polar Coordinates
Write

$$f(re^{i\theta}) = u(r, \theta) + iv(r, \theta), \qquad r > 0.$$

If the relevant first partial derivatives are continuous, then the Cauchy–Riemann equations become

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \qquad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}.$$

Strategy. To test whether a given $f(z) = u(x, y) + iv(x, y)$ can be holomorphic, compute the four first partial derivatives and check the Cauchy–Riemann equations. Failure at one point immediately rules out holomorphicity there.

Fact 2.139. Standard Holomorphic and Nonholomorphic Examples
Every polynomial, e^z, $\sin z$, and $\cos z$ is entire. The function $\tan z$ is meromorphic but not entire because it has poles at

$$z = \frac{\pi}{2} + k\pi, \qquad k \in \mathbb{Z}.$$

The function $z \mapsto \operatorname{Re}(z)$ is not complex differentiable at any point, since its difference quotient depends on the direction of approach.

Proposition 2.140. Holomorphic Functions Have Harmonic Components
If $f = u + iv$ is holomorphic and u, v have continuous second partial derivatives, then

$$\Delta u = 0, \qquad \Delta v = 0.$$

Thus the real and imaginary parts of a holomorphic function are harmonic conjugates locally.

Formula 2.141. Complex Exponential, Trigonometric Functions, and Logarithm
For $z = x + iy$,

$$e^z = e^x(\cos y + i \sin y).$$

The complex sine and cosine satisfy

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \qquad \cos z = \frac{e^{iz} + e^{-iz}}{2}.$$

For $z \neq 0$, the complex logarithm is multivalued:

$$\log z = \ln |z| + i\big(\arg z + 2\pi k\big), \qquad k \in \mathbb{Z}.$$

A single-valued holomorphic logarithm requires a choice of branch on a domain that does not wind around the origin.

Theorem 2.142. Cauchy–Goursat Integral Theorem
If f is holomorphic on a simply connected domain $U \subseteq \mathbb{C}$, then for every closed piecewise smooth curve γ in U,

$$\oint_\gamma f(z)\,\mathrm{d}z = 0.$$

Equivalently, contour integrals of f in U are path-independent.

Theorem 2.143. Cauchy Integral Formula
Let f be holomorphic on an open set containing a positively oriented simple closed contour γ and its interior. If a lies inside γ, then

$$f(a) = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{z-a}\,\mathrm{d}z.$$

More generally, for every integer $n \geq 0$,

$$f^{(n)}(a) = \frac{n!}{2\pi i} \oint_\gamma \frac{f(z)}{(z-a)^{n+1}}\,\mathrm{d}z.$$

Strategy. When a contour integral contains $(z-a)^{-m}$ and the remaining factor is holomorphic, compare it directly with Cauchy's integral formula before attempting a parametrization of the contour.

Theorem 2.144. Maximum Modulus Principle
A nonconstant holomorphic function on a connected open set cannot attain a maximum of $|f|$ at an interior point. Consequently, if f is holomorphic on a bounded domain and continuous on its closure, then the maximum of $|f|$ occurs on the boundary.

Theorem 2.145. Liouville's Theorem
Every bounded entire function is constant.

Remark. Liouville's Theorem gives a short proof of the Fundamental Theorem of Algebra: if a nonconstant complex polynomial had no zero, then its reciprocal would be a bounded entire function.

Theorem 2.146. Taylor Series for Holomorphic Functions
If f is holomorphic on the open disk $|z-a| < R$, then

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(z-a)^n, \qquad |z-a| < R.$$

Thus a holomorphic function is analytic: complex differentiability implies local representability by a convergent power series.

Theorem 2.147. Laurent Series
If f is holomorphic on an annulus

$$r < |z-a| < R,$$

then it has a unique Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n$$

converging throughout the annulus. The coefficient c_{-1} is the residue of f at a.

Definition 2.148. Isolated Singularities and Residues
Let a be an isolated singularity of f. In the Laurent expansion about a, the residue is

$$\operatorname{Res}(f; a) = c_{-1}.$$

The singularity is removable if every negative-power coefficient is zero, a pole if only finitely many negative-power coefficients are nonzero, and essential if infinitely many are nonzero.

Formula 2.149. Residues at Poles
If a is a simple pole of f, then

$$\operatorname{Res}(f;a)=\lim_{z\to a}(z-a)f(z).$$

More generally, if a is a pole of order m, then

$$\operatorname{Res}(f;a)=\frac{1}{(m-1)!}\lim_{z\to a}\frac{\mathrm{d}^{m-1}}{\mathrm{d}z^{m-1}}\left((z-a)^m f(z)\right).$$

If $f(z)=g(z)/h(z)$, where g and h are holomorphic, $h(a)=0$, and $h'(a)\neq 0$, then

$$\operatorname{Res}(f;a)=\frac{g(a)}{h'(a)}.$$

Theorem 2.150. Residue Theorem
Let f be holomorphic inside and on a positively oriented simple closed contour γ, except for isolated singularities $a_1,\dots,a_m$ inside γ. Then

$$\oint_\gamma f(z)\,\mathrm{d}z=2\pi i\sum_{k=1}^{m}\operatorname{Res}(f;a_k).$$

Strategy. Real Integrals by Residues. For a rational integral over $\mathbb{R}$, extend the integrand to a complex rational function, choose a semicircular contour, sum the residues in the appropriate half-plane, and verify that the contribution from the large arc vanishes. Trigonometric integrals over $[0,2\pi]$ often become contour integrals under the substitution $z=e^{i\theta}$ and

$$\mathrm{d}\theta=\frac{\mathrm{d}z}{iz}.$$

Theorem 2.151. Rouché's Theorem
Let f and g be holomorphic inside and on a simple closed contour γ. If

$$|g(z)|<|f(z)|\qquad \textit{for every } z\in\gamma,$$

then f and $f+g$ have the same number of zeros inside γ, counted with multiplicity.

Definition 2.152. Conformal Map
A holomorphic function f is conformal at z_0 if $f'(z_0)\neq 0$. At such a point, f preserves oriented angles and infinitesimal shapes, although it may change scale.

Definition 2.153. Möbius Transformation; Linear Fractional Transformation
A Möbius transformation is a map of the extended complex plane of the form

$$T(z)=\frac{az+b}{cz+d},\qquad ad-bc\neq 0.$$

It is conformal wherever its derivative is finite and nonzero, and it maps generalized circles—circles and lines—to generalized circles.

Formula 2.154. Standard Conformal Maps
The following transformations are useful to recognize quickly:

$$z\mapsto az+b\quad\text{(rotation, dilation, and translation)},$$

$$z\mapsto\frac{1}{z}\quad\text{(inversion)},\qquad z\mapsto z^n\quad\text{(angle multiplication)},$$

and

$$z\mapsto\frac{z-i}{z+i},$$

which maps the upper half-plane conformally onto the unit disk.

Chapter 3

Linear Algebra

Related POSTECH Courses: (MATH203) Applied Linear Algebra
Related Books: Linear Algebra and Its Applications (Gilbert Strang)

Definition 3.1. Vector Space
A vector space V over a field F is a set equipped with vector addition and scalar multiplication satisfying the standard vector-space axioms. The scalars belong to F; in this book the principal cases are $F = \mathbb{R}$ and $F = \mathbb{C}$.

Definition 3.2. Linear Combination, Span, and Linear Independence
A vector of the form

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$$

is a linear combination of $\mathbf{v}_1, \ldots, \mathbf{v}_k$. Their span is

$$\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_k) = \{c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k : c_1, \ldots, c_k \in F\}.$$

The vectors are linearly independent if

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0}$$

implies $c_1 = \cdots = c_k = 0$.

Definition 3.3. Basis and Dimension
A list

$$\mathbf{v}_1, \ldots, \mathbf{v}_n$$

is a basis of V if it is linearly independent and spans V. If V has a finite basis, every basis has the same number of vectors; this number is the dimension of V, denoted by $\dim V$.

Theorem 3.4. Basis Extension and Reduction
Every linearly independent list in a finite-dimensional vector space can be extended to a basis, and every spanning list contains a basis. More generally, the assertion that every vector space has a basis is equivalent to a standard form of the axiom of choice.

Definition 3.5. Linear Map; Linear Transformation
Let V and W be vector spaces over the same field F. A map $T : V \to W$ is linear if

$$\forall \alpha, \beta \in F, \quad \forall \mathbf{u}, \mathbf{v} \in V, \qquad T(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}).$$

Equivalently, T preserves vector addition and scalar multiplication. Its kernel and image are

$$\ker T = \{\mathbf{v} \in V \mid T(\mathbf{v}) = \mathbf{0}\}, \qquad \operatorname{im} T = \{T(\mathbf{v}) \mid \mathbf{v} \in V\}.$$

Formula 3.6. Coordinate Vectors, Matrix Representation, and Change of Basis
Let

$$\mathcal{B} = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$$

be an ordered basis of V. The coordinate vector of $\mathbf{v} \in V$ relative to $\mathcal{B}$ is the unique column vector $[\mathbf{v}]_{\mathcal{B}}$ satisfying

$$\mathbf{v} = \begin{pmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{pmatrix} [\mathbf{v}]_{\mathcal{B}}.$$

If $T : V \to W$ is linear and $\mathcal{C}$ is an ordered basis of W, then the matrix $[T]_{\mathcal{C}\leftarrow\mathcal{B}}$ is characterized by

$$[T(\mathbf{v})]_{\mathcal{C}} = [T]_{\mathcal{C}\leftarrow\mathcal{B}}[\mathbf{v}]_{\mathcal{B}}.$$

If $\mathcal{B}$ and $\mathcal{B}'$ are ordered bases of the same n-dimensional space and

$$[\mathbf{v}]_{\mathcal{B}} = \mathbf{Q}[\mathbf{v}]_{\mathcal{B}'},$$

then $\mathbf{Q}$ is invertible. For a linear operator $T : V \to V$,

$$[T]_{\mathcal{B}'} = \mathbf{Q}^{-1}[T]_{\mathcal{B}}\mathbf{Q}.$$

Thus matrices representing the same linear operator in different bases are similar.

Fact 3.7. Translation Is Affine, Not Linear
For a fixed vector $\mathbf{v} \in \mathbb{R}^n$, the translation

$$T_{\mathbf{v}}(\mathbf{x}) = \mathbf{x} + \mathbf{v}$$

is bijective, with inverse $T_{-\mathbf{v}}$. It is a linear map if and only if $\mathbf{v} = \mathbf{0}$.

Formula 3.8. Inner Product and Adjoint Transfer
For real column vectors,

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^{\mathsf{T}}\mathbf{v}.$$

For a real matrix $\mathbf{A}$ of compatible size,

$$\langle \mathbf{x}, \mathbf{A}\mathbf{y} \rangle = \langle \mathbf{A}^{\mathsf{T}}\mathbf{x}, \mathbf{y} \rangle.$$

Hence, if $\mathbf{A}$ is symmetric,

$$\langle \mathbf{x}, \mathbf{A}\mathbf{y} \rangle = \langle \mathbf{A}\mathbf{x}, \mathbf{y} \rangle.$$

Over $\mathbb{C}$, use

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^{\mathsf{H}}\mathbf{v}$$

and replace $\mathbf{A}^{\mathsf{T}}$ by $\mathbf{A}^{\mathsf{H}}$.

Theorem 3.9. Rank–Nullity Theorem
Let $T : V \to W$ be a linear map between finite-dimensional vector spaces. Then

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

For a matrix $\mathbf{A} \in F^{m\times n}$, this becomes

$$n = \operatorname{rk}(\mathbf{A}) + \dim \operatorname{Null}(\mathbf{A}).$$

Strategy. Rank–Nullity connects variables, pivots, and free variables. It is the linear algebra version of accounting: nothing disappears; dimensions are redistributed.

Theorem 3.10. Row Rank Equals Column Rank
For every matrix $\mathbf{A}$,

$$\dim \operatorname{Row}(\mathbf{A}) = \dim \operatorname{Col}(\mathbf{A}).$$

This common dimension is called the rank of $\mathbf{A}$ and is denoted by $\operatorname{rk}(\mathbf{A})$.

Definition 3.11. Determinant
The determinant is a scalar associated to a square matrix $\mathbf{A} \in F^{n\times n}$, denoted by $\det(\mathbf{A})$. Geometrically, for real matrices, $|\det(\mathbf{A})|$ is the volume scaling factor of the linear transformation $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$.

Definition 3.12. Minor, Cofactor, and Adjugate
Let $\mathbf{A} = (a_{ij}) \in F^{n\times n}$. The minor M_{ij} is the determinant of the $(n-1)\times(n-1)$ matrix obtained by deleting row i and column j. The corresponding cofactor is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

The cofactor matrix is (C_{ij}), and the adjugate is its transpose:

$$\operatorname{adj}(\mathbf{A}) = (C_{ij})^{\mathsf{T}}.$$

Cofactors are the coefficients in Laplace expansion and are therefore the bridge between determinants and the adjugate formula for an inverse.

Theorem 3.13. Invertible Matrix Theorem
For a square matrix $\mathbf{A} \in F^{n\times n}$, the following statements are equivalent:

$$\mathbf{A} \text{ is invertible}, \qquad \det(\mathbf{A}) \neq 0, \qquad \operatorname{rk}(\mathbf{A}) = n,$$

$$\operatorname{Null}(\mathbf{A}) = \{\mathbf{0}\},$$

and

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

has a unique solution for every $\mathbf{b} \in F^n$.

Remark. This theorem is a dictionary. It translates between determinants, rank, null spaces, and solvability of linear systems.

Proposition 3.14. One-Sided Inverse for Square Matrices
If $\mathbf{A}, \mathbf{B} \in F^{n\times n}$ satisfy

$$\mathbf{A}\mathbf{B} = \mathbf{I}_n,$$

then both matrices are invertible and

$$\mathbf{B}\mathbf{A} = \mathbf{I}_n.$$

Thus, for square matrices, a right inverse is automatically a left inverse, and conversely.

Formula 3.15. Inverse of a 2×2 Matrix
If

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

and $ad - bc \neq 0$, then

$$\mathbf{A}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Theorem 3.16. Cramer's Rule
Let $\mathbf{A} \in F^{n\times n}$ be invertible. The unique solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ is given by

$$x_i = \frac{\det(\mathbf{A}_i)}{\det(\mathbf{A})},$$

where $\mathbf{A}_i$ is obtained from $\mathbf{A}$ by replacing its ith column with $\mathbf{b}$.

Formula 3.17. Adjugate Formula for the Inverse
For every square matrix $\mathbf{A}$,

$$\mathbf{A}\,\mathrm{adj}(\mathbf{A}) = \mathrm{adj}(\mathbf{A})\mathbf{A} = \det(\mathbf{A})\mathbf{I}_n.$$

Hence, if $\det(\mathbf{A}) \neq 0$, then

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})}\,\mathrm{adj}(\mathbf{A}).$$

Formula 3.18. Determinant as Volume
If $\mathbf{v}_1, \ldots, \mathbf{v}_n \in \mathbb{R}^n$, then the n-dimensional volume of the parallelepiped spanned by these vectors is

$$\left|\det\begin{pmatrix}\mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n\end{pmatrix}\right|.$$

In $\mathbb{R}^3$, this is the absolute value of the scalar triple product

$$|(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}|.$$

Formula 3.19. Scalar and Vector Triple Products
For $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$,

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = \det(\mathbf{a}, \mathbf{b}, \mathbf{c}).$$

The vector triple-product identities are

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c},$$

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a}.$$

Formula 3.20. Cross-Product Determinant Identity
For $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$,

$$\det(\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}) = \det(\mathbf{a}, \mathbf{b}, \mathbf{c})^2.$$

Therefore, if $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are linearly independent, then

$$\mathbf{a} \times \mathbf{b}, \qquad \mathbf{b} \times \mathbf{c}, \qquad \mathbf{c} \times \mathbf{a}$$

are also linearly independent.

Proposition 3.21. Basic Properties of Determinants
For square matrices $\mathbf{A}, \mathbf{B} \in F^{n \times n}$, we have

$$\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B}),$$

$$\det(\mathbf{A}^{\mathsf{T}}) = \det(\mathbf{A}),$$

and, if $\mathbf{A}$ is invertible,

$$\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}.$$

Formula 3.22. Determinant of a Triangular Matrix
If $\mathbf{A}$ is upper triangular or lower triangular with diagonal entries $a_{11}, a_{22}, \ldots, a_{nn}$, then

$$\det(\mathbf{A}) = a_{11}a_{22}\cdots a_{nn}.$$

Formula 3.23. Determinant of a Block Triangular Matrix
If the diagonal blocks are square, then

$$\det\begin{pmatrix}\mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{D}\end{pmatrix} = \det(\mathbf{A})\det(\mathbf{D}),$$

and the same formula holds for a lower block triangular matrix.

Proposition 3.24. Multilinearity and Alternation of the Determinant
The determinant is linear in each row separately and also linear in each column separately. If two rows or two columns are interchanged, the sign of the determinant changes. In particular, if the rows or columns of $\mathbf{A}$ *are linearly dependent, then*

$$\det(\mathbf{A}) = 0.$$

Strategy. In short determinant problems, first look for dependent rows or columns, row or column sums, repeated patterns, triangular form, and easy row operations. Direct expansion should usually be the last resort, except for 2×2 and 3×3 matrices.

Proposition 3.25. Characterization of the Determinant
Let

$$f : (F^n)^n \to F$$

be alternating and multilinear. Then

$$f(\mathbf{v}_1, \dots, \mathbf{v}_n) = f(\mathbf{e}_1, \dots, \mathbf{e}_n) \det \begin{pmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{pmatrix}.$$

In particular, the determinant is the unique alternating multilinear function satisfying

$$\det(\mathbf{I}_n) = 1.$$

Formula 3.26. Laplace Expansion; Cofactor Expansion
Let $\mathbf{A} = (a_{ij}) \in F^{n\times n}$, and let C_{ij} be the cofactor of a_{ij}. Expanding along row i gives

$$\det(\mathbf{A}) = \sum_{j=1}^{n} a_{ij} C_{ij}.$$

Expanding along column j gives

$$\det(\mathbf{A}) = \sum_{i=1}^{n} a_{ij} C_{ij}.$$

Formula 3.27. Sarrus' Rule
For a 3×3 matrix,

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aei + bfg + cdh - ceg - bdi - afh.$$

Formula 3.28. Determinant and Elementary Row Operations
The determinant changes under elementary row operations as follows.

- Swapping two rows multiplies the determinant by -1.
- Multiplying one row by c multiplies the determinant by c.
- Adding a multiple of one row to another row does not change the determinant.

The same rules hold for columns.

Formula 3.29. Matrix Determinant Lemma
If $\mathbf{A} \in F^{n\times n}$ is invertible and $\mathbf{u}, \mathbf{v} \in F^n$, then

$$\det(\mathbf{A} + \mathbf{u}\mathbf{v}^\mathsf{T}) = \det(\mathbf{A}) \left(1 + \mathbf{v}^\mathsf{T}\mathbf{A}^{-1}\mathbf{u}\right).$$

In particular,

$$\det(\mathbf{I}_n + \mathbf{u}\mathbf{v}^\mathsf{T}) = 1 + \mathbf{v}^\mathsf{T}\mathbf{u}.$$

Theorem 3.30. Sylvester's Determinant Theorem
If $\mathbf{A} \in F^{m\times n}$ *and* $\mathbf{B} \in F^{n\times m}$*, then*

$$\det(\mathbf{I}_m + \mathbf{AB}) = \det(\mathbf{I}_n + \mathbf{BA}).$$

This allows a determinant involving a large product to be replaced by a determinant of the smaller dimension.

Formula 3.31. Sherman–Morrison Formula
If $\mathbf{A}$ is invertible and $1 + \mathbf{v}^\mathsf{T}\mathbf{A}^{-1}\mathbf{u} \neq 0$, then

$$(\mathbf{A} + \mathbf{u}\mathbf{v}^\mathsf{T})^{-1} = \mathbf{A}^{-1} - \frac{\mathbf{A}^{-1}\mathbf{u}\mathbf{v}^\mathsf{T}\mathbf{A}^{-1}}{1 + \mathbf{v}^\mathsf{T}\mathbf{A}^{-1}\mathbf{u}}.$$

It is the inverse counterpart of the Matrix Determinant Lemma.

Formula 3.32. Schur Complement Determinant Formula
If $\mathbf{A}$ is invertible, then

$$\det\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix} = \det(\mathbf{A})\det(\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}).$$

The matrix $\mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}$ is the Schur complement of $\mathbf{A}$.

Formula 3.33. Symmetric Block Determinant
If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same size, then

$$\det\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix} = \det(\mathbf{A} + \mathbf{B})\det(\mathbf{A} - \mathbf{B}).$$

This follows by changing to the invariant subspaces consisting of vectors of the form $(\mathbf{x}, \mathbf{x})$ and $(\mathbf{x}, -\mathbf{x})$.

Definition 3.34. Eigenvalue and Characteristic Polynomial
Let $\mathbf{A} \in F^{n\times n}$. A scalar $\lambda \in F$ is an eigenvalue of $\mathbf{A}$ if there exists a nonzero vector $\mathbf{v} \in F^n$ s.t.

$$\mathbf{Av} = \lambda\mathbf{v}.$$

The characteristic polynomial of $\mathbf{A}$ is

$$p_{\mathbf{A}}(\lambda) = \det(\lambda\mathbf{I}_n - \mathbf{A}).$$

Definition 3.35. Conjugate Transpose; Hermitian Matrix
For a complex matrix $\mathbf{A} = (a_{ij})$, its conjugate transpose, or Hermitian transpose, is

$$\mathbf{A}^\mathsf{H} = \overline{\mathbf{A}}^\mathsf{T} = (\overline{a_{ji}}).$$

A complex square matrix is Hermitian if

$$\mathbf{A}^\mathsf{H} = \mathbf{A}.$$

For real matrices, the Hermitian transpose is the ordinary transpose.

Proposition 3.36. Basic Rules for the Hermitian Transpose
For complex matrices of compatible sizes,

$$(\mathbf{A} + \mathbf{B})^\mathsf{H} = \mathbf{A}^\mathsf{H} + \mathbf{B}^\mathsf{H}, \qquad (c\mathbf{A})^\mathsf{H} = \overline{c}\,\mathbf{A}^\mathsf{H},$$

$$(\mathbf{AB})^\mathsf{H} = \mathbf{B}^\mathsf{H}\mathbf{A}^\mathsf{H}, \qquad (\mathbf{A}^\mathsf{H})^\mathsf{H} = \mathbf{A}.$$

Definition 3.37. Orthogonal and Unitary Matrices
A real square matrix $\mathbf{Q}$ is orthogonal if

$$\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}_n.$$

A complex square matrix $\mathbf{U}$ is unitary if

$$\mathbf{U}^\mathsf{H}\mathbf{U} = \mathbf{I}_n.$$

Equivalently, their columns form orthonormal bases over $\mathbb{R}$ and $\mathbb{C}$, respectively.

Formula 3.38. Special Matrix Recognition Table
For real or complex square matrices, the following implications permit rapid determinant, inverse, and eigenvalue calculations.

Matrix type	Defining relation	Immediate consequences
Diagonal	$\mathbf{D} = \operatorname{diag}(d_1, \dots, d_n)$	eigenvalues d_i; $\det(\mathbf{D}) = \prod_i d_i$
Triangular	$a_{ij} = 0$ for $i > j$ or for $i < j$	eigenvalues are diagonal entries; determinant is their product
Orthogonal	$\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}_n$	$\mathbf{Q}^{-1} = \mathbf{Q}^\mathsf{T}$; $\lvert\det(\mathbf{Q})\rvert = 1$
Unitary	$\mathbf{U}^\mathsf{H}\mathbf{U} = \mathbf{I}_n$	$\mathbf{U}^{-1} = \mathbf{U}^\mathsf{H}$; every eigenvalue has modulus 1
Idempotent	$\mathbf{P}^2 = \mathbf{P}$	eigenvalues belong to $\{0, 1\}$; $\operatorname{rk}(\mathbf{P}) = \operatorname{tr}(\mathbf{P})$
Involutory	$\mathbf{A}^2 = \mathbf{I}_n$	$\mathbf{A}^{-1} = \mathbf{A}$; eigenvalues belong to $\{-1, 1\}$
Nilpotent	$\mathbf{N}^k = \mathbf{0}$ for some k	every eigenvalue is 0; $\det(\mathbf{N}) = 0$
Permutation	one 1 in each row and column	$\mathbf{P}^{-1} = \mathbf{P}^\mathsf{T}$; determinant is ± 1
Orthogonal projection	$\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^\mathsf{T}$	$\mathbb{R}^n = \operatorname{Col}(\mathbf{P}) \oplus \operatorname{Null}(\mathbf{P})$

Formula 3.39. Vandermonde Determinant
For scalars $x_1, x_2, \dots, x_n$,

$$\det\begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix} = \prod_{1 \le i < j \le n} (x_j - x_i).$$

Fact 3.40. Pascal Matrices
Define the lower triangular Pascal matrix by

$$\mathbf{P}_n = (p_{ij})_{0 \le i,j \le n}, \qquad p_{ij} = \begin{cases} \binom{i}{j}, & j \le i, \\ 0, & j > i. \end{cases}$$

Since all diagonal entries are equal to 1, we have

$$\det(\mathbf{P}_n) = 1.$$

Define the symmetric Pascal matrix by

$$\mathbf{S}_n = (s_{ij})_{0 \le i,j \le n}, \qquad s_{ij} = \binom{i+j}{i}.$$

This matrix also has determinant 1.

Remark. Pascal matrices have combinatorial entries while their determinants admit simple closed forms.

Formula 3.41. Cauchy Determinant
Let

$$c_{ij} = \frac{1}{x_i + y_j},$$

where every denominator is nonzero and the x_i and y_j are pairwise distinct within their respective families. Then

$$\det(c_{ij})_{1\le i,j\le n} = \frac{\displaystyle\prod_{1\le i<j\le n} (x_j - x_i)(y_j - y_i)}{\displaystyle\prod_{i=1}^{n}\prod_{j=1}^{n}(x_i + y_j)}.$$

The Hilbert matrix is a famous special case of a Cauchy matrix.

Formula 3.42. Hilbert Matrix Determinant
The $n \times n$ Hilbert matrix is

$$\mathbf{H}_n = \left(\frac{1}{i+j-1}\right)_{1\le i,j\le n}.$$

As a special Cauchy matrix, it satisfies

$$\det(\mathbf{H}_n) = \frac{\displaystyle\prod_{k=1}^{n-1}(k!)^4}{\displaystyle\prod_{k=1}^{2n-1} k!}.$$

Definition 3.43. Hadamard Matrix
A Hadamard matrix of order n is an $n \times n$ matrix $\mathbf{H}$ whose entries are all $+1$ or -1 and whose rows are mutually orthogonal. Equivalently,

$$\mathbf{H}\mathbf{H}^{\mathsf{T}} = n\mathbf{I}_n.$$

Formula 3.44. Hadamard Determinant
If $\mathbf{H}$ is a Hadamard matrix of order n, then

$$|\det(\mathbf{H})| = n^{n/2}.$$

Formula 3.45. Permutation Matrices
A permutation matrix $\mathbf{P}_\sigma$ has exactly one entry equal to 1 in each row and each column. It satisfies

$$\mathbf{P}_\sigma^{-1} = \mathbf{P}_\sigma^{\mathsf{T}}, \qquad \det(\mathbf{P}_\sigma) = \operatorname{sgn}(\sigma).$$

Its powers are determined by the cycle structure of σ.

Formula 3.46. Idempotent, Involutory, and Nilpotent Matrices
If $\mathbf{P}^2 = \mathbf{P}$, then $\mathbf{P}$ is idempotent and its eigenvalues belong to $\{0, 1\}$. If $\mathbf{A}^2 = \mathbf{I}_n$, then $\mathbf{A}$ is involutory and its eigenvalues belong to $\{-1, 1\}$. If $\mathbf{N}^k = \mathbf{0}$ for some k, then $\mathbf{N}$ is nilpotent and every eigenvalue is 0.

Formula 3.47. Eigenvalues of a Circulant Matrix
Let $\mathbf{C}$ be the circulant matrix whose first row is

$$(c_0, c_1, \ldots, c_{n-1}).$$

With $\omega_n = e^{-2\pi i/n}$, its eigenvalues are

$$\lambda_j = \sum_{k=0}^{n-1} c_k \omega_n^{jk}, \qquad j = 0, 1, \ldots, n-1.$$

Hence $\det(\mathbf{C}) = \prod_{j=0}^{n-1} \lambda_j$.

Formula 3.48. Discrete Fourier Matrix
Let $\omega_n = e^{2\pi i/n}$. The normalized discrete Fourier matrix is

$$\mathbf{F}_n = \frac{1}{\sqrt{n}} \left(\omega_n^{-jk}\right)_{0 \le j,k \le n-1}.$$

It is unitary:

$$\mathbf{F}_n^{\mathsf{H}} \mathbf{F}_n = \mathbf{I}_n.$$

Every circulant matrix is diagonalized by $\mathbf{F}_n$.

Formula 3.49. Symmetric Tridiagonal Toeplitz Matrix
For

$$\mathbf{T}_n = \begin{pmatrix} a & b & 0 & \cdots & 0 \\ b & a & b & \ddots & \vdots \\ 0 & b & a & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & b \\ 0 & \cdots & 0 & b & a \end{pmatrix},$$

the eigenvalues are

$$a + 2b\cos\frac{k\pi}{n+1}, \qquad k = 1, 2, \ldots, n.$$

Formula 3.50. Continuant Recurrence for Tridiagonal Determinants
Let

$$\mathbf{T}_n = \begin{pmatrix} a_1 & b_1 & 0 & \cdots & 0 \\ c_1 & a_2 & b_2 & \ddots & \vdots \\ 0 & c_2 & a_3 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & b_{n-1} \\ 0 & \cdots & 0 & c_{n-1} & a_n \end{pmatrix}.$$

If $D_n = \det(\mathbf{T}_n)$, then

$$D_0 = 1, \qquad D_1 = a_1, \qquad D_n = a_n D_{n-1} - b_{n-1} c_{n-1} D_{n-2}.$$

Formula 3.51. Householder Reflection
For a nonzero vector $\mathbf{v}$,

$$\mathbf{H} = \mathbf{I}_n - 2\frac{\mathbf{v}\mathbf{v}^{\mathsf{T}}}{\mathbf{v}^{\mathsf{T}}\mathbf{v}}$$

is symmetric, orthogonal, and involutory:

$$\mathbf{H}^{\mathsf{T}} = \mathbf{H}, \qquad \mathbf{H}^{\mathsf{T}}\mathbf{H} = \mathbf{I}_n, \qquad \mathbf{H}^2 = \mathbf{I}_n.$$

It reflects vectors across the hyperplane perpendicular to $\mathbf{v}$.

Formula 3.52. Rotation and Reflection Matrices
Rotation by angle θ in the plane is represented by

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

Reflection across the line making angle θ with the positive x-axis is represented by

$$\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}.$$

Formula 3.53. Eigenvalues of a Three-Dimensional Rotation Matrix
A rotation of $\mathbb{R}^3$ by angle θ around an axis through the origin has eigenvalues

$$1, \qquad e^{i\theta}, \qquad e^{-i\theta}.$$

The eigenvalue 1 corresponds to the rotation axis. Therefore, if $\theta \not\equiv 0, \pi \pmod{2\pi}$, the rotation matrix has exactly one real eigenvalue.

Proposition 3.54. Trace, Determinant, and Eigenvalues
Let $\mathbf{A} \in \mathbb{C}^{n\times n}$ *have eigenvalues* $\lambda_1, \dots, \lambda_n$*, counted with algebraic multiplicity. Then*

$$\operatorname{tr}(\mathbf{A}) = \lambda_1 + \cdots + \lambda_n$$

and

$$\det(\mathbf{A}) = \lambda_1 \lambda_2 \cdots \lambda_n.$$

Proposition 3.55. Cyclic Property of the Trace
Whenever the products are defined and square,

$$\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA}).$$

More generally,

$$\operatorname{tr}(\mathbf{ABC}) = \operatorname{tr}(\mathbf{BCA}) = \operatorname{tr}(\mathbf{CAB}).$$

The factors may be cyclically permuted, but they may not in general be rearranged arbitrarily.

Proposition 3.56. Eigenvalues and the Characteristic Polynomial
A scalar λ *is an eigenvalue of* $\mathbf{A}$ *if and only if*

$$\det(\lambda \mathbf{I}_n - \mathbf{A}) = 0.$$

Equivalently, the eigenvalues of $\mathbf{A}$ *are the roots of the characteristic polynomial* $p_{\mathbf{A}}(\lambda)$*.*

Formula 3.57. All-Ones Matrix
Let $\mathbf{J}_n$ be the $n \times n$ matrix whose entries are all 1. Then

$$\mathbf{J}_n \mathbf{1} = n\mathbf{1}, \qquad \mathbf{J}_n \mathbf{v} = \mathbf{0} \quad \text{if} \quad v_1 + \cdots + v_n = 0.$$

Hence the eigenvalues of $\mathbf{J}_n$ are n with multiplicity 1 and 0 with multiplicity $n-1$. The matrix $\mathbf{J}_n - \mathbf{I}_n$ has eigenvalues $n-1$ with multiplicity 1 and -1 with multiplicity $n-1$.

Formula 3.58. Constant Diagonal and Off-Diagonal Matrix
Let $\mathbf{A} \in F^{n\times n}$ have diagonal entries a and off-diagonal entries b. Then

$$\mathbf{A} = (a-b)\mathbf{I}_n + b\mathbf{J}_n.$$

Counting algebraic multiplicity, its eigenvalues consist of

$$a + (n-1)b$$

once and

$$a - b$$

$n-1$ times. If these two values coincide, they together give the single eigenvalue of multiplicity n.

Formula 3.59. Rank-One Outer Product
Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ be nonzero and set

$$\mathbf{A} = \mathbf{v}\mathbf{w}^\mathsf{T}.$$

Then $\mathrm{rk}(\mathbf{A}) = 1$ and

$$\mathbf{A}\mathbf{v} = (\mathbf{w}^\mathsf{T}\mathbf{v})\mathbf{v}.$$

The characteristic polynomial is

$$\lambda^{n-1}(\lambda - \mathbf{w}^\mathsf{T}\mathbf{v}).$$

Thus the eigenvalues, counted with algebraic multiplicity, are $\mathbf{w}^\mathsf{T}\mathbf{v}$ and 0 repeated $n-1$ times. If $\mathbf{w}^\mathsf{T}\mathbf{v} = 0$, every eigenvalue is 0 and $\mathbf{A}^2 = \mathbf{0}$.

Theorem 3.60. Cayley–Hamilton Theorem
Every square matrix satisfies its own characteristic polynomial. That is, if

$$p_\mathbf{A}(\lambda) = \det(\lambda \mathbf{I}_n - \mathbf{A}),$$

then

$$p_\mathbf{A}(\mathbf{A}) = \mathbf{0}.$$

Remark. Cayley–Hamilton is a famous named theorem. It is especially useful for reducing high powers of a matrix to lower powers.

Formula 3.61. Companion Matrix and Linear Recurrences
For

$$p(x) = x^m - c_1 x^{m-1} - \cdots - c_m,$$

the companion matrix

$$\mathbf{C} = \begin{pmatrix} c_1 & c_2 & \cdots & c_{m-1} & c_m \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}$$

has characteristic polynomial p. The recurrence

$$a_n = c_1 a_{n-1} + \cdots + c_m a_{n-m}$$

is encoded by multiplication by $\mathbf{C}$, so fast matrix exponentiation computes distant terms efficiently.

Formula 3.62. Fibonacci Q-Matrix
For

$$\mathbf{Q} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},$$

we have

$$\mathbf{Q}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} \qquad (n \geq 1).$$

Taking determinants gives Cassini's identity

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n.$$

Formula 3.63. Matrix Geometric Series
Let $\mathbf{A}$ be an $n \times n$ matrix. If $\mathbf{I}_n - \mathbf{A}$ is invertible, then

$$\sum_{k=0}^{N-1} \mathbf{A}^k = (\mathbf{I}_n - \mathbf{A}^N)(\mathbf{I}_n - \mathbf{A})^{-1}.$$

If the spectral radius of $\mathbf{A}$ is less than 1, then

$$\sum_{k=0}^{\infty} \mathbf{A}^k = (\mathbf{I}_n - \mathbf{A})^{-1}.$$

Strategy. Fast Matrix Powers. To compute a large power $\mathbf{A}^N$, first look for one of the following: diagonalization, a short minimal polynomial, idempotence, involution, nilpotence, rank one, a circulant structure, or a decomposition into $a\mathbf{I}_n + b\mathbf{J}_n$. Cayley–Hamilton then reduces every sufficiently high power to a linear combination of lower powers.

Definition 3.64. Algebraic and Geometric Multiplicity
Let λ be an eigenvalue of $\mathbf{A} \in F^{n\times n}$. Its algebraic multiplicity $m_a(\lambda)$ is its multiplicity as a root of $p_{\mathbf{A}}(t)$, while its geometric multiplicity is

$$m_g(\lambda) = \dim \operatorname{Null}(\mathbf{A} - \lambda \mathbf{I}_n).$$

For every eigenvalue,

$$1 \le m_g(\lambda) \le m_a(\lambda).$$

Definition 3.65. Diagonalizable Matrix
A matrix $\mathbf{A} \in F^{n\times n}$ is diagonalizable over F if there exist an invertible matrix $\mathbf{P} \in F^{n\times n}$ and a diagonal matrix $\mathbf{D}$ s.t.

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}.$$

Equivalently, the columns of $\mathbf{P}$ can be chosen as a basis of eigenvectors of $\mathbf{A}$. Diagonalization is useful because powers and polynomial functions reduce to entrywise calculations on $\mathbf{D}$:

$$\mathbf{A}^k = \mathbf{P}\mathbf{D}^k\mathbf{P}^{-1}.$$

Theorem 3.66. Diagonalization Criteria
Suppose the characteristic polynomial of $\mathbf{A} \in F^{n\times n}$ splits over F. The following are equivalent:

$$\mathbf{A} \text{ is diagonalizable over } F,$$

$$\sum_{\lambda} m_g(\lambda) = n,$$

and

$$m_g(\lambda) = m_a(\lambda) \quad \text{for every eigenvalue } \lambda.$$

Equivalently, the minimal polynomial of $\mathbf{A}$ splits over F into distinct linear factors.

Corollary 3.67. Distinct Eigenvalues Imply Diagonalizability
If an $n \times n$ matrix $\mathbf{A}$ has n distinct eigenvalues in F, then $\mathbf{A}$ is diagonalizable over F.

Remark. The field matters. A real matrix can fail to be diagonalizable over $\mathbb{R}$ because its characteristic polynomial has nonreal roots, while the same matrix may be diagonalizable over $\mathbb{C}$. Over $\mathbb{C}$, every characteristic polynomial splits, so only the multiplicity conditions remain. Stronger diagonalization results for symmetric, Hermitian, and normal matrices appear below.

Proposition 3.68. Polynomials of a Diagonalizable Matrix
Let $\mathbf{A} \in F^{n\times n}$ be diagonalizable over F, and suppose

$$\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}, \qquad \mathbf{D} = \operatorname{diag}(\lambda_1, \dots, \lambda_n).$$

For a polynomial $p(x) \in F[x]$, we have

$$p(\mathbf{A}) = \mathbf{S}p(\mathbf{D})\mathbf{S}^{-1},$$

where

$$p(\mathbf{D}) = \operatorname{diag}\big(p(\lambda_1), \dots, p(\lambda_n)\big).$$

In particular,

$$\det(p(\mathbf{A})) = p(\lambda_1)p(\lambda_2)\cdots p(\lambda_n).$$

Idea. Similar matrices represent the same linear transformation in different bases. Therefore, applying a polynomial to a diagonalizable matrix is the same as applying that polynomial to each diagonal entry after diagonalization.

Strategy. Distinct eigenvalues are an easy sufficient condition for diagonalizability. They are not necessary: a matrix can be diagonalizable even when some eigenvalues are repeated.

Theorem 3.69. Gershgorin Circle Theorem
Let $\mathbf{A} = (a_{ij}) \in \mathbb{C}^{n\times n}$. Every eigenvalue of $\mathbf{A}$ lies in at least one disk

$$\left\{ z \in \mathbb{C} \mid |z - a_{ii}| \le \sum_{j\ne i} |a_{ij}| \right\}.$$

Theorem 3.70. Perron–Frobenius Theorem, Positive-Matrix Form
If every entry of a real square matrix $\mathbf{A}$ is positive, then $\rho(\mathbf{A})$ is a positive simple eigenvalue with a positive eigenvector, and every other eigenvalue λ satisfies

$$|\lambda| < \rho(\mathbf{A}).$$

Theorem 3.71. Spectral Theorem for Real Symmetric Matrices
If $\mathbf{A} \in \mathbb{R}^{n\times n}$ is symmetric, then all eigenvalues of $\mathbf{A}$ are real, and there exists an orthogonal matrix $\mathbf{Q}$ and a diagonal matrix $\mathbf{D}$ s.t.

$$\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}}.$$

Equivalently, $\mathbb{R}^n$ has an orthonormal basis consisting of eigenvectors of $\mathbf{A}$.

Remark. A real symmetric matrix has real eigenvalues and admits orthogonal diagonalization.

Definition 3.72. Normal Matrix
A complex square matrix $\mathbf{A}$ is normal if

$$\mathbf{A}^{\mathsf{H}}\mathbf{A} = \mathbf{A}\mathbf{A}^{\mathsf{H}}.$$

Hermitian, skew-Hermitian, and unitary matrices are normal.

Theorem 3.73. Spectral Theorem for Normal Matrices
A complex square matrix is normal if and only if it is unitarily diagonalizable. Equivalently, $\mathbf{A}$ is normal if and only if there exist a unitary matrix $\mathbf{U}$ and a diagonal matrix $\mathbf{D}$ s.t.

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^{\mathsf{H}}.$$

Proposition 3.74. Hermitian Matrix Quick Facts
If $\mathbf{A}$ *is Hermitian, then*

$$\mathbf{x}^{\mathsf{H}}\mathbf{A}\mathbf{x} \in \mathbb{R}$$

for every $\mathbf{x} \in \mathbb{C}^n$*. All eigenvalues of* $\mathbf{A}$ *are real, and eigenvectors corresponding to distinct eigenvalues are orthogonal.*

Theorem 3.75. Spectral Theorem for Hermitian Matrices
If $\mathbf{A} \in \mathbb{C}^{n\times n}$ *is Hermitian, then there exists a unitary matrix* $\mathbf{U}$ *and a real diagonal matrix* $\mathbf{D}$ *s.t.*

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^{\mathsf{H}}.$$

Equivalently, $\mathbb{C}^n$ *has an orthonormal basis consisting of eigenvectors of* $\mathbf{A}$*.*

Remark. The real analogue of a Hermitian matrix is a real symmetric matrix. The real analogue of a unitary matrix is an orthogonal matrix.

Definition 3.76. Markov Matrix; Stochastic Matrix
A column-stochastic matrix is a square matrix with nonnegative entries whose columns each sum to 1. Thus

$$\mathbf{1}^{\mathsf{T}}\mathbf{A} = \mathbf{1}^{\mathsf{T}}.$$

Consequently, 1 is an eigenvalue of $\mathbf{A}^{\mathsf{T}}$ and therefore also of $\mathbf{A}$. A probability column vector evolves by

$$\mathbf{p}_{k+1} = \mathbf{A}\mathbf{p}_k.$$

A stationary distribution is a probability vector satisfying $\mathbf{A}\mathbf{p} = \mathbf{p}$.

Remark. Markov matrices are linear algebra objects behind finite-state Markov chains. A stationary distribution is an eigenvector corresponding to the eigenvalue 1.

Theorem 3.77. Cauchy–Schwarz Inequality
For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$*, we have*

$$|\langle \mathbf{u}, \mathbf{v}\rangle| \le \|\mathbf{u}\|\,\|\mathbf{v}\|.$$

Equivalently, for real numbers $a_1, \ldots, a_n$ *and* $b_1, \ldots, b_n$*,*

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \le \left(\sum_{i=1}^n a_i^2\right)\left(\sum_{i=1}^n b_i^2\right).$$

Strategy. The expression $\sum a_i b_i$ is a strong signal for Cauchy–Schwarz. The vector form is often the most efficient method to recognize the inequality.

Corollary 3.78. Triangle Inequality
For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$*, we have*

$$\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Proposition 3.79. Properties of Orthogonal Matrices
If $\mathbf{Q}$ *is orthogonal, then*

$$\|\mathbf{Q}\mathbf{v}\| = \|\mathbf{v}\|, \qquad \langle \mathbf{Q}\mathbf{u}, \mathbf{Q}\mathbf{v}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle, \qquad \det(\mathbf{Q}) = \pm 1.$$

Definition 3.80. Positive Definite Matrix
A Hermitian matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$ is positive definite if

$$\mathbf{x}^{\mathsf{H}}\mathbf{A}\mathbf{x} > 0$$

for every nonzero $\mathbf{x} \in \mathbb{C}^n$. Over $\mathbb{R}$, this is the familiar condition that $\mathbf{A}$ is symmetric and

$$\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} > 0$$

for every nonzero $\mathbf{x} \in \mathbb{R}^n$.

Definition 3.81. Singular Values
Let $\mathbf{A} \in \mathbb{C}^{m\times n}$. The singular values of $\mathbf{A}$ are the nonnegative square roots of the eigenvalues of

$$\mathbf{A}^{\mathsf{H}}\mathbf{A}.$$

For real matrices, $\mathbf{A}^{\mathsf{H}} = \mathbf{A}^{\mathsf{T}}$.

Formula 3.82. Matrix Decomposition Reference Table
The most useful matrix decompositions are summarized by their hypotheses and the computation they simplify.

Decomposition	Form	Typical condition	Main use
Rank factorization	$\mathbf{A} = \mathbf{UV}$	$\mathrm{rk}(\mathbf{A}) = r$, with $\mathbf{U} \in F^{m\times r}$, $\mathbf{V} \in F^{r\times n}$ of rank r	expose rank and column/row spaces
LU	$\mathbf{A} = \mathbf{LU}$	elimination without row exchanges; with pivoting, $\mathbf{PA} = \mathbf{LU}$	repeated linear solves, determinants
QR	$\mathbf{A} = \mathbf{QR}$	full column rank, with orthonormal columns of $\mathbf{Q}$	least squares, orthogonalization
Cholesky	$\mathbf{A} = \mathbf{LL}^{\mathsf{H}}$	Hermitian positive definite; over $\mathbb{R}$, this is $\mathbf{LL}^{\mathsf{T}}$	fast stable linear solves
Diagonalization	$\mathbf{A} = \mathbf{PDP}^{-1}$	a basis of eigenvectors	powers and functions of $\mathbf{A}$
Orthogonal spectral	$\mathbf{A} = \mathbf{QDQ}^{\mathsf{T}}$	real symmetric	real orthonormal eigenbasis
Unitary spectral	$\mathbf{A} = \mathbf{\Omega\Lambda\Omega}^{\mathsf{H}}$	complex normal	orthonormal eigenbasis over $\mathbb{C}$
Schur	$\mathbf{A} = \mathbf{QTQ}^{\mathsf{H}}$	every complex square matrix	triangularize without assuming diagonalizability
SVD	$\mathbf{A} = \mathbf{U\Sigma V}^{\mathsf{H}}$	every real or complex matrix	rank, least squares, conditioning, best low-rank approximation

For real matrices, $\mathbf{V}^{\mathsf{H}} = \mathbf{V}^{\mathsf{T}}$. The displayed letters are matrices and therefore follow the book's bold-uppercase convention.

Algorithm 3.83. Gaussian Elimination
Gaussian elimination uses elementary row operations to transform a matrix into an upper triangular or row echelon form. It is the standard computational method for solving linear systems, finding ranks, and computing inverses.

Theorem 3.84. LU Decomposition
When Gaussian elimination can be performed without row exchanges, a square matrix $\mathbf{A}$ *can be written as*

$$\mathbf{A} = \mathbf{LU},$$

where $\mathbf{L}$ *is lower triangular and* $\mathbf{U}$ *is upper triangular.*

Theorem 3.85. Gram–Schmidt Process; QR Decomposition
From any linearly independent list

$$\mathbf{v}_1, \ldots, \mathbf{v}_k$$

in an inner product space, one can construct an orthonormal list

$$\mathbf{q}_1, \dots, \mathbf{q}_k$$

with the same span. In matrix form, this leads to the QR decomposition

$$\mathbf{A} = \mathbf{Q}\mathbf{R},$$

where the columns of $\mathbf{Q}$ *are orthonormal and* $\mathbf{R}$ *is upper triangular.*

Remark. Gram–Schmidt is the standard procedure for turning an arbitrary basis into an orthonormal basis.

Formula 3.86. Orthogonal Projection Matrices
For a nonzero vector $\mathbf{v} \in \mathbb{R}^n$,

$$\operatorname{proj}_{\mathbf{v}} \mathbf{x} = \frac{\mathbf{v}^\mathsf{T}\mathbf{x}}{\mathbf{v}^\mathsf{T}\mathbf{v}}\mathbf{v},$$

and the projection matrix onto $\operatorname{span}(\mathbf{v})$ is

$$\mathbf{P}_{\mathbf{v}} = \frac{\mathbf{v}\mathbf{v}^\mathsf{T}}{\mathbf{v}^\mathsf{T}\mathbf{v}}.$$

If $\|\mathbf{v}\| = 1$, then

$$\mathbf{P}_{\mathbf{v}} = \mathbf{v}\mathbf{v}^\mathsf{T}, \qquad \mathbf{I}_n - \mathbf{v}\mathbf{v}^\mathsf{T}$$

projects onto the hyperplane $\mathbf{v}^\perp$.

If the columns of $\mathbf{Q} \in \mathbb{R}^{n\times k}$ are orthonormal, then the orthogonal projection matrix onto $\operatorname{Col}(\mathbf{Q})$ is

$$\mathbf{P} = \mathbf{Q}\mathbf{Q}^\mathsf{T}.$$

More generally, if the columns of $\mathbf{A} \in \mathbb{R}^{n\times k}$ are linearly independent, then

$$\mathbf{P} = \mathbf{A}(\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}.$$

Over $\mathbb{C}$, replace every transpose by the Hermitian transpose.

Theorem 3.87. Least-Squares Normal Equations
Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ *and* $\mathbf{b} \in \mathbb{R}^m$. *A vector* $\widehat{\mathbf{x}}$ *minimizes*

$$\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$$

if and only if the residual is orthogonal to $\operatorname{Col}(\mathbf{A})$, *equivalently,*

$$\mathbf{A}^\mathsf{T}\mathbf{A}\widehat{\mathbf{x}} = \mathbf{A}^\mathsf{T}\mathbf{b}.$$

If the columns of $\mathbf{A}$ *are linearly independent, then the minimizer is unique and*

$$\widehat{\mathbf{x}} = (\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}\mathbf{b}.$$

The fitted vector is the orthogonal projection

$$\widehat{\mathbf{b}} = \mathbf{A}\widehat{\mathbf{x}} = \mathbf{P}\mathbf{b}, \qquad \mathbf{P} = \mathbf{A}(\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}.$$

Proposition 3.88. Projection Matrix Criterion
A matrix $\mathbf{P}$ *is a projection matrix if*

$$\mathbf{P}^2 = \mathbf{P}.$$

Over $\mathbb{R}$, *it is an orthogonal projection matrix if, in addition,*

$$\mathbf{P}^\mathsf{T} = \mathbf{P}.$$

Over $\mathbb{C}$, *the corresponding condition is*

$$\mathbf{P}^\mathsf{H} = \mathbf{P}.$$

Theorem 3.89. Sylvester's Criterion
A real symmetric matrix is positive definite if and only if all of its leading principal minors are positive.

Theorem 3.90. Singular Value Decomposition; SVD
Every real or complex matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$ can be written as

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{H}},$$

where $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices and $\mathbf{\Sigma}$ is a rectangular diagonal matrix whose diagonal entries are the singular values of $\mathbf{A}$. For real $\mathbf{A}$, the factors may be chosen real and orthogonal, so $\mathbf{V}^{\mathsf{H}} = \mathbf{V}^{\mathsf{T}}$.

Remark. Singular Value Decomposition applies to every real or complex matrix, not only to square or diagonalizable matrices. This is why SVD is one of the most useful decompositions in applied linear algebra.

Definition 3.91. Generalized Eigenvectors and Jordan Chains
Let λ be an eigenvalue of a square matrix $\mathbf{A}$. A nonzero vector $\mathbf{v}$ is a generalized eigenvector associated with λ if

$$(\mathbf{A} - \lambda\mathbf{I}_n)^k\mathbf{v} = \mathbf{0}$$

for some positive integer k. A Jordan chain $\mathbf{v}_1, \dots, \mathbf{v}_r$ satisfies

$$(\mathbf{A} - \lambda\mathbf{I}_n)\mathbf{v}_1 = \mathbf{0}, \qquad (\mathbf{A} - \lambda\mathbf{I}_n)\mathbf{v}_{j+1} = \mathbf{v}_j.$$

Ordinary eigenvectors are precisely the generalized eigenvectors of rank 1.

Theorem 3.92. Jordan Canonical Form
Over the complex numbers, every square matrix is similar to a Jordan matrix. That is, for every $\mathbf{A} \in \mathbb{C}^{n\times n}$, there exists an invertible matrix $\mathbf{P}$ s.t.

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$$

is in Jordan canonical form.

Remark. Jordan canonical form describes the structure that remains when a matrix cannot be diagonalized.

Chapter 4

Differential Equations

Related POSTECH Courses: (MATH200) Differential Equations
Related Books: Elementary Differential Equations and Boundary Value Problems (William E. Boyce, Richard C. DiPrima, Douglas B. Meade)

Definition 4.1. Differential Equation, Order, and Initial Value Problem
An ordinary differential equation (ODE) relates an unknown function of one independent variable to its derivatives. Its order is the highest derivative that occurs. An nth-order linear ODE has the form

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)y' + a_0(t)y = g(t),$$

with a_n not identically zero. It is homogeneous when $g = 0$. An initial value problem (IVP) supplements the equation with enough values such as $y(t_0), y'(t_0), \ldots$ to select a particular solution.

Formula 4.2. First-Order ODE Recognition Table
Before calculating, identify the structural form.

Type	Equation	Reduction or solution
Separable	$y' = g(t)h(y)$	$\displaystyle\int \frac{\mathrm{d}y}{h(y)} = \int g(t)\,\mathrm{d}t + C$ on intervals where $h(y) \neq 0$; also check equilibrium solutions $h(y) = 0$
Linear	$y' + p(t)y = q(t)$	integrating factor $\mu(t) = e^{\int p(t)\,\mathrm{d}t}$
Exact	$M(t,y)\,\mathrm{d}t + N(t,y)\,\mathrm{d}y = 0$	find F with $F_t = M$, $F_y = N$, then $F(t,y) = C$
Bernoulli	$y' + p(t)y = q(t)y^n$, $n \neq 0, 1$	substitute $u = y^{1-n}$ to obtain a linear equation
Autonomous	$y' = f(y)$	equilibria satisfy $f(y) = 0$; away from equilibria the equation is separable

Theorem 4.3. Picard–Lindelöf Existence and Uniqueness Theorem
Consider the initial value problem

$$y' = f(t,y), \qquad y(t_0) = y_0.$$

If f and $\partial f/\partial y$ are continuous near (t_0, y_0), then there exists a unique solution near $t = t_0$.

Remark. The decisive conclusion is existence and uniqueness. It tells us when an initial condition determines exactly one solution.

Theorem 4.4. Superposition Principle
If $y_1, \ldots, y_k$ are solutions of a homogeneous linear differential equation, then any linear combination

$$c_1 y_1 + \cdots + c_k y_k$$

is also a solution.

Remark. The Superposition Principle is the differential-equation version of linear algebra. It is one of the main reasons linear equations are much easier to handle than nonlinear equations.

Proposition 4.5. General Solution of a Nonhomogeneous Linear Equation
For a nonhomogeneous linear differential equation, the general solution has the form

$$y = y_h + y_p,$$

where y_h is the general solution of the associated homogeneous equation and y_p is one particular solution of the nonhomogeneous equation.

Formula 4.6. Separable First-Order Equation
A first-order equation of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = g(x)h(y)$$

is separable. When $h(y) \neq 0$, rewrite it as

$$\frac{1}{h(y)}\,\mathrm{d}y = g(x)\,\mathrm{d}x$$

and integrate both sides.

Formula 4.7. First-Order Linear Equation; Integrating Factor
The equation

$$y' + p(t)y = q(t)$$

has integrating factor

$$\mu(t) = e^{\int p(t)\,\mathrm{d}t}.$$

Then

$$(\mu y)' = \mu q,$$

so

$$y(t) = \frac{1}{\mu(t)}\left(\int \mu(t)q(t)\,\mathrm{d}t + C\right).$$

Formula 4.8. Exact Differential Equation
On a domain $D \subseteq \mathbb{R}^2$, an equation

$$M(x,y)\,\mathrm{d}x + N(x,y)\,\mathrm{d}y = 0$$

is exact if there is a potential function $F \in C^1(D)$ s.t.

$$F_x = M, \qquad F_y = N.$$

If $M, N \in C^1(D)$ and D is open and simply connected, then the equation is exact if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

In that case, the solution is given implicitly by

$$F(x,y) = C.$$

Formula 4.9. Bernoulli Equation
The Bernoulli equation

$$y' + p(t)y = q(t)y^n \qquad (n \neq 0, 1)$$

becomes a first-order linear equation after the substitution

$$u = y^{1-n}.$$

Tactic. For a first-order ODE, first check whether it is separable, linear, exact, or Bernoulli. These four patterns cover many quick undergraduate differential-equation questions.

Formula 4.10. Second-Order Constant-Coefficient Equation
Consider the homogeneous equation

$$ay'' + by' + cy = 0, \qquad a \neq 0.$$

Its characteristic equation is

$$ar^2 + br + c = 0.$$

If the roots are distinct real numbers r_1, r_2, then

$$y_h = c_1 e^{r_1 t} + c_2 e^{r_2 t}.$$

If the equation has a repeated real root r, then

$$y_h = (c_1 + c_2 t)e^{rt}.$$

If the roots are $\alpha \pm i\beta$ with $\beta \neq 0$, then

$$y_h = e^{\alpha t}\left(c_1 \cos \beta t + c_2 \sin \beta t\right).$$

Remark. The characteristic equation converts a constant-coefficient linear ODE into an algebraic equation. This is the key trick.

Definition 4.11. Wronskian
For two differentiable functions y_1 and y_2, their Wronskian is

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} = y_1(t)y_2'(t) - y_1'(t)y_2(t).$$

Proposition 4.12. Wronskian and Linear Independence
If

$$W(y_1, y_2)(t_0) \neq 0$$

for some t_0, then y_1 and y_2 are linearly independent.

Formula 4.13. Abel's Identity
For

$$y'' + p(t)y' + q(t)y = 0,$$

the Wronskian of two solutions satisfies

$$W(t) = Ce^{-\int p(t)\,\mathrm{d}t}.$$

In particular, the Wronskian is either always zero or never zero on an interval where p is continuous.

Formula 4.14. Euler–Cauchy Equation
On an interval with $x > 0$, the equation

$$ax^2y'' + bxy' + cy = 0, \qquad a \neq 0,$$

suggests the trial solution $y = x^r$, giving the indicial equation

$$ar(r-1) + br + c = 0.$$

If its roots are distinct real numbers r_1, r_2, then

$$y = c_1x^{r_1} + c_2x^{r_2}.$$

For a repeated root r,

$$y = x^r(c_1 + c_2 \ln x).$$

For roots $\alpha \pm i\beta$ with $\beta \neq 0$,

$$y = x^\alpha\big(c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)\big).$$

On intervals with $x < 0$, the analogous real formulas use $|x|$.

Algorithm 4.15. Undetermined Coefficients
For a constant-coefficient equation with forcing term made from polynomials, exponentials, sines, or cosines, guess a particular solution of the same type. If the guess overlaps with y_h, multiply the guess by the smallest power of t that removes the overlap.

Formula 4.16. Variation of Parameters
For

$$y'' + p(t)y' + q(t)y = g(t)$$

with fundamental solutions y_1, y_2 of the homogeneous equation, a particular solution is

$$y_p = -y_1 \int \frac{y_2 g}{W}\, \mathrm{d}t + y_2 \int \frac{y_1 g}{W}\, \mathrm{d}t,$$

where

$$W = y_1y_2' - y_1'y_2.$$

Formula 4.17. Exponential Growth and Decay
The equation

$$y' = ky$$

has solution

$$y(t) = Ce^{kt}.$$

If $k > 0$, this models exponential growth. If $k < 0$, this models exponential decay.

Formula 4.18. Logistic Equation
The logistic equation is

$$y' = ky\left(1 - \frac{y}{K}\right), \qquad k > 0, \quad K > 0.$$

Its equilibrium solutions are $y = 0$ and $y = K$. Every non-equilibrium solution on an interval on which it is defined has the form

$$y(t) = \frac{K}{1 + Ce^{-kt}},$$

where C is determined by the initial condition. For $0 < y(0) < K$, one has $C > 0$ and $y(t) \to K$ as $t \to \infty$.

Remark. The logistic equation is the standard model for population growth with limited resources. The parameter K represents the limiting population.

Formula 4.19. Simple Harmonic Oscillator
The equation

$$x'' + \omega^2 x = 0$$

has general solution

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t.$$

Equivalently,

$$x(t) = A \cos(\omega t - \delta)$$

for suitable constants A and δ.

Remark. This is the equation behind ideal mass–spring motion and many small oscillation models in physics.

Formula 4.20. Damped Harmonic Oscillator
For $m > 0$, $c \geq 0$, and $k > 0$, the equation

$$mx'' + cx' + kx = 0$$

has characteristic equation $mr^2 + cr + k = 0$. Put

$$\alpha = \frac{c}{2m}.$$

The three regimes and solution forms are

$$c^2 < 4mk: \quad x(t) = e^{-\alpha t}\big(c_1 \cos(\omega_d t) + c_2 \sin(\omega_d t)\big), \quad \omega_d = \sqrt{\frac{k}{m} - \alpha^2},$$

$$c^2 = 4mk: \quad x(t) = (c_1 + c_2 t)e^{-\alpha t},$$

and, if $c^2 > 4mk$,

$$x(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}, \qquad r_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}.$$

These are called underdamped, critically damped, and overdamped, respectively.

Definition 4.21. Laplace Transform
The Laplace transform of a function $f(t)$ is

$$\mathcal{L}\{f(t)\}(s) = \int_0^\infty e^{-st} f(t)\,\mathrm{d}t.$$

Formula 4.22. Laplace Transform Table
Let $F(s) = \mathcal{L}\{f(t)\}(s)$. The basic transforms are

$f(t)$	$\mathcal{L}\{f(t)\}(s)$	Conditions
1	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
t^n	$\frac{n!}{s^{n+1}}$	$n \in \mathbb{N}_0$
e^{at}	$\frac{1}{s-a}$	$\mathrm{Re}(s) > a$
$t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}$	$n \in \mathbb{N}_0,\ s > a$

$f(t)$	$\mathcal{L}\{f(t)\}(s)$	Conditions
$\sin at$	$\dfrac{a}{s^2+a^2}$	$\mathrm{Re}(s) > 0$
$\cos at$	$\dfrac{s}{s^2+a^2}$	$\mathrm{Re}(s) > 0$
$e^{at}\sin bt$	$\dfrac{b}{(s-a)^2+b^2}$	$\mathrm{Re}(s) > a$
$e^{at}\cos bt$	$\dfrac{s-a}{(s-a)^2+b^2}$	$\mathrm{Re}(s) > a$
$\sinh at$	$\dfrac{a}{s^2-a^2}$	$\mathrm{Re}(s) > \lvert a\rvert$
$\cosh at$	$\dfrac{s}{s^2-a^2}$	$\mathrm{Re}(s) > \lvert a\rvert$
$f'(t)$	$sF(s) - f(0)$	
$f''(t)$	$s^2F(s) - sf(0) - f'(0)$	
$tf(t)$	$-F'(s)$	
$\int_0^t f(\tau)\,\mathrm{d}\tau$	$\dfrac{F(s)}{s}$	
$u(t-a)f(t-a)$	$e^{-as}F(s)$	$a > 0$
$f(t+T) = f(t)$	$\dfrac{\int_0^T e^{-st}f(t)\,\mathrm{d}t}{1-e^{-sT}}$	$T > 0,\ s > 0$

Proposition 4.23. Laplace Shift and Convolution
If $F(s) = \mathcal{L}\{f(t)\}(s)$, then

$$\mathcal{L}\{e^{at}f(t)\}(s) = F(s-a).$$

If

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\,\mathrm{d}\tau,$$

then

$$\mathcal{L}\{f * g\} = \mathcal{L}\{f\}\mathcal{L}\{g\}.$$

Formula 4.24. Fourier Series
For a piecewise smooth function f on $[-L, L]$,

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right),$$

where

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,\mathrm{d}x, \qquad b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,\mathrm{d}x.$$

If f is even, then $b_n = 0$; if f is odd, then $a_n = 0$. At a jump discontinuity, the series converges to the midpoint of the left and right limits.

Formula 4.25. Classical Fourier Series
For $-\pi < x < \pi$,

$$x = 2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin(nx),$$

$$x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos(nx),$$

and, for $0 < x < \pi$,

$$1 = \frac{4}{\pi}\sum_{k=0}^{\infty}\frac{\sin((2k+1)x)}{2k+1}.$$

Definition 4.26. Fourier Transform
For an integrable function $f : \mathbb{R} \to \mathbb{C}$, define

$$\widehat{f}(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}\,\mathrm{d}t.$$

Under suitable hypotheses, the inversion formula is

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f}(\omega)e^{i\omega t}\,\mathrm{d}\omega.$$

Formula 4.27. Fourier Transform Table
With the convention above, the standard identities are

Function or operation	Fourier transform
$f'(t)$	$i\omega\widehat{f}(\omega)$
$f(t-a)$	$e^{-i\omega a}\widehat{f}(\omega)$
$e^{iat}f(t)$	$\widehat{f}(\omega - a)$
$f(at)$, $a \neq 0$	$\frac{1}{\lvert a\rvert}\widehat{f}\left(\frac{\omega}{a}\right)$
$tf(t)$	$i\frac{\mathrm{d}}{\mathrm{d}\omega}\widehat{f}(\omega)$
$(f * g)(t)$	$\widehat{f}(\omega)\widehat{g}(\omega)$
e^{-at^2}, $a > 0$	$\sqrt{\frac{\pi}{a}}e^{-\omega^2/(4a)}$

Formula 4.28. Linear Systems and Matrix Exponential
The homogeneous linear system

$$\mathbf{x}' = \mathbf{A}\mathbf{x}$$

has solution

$$\mathbf{x}(t) = e^{t\mathbf{A}}\mathbf{x}(0).$$

If $\mathbf{A}$ is diagonalizable as $\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$, then

$$e^{t\mathbf{A}} = \mathbf{S}e^{t\mathbf{D}}\mathbf{S}^{-1}, \qquad e^{t\mathbf{D}} = \mathrm{diag}(e^{\lambda_1 t}, \ldots, e^{\lambda_n t}).$$

Strategy. For a constant-coefficient linear system, the eigenvalues of $\mathbf{A}$ control the qualitative behavior. Negative real parts indicate decay, positive real parts indicate growth, and nonzero imaginary parts indicate rotation or oscillation.

Remark. The Laplace transform turns many differential equations into algebraic equations. This is its main computational advantage.

Definition 4.29. Heat Equation
The heat equation is the partial differential equation

$$u_t = \alpha u_{xx}, \qquad \alpha > 0.$$

It models diffusion, especially the flow of heat in a one-dimensional medium.

Definition 4.30. Wave Equation
The one-dimensional wave equation is

$$u_{tt} = c^2 u_{xx}.$$

It models waves traveling with speed c.

Theorem 4.31. d'Alembert's Formula
Let $c > 0$, let $f \in C^2(\mathbb{R})$, and let $g \in C^1(\mathbb{R})$. The classical solution of

$$u_{tt} = c^2 u_{xx}, \qquad u(x,0) = f(x), \qquad u_t(x,0) = g(x),$$

is

$$u(x,t) = \frac{f(x-ct) + f(x+ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s)\,\mathrm{d}s.$$

Remark. d'Alembert's formula shows that waves move along the characteristic lines $x - ct =$ constant and $x + ct =$ constant.

Definition 4.32. Laplace Equation
Laplace's equation is

$$\Delta u = 0.$$

In two variables, this means

$$u_{xx} + u_{yy} = 0.$$

A function satisfying Laplace's equation is called harmonic.

Definition 4.33. Poisson Equation
Poisson's equation is

$$\Delta u = f.$$

It is an inhomogeneous version of Laplace's equation.

Remark. The heat equation, wave equation, and Laplace equation are three fundamental partial differential equations with distinct physical interpretations.

Chapter 5

Combinatorics and Discrete Mathematics

Related POSTECH Courses: (MATH261) Discrete Mathematics
Related Books: Introductory Combinatorics (Richard A. Brualdi), Invitation to Discrete Mathematics (Jiří Matoušek, Jaroslav Nešetřil)

Definition 5.1. Permutation, Combination, and Multiset Selection
An ordered selection of k distinct objects from n distinct objects is a k-permutation, counted by

$$P(n,k) = \frac{n!}{(n-k)!}.$$

An unordered k-element selection is a k-combination, counted by

$$\binom{n}{k}.$$

If repetition is allowed and order is ignored, the number of size-k multisets from n types is

$$\left(\!\!\binom{n}{k}\!\!\right) = \binom{n+k-1}{k}.$$

These three models should be distinguished before applying a formula.

Formula 5.2. Binomial Theorem; Newton's Binomial Formula
For every nonnegative integer n,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

In particular,

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k.$$

Remark. The coefficients $\binom{n}{k}$ are the entries of Pascal's triangle. This is one of the most recognizable bridges between algebra and counting.

Strategy. Bijection, Double Counting, and Complementary Counting. Before expanding a complicated sum, try to count the same objects in a second way. Bijections, double counting, and counting the complement are among the fastest and most reusable contest methods.

Formula 5.3. Binomial Inversion
The relations
$$b_n = \sum_{k=0}^{n} \binom{n}{k} a_k$$
and
$$a_n = \sum_{k=0}^{n} (-1)^{n-k} \binom{n}{k} b_k$$
are equivalent.

Formula 5.4. Pascal's Identity and Vandermonde's Identity
Pascal's identity is
$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$
Vandermonde's identity is
$$\sum_{k=0}^{r} \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}.$$

Formula 5.5. Multinomial Theorem
For nonnegative integers $k_1, \dots, k_m$ with $k_1 + \cdots + k_m = n$,
$$(x_1 + \cdots + x_m)^n = \sum_{k_1+\cdots+k_m=n} \frac{n!}{k_1! \cdots k_m!} x_1^{k_1} \cdots x_m^{k_m}.$$
The coefficient
$$\frac{n!}{k_1! \cdots k_m!}$$
is called a multinomial coefficient.

Formula 5.6. Stars and Bars
The number of nonnegative integer solutions of
$$x_1 + x_2 + \cdots + x_n = k$$
is
$$\binom{n+k-1}{k} = \left(\!\binom{n}{k}\!\right).$$
Equivalently, the expansion of $(x_1 + x_2 + \cdots + x_n)^k$ has
$$\binom{n+k-1}{k}$$
distinct monomial terms.

Formula 5.7. Lattice Path Counting
The number of shortest lattice paths from $(0, 0)$ to (m, n) using only unit steps to the right and upward is
$$\binom{m+n}{m} = \binom{m+n}{n}.$$

Theorem 5.8. Reflection Principle and Ballot-Path Formula
The reflection principle converts a path that first crosses a boundary into an unrestricted reflected

path. In particular, for integers $a \geq b \geq 0$, the number of paths from $(0,0)$ to (a,b) using right and up steps and never entering the region $y > x$ is

$$\binom{a+b}{b} - \binom{a+b}{b-1} = \frac{a-b+1}{a+1}\binom{a+b}{b}.$$

When $a = b = n$, this becomes the Catalan number

$$\frac{1}{n+1}\binom{2n}{n}.$$

Strategy. Tail Switching for Intersecting Paths. When two directed lattice paths meet, exchange their tails after the first intersection. This produces a bijection with a pair of paths whose endpoints have been crossed. The method is a standard way to count or cancel intersecting path pairs.

Formula 5.9. Ordinary Generating Function Table
For a sequence $(a_n)_{n\geq 0}$, its ordinary generating function is

$$A(x) = \sum_{n=0}^{\infty} a_n x^n.$$

The basic examples are

Sequence a_n	Ordinary generating function $A(x)$	Conditions
1	$\frac{1}{1-x}$	$\lvert x\rvert < 1$
r^n	$\frac{1}{1-rx}$	$\lvert rx\rvert < 1$
n	$\frac{x}{(1-x)^2}$	$\lvert x\rvert < 1$
n^2	$\frac{x(1+x)}{(1-x)^3}$	$\lvert x\rvert < 1$
$\binom{n}{k}$	$\frac{x^k}{(1-x)^{k+1}}$	$k \in \mathbb{N}_0,\ \lvert x\rvert < 1$
$\binom{n+r}{r}$	$\frac{1}{(1-x)^{r+1}}$	$r \in \mathbb{N}_0,\ \lvert x\rvert < 1$
F_n, where $F_0 = 0$ and $F_1 = 1$	$\frac{x}{1-x-x^2}$	$\lvert x\rvert < \frac{\sqrt{5}-1}{2}$
$\frac{1}{n+1}\binom{2n}{n}$	$\frac{1-\sqrt{1-4x}}{2x}$	$\lvert x\rvert < \frac{1}{4}$

Tactic. Generating functions are useful when a recurrence or counting problem has a repeated structure; a known generating function can replace a long derivation.

Formula 5.10. Stirling Numbers and Bell Numbers
The Stirling number of the second kind

$$\left\{ n \atop k \right\}$$

counts partitions of an n-element set into k nonempty blocks and satisfies

$$\left\{ n \atop k \right\} = k\left\{ n-1 \atop k \right\} + \left\{ n-1 \atop k-1 \right\}.$$

The Bell number

$$B_n = \sum_{k=0}^{n} \left\{ n \atop k \right\}$$

counts all set partitions of an n-element set.

Formula 5.11. Fibonacci Numbers; Binet's Formula
If $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$, then

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \qquad \psi = \frac{1-\sqrt{5}}{2}.$$

In particular,

$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n} = \varphi.$$

Formula 5.12. Roots-of-Unity Filter
Let $\omega_m = e^{2\pi i/m}$. Then

$$\frac{1}{m}\sum_{j=0}^{m-1} \omega_m^{j(n-r)} = \begin{cases} 1, & n \equiv r \pmod{m}, \\ 0, & n \not\equiv r \pmod{m}. \end{cases}$$

Consequently, if $A(x) = \sum_{n\ge 0} a_n x^n$, then the terms with indices congruent to r modulo m are extracted by

$$\sum_{n\equiv r \,(\mathrm{mod}\, m)} a_n x^n = \frac{1}{m}\sum_{j=0}^{m-1} \omega_m^{-rj} A(\omega_m^j x).$$

Formula 5.13. Tower of Hanoi
Let T_n be the minimum number of moves needed to move n disks in the Tower of Hanoi puzzle. Then

$$T_n = 2T_{n-1} + 1, \qquad T_1 = 1,$$

and hence

$$T_n = 2^n - 1.$$

Remark. The recurrence comes from moving the top $n-1$ disks away, moving the largest disk once, and then moving the $n-1$ disks back on top of it.

Strategy. State Compression and Transfer Matrices. When a counting process depends only on finitely many recent states, place the transition counts in a matrix $\mathbf{T}$. Counts of length n then have the form

$$\mathbf{u}^{\mathsf{T}}\mathbf{T}^n\mathbf{v}.$$

This is effective for forbidden substrings, tilings, walks, and finite automata.

Theorem 5.14. Pigeonhole Principle; Dirichlet's Box Principle
If $n+1$ objects are placed into n boxes, then at least one box contains at least two objects. More generally, if N objects are placed into k boxes, then at least one box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects.

Strategy. The Pigeonhole Principle is often hidden in problems asking us to prove that two objects must share a property. The hard part is usually choosing the right "boxes."

Theorem 5.15. Consecutive-Sum Divisibility Lemma
For any integers $a_1, \ldots, a_n$, there exists a nonempty consecutive block whose sum is divisible by n. Define prefix sums

$$S_0 = 0, \qquad S_k = \sum_{i=1}^{k} a_i.$$

If some $S_k \equiv 0 \pmod{n}$, use the first k terms. Otherwise, two of $S_1, \ldots, S_n$ have the same residue modulo n, and their difference is the required block sum.

Definition 5.16. Graph Walks, Paths, Trails, Cycles, and Trees
In a graph $G = (V, E)$, a walk is a sequence of adjacent vertices. A trail repeats no edge, while a path repeats no vertex. A closed trail is a circuit, and a cycle is a closed path apart from its repeated initial and terminal vertex. A graph is connected if every two vertices are joined by a path. A tree is a connected graph containing no cycle.

An Eulerian trail uses every edge exactly once; an Eulerian circuit is a closed Eulerian trail. A Hamiltonian path visits every vertex exactly once; a Hamiltonian cycle returns to its starting vertex. Thus the one-stroke drawing problem is an Eulerian, not a Hamiltonian, question.

Strategy. Coloring, Parity, and Bipartite Obstructions. Colorings convert geometric or movement constraints into parity data. On a bipartite graph, every path alternates colors. Thus a Hamiltonian path can exist only if the two color classes differ in size by at most 1, and a Hamiltonian cycle requires the two classes to have equal size. Checkerboard colorings are especially effective for tiling and movement problems.

Theorem 5.17. Inclusion–Exclusion Principle; Sieve Formula
For finite sets $A_1, \dots, A_n$,

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1}|A_1 \cap \cdots \cap A_n|.$$

Remark. Inclusion–Exclusion is the standard correction mechanism for overcounting. It is also historically connected to sieve methods.

Formula 5.18. Derangement Formula and Recurrence
Let D_n be the number of derangements of n objects, that is, permutations with no fixed points. Then

$$D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!} = \left\lfloor \frac{n!}{e} + \frac{1}{2} \right\rfloor.$$

The characteristic recurrence is

$$D_n = (n-1)(D_{n-1} + D_{n-2}) \qquad (n \geq 2),$$

with $D_0 = 1$ and $D_1 = 0$. Equivalently,

$$D_n = nD_{n-1} + (-1)^n.$$

The first recurrence comes from choosing the image of a distinguished object and separating the cases in which a resulting transposition is completed or not.

Remark. The approximation $D_n \approx n!/e$ is very memorable. In probability language, a random permutation has no fixed point with probability approximately $1/e$.

Formula 5.19. Catalan Numbers
The nth Catalan number is

$$C_n = \frac{1}{n+1}\binom{2n}{n}.$$

It begins as

$$1,\ 1,\ 2,\ 5,\ 14,\ 42,\ 132, \dots.$$

Remark. Catalan numbers count many different objects, including correct parenthesizations, Dyck paths, triangulations of a polygon, and rooted full binary trees. The number 42 is a standard recognition item.

Remark. Cayley's formula gives a compact and surprising enumeration of labeled trees.

Theorem 5.20. Burnside's Lemma; Cauchy–Frobenius–Burnside Lemma
Suppose a finite group G acts on a finite set X. The number of orbits is

$$\frac{1}{|G|}\sum_{g\in G}|\operatorname{Fix}(g)|,$$

where

$$\operatorname{Fix}(g)=\{x\in X \mid g\cdot x=x\}.$$

Idea. Burnside's Lemma says that the number of essentially different objects is the average number of objects fixed by a symmetry.

Theorem 5.21. Equivalent Characterizations of a Tree
For a finite graph G with n vertices, any two of the following conditions imply the third, and in fact all three are equivalent:

$$G \textit{ is connected and acyclic},$$

$$G \textit{ is connected and } |E|=n-1,$$

and

$$\textit{every two vertices are joined by a unique path.}$$

Equivalently, a tree is minimally connected and maximally acyclic.

Theorem 5.22. Bipartite Graph Criterion
A graph is bipartite if and only if it contains no odd cycle. Therefore a two-coloring obstruction and an odd-cycle obstruction are the same phenomenon.

Lemma 5.23. Handshaking Lemma
For a finite undirected graph $G=(V,E)$,

$$\sum_{v\in V}\deg(v)=2|E|.$$

In particular, the number of vertices of odd degree is even.

Remark. The name comes from the idea that every handshake contributes 2 to the total count of hands shaken.

Theorem 5.24. Euler's Formula for Planar Graphs
If a connected planar graph has V vertices, E edges, and F faces, then

$$V-E+F=2.$$

Remark. This is the graph-theoretic version of Euler's polyhedron formula. It is one of the most famous formulas involving planar graphs.

Theorem 5.25. Kuratowski's Theorem
A finite graph is planar if and only if it contains no subgraph that is a subdivision of K_5 or $K_{3,3}$. Thus the two classical complete graphs K_5 and $K_{3,3}$ are the fundamental obstructions to planarity in the sense of subdivisions.

Formula 5.26. Regions Cut by Hyperplanes in General Position
The maximum number of regions into which n hyperplanes in general position divide d-dimensional space is

$$R_d(n)=\sum_{k=0}^{d}\binom{n}{k}.$$

In particular, n lines divide the plane into at most

$$1 + n + \binom{n}{2}$$

regions, and n planes divide $\mathbb{R}^3$ into at most

$$1 + n + \binom{n}{2} + \binom{n}{3}$$

regions.

Formula 5.27. Chords and Diagonals in Convex Position
If n points lie on a circle and no three interior chords are concurrent, drawing every chord divides the disk into

$$1 + \binom{n}{2} + \binom{n}{4}$$

regions. A convex n-gon has at most

$$\binom{n}{4}$$

interior intersection points of diagonals.

Formula 5.28. Triangles in a Complete Diagonal Drawing
For n vertices in convex position with no three diagonals concurrent, the number of triangles whose sides lie on sides or diagonals of the complete drawing is

$$\binom{n}{3} + 4\binom{n}{4} + 5\binom{n}{5} + \binom{n}{6}.$$

Formula 5.29. Triangulation with Interior Vertices
Suppose a polygonal disk is triangulated using V vertices in total, of which b lie on the boundary. Then the number of edges and triangular faces are

$$E = 3V - b - 3, \qquad F_{\triangle} = 2V - b - 2.$$

These values follow from Euler's formula and double counting the triangle-edge incidences.

Strategy. Double Counting with Euler's Formula. In a planar subdivision, count edge-face incidences first and then use $V - E + F = 2$. Interior edges are counted twice and boundary edges once. This is often faster and safer than trying to enumerate faces directly.

Theorem 5.30. Eulerian Trail Criterion
A connected finite graph has an Eulerian circuit if and only if every vertex has even degree. It has an Eulerian trail but not an Eulerian circuit if and only if exactly two vertices have odd degree.

Remark. This theorem is historically connected to Euler's solution of the Seven Bridges of Königsberg problem, which marks the beginning of graph theory and topology. The historical city Königsberg is now Kaliningrad, Russia.

Theorem 5.31. Hall's Marriage Theorem
Let $G = (X \cup Y, E)$ be a finite bipartite graph. There is a matching that covers every vertex of X if and only if, for every subset $S \subseteq X$,

$$|N(S)| \geq |S|,$$

where $N(S)$ is the set of neighbors of vertices in S.

Strategy. Hall's Marriage Theorem is the standard named theorem for proving that a perfect assignment or matching exists.

Theorem 5.32. Kőnig's Theorem
In every finite bipartite graph, the size of a maximum matching equals the size of a minimum vertex cover.

Theorem 5.33. Max-Flow Min-Cut Theorem
In a finite capacitated network, the maximum value of a source–sink flow equals the minimum capacity of a source–sink cut.

Theorem 5.34. Menger's Theorem, Vertex Form
In a finite graph, for nonadjacent vertices u and v, the maximum number of pairwise internally vertex-disjoint u–v paths equals the minimum number of vertices whose deletion separates u from v.

Theorem 5.35. Kirchhoff's Matrix–Tree Theorem
Let

$$\mathbf{L} = \mathbf{D} - \mathbf{A}$$

be the Laplacian matrix of a finite graph. The number of spanning trees equals any cofactor of $\mathbf{L}$ *obtained by deleting the same row and column.*

Theorem 5.36. Sperner's Theorem
The largest family of subsets of $[n]$ in which no member contains another has size

$$\binom{n}{\lfloor n/2 \rfloor}.$$

Theorem 5.37. Dilworth's Theorem
In a finite partially ordered set, the maximum size of an antichain equals the minimum number of chains covering the poset.

Theorem 5.38. Turán's Theorem
Among all n-vertex graphs containing no K_{r+1}, the maximum number of edges is attained by the complete r-partite graph whose part sizes differ by at most 1.

Theorem 5.39. Dirac's and Ore's Hamiltonian Criteria
Let G be a simple graph on $n \geq 3$ vertices. If every vertex has degree at least $n/2$, then G is Hamiltonian. More generally, it is enough that

$$\deg(u) + \deg(v) \geq n$$

for every pair of nonadjacent vertices u, v.

Theorem 5.40. Ramsey's Theorem
For any positive integers r and s, there exists an integer $R(r, s)$ s.t. every red-blue coloring of the edges of a sufficiently large complete graph contains either a red K_r or a blue K_s.

Formula 5.41. The Party Problem
The classical Ramsey number

$$R(3, 3) = 6$$

says that among any six people, there are either three mutual acquaintances or three mutual strangers.

Remark. Ramsey theory is the mathematics of unavoidable structure. Its slogan is that complete disorder is impossible in a large enough system.

Theorem 5.42. Erdős–Szekeres Theorem
Any sequence of

$$(r-1)(s-1)+1$$

distinct real numbers contains either an increasing subsequence of length r or a decreasing subsequence of length s.

Remark. This is a beautiful pigeonhole-style result. It is often remembered as the monotone subsequence theorem.

Chapter 6

Probability and Statistics

Related POSTECH Courses: (MATH230) Probability & Statistics
Related Books: Probability & Statistics for Engineers & Scientists (Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers, Keying Ye)

Definition 6.1. Probability Space, Event, and Conditional Probability
A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the sample space, $\mathcal{F}$ is a σ-algebra of events, and $\mathbb{P} : \mathcal{F} \to [0,1]$ satisfies

$$\mathbb{P}(\Omega) = 1, \qquad \mathbb{P}\left(\bigsqcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

for pairwise disjoint events $A_i \in \mathcal{F}$. If $\mathbb{P}(B) > 0$, the conditional probability of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Definition 6.2. Random Variable, Distribution, and Distribution Function
A random variable is a measurable function $X : \Omega \to \mathbb{R}$. Its distribution is the probability law of its values. For a discrete random variable, the probability mass function is

$$p_X(x) = \mathbb{P}(X = x), \qquad \sum_x p_X(x) = 1.$$

For a continuous random variable with density f_X,

$$\mathbb{P}(a < X \le b) = \int_a^b f_X(x)\,\mathrm{d}x, \qquad \int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1.$$

In either case the cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \le x).$$

Definition 6.3. Expectation and Variance
If the relevant sum or integral converges absolutely, then

$$\mathbb{E}[X] = \begin{cases} \displaystyle\sum_x x\,\mathbb{P}(X = x), & X \text{ discrete}, \\ \displaystyle\int_{-\infty}^{\infty} x f_X(x)\,\mathrm{d}x, & X \text{ continuous}, \end{cases}$$

and, whenever the second moment is finite,

$$\mathrm{Var}(X) = \mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big].$$

Theorem 6.4. Law of Total Probability
If $B_1, \dots, B_n$ form a partition of the sample space and $\mathbb{P}(B_i) > 0$ for all i, then

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Theorem 6.5. Law of Total Expectation; Tower Property
If X is integrable and Y is a random variable, then

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]].$$

When Y is discrete, this becomes

$$\mathbb{E}[X] = \sum_{y} \mathbb{E}[X \mid Y = y]\mathbb{P}(Y = y),$$

where the sum is taken over values y with positive probability.

Formula 6.6. Law of Total Variance
If X has finite second moment, then

$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(\mathbb{E}[X \mid Y]).$$

Theorem 6.7. Bayes' Theorem
If $B_1, \dots, B_n$ form a partition of the sample space and $\mathbb{P}(A) > 0$, then

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_{i=1}^{n} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

In the two-event form,

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

Remark. Bayes' Theorem is the standard formula for reversing conditional probability. It updates a prior probability after observing evidence.

Remark. In general,

$$\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A).$$

The two quantities reverse the roles of evidence and hypothesis. Bayes' Theorem relates them; it does not permit the conditioning bar to be reversed without the corresponding prior probabilities.

Definition 6.8. Independence
Events A and B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Equivalently, if $\mathbb{P}(B) > 0$, then

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

Random variables X and Y are independent if events determined by X and events determined by Y are independent. A family of events is mutually independent if every finite subfamily has intersection probability equal to the product of its individual probabilities. In general,

$$\text{pairwise independence} \not\Rightarrow \text{mutual independence}.$$

Definition 6.9. Conditional Distribution
For discrete random variables X and Y, whenever $\mathbb{P}(Y = y) > 0$,

$$\mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}.$$

If (X, Y) has joint density $f_{X,Y}$ and Y has marginal density f_Y, then, whenever $f_Y(y) > 0$, the conditional density of X given $Y = y$ is

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

Conditional mass and density functions sum or integrate to 1 in the variable being conditioned.

Tactic. "Disjoint" and "independent" are different. If two nonempty events are disjoint, then knowing that one occurred makes the other impossible; they are usually not independent.

Formula 6.10. Complement Rule and Inclusion–Exclusion for Probability
The complement rule is

$$\mathbb{P}(A^{\mathsf{c}}) = 1 - \mathbb{P}(A).$$

For two events,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

For three events,

$$\begin{aligned}\mathbb{P}(A \cup B \cup C) = {} & \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) \\ & - \mathbb{P}(A \cap B) - \mathbb{P}(B \cap C) - \mathbb{P}(C \cap A) \\ & + \mathbb{P}(A \cap B \cap C).\end{aligned}$$

Proposition 6.11. Linearity of Expectation
If $X_1, \ldots, X_n$ are integrable random variables and $a_1, \ldots, a_n$ are scalars, then

$$\mathbb{E}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i].$$

Independence is not required.

Strategy. Linearity of expectation is often the most efficient method for counting an expected number of objects. Introduce indicator random variables and add their expectations.

Proposition 6.12. Expectation of a Product of Independent Random Variables
If $X_1, \ldots, X_n$ are independent integrable random variables, then their product is integrable and

$$\mathbb{E}[X_1 \cdots X_n] = \mathbb{E}[X_1] \cdots \mathbb{E}[X_n].$$

In particular, if they are also identically distributed, the right-hand side is $(\mathbb{E}[X_1])^n$.

Formula 6.13. Common Probability Distributions
The most frequently used distributions are summarized below. Unless otherwise indicated, $0 < p < 1$, $n, r \in \mathbb{N}$, $\lambda > 0$, $\alpha, \beta > 0$, $a < b$, $\sigma > 0$, and $\nu \in \mathbb{N}$. For the negative binomial row, X is the trial number on which the rth success occurs; the Gamma distribution uses the rate parameter λ.

Distribution	Mass or density function	Mean	Variance
Bernoulli(p)	$\mathbb{P}(X = x) = p^x(1-p)^{1-x}$, $x \in \{0, 1\}$	p	$p(1-p)$

Distribution	Mass or density function	Mean	Variance
Binomial(n, p)	$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
Geometric(p)	$\mathbb{P}(X = k) = (1-p)^{k-1} p$, $k \geq 1$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative binomial(r, p)	$\mathbb{P}(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$, $k \geq r$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$
Poisson(λ)	$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$	λ	λ
Uniform(a, b)	$f(x) = \frac{1}{b-a}$, $a \leq x \leq b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ)	$f(x) = \lambda e^{-\lambda x}$, $x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Gamma(α, λ)	$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$, $x > 0$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
Beta(α, β)	$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$, $0 < x < 1$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
$\chi^2(\nu)$	$f(x) = \frac{x^{\nu/2-1} e^{-x/2}}{2^{\nu/2}\Gamma(\nu/2)}$, $x > 0$	ν	2ν
Normal(μ, σ^2)	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2

Formula 6.14. Classical Sampling-Distribution Constructions
Let $Z, Z_1, \ldots, Z_\nu \sim N(0, 1)$ be independent. Then

$$\sum_{i=1}^{\nu} Z_i^2 \sim \chi^2(\nu).$$

If $U \sim \chi^2(\nu)$ is independent of Z, then

$$\frac{Z}{\sqrt{U/\nu}} \sim t(\nu).$$

If $U_1 \sim \chi^2(d_1)$ and $U_2 \sim \chi^2(d_2)$ are independent, then

$$\frac{U_1/d_1}{U_2/d_2} \sim F(d_1, d_2).$$

These three constructions are the standard recognition rules for the chi-square, Student t, and F distributions.

Strategy. Waiting-until questions usually start with a geometric distribution. If a running sum is also involved, combine the expected number of trials with the expected contribution per trial, or condition on the first step.

Formula 6.15. Variance, Covariance, and Correlation
For random variables with finite second moments,

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$

$$\operatorname{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y],$$

and

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y).$$

If $\operatorname{Var}(X) > 0$ and $\operatorname{Var}(Y) > 0$, their correlation coefficient is

$$\rho_{X,Y} = \frac{\operatorname{Cov}(X, Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}}, \qquad -1 \leq \rho_{X,Y} \leq 1.$$

If X and Y are independent, then $\mathrm{Cov}(X,Y)=0$, so their variances add. In general,

$$\mathrm{Cov}(X,Y)=0 \not\Rightarrow X \text{ and } Y \text{ are independent.}$$

Formula 6.16. Hypergeometric Distribution
Suppose a population has N objects, of which K are successes. If n objects are chosen without replacement and X is the number of successes chosen, then

$$\mathbb{P}(X=k)=\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}.$$

Its mean and variance are

$$\mathbb{E}[X]=n\frac{K}{N},$$
$$\mathrm{Var}(X)=n\frac{K}{N}\left(1-\frac{K}{N}\right)\frac{N-n}{N-1}.$$

The final factor is the finite-population correction caused by sampling without replacement.

Theorem 6.17. Poisson Limit Theorem; Law of Small Numbers
Let $p_n\in[0,1]$ satisfy $np_n\to\lambda>0$. For each fixed nonnegative integer k,

$$\binom{n}{k}p_n^k(1-p_n)^{n-k}\longrightarrow e^{-\lambda}\frac{\lambda^k}{k!}.$$

Thus $\mathrm{Bin}(n,p_n)$ *converges in distribution to* $\mathrm{Poi}(\lambda)$.

Remark. This theorem explains why the Poisson distribution models rare events: many trials, small success probability, and finite expected count.

Formula 6.18. Empirical Rule; 68–95–99.7 Rule
For a normal distribution, approximately

$$68\%$$

of the mass lies within one standard deviation of the mean,

$$95\%$$

lies within two standard deviations, and

$$99.7\%$$

lies within three standard deviations.

Definition 6.19. Modes of Convergence for Random Variables
Let X_n and X be random variables on the same probability space. The principal modes of convergence used in probability are:

(i) $X_n \xrightarrow{\text{a.s.}} X$ if
$$\mathbb{P}\big(\{\omega \mid X_n(\omega)\to X(\omega)\}\big)=1;$$

(ii) $X_n \xrightarrow{\mathbb{P}} X$ if
$$\forall\varepsilon>0,\qquad \mathbb{P}\big(|X_n-X|>\varepsilon\big)\to 0;$$

(iii) $X_n \xrightarrow{L^p} X$ for $p\geq 1$ if
$$\mathbb{E}\big[|X_n-X|^p\big]\to 0;$$

(iv) $X_n \xrightarrow{\mathrm{d}} X$ if

$$F_{X_n}(x) \to F_X(x)$$

at every continuity point x of the distribution function F_X.

Almost-sure convergence implies convergence in probability, and convergence in probability implies convergence in distribution.

Theorem 6.20. Borel–Cantelli Lemmas
If

$$\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty,$$

then $\mathbb{P}(A_n \textit{ infinitely often}) = 0$*. If the* A_n *are independent and the series diverges, then*

$$\mathbb{P}(A_n \textit{ infinitely often}) = 1.$$

Theorem 6.21. Weak Law of Large Numbers
Let $X_1, X_2, \ldots$ *be independent and identically distributed random variables with finite mean* μ *and finite variance* σ^2*. Then*

$$\frac{X_1 + \cdots + X_n}{n} \xrightarrow{\mathbb{P}} \mu.$$

This follows directly from Chebyshev's inequality because the variance of the sample mean is σ^2/n*.*

Idea. The Law of Large Numbers says that the sample average stabilizes near the true mean when the number of trials becomes large.

Theorem 6.22. Central Limit Theorem
Let $X_1, X_2, \ldots$ *be independent identically distributed random variables with mean* μ *and variance* $\sigma^2 > 0$*. Then*

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{\mathrm{d}} Z, \qquad Z \sim \mathrm{N}(0, 1).$$

Thus the standardized sum converges in distribution to the standard normal distribution.

Remark. The Central Limit Theorem explains why the normal distribution appears so often in nature and statistics.

Theorem 6.23. Markov's Inequality
If X *is a nonnegative random variable and* $a > 0$*, then*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Theorem 6.24. Chebyshev's Inequality
If X *has finite mean* μ *and finite variance* $\sigma^2 > 0$*, then for every* $k > 0$*,*

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Equivalently, for every $\varepsilon > 0$*,*

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\mathrm{Var}(X)}{\varepsilon^2}.$$

Remark. Markov's inequality uses only the mean, while Chebyshev's inequality uses the variance. Both are probability bounds that work without assuming a normal distribution.

Theorem 6.25. Chernoff Bound for a Binomial Random Variable
If $X \sim \mathrm{Bin}(n,p)$ *and* $\mu = np$*, then for* $\delta > 0$*,*

$$\mathbb{P}(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu}.$$

For $0 < \delta < 1$*,*

$$\mathbb{P}(|X-\mu| \geq \delta\mu) \leq 2e^{-\delta^2\mu/3}.$$

Theorem 6.26. Jensen's Inequality
Let X *be integrable and take values in an interval* I*, and let* $\varphi : I \to \mathbb{R}$ *be convex with* $\varphi(X)$ *integrable. Then*

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

If φ *is concave under the analogous integrability assumptions, the inequality is reversed.*

Idea. Jensen's inequality says that, for a convex function, applying the function after averaging gives a value no larger than averaging after applying the function.

Formula 6.27. Exact Birthday Problem
Under the standard model of 365 equally likely birthdays and no leap day, the probability that at least two people among n share a birthday is

$$1 - \frac{365 \cdot 364 \cdots (365-n+1)}{365^n} \qquad (0 \leq n \leq 365).$$

For $n \geq 366$, the probability is 1 by the pigeonhole principle.

Formula 6.28. Birthday Problem Approximation
In a group of n people, assuming 365 equally likely birthdays, the probability that at least two people share a birthday is approximately

$$1 - \exp\left(-\frac{n(n-1)}{730}\right).$$

For $n = 23$, this probability is already greater than $1/2$.

Remark. The birthday problem is famous because the answer is surprisingly small: only 23 people are needed for a better-than-even chance of a shared birthday.

Formula 6.29. Expected Number of Runs in Coin Tosses
In n tosses of a fair coin, a run is a maximal consecutive block of identical outcomes. If R_n is the number of runs, then

$$\mathbb{E}[R_n] = 1 + \frac{n-1}{2} = \frac{n+1}{2}.$$

Idea. The first toss starts one run. For each of the remaining $n-1$ positions, a new run begins exactly when the current toss differs from the previous toss, which has probability $1/2$.

Formula 6.30. Monty Hall Problem
There are three doors, exactly one of which hides a prize. The contestant first chooses a door. A host who knows the prize location must then open one of the other doors that hides no prize; whenever two such doors are available, the host chooses between them by a rule independent of the prize location. The contestant is then offered the unopened door. Under this protocol, staying wins with probability $1/3$, whereas switching wins with probability $2/3$.

Remark. The host's action is not random information: the host always reveals a losing door. This is why the original $2/3$ probability of having chosen incorrectly transfers to the remaining unopened door.

Formula 6.31. Coupon Collector

If there are n equally likely coupon types, the expected number of trials needed to see all n types is

$$nH_n = n\left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right).$$

Strategy. For a fair die, the expected number of rolls needed to see all six faces is $6H_6$. If a question asks about the order in which the first appearances occur, ignore repeated rolls and look only at the random ordering of the six faces.

Formula 6.32. Gambler's Ruin

Suppose a gambler starts with i dollars and stops when reaching either 0 dollars or N dollars. If the probability of winning each round is p and $q = 1 - p$, then the probability of reaching N before 0 is

$$\frac{1 - (q/p)^i}{1 - (q/p)^N} \qquad (p \neq q).$$

In the fair case $p = q = 1/2$, the probability is

$$\frac{i}{N}.$$

Remark. Gambler's ruin is a classic random walk result. It is a good example where the fair case has a very simple answer.

Chapter 7

Geometry

Synthetic Geometry

Fact 7.1. Euclid's *Elements* and the Axiomatic Method
Euclid's *Elements* consists of thirteen books and organizes geometry and number theory through definitions, common notions, postulates, propositions, and proofs. Book I begins with five postulates; the fifth is the parallel postulate.

Fact 7.2. Euclid's Fifth Postulate and Playfair's Axiom
Euclid's fifth postulate states that if a transversal makes the two interior angles on one side sum to less than two right angles, then the two lines meet on that side when extended. In the usual axiomatic setting for Euclidean plane geometry, this is equivalent to Playfair's axiom: through a point not on a given line, there is exactly one line parallel to the given line.

Definition 7.3. Euclidean, Hyperbolic, and Elliptic Geometry
The three constant-curvature geometries are distinguished by their parallel behavior and triangle angle sums.

Geometry	Lines through a point off a given line	Triangle angle sum
Euclidean	Exactly one parallel	Equal to π
Hyperbolic	Infinitely many disjoint lines	Less than π
Elliptic / spherical	No disjoint geodesic line	Greater than π

Theorem 7.4. Parallel Postulate and Triangle Angle Sum
Within standard neutral geometry, the Euclidean parallel postulate is equivalent to the assertion that every triangle has angle sum

$$\pi.$$

Consequently, the familiar triangle angle-sum theorem is not a consequence of Euclid's first four postulates alone.

Formula 7.5. Geodesic Triangle Angle Sum on Constant Curvature
Let a geodesic triangle lie on a surface of constant Gaussian curvature K. If its angles are α, β, γ, then the Gauss–Bonnet formula gives

$$\alpha + \beta + \gamma - \pi = K\,\mathrm{area}(\triangle ABC).$$

Thus positive curvature produces angle excess, zero curvature gives the Euclidean angle sum, and negative curvature produces angle defect.

Formula 7.6. Basic Plane and Solid Geometry Table
The basic measurement formulas are

Figure	Perimeter or surface area	Area or volume
Triangle	$a+b+c$	$\frac{1}{2}bh$
Parallelogram	$2(a+b)$	bh
Trapezoid	$a+b+c+d$	$\frac{1}{2}(a+c)h$
Regular n-gon	na	$\frac{1}{2}(na)r = \frac{na^2}{4\tan(\pi/n)}$
Circle	$2\pi r$	πr^2
Circular sector	$r\theta$	$\frac{1}{2}r^2\theta$
Rectangular box	$2(ab+bc+ca)$	abc
Right circular cylinder	$2\pi r(r+h)$	$\pi r^2 h$
Right circular cone	$\pi r(r+\ell)$	$\frac{1}{3}\pi r^2 h$
Sphere	$4\pi r^2$	$\frac{4}{3}\pi r^3$

In the regular-polygon row, r is the apothem; in the cone row, $\ell = \sqrt{r^2+h^2}$ is the slant height. Sector angles are measured in radians.

Theorem 7.7. Triangle Angle Sum and Exterior Angle Theorem
For every triangle $\triangle ABC$,

$$\angle A + \angle B + \angle C = \pi.$$

An exterior angle equals the sum of the two nonadjacent interior angles.

Proposition 7.8. Triangle Congruence and Similarity Criteria
Two triangles are congruent under any of the criteria

$$\text{SSS}, \qquad \text{SAS}, \qquad \text{ASA}, \qquad \text{AAS}, \qquad \text{RHS}.$$

They are similar under any of the criteria

$$\text{AA}, \qquad \text{SAS}, \qquad \text{SSS},$$

where the latter two use proportional corresponding side lengths.

Remark. For right triangles, RHS (right angle–hypotenuse–side), also called HL in some conventions, is a valid congruence criterion. A condition sometimes called RHA (right angle–hypotenuse–acute angle) is also sufficient, but it is already an ASA/AAS case because the two acute angles of a right triangle are complementary. By contrast, SSA is not a congruence criterion in general; the right-triangle hypotheses are essential in the familiar hypotenuse-based special cases.

Theorem 7.9. Midpoint Theorem and Centroid Ratio
The segment joining the midpoints of two sides of a triangle is parallel to the third side and has half its length. The three medians are concurrent at the centroid G, and G divides every median in the ratio

$$2 : 1$$

measured from the vertex.

Strategy. Reverse Engineering in Geometric Constructions. First draw a completed configuration satisfying the desired conditions. Then identify which points, perpendiculars, parallels, angle bisectors, circles, or reflections in the completed picture can be constructed from the original data. Familiar configurations involving triangle centers, similar triangles, and power of a point are especially useful targets.

Definition 7.10. Classical Triangle Centers
For a triangle $\triangle ABC$, the most important classical centers are:

$$O = \text{circumcenter}, \qquad I = \text{incenter}, \qquad H = \text{orthocenter}, \qquad G = \text{centroid}.$$

Here O is the center of the circumcircle, I is the center of the incircle, H is the intersection of the altitudes, and G is the intersection of the medians.

Theorem 7.11. Gergonne Point and Nagel Point
The cevians joining the vertices of a triangle to the contact points of its incircle with the opposite sides are concurrent at the Gergonne point. The cevians joining the vertices to the contact points of the opposite excircles are concurrent at the Nagel point.

Theorem 7.12. Symmedian Lemma and Lemoine Point
Let the A-symmedian of $\triangle ABC$ meet BC at K. Then

$$\frac{BK}{KC} = \frac{AB^2}{AC^2}.$$

The three symmedians are concurrent at the Lemoine point, also called the symmedian point.

Theorem 7.13. Fermat Point; Torricelli Point
If every angle of $\triangle ABC$ is less than 120°, there is a unique point P minimizing

$$PA + PB + PC.$$

It satisfies

$$\angle APB = \angle BPC = \angle CPA = 120^\circ.$$

If one angle is at least 120°, the minimizing point is that vertex.

Formula 7.14. Extended Law of Sines
For a triangle $\triangle ABC$ with side lengths

$$a = BC, \qquad b = CA, \qquad c = AB,$$

and circumradius R, we have

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

Formula 7.15. Law of Cosines
For a triangle $\triangle ABC$ with side lengths

$$a = BC, \qquad b = CA, \qquad c = AB,$$

we have

$$c^2 = a^2 + b^2 - 2ab\cos C.$$

The analogous formulas hold after cyclic permutation of a, b, c.

Formula 7.16. Heron's Formula
Let $\triangle ABC$ have side lengths a, b, c and semiperimeter

$$s = \frac{a+b+c}{2}.$$

Then its area $[ABC]$ is

$$[ABC] = \sqrt{s(s-a)(s-b)(s-c)}.$$

Also,

$$[ABC] = rs \qquad \text{and} \qquad [ABC] = \frac{abc}{4R},$$

where r is the inradius and R is the circumradius.

Strategy. The formulas

$$[ABC] = rs, \qquad [ABC] = \frac{abc}{4R}$$

are often more useful than Heron's formula when incircles or circumcircles appear.

Formula 7.17. Triangle Area Formulas
For a triangle with side lengths a, b, c, angles A, B, C, circumradius R, and inradius r,

$$[ABC] = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B = \frac{1}{2}ab\sin C,$$

and

$$[ABC] = rs = \frac{abc}{4R}.$$

Theorem 7.18. Weitzenböck's Inequality
For a triangle with side lengths a, b, c and area $[ABC]$,

$$a^2 + b^2 + c^2 \geq 4\sqrt{3}\,[ABC],$$

with equality if and only if the triangle is equilateral.

Formula 7.19. Inradius and Exradii
For a triangle $\triangle ABC$ with semiperimeter s, the inradius is

$$r = \frac{[ABC]}{s}.$$

If r_a, r_b, r_c are the exradii opposite A, B, C, respectively, then

$$r_a = \frac{[ABC]}{s-a}, \qquad r_b = \frac{[ABC]}{s-b}, \qquad r_c = \frac{[ABC]}{s-c}.$$

Formula 7.20. Median Length Formula; Apollonius' Theorem
If m_a is the median from A to side $a = BC$, then

$$m_a^2 = \frac{2b^2 + 2c^2 - a^2}{4}.$$

Equivalently, if M is the midpoint of BC, then

$$AB^2 + AC^2 = 2(AM^2 + BM^2).$$

Theorem 7.21. Stewart's Theorem
Let D lie on side BC of a triangle $\triangle ABC$. Write

$$a = BC, \qquad b = CA, \qquad c = AB, \qquad m = BD, \qquad n = DC, \qquad d = AD,$$

so that $a = m + n$. Then

$$b^2m + c^2n = a(d^2 + mn).$$

Equivalently,

$$d^2 = \frac{b^2m + c^2n}{a} - mn.$$

Remark. Stewart's Theorem is the standard general formula for the length of a cevian. The median-length formula is the special case $m = n = a/2$.

Formula 7.22. Internal Angle-Bisector Length
Let the internal angle bisector from A meet BC at D, and denote its length by $\ell_a = AD$. For

$$a = BC, \qquad b = CA, \qquad c = AB,$$

we have

$$\ell_a^2 = bc\left(1 - \frac{a^2}{(b+c)^2}\right) = \frac{4bc\,s(s-a)}{(b+c)^2},$$

where $s = (a+b+c)/2$.

Theorem 7.23. Angle Bisector Theorem
In a triangle $\triangle ABC$, suppose the internal angle bisector of $\angle A$ meets BC at D. Then

$$\frac{BD}{DC} = \frac{AB}{AC}.$$

Theorem 7.24. Ceva's Theorem
Let D, E, F lie on the lines BC, CA, AB, respectively. Then the cevians AD, BE, CF are concurrent if and only if

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1,$$

using directed lengths.

Remark. Ceva's Theorem is the standard theorem for proving concurrence of three lines in a triangle.

Theorem 7.25. Trigonometric Form of Ceva's Theorem
Let D, E, F lie on the lines BC, CA, AB, respectively. The cevians AD, BE, CF are concurrent if and only if

$$\frac{\sin\angle BAD}{\sin\angle DAC} \cdot \frac{\sin\angle CBE}{\sin\angle EBA} \cdot \frac{\sin\angle ACF}{\sin\angle FCB} = 1,$$

with the usual directed-angle convention when points lie on extended sides.

Strategy. Mass Points. In a cevian configuration, assign masses to vertices so that a point dividing a segment balances the endpoint masses inversely to the segment lengths. For example, if $D \in BC$, then

$$\frac{BD}{DC} = \frac{m_C}{m_B}.$$

Consistent masses convert several ratio computations into weighted averages and often recover Ceva-type relations immediately.

Theorem 7.26. Menelaus' Theorem
Let D, E, F lie on the lines BC, CA, AB, respectively. Then the points D, E, F are collinear if and only if

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = -1,$$

using directed lengths.

Remark. Ceva is for concurrence, while Menelaus is for collinearity. This pair is one of the most important toolkits in olympiad geometry.

Theorem 7.27. Euler Line Theorem
In any non-equilateral triangle, the circumcenter O, centroid G, and orthocenter H are collinear. Moreover,

$$OG : GH = 1 : 2.$$

The line passing through these points is called the Euler line.

Theorem 7.28. Nine-Point Circle Theorem; Euler Circle
In any triangle, the following nine points lie on one circle: the three side midpoints, the three feet of the altitudes, and the three midpoints between the orthocenter and the vertices. This circle is called the nine-point circle.

Remark. The center of the nine-point circle is the midpoint of the segment joining the circumcenter O and the orthocenter H.

Theorem 7.29. Feuerbach's Theorem
The nine-point circle of a triangle is internally tangent to the incircle and externally tangent to each of the three excircles.

Theorem 7.30. Euler's Formula for the Incenter and Circumcenter
For a triangle with circumradius R, inradius r, circumcenter O, and incenter I, we have

$$OI^2 = R^2 - 2Rr.$$

In particular,

$$R \geq 2r.$$

Remark. The inequality $R \geq 2r$ is called Euler's inequality for triangles. Equality holds exactly for an equilateral triangle.

Strategy. Spiral Similarity. When two segments must be compared through both an angle and a ratio, look for a center of spiral similarity. It often reveals the required similar triangles immediately.

Theorem 7.31. Inscribed Angle Theorem
An inscribed angle equals half the central angle subtending the same arc. Consequently, angles subtending the same chord are equal.

Theorem 7.32. Cyclic Quadrilateral Criteria
Let A, B, C, D be four distinct points with no three collinear. Using directed angles modulo π, the points are concyclic if and only if

$$\angle BAC \equiv \angle BDC \pmod{\pi}.$$

If $ABCD$ is a convex quadrilateral in cyclic order, this is equivalent to

$$\angle ABC + \angle ADC = \pi.$$

Theorem 7.33. Tangent–Chord Theorem; Alternate Segment Theorem
The angle between a tangent to a circle and a chord through the point of tangency equals the inscribed angle subtending that chord in the opposite arc.

Theorem 7.34. Power of a Point Theorem
Let P be a point and let a line through P meet a circle at A and B. The product

$$PA \cdot PB$$

is independent of the choice of the line through P. It is called the power of P with respect to the circle.

Formula 7.35. Tangent–Secant Theorem
If PT is tangent to a circle at T and a secant through P meets the circle at A and B, then

$$PT^2 = PA \cdot PB.$$

Theorem 7.36. Radical Axis Theorem
For two nonconcentric circles, the set of points having equal powers with respect to the two circles is a line. This line is called the radical axis of the two circles.

Remark. Radical axes are powerful because they turn circle configurations into collinearity statements.

Theorem 7.37. Ptolemy's Theorem
Let A, B, C, D be the vertices of a convex cyclic quadrilateral in this cyclic order. Then

$$AC \cdot BD = AB \cdot CD + BC \cdot DA.$$

Conversely, equality in Ptolemy's inequality for a convex quadrilateral characterizes cyclicity.

Theorem 7.38. Ptolemy's Inequality
For any four points A, B, C, D in the Euclidean plane,

$$AC \cdot BD \leq AB \cdot CD + BC \cdot DA.$$

Equality holds in the cyclic convex case and in the corresponding degenerate collinear case.

Formula 7.39. Brahmagupta's Formula
If a cyclic quadrilateral has side lengths a, b, c, d and semiperimeter

$$s = \frac{a+b+c+d}{2},$$

then its area is

$$\operatorname{area}(ABCD) = \sqrt{(s-a)(s-b)(s-c)(s-d)}.$$

Heron's formula is the limiting triangular case.

Theorem 7.40. Varignon's Theorem
The midpoints of the four sides of any quadrilateral form a parallelogram. Its area is half the area of the original quadrilateral.

Theorem 7.41. Pitot's Theorem
If a convex quadrilateral with consecutive side lengths a, b, c, d has an incircle tangent to all four sides, then

$$a + c = b + d.$$

Theorem 7.42. Euler's Quadrilateral Theorem
If M and N are the midpoints of the diagonals of a quadrilateral with side lengths a, b, c, d and diagonal lengths p, q, then

$$a^2 + b^2 + c^2 + d^2 = p^2 + q^2 + 4MN^2.$$

The parallelogram law is the case $M = N$.

Theorem 7.43. Simson Line Theorem
Let P be a point on the circumcircle of $\triangle ABC$. The feet of the perpendiculars from P to the three lines AB, BC, CA are collinear. This line is called the Simson line of P with respect to $\triangle ABC$.

Theorem 7.44. Miquel's Theorem
In a complete quadrilateral, the circumcircles of the four triangles formed by choosing three of the four lines pass through a common point. This point is called the Miquel point.

Theorem 7.45. British Flag Theorem
For any point P in the plane of a rectangle $ABCD$,

$$PA^2 + PC^2 = PB^2 + PD^2.$$

Theorem 7.46. Van Aubel's Theorem
Construct squares externally on the four sides of a quadrilateral. The segments joining the centers of opposite squares are equal in length and perpendicular.

Theorem 7.47. Viviani's Theorem
For any point P inside an equilateral triangle of altitude h, the sum of the perpendicular distances from P to the three sides is h.

Theorem 7.48. Desargues' Theorem
If two triangles are perspective from a point, then the three intersections of corresponding sides are collinear. Conversely, in the projective plane, collinearity of those intersections implies concurrence of the three lines joining corresponding vertices.

Theorem 7.49. Pappus' Hexagon Theorem
Let A, B, C lie on one line and A', B', C' lie on another. Then the three points

$$AB' \cap A'B, \qquad AC' \cap A'C, \qquad BC' \cap B'C$$

are collinear.

Theorem 7.50. Homothety Centers of Two Circles
For two nonconcentric circles, the two external common tangents meet at the external homothety center, and the two internal common tangents meet at the internal homothety center, when these intersections exist. Each homothety center lies on the line joining the two circle centers.

Formula 7.51. Circle Inversion
Under inversion with center O and radius ρ, a point $P \neq O$ is sent to the point P' on the ray OP satisfying

$$OP \cdot OP' = \rho^2.$$

Inversion preserves angles in magnitude, sends circles through O to lines not through O, and sends such lines back to circles through O.

Formula 7.52. Regular Polygon Tiling Angle Condition
If q regular p-gons meet edge-to-edge at each vertex of a Euclidean tiling, then

$$q\left(1 - \frac{2}{p}\right)\pi = 2\pi,$$

equivalently,

$$(p-2)(q-2) = 4.$$

Thus the only tilings by one type of regular polygon are

$$(p, q) = (3, 6), (4, 4), (6, 3).$$

Formula 7.53. Obtuse Triangles from Equally Spaced Points
Let N equally spaced points lie on a circle. The number of obtuse triangles determined by three of the points is

$$N\binom{k-1}{2} \quad \text{if } N = 2k,$$

and

$$N\binom{k}{2} \quad \text{if } N = 2k+1.$$

For even N, triangles containing a diameter are right rather than obtuse.

Fact 7.54. Alternating-Area Invariant in a Regular Even-Gon
Let $V_1, \dots, V_{2m}$ be the vertices of a regular $2m$-gon in cyclic order, and let P be any point inside it. Then

$$[PV_1V_2] + [PV_3V_4] + \cdots + [PV_{2m-1}V_{2m}]$$

equals

$$[PV_2V_3] + [PV_4V_5] + \cdots + [PV_{2m}V_1].$$

The equality follows from the cancellation of alternating directed edge vectors.

Theorem 7.55. Classification of Platonic Solids
There are exactly five Platonic solids:

tetrahedron, cube, octahedron, dodecahedron, icosahedron.

Analytic Geometry

Formula 7.56. Coordinate Geometry Table
For points $A = (x_1, y_1)$ and $B = (x_2, y_2)$,

Quantity	Formula
Distance	$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Midpoint	$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$
Internal division $AP : PB = m : n$	$\left(\frac{nx_1 + mx_2}{m+n}, \frac{ny_1 + my_2}{m+n}\right)$
Slope	$\frac{y_2 - y_1}{x_2 - x_1}$, when $x_1 \neq x_2$
Point-slope line	$y - y_1 = m(x - x_1)$
General line	$ax + by + c = 0$
Distance from (x_0, y_0) to $ax + by + c = 0$	$\frac{\lvert ax_0 + by_0 + c\rvert}{\sqrt{a^2 + b^2}}$

Formula 7.57. Triangle Centers in Coordinates
For $A = (x_1, y_1)$, $B = (x_2, y_2)$, and $C = (x_3, y_3)$, the centroid is

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right).$$

If the side lengths opposite A, B, C are a, b, c, respectively, then the incenter is

$$I = \frac{aA + bB + cC}{a + b + c}.$$

Formula 7.58. Barycentric Coordinates
Let A, B, C be noncollinear. Every point P in their affine plane has unique coordinates (α, β, γ) satisfying

$$\alpha + \beta + \gamma = 1, \qquad \mathbf{p} = \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}.$$

For P inside $\triangle ABC$,

$$\alpha = \frac{[PBC]}{[ABC]}, \qquad \beta = \frac{[PCA]}{[ABC]}, \qquad \gamma = \frac{[PAB]}{[ABC]}.$$

Formula 7.59. Shoelace Formula; Gauss's Area Formula
For a polygon with vertices

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$$

in cyclic order, its area is

$$\frac{1}{2}\left|\sum_{i=1}^{n}(x_i y_{i+1} - x_{i+1} y_i)\right|,$$

where

$$(x_{n+1}, y_{n+1}) = (x_1, y_1).$$

Remark. The Shoelace Formula is one of the cleanest examples of analytic geometry: a geometric area becomes a determinant-like coordinate calculation.

Formula 7.60. Oriented Area and Collinearity Test
For points $A = (x_1, y_1)$, $B = (x_2, y_2)$, and $C = (x_3, y_3)$, the oriented double area is

$$2[ABC]_{\text{or}} = \det\begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}.$$

Hence A, B, C are collinear if and only if this determinant is 0.

Formula 7.61. Plane Rotation
Counterclockwise rotation through angle θ about the origin sends

$$\begin{pmatrix} x \\ y \end{pmatrix} \longmapsto \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}.$$

In complex notation, the same transformation is simply

$$z \longmapsto e^{i\theta} z.$$

Formula 7.62. Reflection across a Line through the Origin
If a line through the origin makes angle θ with the positive x-axis, reflection across that line is

$$z \longmapsto e^{2i\theta}\overline{z}$$

in complex notation.

Strategy. Affine Normalization. Affine transformations preserve collinearity, parallelism, ratios along one line, concurrence, and ratios of areas. Therefore, when a problem depends only on these properties, one may send a triangle to convenient coordinates such as

$$A = (0, 0), \qquad B = (1, 0), \qquad C = (0, 1).$$

Angles, lengths, and circles are not generally preserved.

Theorem 7.63. Cavalieri's Principle
If two plane regions lie between the same pair of parallel lines and every line parallel to them cuts the regions in segments whose lengths have a constant ratio c, then their areas have ratio c. The three-dimensional analogue replaces segment lengths by cross-sectional areas and concludes the same ratio for volumes.

Strategy. Unfolding Method for Reflection Problems. A broken path reflecting from a line or face becomes a straight segment after reflecting the region across each boundary in succession. Thus a shortest reflected path is found by measuring a straight-line distance in the unfolded picture and then folding the path back.

Formula 7.64. Lattice Segment and Cell Counts
A segment from $(0, 0)$ to (m, n), where $m, n \in \mathbb{N}$, passes through

$$m + n - \gcd(m, n)$$

unit squares. The number of lattice points on the segment, including its endpoints, is

$$\gcd(m, n) + 1.$$

In three dimensions, a segment from $(0, 0, 0)$ to (a, b, c) passes through

$$a + b + c - \gcd(a, b) - \gcd(a, c) - \gcd(b, c) + \gcd(a, b, c)$$

unit cubes.

Formula 7.65. Vector Test for Parallel Segments
Directed segments AB and CD are parallel if and only if

$$\mathbf{b} - \mathbf{a} = \lambda(\mathbf{d} - \mathbf{c})$$

for some scalar λ. Equality with $\lambda = 1$ identifies equal directed segments and is often the most efficient method to detect a hidden parallelogram.

Formula 7.66. Vector Projection
If $\mathbf{u} \neq \mathbf{0}$, then the projection of $\mathbf{v}$ onto the line spanned by $\mathbf{u}$ is

$$\operatorname{proj}_{\mathbf{u}} \mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}.$$

The scalar component of $\mathbf{v}$ in the direction of $\mathbf{u}$ is

$$\frac{\mathbf{v} \cdot \mathbf{u}}{\|\mathbf{u}\|}.$$

Formula 7.67. Vector Equations of Lines and Planes
A line through a point $\mathbf{p}$ with direction vector $\mathbf{v} \neq \mathbf{0}$ is

$$\mathbf{x} = \mathbf{p} + t\mathbf{v} \qquad (t \in \mathbb{R}).$$

A plane through a point $\mathbf{p}$ with normal vector $\mathbf{n} \neq \mathbf{0}$ is

$$(\mathbf{x} - \mathbf{p}) \cdot \mathbf{n} = 0.$$

Formula 7.68. Angles and Volumes in $\mathbb{R}^3$
If nonzero vectors $\mathbf{u}, \mathbf{v}$ are directions of two lines, their acute angle θ satisfies

$$\cos\theta = \frac{|\mathbf{u} \cdot \mathbf{v}|}{\|\mathbf{u}\| \, \|\mathbf{v}\|}.$$

If $\mathbf{n}_1, \mathbf{n}_2$ are normals to two planes, the acute angle between the planes is given by the same formula with the normals. If $\mathbf{v}$ is a line direction and $\mathbf{n}$ is a plane normal, the acute line–plane angle α satisfies

$$\sin\alpha = \frac{|\mathbf{v} \cdot \mathbf{n}|}{\|\mathbf{v}\| \, \|\mathbf{n}\|}.$$

The volume of the parallelepiped spanned by $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is

$$\big|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})\big|,$$

and the tetrahedron with the same three edge vectors from one vertex has one sixth of this volume.

Formula 7.69. Distance between Skew Lines
Let

$$L_1 : \mathbf{x} = \mathbf{p}_1 + t\mathbf{u}, \qquad L_2 : \mathbf{x} = \mathbf{p}_2 + s\mathbf{v},$$

with $\mathbf{u} \times \mathbf{v} \neq \mathbf{0}$. Then

$$\operatorname{dist}(L_1, L_2) = \frac{|(\mathbf{p}_2 - \mathbf{p}_1) \cdot (\mathbf{u} \times \mathbf{v})|}{\|\mathbf{u} \times \mathbf{v}\|}.$$

Formula 7.70. Distance from a Point to a Line
In $\mathbb{R}^3$, the distance from a point $\mathbf{p}_0$ to the line

$$\mathbf{x} = \mathbf{p} + t\mathbf{v}$$

is

$$\frac{\|(\mathbf{p}_0 - \mathbf{p}) \times \mathbf{v}\|}{\|\mathbf{v}\|}.$$

In $\mathbb{R}^2$, the distance from (x_0, y_0) to the line $ax + by + c = 0$ is

$$\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}.$$

Formula 7.71. Distance from a Point to a Plane
The distance from a point $\mathbf{p}_0 = (x_0, y_0, z_0)$ to the plane

$$ax + by + cz + d = 0$$

is

$$\frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Formula 7.72. Sphere Equation
A sphere with center $\mathbf{c} = (a, b, c)$ and radius r has equation

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2.$$

Equivalently,

$$\|\mathbf{x} - \mathbf{c}\| = r.$$

Formula 7.73. Standard Forms of Common Quadrics in $\mathbb{R}^3$
With $a, b, c > 0$, the principal-axis standard forms include:

Surface	Standard form
Ellipsoid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
Elliptic paraboloid	$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$
Hyperbolic paraboloid	$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$
One-sheet hyperboloid	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$
Two-sheet hyperboloid	$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
Elliptic cone	$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$
Elliptic cylinder	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

More generally, a quadric can be written

$$\mathbf{x}^{\mathsf{T}}\mathbf{A}\mathbf{x} + \mathbf{b}^{\mathsf{T}}\mathbf{x} + c_0 = 0,$$

with $\mathbf{A}$ real symmetric. Translation and orthogonal diagonalization of $\mathbf{A}$ reduce the quadratic part to principal axes, linking the geometry directly to the spectral theorem.

Strategy. Line–Circle Intersection by a Discriminant. Substitute a parametric line into a circle equation. The resulting quadratic equation has two, one, or no real solutions according as its discriminant is positive, zero, or negative. The zero-discriminant case is the tangency condition.

Formula 7.74. Circle Equation
The circle with center (a, b) and radius r has equation

$$(x-a)^2 + (y-b)^2 = r^2.$$

The general equation

$$x^2 + y^2 + Dx + Ey + F = 0$$

can be converted to center-radius form by completing the square.

Theorem 7.75. Three Perpendiculars Theorem
Let Π be a plane, let $P \notin \Pi$, and let H be the orthogonal projection of P onto Π. Let $Q \in \Pi$ with $Q \neq H$, and let $\ell \subseteq \Pi$ be a line through Q. If $HQ \perp \ell$, then

$$PQ \perp \ell.$$

Equivalently, a line in a plane that is perpendicular to the orthogonal projection of an oblique line is perpendicular to the oblique line itself.

Remark. In coordinate geometry, the three perpendiculars theorem is often replaced by dot products and projections. Still, it is a standard spatial-geometry fact from the classical school curriculum.

Definition 7.76. Conic Section
A conic section is a curve obtained by intersecting a plane with a double cone. The nondegenerate conics are:

ellipse, parabola, hyperbola.

Equivalently, a conic can be described using a focus, a directrix, and an eccentricity e.

Formula 7.77. Focus–Directrix Definition of Conics
A conic is the set of points P satisfying

$$\frac{\operatorname{dist}(P, \text{focus})}{\operatorname{dist}(P, \text{directrix})} = e.$$

It is an ellipse if

$$0 < e < 1,$$

a parabola if

$$e = 1,$$

and a hyperbola if

$$e > 1.$$

Formula 7.78. Standard Forms of Conics
After translation and, when necessary, rotation of coordinates, every nondegenerate real conic can be put into one of the following standard forms.

Conic	Standard form	Basic condition and feature
Circle	$(x-h)^2 + (y-k)^2 = r^2$	center (h, k), radius $r > 0$
Ellipse	$\dfrac{(x-h)^2}{a^2} + \dfrac{(y-k)^2}{b^2} = 1$, $a \geq b > 0$	$c^2 = a^2 - b^2$, foci at distance c from the center

Conic	Standard form	Basic condition and feature
Parabola	$(y-k)^2 = 4p(x-h)$ or $(x-h)^2 = 4p(y-k)$	focus–directrix distance parameter $p \neq 0$
Hyperbola	$\dfrac{(x-h)^2}{a^2} - \dfrac{(y-k)^2}{b^2} = 1$ or with the signs reversed	$c^2 = a^2 + b^2$, foci at distance c from the center

The general second-degree equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

is classified, in the nondegenerate case, by the quadratic part; in particular $B^2 - 4AC < 0, = 0,$ and > 0 correspond to ellipse-type, parabola-type, and hyperbola-type conics, respectively.

Formula 7.79. Eccentricity of an Ellipse and a Hyperbola
For an ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad (a \geq b > 0),$$

the eccentricity is

$$e = \frac{c}{a}, \qquad c^2 = a^2 - b^2.$$

For a hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

the eccentricity is

$$e = \frac{c}{a}, \qquad c^2 = a^2 + b^2.$$

Formula 7.80. Tangent Lines to Standard Conics
At a point (x_1, y_1) on the indicated conic, the tangent line is

$$xx_1 + yy_1 = r^2 \qquad \text{for } x^2 + y^2 = r^2,$$

$$yy_1 = 2a(x + x_1) \qquad \text{for } y^2 = 4ax,$$

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \qquad \text{for } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

and

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1 \qquad \text{for } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

Theorem 7.81. Reflection Property of Conics
At a point of a nondegenerate conic, the tangent makes equal angles with the two relevant focal directions. Consequently, a ray emitted from one focus of an ellipse reflects toward the other focus; a ray parallel to the axis of a parabola reflects through its focus; and a ray directed toward one focus of a hyperbola reflects along a ray whose backward extension passes through the other focus.

Remark. The reflection property gives a geometric reason why conics appear in optics, satellite dishes, whispering galleries, and orbital mechanics.

Theorem 7.82. Quadratic Classification of Conics
Consider a real quadratic equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

The discriminant

$$B^2 - 4AC$$

determines the type of the nondegenerate conic:

$$B^2 - 4AC < 0 \quad \Rightarrow \quad \textit{ellipse-type},$$

$$B^2 - 4AC = 0 \quad \Rightarrow \quad \textit{parabola-type},$$

and

$$B^2 - 4AC > 0 \quad \Rightarrow \quad \textit{hyperbola-type}.$$

Remark. This is the analytic-geometry counterpart of classifying conic sections by their shapes.

Formula 7.83. Polar Equation of a Conic
With a focus at the pole, a conic has polar equation

$$r = \frac{\ell}{1 + e\cos\theta}$$

after choosing a suitable axis. Here e is the eccentricity and ℓ is the semi-latus rectum.

Remark. The same equation describes ellipses, parabolas, and hyperbolas depending on whether $e < 1$, $e = 1$, or $e > 1$.

Definition 7.84. Apollonius Circle
Given two fixed points A and B and a positive constant $k \neq 1$, the locus of points P satisfying

$$\frac{PA}{PB} = k$$

is a circle. This circle is called an Apollonius circle.

Remark. Apollonius circles turn a ratio of distances into a concrete circle, which is a common hidden structure in geometry problems.

Theorem 7.85. Descartes' Circle Theorem; Soddy Circle Theorem
Suppose four circles are mutually tangent, and let their curvatures be

$$k_1, k_2, k_3, k_4,$$

where curvature means the reciprocal of the radius, with sign convention for an enclosing circle. Then

$$(k_1 + k_2 + k_3 + k_4)^2 = 2(k_1^2 + k_2^2 + k_3^2 + k_4^2).$$

Remark. Descartes' Circle Theorem is a striking formula: tangency of four circles becomes one quadratic equation in their curvatures.

Formula 7.86. Boundary Lattice Points on a Segment
If a lattice segment has displacement $(\Delta x, \Delta y)$, then the number of lattice points on it, including both endpoints, is

$$\gcd(|\Delta x|, |\Delta y|) + 1.$$

Summing the gcd contributions over polygon edges gives the number B of boundary lattice points without double-counting vertices.

Theorem 7.87. Pick's Theorem
Let P be a simple polygon whose vertices lie on lattice points in the plane. If I is the number of lattice points strictly inside P and B is the number of lattice points on the boundary of P, then

$$\operatorname{area}(P) = I + \frac{B}{2} - 1.$$

Remark. Pick's Theorem is a beautiful bridge between geometry and arithmetic: area is determined by counting lattice points.

Chapter 8

Number Theory

Related POSTECH Courses: (MATH304) Introduction to Number Theory
Related Books: Introduction to Number Theory (William W. Adams, Larry Joel Goldstein)

Definition 8.1. Divisibility, Prime, Greatest Common Divisor, and Congruence
For $a, b \in \mathbb{Z}$, write $a \mid b$ if

$$\exists k \in \mathbb{Z}, \qquad b = ak.$$

An integer $p > 1$ is prime if its only positive divisors are 1 and p. For integers a, b, not both zero, $\gcd(a, b)$ is their greatest positive common divisor. For $n \in \mathbb{N}$,

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

Theorem 8.2. Fundamental Theorem of Arithmetic
Every integer $n > 1$ can be written as a product of prime numbers. Moreover, this factorization is unique up to the order of the factors. Equivalently,

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k},$$

where $p_1, \ldots, p_k$ are distinct primes and $\alpha_1, \ldots, \alpha_k$ are positive integers, and this expression is unique up to reordering.

Remark. This theorem is often called unique prime factorization. It is the reason prime numbers are treated as the basic building blocks of integers.

Theorem 8.3. Euclid's Theorem; Infinitude of Primes
There are infinitely many prime numbers.

Idea. If $p_1, \ldots, p_n$ are finitely many primes, consider

$$p_1 p_2 \cdots p_n + 1.$$

This number is not divisible by any prime on the list.

Theorem 8.4. Bézout's Identity; Bézout's Lemma
Let $a, b \in \mathbb{Z}$, not both zero. Then there exist integers $x, y \in \mathbb{Z}$ s.t.

$$ax + by = \gcd(a, b).$$

In particular, if $\gcd(a, b) = 1$, then there exist integers $x, y \in \mathbb{Z}$ s.t.

$$ax + by = 1.$$

Remark. Bézout's identity is the algebraic heart of the Euclidean algorithm and modular inverses.

Algorithm 8.5. Euclidean Algorithm
To compute $\gcd(a, b)$ for integers $a, b \in \mathbb{Z}$, repeatedly divide the larger number by the smaller number and replace the pair by

$$(b, r),$$

where r is the remainder. Continue until the remainder becomes 0. The last nonzero remainder is $\gcd(a, b)$.

Proposition 8.6. Basic Rules of Congruences
If

$$a \equiv b \pmod{n}, \qquad c \equiv d \pmod{n},$$

then

$$a + c \equiv b + d \pmod{n}, \qquad ac \equiv bd \pmod{n}.$$

If $\gcd(c, n) = 1$*, then*

$$ac \equiv bc \pmod{n} \implies a \equiv b \pmod{n}.$$

Formula 8.7. Fast Residue Filters for Powers
The following small-modulus facts are useful impossibility tests:

$$x^2 \equiv 0, 1 \pmod{4}, \qquad x^2 \equiv 0, 1, 4 \pmod{8},$$

$$x^3 \equiv 0, \pm 1 \pmod{9}, \qquad x^4 \equiv 0, 1 \pmod{16}.$$

In particular,

$$x \text{ odd} \implies x^2 \equiv 1 \pmod{8}.$$

Thus a proposed square, cube, or fourth power can often be ruled out before any large calculation is attempted.

Theorem 8.8. Linear Congruence Criterion
The congruence

$$ax \equiv b \pmod{n}$$

has a solution if and only if

$$\gcd(a, n) \mid b.$$

If $d = \gcd(a, n)$ *and* $d \mid b$*, then there are exactly* d *solutions modulo* n*.*

Formula 8.9. Least Common Multiple from Prime Factorization
If

$$a = \prod_p p^{\alpha_p}, \qquad b = \prod_p p^{\beta_p},$$

then

$$\gcd(a, b) = \prod_p p^{\min(\alpha_p, \beta_p)}, \qquad \operatorname{lcm}(a, b) = \prod_p p^{\max(\alpha_p, \beta_p)}.$$

More generally, the least common multiple of several integers is found by taking the largest exponent of each prime appearing in the list.

Formula 8.10. Basic Divisibility Tests
For a decimal integer N, the following tests are useful.

- Divisibility by 2, 4, and 8 depends on the last 1, 2, and 3 digits, respectively.
- Divisibility by 3 and 9 depends on the digit sum.
- Divisibility by 5 depends on the last digit.

- Divisibility by 11 depends on the alternating sum of the digits.

Strategy. Digit problems often combine an lcm condition, a digit-sum condition, and a last-digits condition. First minimize the number of digits, then arrange the final few digits to satisfy the strongest divisibility conditions.

Definition 8.11. Complete Residue System
A set of n integers is a complete residue system modulo n if its elements represent every residue class modulo n exactly once. To prove that $a_0, \dots, a_{n-1}$ form such a system, it is enough to prove

$$a_i \equiv a_j \pmod{n} \implies i = j.$$

Strategy. Modular Casework and Base Representation. When a problem repeats division by N, rewrite the integer in base N. The quotient deletes the final digit and the remainder reveals it. For divisibility or existence problems, first split integers into residue classes and look for a complete residue system rather than checking values individually.

Strategy. Infinite Descent. Assume a positive-integer solution exists and choose one minimizing a natural positive quantity. Construct a strictly smaller solution of the same type, contradicting well-ordering.

Strategy. Vieta Jumping. If an integer equation is quadratic in one variable, regard that variable as one root of a monic quadratic. Viète's formulas give the other integral root. Prove that it yields a smaller positive solution and apply infinite descent.

Theorem 8.12. Frobenius Coin Theorem for Two Denominations
If a and b are coprime positive integers, the largest integer not representable as

$$ax + by, \qquad x, y \in \mathbb{N}_0,$$

is $ab - a - b$. Exactly

$$\frac{(a-1)(b-1)}{2}$$

positive integers are not representable.

Theorem 8.13. Chinese Remainder Theorem; CRT
Let $m_1, \dots, m_k$ be pairwise relatively prime positive integers. Then the system

$$x \equiv a_1 \pmod{m_1}, \qquad x \equiv a_2 \pmod{m_2}, \qquad \dots, \qquad x \equiv a_k \pmod{m_k}$$

has a solution. Moreover, the solution is unique modulo

$$M = m_1 m_2 \cdots m_k.$$

Strategy. CRT is the standard theorem for combining several congruence conditions into one congruence.

Definition 8.14. Arithmetic and Multiplicative Functions
An arithmetic function is a function $f : \mathbb{N} \to \mathbb{C}$. It is multiplicative if

$$\gcd(m, n) = 1 \implies f(mn) = f(m)f(n).$$

Important multiplicative arithmetic functions include Euler's totient $\varphi(n)$, the divisor-counting function $\tau(n)$, the divisor-sum function $\sigma(n)$, and the Möbius function $\mu(n)$. Multiplicativity reduces their evaluation to prime powers.

Formula 8.15. Euler's Phi Function Formula
If
$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$$
is the prime factorization of n, then
$$\varphi(n) = n \prod_{i=1}^{k} \left(1 - \frac{1}{p_i}\right).$$
In particular, for a prime power,
$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^{\alpha-1}(p-1).$$

Formula 8.16. Sum-of-Divisors Function
If
$$n = p_1^{\alpha_1} \cdots p_k^{\alpha_k},$$
then the number of positive divisors of n is
$$\tau(n) = (\alpha_1 + 1) \cdots (\alpha_k + 1),$$
and the sum of positive divisors of n is
$$\sigma(n) = \prod_{i=1}^{k} \frac{p_i^{\alpha_i+1} - 1}{p_i - 1}.$$

Formula 8.17. p-Adic Valuation of a Factorial; Legendre's Formula
For a prime p and a positive integer n,
$$\nu_p(n!) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor.$$
Only finitely many terms are nonzero. This formula gives the exponent of p in $n!$ and is especially useful for trailing-zero and divisibility questions.

Definition 8.18. p-Adic Absolute Value and p-Adic Numbers
Fix a prime p. For a nonzero rational number x, write
$$x = p^{\nu_p(x)} \frac{a}{b}, \qquad p \nmid ab,$$
and define
$$|x|_p = p^{-\nu_p(x)}, \qquad |0|_p = 0.$$
The p-adic absolute value satisfies the strong triangle inequality
$$|x + y|_p \le \max\{|x|_p, |y|_p\}.$$
The field $\mathbb{Q}_p$ of p-adic numbers is the completion of $\mathbb{Q}$ with respect to this absolute value, just as $\mathbb{R}$ is a completion of $\mathbb{Q}$ with respect to the usual absolute value.

Definition 8.19. Möbius Function
The Möbius function is
$$\mu(n) = \begin{cases} 1, & n = 1, \\ 0, & p^2 \mid n \text{ for some prime } p, \\ (-1)^k, & n \text{ is a product of } k \text{ distinct primes.} \end{cases}$$
It satisfies
$$\sum_{d \mid n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

Theorem 8.20. Möbius Inversion Formula
If arithmetic functions f and g satisfy

$$g(n) = \sum_{d|n} f(d),$$

then

$$f(n) = \sum_{d|n} \mu(d) g\left(\frac{n}{d}\right).$$

Theorem 8.21. Fermat's Little Theorem
Let p be a prime number. If $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, for every integer a,

$$a^p \equiv a \pmod{p}.$$

Theorem 8.22. Euler's Theorem; Euler–Fermat Theorem
If $\gcd(a, n) = 1$, *then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Remark. Fermat's Little Theorem is the special case of Euler's Theorem when $n = p$ is prime, since $\varphi(p) = p - 1$.

Proposition 8.23. Multiplicative Order
If $\gcd(a, n) = 1$, *then $[a]_n$ lies in the finite group $(\mathbb{Z}/n\mathbb{Z})^\times$. Its multiplicative order* $\operatorname{ord}_n(a)$ *divides*

$$|(\mathbb{Z}/n\mathbb{Z})^\times| = \varphi(n).$$

Consequently, for every positive integer k,

$$a^k \equiv 1 \pmod{n} \iff \operatorname{ord}_n(a) \mid k.$$

Definition 8.24. Primitive Root Modulo n
If $\gcd(g, n) = 1$, then g is a primitive root modulo n if

$$\operatorname{ord}_n(g) = \varphi(n).$$

Equivalently, the residue class $[g]_n$ generates the group $(\mathbb{Z}/n\mathbb{Z})^\times$.

Theorem 8.25. Primitive Root Theorem; Gauss's Theorem on Primitive Roots
The multiplicative group $(\mathbb{Z}/n\mathbb{Z})^\times$ is cyclic if and only if

$$n \in \{1, 2, 4, p^k, 2p^k\},$$

where p is an odd prime and $k \geq 1$.

Definition 8.26. Finite Field
A finite field is a field containing finitely many elements. Its number of elements is called its order.

Theorem 8.27. Existence and Uniqueness of Finite Fields
Every finite field has $q = p^n$ elements for some prime p and positive integer n. Conversely, for each prime power $q = p^n$, there exists a field $\mathbb{F}_q$ with q elements, unique up to isomorphism. Its multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$.

Algorithm 8.28. Modular Exponentiation by Period Reduction
To compute $a^n \pmod m$, first reduce a modulo m. If $\gcd(a, m) = 1$, determine a convenient exponent $t > 0$ s.t.

$$a^t \equiv 1 \pmod m,$$

for example from the multiplicative order of a, Euler's theorem, or Fermat's little theorem when applicable. Write

$$n = qt + r, \qquad 0 \le r < t,$$

and reduce

$$a^n \equiv a^r \pmod m.$$

If a is not a unit modulo m, use repeated squaring directly or analyze prime-power factors separately rather than applying a unit-group period without justification.

Strategy. Last-digit and last-two-digit questions are modular arithmetic problems. Work modulo 10 or 100, not with the full number.

Theorem 8.29. Wilson's Theorem
A positive integer $p > 1$ is prime if and only if

$$(p-1)! \equiv -1 \pmod p.$$

Remark. Wilson's theorem is famous because it gives a clean factorial characterization of primes, even though it is not efficient as a primality test.

Theorem 8.30. Lucas's Theorem
Let p be prime and write

$$n = \sum_{i=0}^{r} n_i p^i, \qquad k = \sum_{i=0}^{r} k_i p^i$$

in base p. Then

$$\binom{n}{k} \equiv \prod_{i=0}^{r} \binom{n_i}{k_i} \pmod p.$$

Theorem 8.31. Kummer's Theorem
The valuation

$$\nu_p\left(\binom{n}{k}\right)$$

equals the number of carries when k and $n-k$ are added in base p.

Theorem 8.32. Lifting the Exponent Lemma; LTE
Let p be an odd prime, and suppose that $p \mid x - y$ while $p \nmid xy$. Then, for every positive integer n,

$$\nu_p(x^n - y^n) = \nu_p(x-y) + \nu_p(n).$$

A standard companion form is: if n is odd, $p \mid x + y$, and $p \nmid xy$, then

$$\nu_p(x^n + y^n) = \nu_p(x+y) + \nu_p(n).$$

Fact 8.33. Quadratic Residues Modulo Small Moduli
Squares have very restricted residues modulo small integers:

$$x^2 \equiv 0, 1 \pmod 4, \qquad x^2 \equiv 0, 1, 4 \pmod 8.$$

In particular, an integer congruent to 2 or 3 modulo 4 cannot be a square, and an integer congruent to $2, 3, 5, 6, 7$ modulo 8 cannot be a square.

Tactic. For Diophantine equations, first reduce modulo 4, 8, 3, 5, or 9. Many impossible equations are eliminated by the small list of possible square residues.

Definition 8.34. Legendre Symbol
Let p be an odd prime. The Legendre symbol is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 1, & \text{if } a \text{ is a nonzero quadratic residue modulo } p, \\ -1, & \text{if } a \text{ is a quadratic nonresidue modulo } p, \\ 0, & \text{if } p \mid a. \end{cases}$$

Theorem 8.35. Euler's Criterion
Let p be an odd prime. For any integer a,

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}.$$

Equivalently, if $p \nmid a$, then

$$a^{(p-1)/2} \equiv \begin{cases} 1 \pmod{p}, & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 \pmod{p}, & \text{if } a \text{ is a quadratic nonresidue modulo } p. \end{cases}$$

Theorem 8.36. Law of Quadratic Reciprocity; Gauss's Theorem of Quadratic Reciprocity
Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}.$$

Equivalently,

$$\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$$

unless both p and q are congruent to 3 modulo 4.

Remark. Quadratic reciprocity is one of the crown jewels of elementary number theory. Gauss called it the golden theorem.

Theorem 8.37. Fermat's Two-Square Theorem; Fermat's Theorem on Sums of Two Squares
An odd prime p can be written as a sum of two squares,

$$p = a^2 + b^2$$

for some integers $a, b \in \mathbb{Z}$, if and only if

$$p \equiv 1 \pmod{4}.$$

Also,

$$2 = 1^2 + 1^2.$$

Theorem 8.38. Sum of Two Squares Criterion
A positive integer

$$n = 2^a \prod_{p \equiv 1 \pmod{4}} p^{\alpha_p} \prod_{q \equiv 3 \pmod{4}} q^{\beta_q}$$

is representable as $n = x^2 + y^2$ with integers x, y if and only if every exponent β_q is even.

Formula 8.39. Brahmagupta–Fibonacci Identity

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2 = (ac + bd)^2 + (ad - bc)^2.$$

Formula 8.40. Sophie Germain's Identity

$$a^4 + 4b^4 = (a^2 - 2ab + 2b^2)(a^2 + 2ab + 2b^2).$$

Theorem 8.41. Lagrange's Four-Square Theorem
Every nonnegative integer can be written as a sum of four integer squares. That is, for every $n \in \mathbb{N}_0$, there exist integers $a, b, c, d \in \mathbb{Z}$ s.t.

$$n = a^2 + b^2 + c^2 + d^2.$$

Remark. The two-square theorem has a congruence condition. The four-square theorem has no exception at all.

Theorem 8.42. Continued-Fraction Best Approximation Theorem
Let p_k/q_k be the convergents of the simple continued fraction of an irrational number α. Then

$$\left|\alpha - \frac{p_k}{q_k}\right| < \frac{1}{q_k q_{k+1}}.$$

Moreover, the convergents are best approximations of the second kind: if $p, q \in \mathbb{Z}$, $0 < q < q_{k+1}$, and $p/q \neq p_k/q_k$, then

$$|q_k\alpha - p_k| < |q\alpha - p|.$$

Strategy. Problems asking for an unusually accurate fraction with the smallest possible denominator often signal a continued-fraction expansion.

Definition 8.43. Pell Equation
A Pell equation has the form

$$x^2 - Dy^2 = 1,$$

where D is a positive nonsquare integer. If (x_1, y_1) is the fundamental positive solution, then all positive solutions satisfy

$$x_n + y_n\sqrt{D} = (x_1 + y_1\sqrt{D})^n.$$

Continued fractions of $\sqrt{D}$ produce the fundamental solution.

Theorem 8.44. Dirichlet's Theorem on Arithmetic Progressions
If a and d are relatively prime positive integers, then the arithmetic progression

$$a,\ a+d,\ a+2d,\ a+3d, \ldots$$

contains infinitely many primes.

Remark. Dirichlet's theorem is a major result connecting primes and arithmetic progressions. The condition $\gcd(a, d) = 1$ is necessary.

Theorem 8.45. Bertrand's Postulate; Chebyshev's Theorem
For every integer $n > 1$, there exists a prime p s.t.

$$n < p < 2n.$$

Theorem 8.46. Prime Number Theorem
Let $\pi(x)$ be the number of primes less than or equal to x. Then

$$\pi(x) \sim \frac{x}{\ln x} \qquad \text{as } x \to \infty.$$

Idea. The Prime Number Theorem says that the "probability" that a large integer near x is prime is roughly $1/\ln x$. This is a mental model, not a literal probability statement.

Definition 8.47. Riemann Zeta Function
For $\mathrm{Re}(s) > 1$, the Riemann zeta function is defined by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

It also has the Euler product

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}.$$

Remark. The Euler product is the bridge between the zeta function and prime numbers. The Riemann Hypothesis asserts that the nontrivial zeros of $\zeta(s)$ all have real part $1/2$.

Definition 8.48. Perfect Number and Mersenne Prime
A positive integer N is perfect if it equals the sum of its positive proper divisors, equivalently

$$\sigma(N) = 2N.$$

A Mersenne prime is a prime of the form

$$2^p - 1.$$

If $2^p - 1$ is prime, then p must be prime.

Theorem 8.49. Euclid–Euler Theorem on Perfect Numbers
An even positive integer N is perfect if and only if

$$N = 2^{p-1}(2^p - 1),$$

where $2^p - 1$ is a Mersenne prime.

Theorem 8.50. Fermat's Last Theorem
For every integer $n \geq 3$, there do not exist positive integers a, b, c s.t.

$$a^n + b^n = c^n.$$

Remark. Fermat's Last Theorem was proved by Andrew Wiles. It is one of the most famous theorems in the history of mathematics.

Chapter 9

Algebra

Related POSTECH Courses: (MATH301) Modern Algebra I, (MATH302) Modern Algebra II
Related Books: Abstract Algebra (David S. Dummit, Richard M. Foote)

Definition 9.1. Group
A group is a set G with a binary operation $* : G \times G \to G$ s.t.

$$\forall a, b, c \in G, \qquad (a * b) * c = a * (b * c),$$

$$\exists e \in G \text{ s.t. } \quad \forall a \in G, \qquad e * a = a * e = a,$$

and

$$\forall a \in G, \qquad \exists a^{-1} \in G \text{ s.t. } \quad a * a^{-1} = a^{-1} * a = e.$$

The group is abelian if

$$\forall a, b \in G, \qquad a * b = b * a.$$

Definition 9.2. Subgroup, Homomorphism, and Kernel
A subset $H \subseteq G$ is a subgroup, written $H \leq G$, if it is a group under the operation of G. A map $\varphi : G \to H$ is a group homomorphism if

$$\varphi(ab) = \varphi(a)\varphi(b) \qquad \forall a, b \in G.$$

Its kernel and image are

$$\ker \varphi = \{g \in G \mid \varphi(g) = e_H\}, \qquad \operatorname{im} \varphi = \{\varphi(g) \mid g \in G\}.$$

Definition 9.3. Normal Subgroup and Simple Group
A subgroup $N \leq G$ is normal, written $N \trianglelefteq G$, if

$$gNg^{-1} = N \qquad \text{for every } g \in G.$$

A nontrivial group G is simple if its only normal subgroups are $\{e\}$ and G.

Definition 9.4. Quotient Group
If $N \trianglelefteq G$, the quotient group G/N is the set of cosets

$$G/N = \{gN \mid g \in G\},$$

with multiplication

$$(gN)(hN) = ghN.$$

Normality is exactly the condition that makes this multiplication well-defined.

Fact 9.5. Smallest Nonabelian Simple Group
The alternating group A_5 is simple and has order

$$|A_5| = 60.$$

It is the smallest nonabelian simple group. This fact, together with the Sylow theorems, is a standard shortcut in questions asking which small integers can occur as the order of a simple group.

Theorem 9.6. Disjoint-Cycle Decomposition of a Permutation
Every permutation $\sigma \in S_n$ can be written as a product of disjoint cycles, uniquely up to the order of the cycles and cyclic rotation within each cycle. If $c(\sigma)$ is the number of cycles, including fixed points, then the minimum number of transpositions needed to express σ is

$$n - c(\sigma).$$

Formula 9.7. Inversions and the Sign of a Permutation
The inversion number is

$$\operatorname{inv}(\sigma) = \left|\{(i,j) \mid i < j,\ \sigma(i) > \sigma(j)\}\right|.$$

The sign satisfies

$$\operatorname{sgn}(\sigma) = (-1)^{\operatorname{inv}(\sigma)} = (-1)^{n-c(\sigma)}.$$

Thus any transposition reverses parity.

Fact 9.8. Dihedral Group
The symmetries of a regular n-gon form the dihedral group

$$D_{2n} = \langle r, s \mid r^n = e,\ s^2 = e,\ srs = r^{-1}\rangle.$$

Its elements are

$$e, r, r^2, \ldots, r^{n-1}, \qquad s, rs, r^2s, \ldots, r^{n-1}s.$$

This presentation is the standard algebraic model for rotation and reflection counting with Burnside's Lemma.

Definition 9.9. Cyclic Group
A group G is cyclic if there exists an element $g \in G$ s.t. every element of G is a power of g. In this case, we write

$$G = \langle g\rangle.$$

Theorem 9.10. Lagrange's Theorem
If G is a finite group and $H \leq G$, then

$$|H| \mid |G|.$$

Equivalently, the order of every subgroup divides the order of the group.

Corollary 9.11. Order of an Element
If G is a finite group and $g \in G$, then the order of g divides $|G|$. In particular,

$$g^{|G|} = e.$$

Theorem 9.12. Cauchy's Theorem
If G is a finite group and a prime p divides $|G|$, then G contains an element of order p.

Theorem 9.13. First Isomorphism Theorem for Groups
If $\varphi : G \to H$ is a group homomorphism, then

$$G/\ker\varphi \cong \operatorname{im}\varphi.$$

Definition 9.14. Group Action, Orbit, and Stabilizer
A group action of G on a set X is a map

$$G \times X \to X, \qquad (g, x) \mapsto g \cdot x,$$

satisfying $e \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$. For $x \in X$, the orbit and stabilizer are

$$G \cdot x = \{g \cdot x \mid g \in G\}, \qquad G_x = \{g \in G \mid g \cdot x = x\}.$$

Group actions translate algebraic symmetry into permutations of a set.

Theorem 9.15. Cayley's Theorem
Every group is isomorphic to a subgroup of a symmetric group. In particular, every finite group of order n is isomorphic to a subgroup of S_n.

Theorem 9.16. Orbit–Stabilizer Theorem
If a finite group G acts on a set X and $x \in X$, then

$$|G \cdot x| = [G : G_x] = \frac{|G|}{|G_x|},$$

where $G \cdot x$ is the orbit of x and G_x is its stabilizer.

Theorem 9.17. Class Equation
Let G be a finite group, and choose one representative x_i from each conjugacy class not contained in the center $Z(G)$. Then

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)].$$

Theorem 9.18. Sylow Theorems
Let G be a finite group and suppose

$$|G| = p^k m, \qquad p \nmid m,$$

where p is prime. Then G has a subgroup of order p^k, called a Sylow p-subgroup. Moreover, all Sylow p-subgroups are conjugate, and the number n_p of Sylow p-subgroups satisfies

$$n_p \equiv 1 \pmod{p}, \qquad n_p \mid m.$$

Strategy. For a finite group, subgroup orders and element orders must divide the group order. When a prime divides $|G|$, Cauchy's theorem guarantees an element of that prime order.

Fact 9.19. Quaternion Group
The quaternion group is

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\},$$

with

$$i^2 = j^2 = k^2 = ijk = -1.$$

It is nonabelian, but every subgroup of Q_8 is normal.

Remark. The quaternion group is a standard small group with behavior that is less intuitive than cyclic or dihedral groups, so it is a natural source of quick algebra questions.

Theorem 9.20. Fundamental Theorem of Finite Abelian Groups
Every finite abelian group is isomorphic to a direct product of cyclic groups of prime-power order:

$$G \cong \mathbb{Z}_{p_1^{a_1}} \times \cdots \times \mathbb{Z}_{p_r^{a_r}}.$$

The multiset of prime powers is uniquely determined by G, up to reordering. Equivalently, G admits a unique invariant-factor decomposition

$$G \cong \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_k}, \qquad n_1 \mid n_2 \mid \cdots \mid n_k,$$

up to isomorphism.

Definition 9.21. Ring and Field
A ring is a set R with operations $+$ and $\cdot$ s.t. $(R, +)$ is an abelian group, multiplication is associative, and multiplication distributes over addition on both sides. Throughout this book, rings are assumed to have a multiplicative identity 1_R unless stated otherwise. A field is a commutative ring in which $1_R \neq 0_R$ and every nonzero element has a multiplicative inverse.

Definition 9.22. Ideal, Ring Homomorphism, and Quotient Ring
An ideal I of a commutative ring R is an additive subgroup s.t.

$$r \in R \wedge a \in I \quad \Longrightarrow \quad ra \in I.$$

A ring homomorphism $\varphi : R \to S$ preserves addition, multiplication, and the multiplicative identity under the conventions of this book. Its kernel is an ideal. If I is an ideal of R, the quotient ring R/I consists of additive cosets with the induced ring operations.

Definition 9.23. Integral Domain
An integral domain is a commutative ring with identity in which there are no zero divisors. In other words, if $ab = 0$, then $a = 0$ or $b = 0$.

Definition 9.24. UFD, PID, and Euclidean Domain
A unique factorization domain, abbreviated UFD, is an integral domain in which every nonzero nonunit factors into irreducibles and this factorization is unique up to order and multiplication by units.

A principal ideal domain, abbreviated PID, is an integral domain in which every ideal has the form

$$(a) = \{ra \mid r \in R\}$$

for some $a \in R$.

A Euclidean domain is an integral domain equipped with a function

$$\delta : R \setminus \{0\} \to \mathbb{N}_0$$

s.t. for all $a, b \in R$ with $b \neq 0$, there exist $q, r \in R$ satisfying

$$a = bq + r, \qquad r = 0 \vee \delta(r) < \delta(b).$$

Theorem 9.25. Factorization-Domain Hierarchy
The standard implications are

$$\textit{field} \Longrightarrow \textit{Euclidean domain} \Longrightarrow \textit{PID} \Longrightarrow \textit{UFD} \Longrightarrow \textit{integral domain.}$$

None of the reverse implications holds in general.

Definition 9.26. Quaternion Algebra
The quaternions form the real algebra

$$\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\},$$

where

$$i^2 = j^2 = k^2 = ijk = -1.$$

Quaternion multiplication is associative but not commutative; for example,

$$ij = k, \qquad ji = -k.$$

If

$$q = a + bi + cj + dk,$$

then

$$\overline{q} = a - bi - cj - dk, \qquad |q|^2 = q\overline{q} = a^2 + b^2 + c^2 + d^2.$$

Every nonzero quaternion has inverse

$$q^{-1} = \frac{\overline{q}}{|q|^2}.$$

Thus $\mathbb{H}$ is a noncommutative division ring, not a field under the convention that fields are commutative.

Formula 9.27. Basic Factorization Identities
The most frequently used polynomial factorizations include

$$a^2 - b^2 = (a-b)(a+b),$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2), \qquad a^3 + b^3 = (a+b)(a^2 - ab + b^2).$$

More generally,

$$a^n - b^n = (a-b)\sum_{k=0}^{n-1} a^{n-1-k}b^k,$$

and, when n is odd,

$$a^n + b^n = (a+b)\sum_{k=0}^{n-1} (-1)^k a^{n-1-k}b^k.$$

Formula 9.28. Cubic Factorization Identity
For all a, b, c,

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca),$$

or equivalently,

$$a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a+b+c)\left((a-b)^2 + (b-c)^2 + (c-a)^2\right).$$

In particular, if $a + b + c = 0$, then

$$a^3 + b^3 + c^3 = 3abc.$$

Algorithm 9.29. Polynomial Division Algorithm
Let F be a field. If $f(x), g(x) \in F[x]$ and $g(x) \neq 0$, then there exist unique polynomials $q(x), r(x) \in F[x]$ s.t.

$$f(x) = q(x)g(x) + r(x), \qquad r(x) = 0 \vee \deg r < \deg g.$$

Algorithm 9.30. Euclidean Algorithm for Polynomials
Over a field F, repeatedly apply polynomial division

$$f = qg + r, \qquad r = 0 \vee \deg r < \deg g,$$

and replace (f, g) by (g, r). The last nonzero remainder, normalized to be monic, is $\gcd(f, g)$. Back-substitution expresses the gcd as

$$af + bg = \gcd(f, g)$$

for some $a, b \in F[x]$.

Theorem 9.31. Factor Theorem
For a polynomial $f(x) \in F[x]$ and an element $a \in F$,

$$f(a) = 0 \iff x - a \text{ divides } f(x).$$

Proposition 9.32. Root-Counting Principle for Polynomials
A nonzero polynomial of degree n over a field has at most n roots. Hence two polynomials of degree at most n that agree at $n+1$ distinct points are identical.

Formula 9.33. Lagrange Interpolation Formula
For distinct $x_0, \dots, x_n$, the unique polynomial of degree at most n satisfying $P(x_i) = y_i$ is

$$P(x) = \sum_{i=0}^{n} y_i \prod_{\substack{0 \le j \le n \\ j \ne i}} \frac{x - x_j}{x_i - x_j}.$$

Formula 9.34. Viète's Formulas
If

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 = a_n \prod_{i=1}^{n} (x - r_i),$$

where the roots are counted with multiplicity, then

$$\sum_{i=1}^{n} r_i = -\frac{a_{n-1}}{a_n}, \qquad \prod_{i=1}^{n} r_i = (-1)^n \frac{a_0}{a_n}.$$

More generally, the elementary symmetric sums of the roots are the coefficients with alternating signs, divided by a_n.

Formula 9.35. Newton's Identities; Newton–Girard Formulas
Let e_k be the elementary symmetric polynomials in roots $r_1, \dots, r_n$, and let

$$p_k = r_1^k + \cdots + r_n^k.$$

With $e_0 = 1$, for $1 \le k \le n$,

$$p_k - e_1 p_{k-1} + e_2 p_{k-2} - \cdots + (-1)^{k-1} e_{k-1} p_1 + (-1)^k k e_k = 0.$$

Formula 9.36. Polynomial Discriminant
If

$$f(x) = a_n \prod_{i=1}^{n} (x - r_i),$$

then

$$\mathrm{Disc}(f) = a_n^{2n-2} \prod_{1 \le i < j \le n} (r_i - r_j)^2.$$

It vanishes if and only if f has a repeated root.

Theorem 9.37. Rational Root Theorem
Let

$$f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x].$$

If a rational number p/q in lowest terms is a root of f, then

$$p \mid a_0, \qquad q \mid a_n.$$

Theorem 9.38. Cauchy Root Bound
If

$$f(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0, \qquad a_n \neq 0,$$

then every complex root z of f satisfies

$$|z| \leq 1 + \max_{0 \leq k < n} \left| \frac{a_k}{a_n} \right|.$$

Thus all roots lie in an explicitly computable disk before any root is solved for exactly.

Theorem 9.39. Eisenstein's Irreducibility Criterion
Let

$$f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x].$$

If there exists a prime p s.t.

$$p \nmid a_n, \qquad p \mid a_i \quad (0 \leq i < n), \qquad p^2 \nmid a_0,$$

then f is irreducible over $\mathbb{Q}$.

Theorem 9.40. Fundamental Theorem of Symmetric Polynomials
Every symmetric polynomial can be expressed uniquely as a polynomial in the elementary symmetric polynomials.

Theorem 9.41. Combinatorial Nullstellensatz, Coefficient Form
Let $P \in F[x_1, \ldots, x_n]$ have total degree $t_1 + \cdots + t_n$, and suppose the coefficient of $x_1^{t_1} \cdots x_n^{t_n}$ is nonzero. If

$$|S_i| > t_i$$

for every finite set $S_i \subseteq F$, then some $(s_1, \ldots, s_n) \in S_1 \times \cdots \times S_n$ satisfies

$$P(s_1, \ldots, s_n) \neq 0.$$

Strategy. Polynomial Method. Encode a forbidden configuration as zeros of a polynomial, then use a degree bound, interpolation, factorization, or a nonzero coefficient to prove that the polynomial cannot vanish everywhere.

Theorem 9.42. Fundamental Theorem of Algebra
Every nonconstant complex polynomial has at least one complex root. Equivalently, every polynomial of degree $n \geq 1$ over $\mathbb{C}$ splits into n linear factors over $\mathbb{C}$, counted with multiplicity.

Definition 9.43. Algebraically Closed Field
A field F is algebraically closed if every nonconstant polynomial in $F[x]$ has a root in F.

Definition 9.44. Field Extension and Galois Group
If $F \subseteq K$ are fields, then K/F is a field extension and its degree is

$$[K : F] = \dim_F K.$$

The extension is finite if this dimension is finite. The group

$$\mathrm{Gal}(K/F) = \{\sigma \in \mathrm{Aut}(K) \mid \sigma(a) = a \ \forall a \in F\}$$

is the Galois group. For a finite Galois extension, intermediate fields and subgroups of the Galois group are related by the Galois correspondence.

Theorem 9.45. Fundamental Theorem of Galois Theory; Galois Correspondence
For a finite Galois extension L/K, there is an inclusion-reversing correspondence between intermediate fields $K \subseteq E \subseteq L$ and subgroups $H \leq \mathrm{Gal}(L/K)$, given by

$$E \longmapsto \mathrm{Gal}(L/E), \qquad H \longmapsto L^H.$$

Theorem 9.46. Abel–Ruffini Theorem
There is no formula for the roots of a general polynomial of degree 5 *or higher using only the field operations and radicals.*

Remark. The Abel–Ruffini theorem does not say that every quintic equation is unsolvable. It says that no universal radical formula exists for the general quintic.

Chapter 10

Topology

Related POSTECH Courses: (MATH321) General Topology
Related Books: Topology (James R. Munkres)

Definition 10.1. Topology
A topology on a set X is a collection $\mathcal{T}$ of subsets of X whose elements are called open sets, satisfying:

(i) $\varnothing, X \in \mathcal{T}$;

(ii) arbitrary unions of sets in $\mathcal{T}$ belong to $\mathcal{T}$;

(iii) finite intersections of sets in $\mathcal{T}$ belong to $\mathcal{T}$.

Definition 10.2. Basis for a Topology
A collection $\mathcal{B}$ of subsets of X is a basis if every $x \in X$ lies in some $B \in \mathcal{B}$ and, whenever $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$, there exists $B_3 \in \mathcal{B}$ s.t.

$$x \in B_3 \subseteq B_1 \cap B_2.$$

The topology generated by $\mathcal{B}$ consists of arbitrary unions of basis elements. Open balls form the standard basis for the metric topology of a metric space.

Definition 10.3. First- and Second-Countability
A topological space is first-countable if every point has a countable local neighborhood basis. It is second-countable if its topology has a countable basis. Every second-countable space is first-countable, but not conversely in general.

Definition 10.4. Open and Closed Sets
If $(X, \mathcal{T})$ is a topological space, a subset $U \subseteq X$ is open if $U \in \mathcal{T}$. A subset $F \subseteq X$ is closed if $X \setminus F \in \mathcal{T}$.

Definition 10.5. Compactness
A topological space X is compact if every open cover has a finite subcover. Equivalently, for every family $\mathcal{U} \subseteq \mathcal{T}$,

$$X \subseteq \bigcup_{U \in \mathcal{U}} U \Rightarrow \exists U_1, \dots, U_n \in \mathcal{U} \text{ s.t. } X \subseteq U_1 \cup \cdots \cup U_n.$$

Definition 10.6. Connectedness
A topological space X is connected if it cannot be written as the union of two disjoint nonempty open sets. Equivalently, the only subsets of X that are both open and closed are $\varnothing$ and X.

Definition 10.7. Continuity in Topological Spaces
A map $f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ is continuous if the inverse image of every open set is open:

$$\forall V \in \mathcal{T}_Y, \qquad f^{-1}(V) \in \mathcal{T}_X.$$

Definition 10.8. Homeomorphism
A homeomorphism is a bijective continuous map whose inverse is also continuous. Equivalently, $f : X \xrightarrow{\sim} Y$ is a homeomorphism if f and f^{-1} are continuous. Two spaces are homeomorphic, written $X \cong Y$, if such a map exists.

Theorem 10.9. Basic Topological Invariants under Continuous Maps
A continuous image of a compact space is compact, and a continuous image of a connected space is connected. Consequently, compactness and connectedness are preserved by homeomorphism.

Definition 10.10. Hausdorff Space
A topological space X is Hausdorff if distinct points can be separated by disjoint open neighborhoods:

$$x \neq y \quad \Longrightarrow \quad \exists U, V \text{ open s.t. } x \in U,\ y \in V,\ U \cap V = \varnothing.$$

Every metric space is Hausdorff.

Theorem 10.11. Compact Subsets of Hausdorff Spaces
Every compact subset of a Hausdorff space is closed. In particular, a continuous bijection from a compact space onto a Hausdorff space is automatically a homeomorphism.

Definition 10.12. Path, Path-Connectedness, and Homotopy
A path from x_0 to x_1 in X is a continuous map

$$\gamma : [0, 1] \to X, \qquad \gamma(0) = x_0, \quad \gamma(1) = x_1.$$

The space X is path-connected if every two points can be joined by a path. Two maps $f, g : X \to Y$ are homotopic, written $f \simeq g$, if there exists a continuous map

$$H : X \times [0, 1] \to Y$$

with $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$. A loop is a path with the same initial and terminal point.

Theorem 10.13. Path-Connectedness Implies Connectedness
Every path-connected space is connected. The converse is false in general, so connectedness and path-connectedness should not be interchanged without additional hypotheses.

Theorem 10.14. Jordan Curve Theorem
Every simple closed curve in the plane separates the plane into exactly two connected components, an interior and an exterior, and is the common boundary of both.

Theorem 10.15. Brouwer Fixed-Point Theorem
Every continuous map from a closed ball in $\mathbb{R}^n$ to itself has at least one fixed point.

Theorem 10.16. Borsuk–Ulam Theorem
Every continuous map

$$f : S^n \to \mathbb{R}^n$$

takes some pair of antipodal points to the same value.

Corollary 10.17. Ham Sandwich Theorem
Any n bounded measurable subsets of $\mathbb{R}^n$ can be bisected simultaneously by one hyperplane.

Theorem 10.18. Hairy Ball Theorem
Every continuous tangent vector field on an even-dimensional sphere S^{2m} vanishes at some point.

Theorem 10.19. Invariance of Domain
If $U \subseteq \mathbb{R}^n$ is open and $f : U \to \mathbb{R}^n$ is continuous and injective, then $f(U)$ is open and f is a homeomorphism from U onto $f(U)$.

Definition 10.20. Manifold
An n-dimensional topological manifold is a Hausdorff, second-countable topological space in which every point has a neighborhood homeomorphic to an open subset of $\mathbb{R}^n$.

Definition 10.21. Fundamental Group
Fix a base point $x_0 \in X$. The fundamental group $\pi_1(X, x_0)$ is the set of endpoint-preserving homotopy classes of loops based at x_0, with multiplication induced by concatenation of loops. It is a basic algebraic invariant of a topological space. A path-connected space is simply connected if

$$\pi_1(X, x_0) \cong \{e\},$$

equivalently, every loop can be contracted to a point.

Fact 10.22. Fundamental Groups of Familiar Spaces
The following basic examples are worth recognizing:

$$\pi_1(S^1) \cong \mathbb{Z}, \qquad \pi_1(S^n) \cong \{e\} \quad (n \geq 2), \qquad \pi_1(\mathbb{T}^2) \cong \mathbb{Z}^2.$$

Strategy. For introductory topology questions, the most common quick checks are: compactness means every open cover has a finite subcover, connectedness means the space cannot be separated into two disjoint nonempty open sets, and a homeomorphism is a continuous bijection with continuous inverse.

Formula 10.23. Euler Characteristic of a Surface
For a triangulated closed surface, the Euler characteristic is

$$\chi = V - E + F.$$

For an orientable closed surface of genus g,

$$\chi = 2 - 2g.$$

Theorem 10.24. Classification of Closed Surfaces
Every connected closed surface is homeomorphic to either an orientable surface of genus $g \geq 0$ or a nonorientable surface obtained as a connected sum of $k \geq 1$ projective planes.

Theorem 10.25. Poincaré Conjecture
Every closed simply connected 3-manifold is homeomorphic to the 3-sphere S^3.

Remark. The Poincaré conjecture was proved by Grigori Perelman using Ricci flow. The fundamental correspondence is

$$\text{Poincaré conjecture} \longleftrightarrow \text{Perelman} \longleftrightarrow \text{Ricci flow}.$$

Theorem 10.26. Thurston Geometrization Theorem
Every closed orientable 3-manifold can be decomposed along embedded 2-spheres and incompressible tori into pieces whose interiors admit geometric structures modeled on one of the eight Thurston geometries:

$$S^3,\ E^3,\ H^3,\ S^2 \times \mathbb{R},\ H^2 \times \mathbb{R},\ \widetilde{\mathrm{SL}}_2(\mathbb{R}),\ \mathrm{Nil},\ \mathrm{Sol}.$$

The sphere decomposition is the prime decomposition, and the torus decomposition is the JSJ decomposition of the irreducible pieces.

Fact 10.27. The Eight Thurston Geometries
The eight model geometries are

$$S^3, \quad \mathrm{E}^3, \quad \mathbb{H}^3, \quad S^2 \times \mathbb{R}, \quad \mathbb{H}^2 \times \mathbb{R}, \quad \widetilde{\mathrm{SL}}_2(\mathbb{R}), \quad \mathrm{Nil}, \quad \mathrm{Sol}.$$

Chapter 11

Miscellaneous Topics

Strategy. Symmetry and Without Loss of Generality. Before splitting into cases, identify transformations that preserve the problem. If several cases differ only by a permutation, reflection, sign change, or cyclic relabeling, choose one representative and state explicitly why the remaining cases follow by symmetry. "Without loss of generality" is justified only after this invariance has been identified.

Strategy. Parity and Modular Obstructions. Before searching for a construction or integer solution, reduce the relevant quantities modulo a small modulus. Parity, residues of powers, and invariant congruence classes often rule out entire families of possibilities with almost no calculation.

Strategy. Telescoping and Local Cancellation. For a sum or product indexed consecutively, try to rewrite each term so that neighboring factors cancel. Typical forms are

$$a_k = b_k - b_{k+1} \qquad \text{or} \qquad a_k = \frac{b_{k+1}}{b_k}.$$

Then

$$\sum_{k=m}^{n} a_k = b_m - b_{n+1}, \qquad \prod_{k=m}^{n} a_k = \frac{b_{n+1}}{b_m}.$$

Strategy. Descent and Minimal Counterexample. If a statement about positive integers or finite objects is suspected to hold universally, assume a counterexample exists and choose one minimal under a natural size parameter. Use the defining relations to construct a strictly smaller counterexample. This contradiction is the abstract form of infinite descent and is useful far beyond number theory.

Strategy. Complex Numbers as Plane Geometry. Identify the Euclidean plane with $\mathbb{C}$ when rotations, similarities, roots of unity, or cyclic configurations are prominent. Multiplication by $e^{i\theta}$ is rotation by θ, multiplication by a positive scalar is dilation, and complex conjugation is reflection across the real axis. Translate back to ordinary geometry only after the algebraic relation has been simplified.

Strategy. Polynomial Encoding. Encode a finite set of numbers as roots of a polynomial, or encode a sequence by a generating polynomial or generating function, when sums, products, symmetric expressions, or recurrence relations are involved. Root–coefficient relations, interpolation, and coefficient extraction can then replace case-by-case manipulation.

Tactic. Floor and Fractional-Part Identities. For every real x,

$$x = \lfloor x \rfloor + \{x\}, \qquad 0 \le \{x\} < 1.$$

Moreover,

$$\lfloor x \rfloor + \lfloor -x \rfloor = \begin{cases} 0, & x \in \mathbb{Z}, \\ -1, & x \notin \mathbb{Z}, \end{cases}$$

and for $n \in \mathbb{Z}$,

$$\lfloor x + n \rfloor = \lfloor x \rfloor + n.$$

These identities convert many apparently analytic floor problems into integer casework.

Strategy. Iterated Knowledge Elimination. Let Ω_0 be a finite set of possible states. For each agent i and state $\omega \in \Omega$, let

$$\mathcal{I}_i^{\Omega}(\omega) = \{\omega' \in \Omega \mid \omega' \text{ gives agent } i \text{ the same information as } \omega\}.$$

Relative to the current state space Ω, agent i knows the state precisely when

$$|\mathcal{I}_i^{\Omega}(\omega)| = 1.$$

A truthful public announcement restricts Ω to exactly the states at which that announcement is true. Repeating this restriction models successive statements about knowledge or ignorance. In particular, the statement "I knew that Q" means

$$\forall \omega' \in \mathcal{I}_i^{\Omega}(\omega), \qquad Q(\omega').$$

Strategy. Transform the Given Objects. Replace the original configuration by an equivalent one whose structure is easier to see: partition a region, introduce prefix sums, encode a movement by vectors, pass to residues, diagonalize a matrix, or complete a partial object. The transformed problem should preserve the quantity or property being asked about.

Strategy. Invariant, Monovariant, and Extremal Principles. An invariant remains unchanged under every allowed move. A monovariant moves strictly in one direction and therefore prevents cycles or proves termination. The extremal principle chooses a largest, smallest, first, or last object and exploits the extra restrictions forced by extremality. These are among the most common contest ideas hidden behind elementary statements.

Strategy. Complete the Configuration and Work Backward. First imagine a completed solution, then determine which parts are forced and which choices remain free. This is effective in geometric constructions, permutation completions, lattice paths, tilings, and assignment problems. Fixing one object under a symmetry often removes redundant cases before the remaining count begins.

Strategy. Rotate, Reflect, and Superimpose. A rotation, reflection, or translated copy can convert a broken path into a straight one, expose equal vectors, pair terms, or turn a local pattern into a global invariant. When two configurations overlap, compare what cancels and what remains rather than recomputing each separately.

Strategy. Homogenization and Normalization. If an inequality is homogeneous, scale the variables to impose a simple condition such as $a + b + c = 1$, $abc = 1$, or $x^2 + y^2 = 1$.

Strategy. Equality Cases and Smoothing. Identify the expected equality case before selecting an inequality. For a symmetric expression, replacing two variables by their average often reveals the extremal configuration.

Theorem 11.1. Triangle and Reverse Triangle Inequalities
For real or complex numbers x, y,

$$||x| - |y|| \leq |x - y| \leq |x| + |y|.$$

More generally, every norm satisfies

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

Formula 11.2. Basic Sum-of-Squares Identities
For real a, b, c,

$$a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}\Big((a-b)^2 + (b-c)^2 + (c-a)^2\Big) \geq 0.$$

Consequently,

$$a^2 + b^2 + c^2 \geq ab + bc + ca$$

and

$$(a+b+c)^2 \leq 3(a^2 + b^2 + c^2).$$

Formula 11.3. Frequently Used Two-Variable Inequalities
For real x, y,

$$x^2 + y^2 \geq 2xy.$$

For positive x, y,

$$\frac{x}{y} + \frac{y}{x} \geq 2, \qquad x + \frac{1}{x} \geq 2.$$

Equality holds exactly when the two compared positive quantities are equal.

Theorem 11.4. Lagrange's Identity
For real numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$,

$$\left(\sum_{i=1}^n a_i^2\right)\left(\sum_{i=1}^n b_i^2\right) - \left(\sum_{i=1}^n a_i b_i\right)^2 = \sum_{1 \leq i < j \leq n} (a_i b_j - a_j b_i)^2.$$

The nonnegativity of the right-hand side gives the Cauchy–Schwarz inequality immediately.

Theorem 11.5. QM–AM–GM–HM Inequalities
For positive $x_1, \ldots, x_n$,

$$\sqrt{\frac{x_1^2 + \cdots + x_n^2}{n}} \geq \frac{x_1 + \cdots + x_n}{n} \geq \sqrt[n]{x_1 \cdots x_n} \geq \frac{n}{\frac{1}{x_1} + \cdots + \frac{1}{x_n}}.$$

Equality holds if and only if all variables are equal.

Theorem 11.6. Weighted AM–GM Inequality
If $w_i \geq 0$, $\sum_{i=1}^n w_i = 1$, and $x_i > 0$, then

$$\sum_{i=1}^n w_i x_i \geq \prod_{i=1}^n x_i^{w_i}.$$

Equality holds when all x_i with positive weight are equal.

Theorem 11.7. Power Mean Inequality
For positive $x_1, \ldots, x_n$ and real $r \neq 0$, define

$$M_r = \left(\frac{x_1^r + \cdots + x_n^r}{n}\right)^{1/r},$$

and define $M_0 = (x_1 \cdots x_n)^{1/n}$. If $r < s$, then

$$M_r \leq M_s.$$

The harmonic, geometric, arithmetic, and quadratic means correspond to $r = -1, 0, 1, 2$, respectively.

Theorem 11.8. Young's Inequality
If $x, y \geq 0$, $p, q > 1$, and

$$\frac{1}{p} + \frac{1}{q} = 1,$$

then

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q}.$$

Theorem 11.9. Cauchy–Schwarz Inequality, Engel Form
For real a_i and positive b_i,

$$\sum_{i=1}^{n} \frac{a_i^2}{b_i} \geq \frac{(a_1 + \cdots + a_n)^2}{b_1 + \cdots + b_n}.$$

Theorem 11.10. Hölder's Inequality
If $p, q > 1$ and $1/p + 1/q = 1$, then

$$\sum_{i=1}^{n} |a_i b_i| \leq \left(\sum_{i=1}^{n} |a_i|^p \right)^{1/p} \left(\sum_{i=1}^{n} |b_i|^q \right)^{1/q}.$$

Theorem 11.11. Minkowski's Inequality
For $p \geq 1$,

$$\left(\sum_{i=1}^{n} |a_i + b_i|^p \right)^{1/p} \leq \left(\sum_{i=1}^{n} |a_i|^p \right)^{1/p} + \left(\sum_{i=1}^{n} |b_i|^p \right)^{1/p}.$$

Theorem 11.12. Rearrangement Inequality
If

$$a_1 \leq \cdots \leq a_n, \qquad b_1 \leq \cdots \leq b_n,$$

then for every permutation $\sigma \in S_n$,

$$\sum_{i=1}^{n} a_i b_{n+1-i} \leq \sum_{i=1}^{n} a_i b_{\sigma(i)} \leq \sum_{i=1}^{n} a_i b_i.$$

Theorem 11.13. Chebyshev's Sum Inequality
If $a_1 \leq \cdots \leq a_n$ and $b_1 \leq \cdots \leq b_n$, then

$$\frac{1}{n} \sum_{i=1}^{n} a_i b_i \geq \left(\frac{1}{n} \sum_{i=1}^{n} a_i \right) \left(\frac{1}{n} \sum_{i=1}^{n} b_i \right).$$

Theorem 11.14. Schur's Inequality
For $a, b, c \geq 0$ and $r \geq 0$,

$$\sum_{\text{cyc}} a^r (a-b)(a-c) \geq 0.$$

Formula 11.15. Nesbitt's Inequality
For positive a, b, c,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

Theorem 11.16. Bernoulli's Inequality
If $x \geq -1$ and $r \geq 1$, then

$$(1+x)^r \geq 1 + rx.$$

Strategy. Comparing Powers by Logarithms. To compare two positive numbers such as a^b and b^a, take logarithms:

$$a^b > b^a \iff b \ln a > a \ln b \iff \frac{\ln a}{a} > \frac{\ln b}{b}.$$

The function $\ln x/x$ is decreasing for $x > e$. This quickly handles comparisons such as 11^9 versus 9^{11} or e^π versus π^e.

Formula 11.17. More Useful Series Values
The following values are common in fast calculation problems:

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1, \qquad \sum_{n=1}^{\infty} \frac{1}{n(n+2)} = \frac{3}{4},$$

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}, \qquad \sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}.$$

Strategy. These values often appear after a simple transformation. For example,

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2}\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\right) = \frac{\ln 2}{2}.$$

Part II

Facts to Memorize

Chapter 12

Mathematical Prizes and Congresses

Science Quiz often asks for the short association between a prize, a congress, a laureate, and a mathematical achievement. This chapter records the facts that should be recognized quickly: not full biographies, but the names, years, quiz anchors, and historical landmarks most likely to become quiz answers.

Fact 12.1. Science Quiz Prize Triad
The three mathematics prizes that should be recognized first are

$$\text{Fields Medal}, \qquad \text{Abel Prize}, \qquad \text{Wolf Prize in Mathematics.}$$

Recognize each prize through the association

$$\text{prize} \quad \longleftrightarrow \quad \text{laureate} \quad \longleftrightarrow \quad \text{achievement or area.}$$

Strategy. History questions often ask for a short association rather than a full biography. Current prominence, a recent award, or a connection to the POSTECH–KAIST context makes such an association especially relevant.

Prize	Origin and first award	Namesake	Administration and cycle
Fields Medal	Proposed at ICM Toronto in 1924; first awarded in 1936.	John Charles Fields, Canadian mathematician and organizer of ICM 1924, who donated funds for the medals.	Awarded by the IMU at the ICM, normally every four years; subject to the age condition.
Abel Prize	Established by the Norwegian Parliament in 2002; first awarded in 2003.	Niels Henrik Abel, Norwegian mathematician known for algebra, analysis, and the impossibility of solving the general quintic by radicals.	Awarded annually by the Norwegian Academy of Science and Letters; presented in Oslo; no age restriction.
Wolf Prize in Mathematics	The Wolf Foundation was established in 1975; the prizes were first awarded in 1978.	Ricardo Wolf, a German-born Cuban inventor, diplomat, and philanthropist; he was not a mathematician.	Awarded by the Wolf Foundation in Israel; normally annual, although mathematics is not necessarily awarded every year; no age restriction.

Fields Medal and ICM

The International Congress of Mathematicians, commonly abbreviated as ICM, is the most important international congress in mathematics. The Fields Medal is awarded by the International Mathematical Union at the ICM every four years. At ICM Toronto in 1924, a resolution proposed two gold medals for outstanding mathematical achievement. John Charles Fields later donated funds for the medals that bear his name. A Fields Medalist must satisfy the age condition: the candidate's 40th birthday must not occur before January 1 of the year of the Congress. Since 1966, up to four medals may be awarded at an ICM.

Year	ICM location	Medalists	Quiz anchors
1936	Oslo, Norway	Lars Ahlfors; Jesse Douglas	Complex analysis, Riemann surfaces; minimal surfaces, Plateau problem.
1950	Cambridge, United States	Laurent Schwartz; Atle Selberg	Distribution theory; analytic number theory, sieve methods, zeta functions.
1954	Amsterdam, Netherlands	Kunihiko Kodaira; Jean-Pierre Serre	Complex geometry, algebraic geometry; algebraic topology, sheaf theory.
1958	Edinburgh, United Kingdom	Klaus Roth; René Thom	Diophantine approximation; cobordism, differential topology.
1962	Stockholm, Sweden	Lars Hörmander; John Milnor	Partial differential equations; differential topology, exotic spheres.
1966	Moscow, USSR	Michael Atiyah; Paul Cohen; Alexander Grothendieck; Stephen Smale	K-theory and index theorem; forcing and continuum hypothesis; modern algebraic geometry; generalized Poincaré conjecture.
1970	Nice, France	Alan Baker; Heisuke Hironaka; Serge Novikov; John G. Thompson	Transcendental number theory; resolution of singularities; topology and geometry; finite group theory.
1974	Vancouver, Canada	Enrico Bombieri; David Mumford	Number theory, complex analysis, minimal surfaces; moduli spaces and algebraic geometry.
1978	Helsinki, Finland	Pierre Deligne; Charles Fefferman; Grigory Margulis; Daniel Quillen	Weil conjectures; several complex variables; Lie groups and ergodic theory; algebraic K-theory.
1982	Warsaw, Poland (held in 1983)	Alain Connes; William Thurston; Shing-Tung Yau	Noncommutative geometry; 3-manifolds and low-dimensional topology; Calabi conjecture and geometric analysis.
1986	Berkeley, United States	Simon Donaldson; Gerd Faltings; Michael Freedman	Gauge theory and 4-manifolds; Mordell conjecture and arithmetic geometry; topological 4-dimensional Poincaré conjecture.
1990	Kyoto, Japan	Vladimir Drinfeld; Vaughan Jones; Shigefumi Mori; Edward Witten	Quantum groups and Langlands program; Jones polynomial; minimal model program; mathematical physics and quantum field theory.
1994	Zurich, Switzerland	Jean Bourgain; Pierre-Louis Lions; Jean-Christophe Yoccoz; Efim Zelmanov	Analysis and Banach spaces; partial differential equations; dynamical systems; restricted Burnside problem.

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Year	ICM location	Medalists	Quiz anchors
1998	Berlin, Germany	Richard Borcherds; W. Timothy Gowers; Maxim Kontsevich; Curtis T. McMullen	Moonshine and automorphic forms; functional analysis and combinatorics; algebraic geometry and mathematical physics; complex dynamics.
2002	Beijing, China	Laurent Lafforgue; Vladimir Voevodsky	Langlands correspondence over function fields; motivic cohomology and $\mathbb{A}^1$-homotopy theory.
2006	Madrid, Spain	Andrei Okounkov; Grigori Perelman; Terence Tao; Wendelin Werner	Probability, representation theory, algebraic geometry; Ricci flow; PDE, harmonic analysis, additive combinatorics; SLE and Brownian motion.
2010	Hyderabad, India	Elon Lindenstrauss; Ngo Bao Chau; Stanislav Smirnov; Cédric Villani	Ergodic theory and number theory; Fundamental Lemma; conformal invariance; Boltzmann equation and optimal transport.
2014	Seoul, South Korea	Artur Avila; Manjul Bhargava; Martin Hairer; Maryam Mirzakhani	Dynamical systems; geometry of numbers; stochastic PDE; Riemann surfaces and moduli spaces.
2018	Rio de Janeiro, Brazil	Caucher Birkar; Alessio Figalli; Peter Scholze; Akshay Venkatesh	Fano varieties and minimal model program; optimal transport; p-adic geometry; number theory and homogeneous dynamics.
2022	Originally Saint Petersburg, Russia; virtual congress; awards in Helsinki, Finland	Hugo Duminil-Copin; June Huh; James Maynard; Maryna Viazovska	Phase transitions in statistical physics; Hodge theory in combinatorics and matroids; prime numbers; sphere packing and the E_8 lattice.
2026	Philadelphia, United States	Yu Deng; John Pardon; Jacob Tsimerman; Hong Wang	PDE and kinetic theory; symplectic geometry and topology; arithmetic geometry and o-minimality; harmonic analysis, geometric measure theory, and the Kakeya problem.

Fact 12.2. ICM and Fields Medal landmarks
The following associations are especially useful.

- ICM stands for International Congress of Mathematicians.
- IMU stands for International Mathematical Union.
- The Fields Medal was first awarded in 1936 at ICM Oslo.
- The first two Fields Medalists were Lars Ahlfors and Jesse Douglas.
- The first year with four Fields Medalists was 1966.
- Jean-Pierre Serre is the youngest Fields Medalist.
- Edward Witten was the first physicist to receive the Fields Medal; as an undergraduate, he majored in history rather than mathematics.
- Andrew Wiles received a special IMU silver plaque in 1998 for his proof of Fermat's Last Theorem.

- Grigori Perelman declined the Fields Medal in 2006.
- Maryam Mirzakhani was the first woman Fields Medalist.
- Maryna Viazovska was the second woman Fields Medalist.
- Hong Wang became the third woman Fields Medalist in 2026.
- ICM 2014 was held in Seoul, South Korea.
- ICM 2022 was originally scheduled for Saint Petersburg, Russia. It was held as a virtual congress, and the prize ceremony took place in Helsinki, Finland.
- ICM 2026: Philadelphia, United States, July 23–30, 2026.
- There is no ICM scheduled for 2027: the regular cycle proceeds from 2026 to 2030.
- ICM 2030 is scheduled for Glasgow, United Kingdom; the associated IMU General Assembly is scheduled for Dublin, Ireland.

Fact 12.3. Fields Medal National and Regional Firsts
The following short associations are useful when a quiz asks for a national or regional first. The labels refer to the laureates' country of origin or the national mathematical community with which the record is conventionally associated.

- Kunihiko Kodaira, 1954 $\leftrightarrow$ first Japanese Fields Medalist.
- Ngô Bào Châu, 2010 $\leftrightarrow$ first Vietnamese Fields Medalist.
- Artur Avila, 2014 $\leftrightarrow$ first Brazilian and first Latin American Fields Medalist.
- Maryam Mirzakhani, 2014 $\leftrightarrow$ first Iranian and first woman Fields Medalist.

Because citizenship and institutional affiliation may change during a mathematician's career, the person's name and year are safer quiz anchors than an oversimplified nationality label.

Fact 12.4. Fields Medal design
The obverse of the Fields Medal bears a profile of Archimedes and the Latin inscription

TRANSIRE SUUM PECTUS MUNDOQUE POTIRI.

The reverse shows a sphere inscribed in a cylinder, recalling the result that Archimedes requested for his tomb. For the corresponding sphere and circumscribed cylinder,

$$V_{\text{cylinder}} : V_{\text{sphere}} = 3 : 2,$$

equivalently

$$\frac{V_{\text{sphere}}}{V_{\text{cylinder}}} = \frac{2}{3}.$$

Abel Prize

The Abel Prize is awarded by the Norwegian Academy of Science and Letters. It is named after Niels Henrik Abel. The Norwegian Parliament established the modern prize in 2002, the bicentenary of Abel's birth, and the first award was made in 2003. Unlike the Fields Medal, the Abel Prize has no age restriction. For Science Quiz, the Abel Prize is often asked through the pattern

$$\text{year} \quad \longleftrightarrow \quad \text{laureate} \quad \longleftrightarrow \quad \text{famous theorem or area.}$$

Year	Laureate(s)	Quiz anchors
2003	Jean-Pierre Serre	Algebraic topology, algebraic geometry, number theory; first Abel Prize laureate.
2004	Michael Atiyah; Isadore Singer	Atiyah–Singer index theorem.
2005	Peter D. Lax	Partial differential equations and applied mathematics.
2006	Lennart Carleson	Harmonic analysis; convergence of Fourier series.
2007	Srinivasa S. R. Varadhan	Probability theory; large deviations.
2008	John G. Thompson; Jacques Tits	Finite groups, classification of finite simple groups; buildings and group theory.
2009	Mikhail Gromov	Geometry, geometric group theory, symplectic geometry.
2010	John Tate	Number theory and arithmetic geometry.
2011	John Milnor	Topology, geometry, dynamics; exotic spheres.
2012	Endre Szemerédi	Discrete mathematics and theoretical computer science; Szemerédi's theorem.
2013	Pierre Deligne	Algebraic geometry; Weil conjectures.
2014	Yakov Sinai	Dynamical systems, ergodic theory, mathematical physics.
2015	John F. Nash Jr.; Louis Nirenberg	Nonlinear PDE, geometry, Nash embedding theorem.
2016	Andrew Wiles	Fermat's Last Theorem; modularity of elliptic curves.
2017	Yves Meyer	Wavelet theory; applications from signal compression to gravitational-wave analysis.
2018	Robert P. Langlands	Langlands program; representation theory and number theory.
2019	Karen Uhlenbeck	Geometric analysis and gauge theory; first woman Abel Prize laureate.
2020	Hillel Furstenberg; Gregory Margulis	Probability, dynamics, ergodic theory, Lie groups.
2021	László Lovász; Avi Wigderson	Discrete mathematics, theoretical computer science, complexity theory.
2022	Dennis P. Sullivan	Topology, geometric topology, dynamical systems.
2023	Luis A. Caffarelli	Nonlinear partial differential equations, free-boundary problems, Monge–Ampère equation.
2024	Michel Talagrand	Probability theory, functional analysis, spin glasses, concentration inequalities.
2025	Masaki Kashiwara	Algebraic analysis, D-modules, representation theory, crystal bases.
2026	Gerd Faltings	Arithmetic geometry; Mordell conjecture, Lang's conjectures, Diophantine geometry.

Fact 12.5. Abel Prize landmarks
The following associations summarize the prize's main quiz anchors.

- Abel Prize ↔ Norwegian Academy of Science and Letters.
- Abel Prize ↔ Niels Henrik Abel.
- First Abel Prize laureate ↔ Jean-Pierre Serre.
- First woman Abel Prize laureate ↔ Karen Uhlenbeck.
- Since 2019, the Abel Prize has carried a monetary award of NOK 7.5 million, on the order of one million US dollars.

- John Milnor ↔ Fields Medal 1962 and Abel Prize 2011.
- Andrew Wiles ↔ Abel Prize 2016 ↔ Fermat's Last Theorem.
- Yves Meyer ↔ Abel Prize 2017 ↔ wavelets.
- Robert Langlands ↔ Abel Prize 2018 ↔ Langlands program.
- Gerd Faltings ↔ Fields Medal 1986 and Abel Prize 2026.

Wolf Prize in Mathematics

The Wolf Prize is awarded by the Wolf Foundation, established in Israel in 1975 by Ricardo Wolf. Wolf was a German-born Cuban inventor, diplomat, and philanthropist, not a mathematician. The prizes were first awarded in 1978. In the sciences, the Wolf Prize is awarded in Medicine, Agriculture, Mathematics, Chemistry, and Physics. It is especially important historically because many Wolf Prize in Mathematics laureates later received, or had already received, other major mathematical prizes. Unlike the Fields Medal, it is not restricted by age, and mathematics need not be awarded every year.

Year	Laureate(s)	Quiz anchors
1978	Israel Gelfand; Carl L. Siegel	Representation theory, functional analysis; number theory, several complex variables.
1979	Jean Leray; André Weil	PDE and algebraic topology; algebraic geometry and number theory.
1980	Henri Cartan; Andrey Kolmogorov	Several complex variables and algebraic topology; probability theory and dynamical systems.
1981	Lars Ahlfors; Oscar Zariski	Complex analysis and Riemann surfaces; algebraic geometry.
1982	Mark Krein; Hassler Whitney	Functional analysis; differential topology and Whitney embedding.
1983	Shiing-Shen Chern	Global differential geometry, Chern classes.
1984	Paul Erdős	Combinatorics, number theory, probabilistic method.
1985	Kunihiko Kodaira; Hans Lewy	Complex geometry; partial differential equations.
1986	Samuel Eilenberg; Atle Selberg	Algebraic topology and category theory; analytic number theory and Selberg trace formula.
1987	Kiyoshi Itô; Peter D. Lax	Stochastic calculus; partial differential equations and applied mathematics.
1988	Lars Hörmander; Friedrich Hirzebruch	Partial differential equations; topology, characteristic classes, Hirzebruch–Riemann–Roch.
1989	Alberto Calderón; John Milnor	Harmonic analysis and singular integrals; topology and exotic spheres.
1990	Ennio De Giorgi; Ilya Piatetski-Shapiro	Minimal surfaces and regularity theory; automorphic forms.
1992	Lennart Carleson; John G. Thompson	Harmonic analysis; finite group theory.

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Year	Laureate(s)	Quiz anchors
1993	Mikhail Gromov; Jacques Tits	Geometry and geometric group theory; buildings and group theory.
1995/96	Robert Langlands; Andrew Wiles	Langlands program; Fermat's Last Theorem.
1996/97	Joseph Keller; Yakov Sinai	Applied mathematics and waves; dynamical systems and mathematical physics.
1999	László Lovász; Elias M. Stein	Discrete mathematics and algorithms; harmonic analysis.
2000	Raoul Bott; Jean-Pierre Serre	Topology and geometry; algebraic topology, algebraic geometry, number theory.
2001	Vladimir Arnold; Saharon Shelah	Dynamical systems and singularity theory; mathematical logic and set theory.
2002/03	Mikio Sato; John Tate	Algebraic analysis and hyperfunctions; arithmetic geometry.
2005	Gregory Margulis; Serge Novikov	Lie groups, ergodic theory, discrete subgroups; topology and geometry.
2006/07	Stephen Smale; Hillel Furstenberg	Differential topology and dynamical systems; ergodic theory and probability.
2008/09	Pierre Deligne; Phillip Griffiths; David Mumford	Algebraic geometry, Hodge theory, moduli spaces.
2010	Shing-Tung Yau; Dennis Sullivan	Geometric analysis; topology and dynamics.
2012	Michael Aschbacher; Luis Caffarelli	Finite groups; nonlinear PDE and free-boundary problems.
2013	George Mostow; Michael Artin	Rigidity, geometry, Lie groups; algebraic geometry.
2014	Peter Sarnak	Number theory, analysis, geometry, automorphic forms.
2015	James Arthur	Trace formula, automorphic representations.
2017	Richard Schoen; Charles Fefferman	Geometric analysis; analysis and partial differential equations.
2018	Alexander Beilinson; Vladimir Drinfeld	Geometric representation theory, mathematical physics, Langlands program.
2019	Jean-François Le Gall; Gregory Lawler	Probability theory, random walks, Brownian motion, SLE.
2020	Simon Donaldson; Yakov Eliashberg	Differential geometry, topology, gauge theory; symplectic and contact topology.
2022	George Lusztig	Representation theory and related areas.
2023	Ingrid Daubechies	Wavelet theory, time-frequency analysis, signal processing.
2024	Adi Shamir; Noga Alon	Mathematical cryptography; combinatorics, graph theory, theoretical computer science.

Fact 12.6. Wolf Prize landmarks
The following associations summarize the prize's main quiz anchors.

- Wolf Prize ↔ Wolf Foundation.
- The name honors Ricardo Wolf; it does not refer to a mathematician named Wolf.
- Wolf Prize in Mathematics ↔ no Fields-style age restriction.
- Paul Erdős ↔ Wolf Prize 1984 ↔ combinatorics and probabilistic method.
- John Milnor ↔ Wolf Prize 1989; together with the Fields Medal and Abel Prize, this makes him a standard example connected to all three major mathematics prizes.

- Andrew Wiles and Robert Langlands shared the 1995/96 Wolf Prize in Mathematics.
- Ingrid Daubechies $\leftrightarrow$ Wolf Prize 2023 $\leftrightarrow$ wavelets.
- Adi Shamir and Noga Alon $\leftrightarrow$ Wolf Prize 2024 $\leftrightarrow$ cryptography and combinatorics.

Fact 12.7. Prize recognition
For concise review, remember the following map first.

- Fields Medal $\leftrightarrow$ ICM $\leftrightarrow$ IMU $\leftrightarrow$ every four years $\leftrightarrow$ age condition.
- Abel Prize $\leftrightarrow$ Norwegian Academy of Science and Letters $\leftrightarrow$ no age restriction.
- Wolf Prize $\leftrightarrow$ Wolf Foundation $\leftrightarrow$ no Fields-style age restriction.
- Fields Medal questions often ask for the year, ICM location, laureate, and keyword.
- Abel and Wolf questions often ask for the laureate–area association.

Chapter 13

Mathematicians and Mathematical Institutions

Science Quiz repeatedly asks for a mathematician from a birth year, a named theorem, a field, a book, or a short biographical clue, and it also uses universities, societies, unions, and research institutes as identifying information. The goal is rapid recognition through a few unmistakable associations.

Mathematicians

The following table is ordered by birth. Ancient dates are approximate, and "fl." indicates the period in which a person is known to have worked. Living mathematicians are written with an open-ended life span, such as "1980–". The third column distinguishes birthplace or historical origin, nationality or citizenship when it is stated, and later residence or major place of work. An arrow records chronological movement only; it does not by itself assert a change of citizenship. When a modern national label would be anachronistic, the historical polity is given together with its present geographical location. Botong Wang is included without a birth year because no reliable public birth year was identified.

Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Thales of Miletus	c. 624–c. 546 BCE	Ancient Greece (Miletus; now Türkiye)	Geometry	Early deductive geometry; Thales' theorem; a right angle in a semicircle.
Pythagoras	c. 570–c. 495 BCE	Ancient Greece (Samos)	Geometry, number theory	Pythagorean theorem; Pythagorean school; numerical ratios in music.
Euclid	fl. c. 300 BCE	Hellenistic Egypt	Geometry, foundations	*Elements* in thirteen books; axiomatic method; five postulates and the parallel postulate; Euclidean algorithm; infinitely many primes.
Archimedes	c. 287–212 BCE	Greek Sicily (now Italy)	Geometry, mechanics	Method of exhaustion; bounds for π; lever; buoyancy; sphere and circumscribed cylinder.
Apollonius of Perga	c. 240–c. 190 BCE	Ancient Greece (Perga; now Türkiye)	Geometry	*Conics*; systematic study of ellipses, parabolas, and hyperbolas.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Diophantus	c. 200–c. 284	Roman Egypt	Number theory, algebra	*Arithmetica*; Diophantine equations.
Liu Hui	c. 225–c. 295	China	Arithmetic, geometry	Commentary on *The Nine Chapters*; polygon method for π; elimination procedures.
Hypatia	c. 355–415	Roman Egypt	Geometry, astronomy	Alexandrian teacher and commentator; associated with editions of Diophantus and Apollonius.
Zu Chongzhi	429–500	China	Numerical mathematics, astronomy	$3.1415926 < \pi < 3.1415927$; the approximation $\pi \approx 355/113$.
Aryabhata	476–550	India	Arithmetic, trigonometry, astronomy	*Aryabhatiya*; sine tables; place-value computation and astronomical algorithms.
Brahmagupta	598–c. 668	India	Algebra, number theory	Rules for zero and negative numbers; Brahmagupta's formula; quadratic equations.
Muhammad al-Khwarizmi	c. 780–c. 850	Khwarazm (now Uzbekistan)	Algebra, arithmetic	Systematic algebra; the words *algebra* and *algorithm*.
Omar Khayyam	1048–1131	Persia (now Iran)	Algebra, geometry	Classification of cubic equations and geometric solutions by conic sections.
Bhāskara II	1114–1185	India	Algebra, number theory, astronomy	*Līlāvatī*; indeterminate equations; cyclic methods related to Pell equations.
Fibonacci	c. 1170–after 1240	Italy	Arithmetic, number theory	*Liber Abaci*; spread of Hindu–Arabic numerals in Europe; Fibonacci sequence.
Madhava of Sangamagrama	c. 1340–c. 1425	India	Infinite series, trigonometry	Kerala school; series for π, sine, and cosine.
Jamshid al-Kashi	c. 1380–1429	Persia (now Iran)	Numerical mathematics, astronomy	Decimal fractions; high-precision computation of π; law of cosines.
Niccolò Tartaglia	c. 1499–1557	Republic of Venice (now Italy)	Algebra	A method for depressed cubic equations; the Cardano–Tartaglia priority dispute.
Gerolamo Cardano	1501–1576	Duchy of Milan (now Italy)	Algebra, probability	*Ars Magna*; cubic and quartic equations; early probability.
Robert Recorde	c. 1512–1558	Wales	Arithmetic, algebra	Introduced the equality sign $=$ in *The Whetstone of Witte* (1557).
Lodovico Ferrari	1522–1565	Italy	Algebra	General solution of the quartic equation, published in Cardano's *Ars Magna*.
Rafael Bombelli	1526–1572	Italy	Algebra	*L'Algebra*; systematic rules for complex numbers in solutions of cubic equations.
François Viète	1540–1603	France	Algebra, trigonometry	Systematic symbolic algebra; Viète's formulas.
Simon Stevin	1548–1620	Flanders (now Belgium)	Arithmetic, mechanics	Promoted decimal fractions in Europe; work on statics and hydrostatics.
John Napier	1550–1617	Scotland	Computation, logarithms	Introduced logarithms in print in 1614; Napier's bones.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Thomas Harriot	c. 1560–1621	England	Algebra, astronomy	Introduced the inequality signs $<$ and $>$ in work published posthumously in 1631.
Galileo Galilei	1564–1642	Italy	Mathematical physics	Mathematical description of motion; telescopic astronomy.
Johannes Kepler	1571–1630	Holy Roman Empire (now Germany)	Geometry, astronomy	Laws of planetary motion; Kepler conjecture on sphere packing.
Marin Mersenne	1588–1648	France	Number theory, acoustics	Mersenne primes $2^p - 1$; scientific correspondence network in seventeenth-century Europe.
Girard Desargues	1591–1661	France	Projective geometry	Desargues' theorem; foundational ideas of projective geometry.
René Descartes	1596–1650	France (birthplace or historical origin) → Dutch Republic (later residence/work)	Analytic geometry, algebra	Cartesian coordinates; *La Géométrie*; rule of signs.
Bonaventura Cavalieri	1598–1647	Italy	Geometry	Method of indivisibles, an important precursor to integral calculus.
Pierre de Fermat	c. 1607–1665	France	Number theory, geometry, probability	Fermat's Last Theorem; little theorem; analytic geometry; probability correspondence with Pascal.
John Wallis	1616–1703	England	Analysis, geometry	Introduced the symbol ∞; Wallis product for π; infinitesimal methods.
Blaise Pascal	1623–1662	France	Probability, geometry	Pascal's triangle; foundations of probability with Fermat; projective geometry.
Christiaan Huygens	1629–1695	Dutch Republic (Netherlands)	Probability, mechanics	First printed treatise on probability; pendulum and wave theory.
Isaac Barrow	1630–1677	England	Geometry, analysis	Geometric form of the fundamental theorem of calculus; Newton's teacher at Cambridge.
Isaac Newton	1643–1727	England	Calculus, mechanics	Fluxions; fundamental theorem of calculus; generalized binomial theorem; laws of motion and gravitation.
Gottfried Wilhelm Leibniz	1646–1716	Holy Roman Empire (now Germany)	Calculus, logic	Independent development of calculus; dx and $\int$ notation; binary arithmetic.
Choi Seok-jeong	1646–1715	Joseon Korea	Combinatorics, recreational mathematics	*Gusuryak*; orthogonal Latin squares of order 9; magic squares; Jisuguimundo, or the Hexagonal Tortoise Problem.
Jakob Bernoulli	1655–1705	Switzerland	Probability, analysis	Bernoulli numbers; law of large numbers; *Ars Conjectandi.*
Guillaume de l'Hôpital	1661–1704	France	Analysis	First calculus textbook; l'Hôpital's rule, developed from lessons by Johann Bernoulli.
Abraham de Moivre	1667–1754	France (birthplace or historical origin) → England (later residence/work)	Probability, complex numbers	De Moivre's formula; normal approximation to the binomial distribution.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Johann Bernoulli	1667–1748	Switzerland	Analysis, mechanics	Brachistochrone problem; early calculus; teacher of Euler and l'Hôpital.
William Jones	1675–1749	Wales	Mathematics, navigation	Introduced the symbol π for the circle ratio in 1706; Euler later popularized it.
Brook Taylor	1685–1731	England	Analysis	Taylor's theorem and Taylor series; finite differences.
Christian Goldbach	1690–1764	Prussia (now Russia) (birthplace or historical origin) → Russia (later residence/work)	Number theory	Goldbach conjecture, proposed in correspondence with Euler.
James Stirling	1692–1770	Scotland	Analysis, combinatorics	Stirling's approximation; Stirling numbers.
Colin Maclaurin	1698–1746	Scotland	Analysis, geometry	Maclaurin series; *Treatise of Fluxions*.
Daniel Bernoulli	1700–1782	Netherlands (birthplace or historical origin) → Switzerland (later residence/work)	Probability, fluid mechanics	Bernoulli principle; expected utility and the St. Petersburg paradox.
Thomas Bayes	c. 1701–1761	England	Probability	Bayes' theorem; posthumous 1763 publication.
Leonhard Euler	1707–1783	Switzerland (birthplace or historical origin) → Russia and Prussia (later residence/work)	Analysis, number theory, geometry	Basel problem; Euler formula and identity; graph theory; polyhedron formula; standardized much modern notation.
Jean le Rond d'Alembert	1717–1783	France	Analysis, mechanics	Wave equation; d'Alembert's principle; work on the fundamental theorem of algebra.
Maria Gaetana Agnesi	1718–1799	Italy	Analysis	*Instituzioni analitiche*; one of the earliest comprehensive calculus textbooks; the witch of Agnesi curve.
Joseph-Louis Lagrange	1736–1813	Sardinia (now Italy) (birthplace or historical origin) → Prussia and France (later residence/work)	Analysis, mechanics, number theory	Lagrangian mechanics; multipliers; four-square theorem.
Gaspard Monge	1746–1818	France	Geometry	Descriptive geometry; Monge formulation of optimal transport.
John Playfair	1748–1819	Scotland	Geometry, geology	Playfair's axiom, the standard modern reformulation of Euclid's parallel postulate.
Pierre-Simon Laplace	1749–1827	France	Probability, celestial mechanics	Laplace transform; Laplacian; analytic theory of probability.
Adrien-Marie Legendre	1752–1833	France	Number theory, analysis	Legendre symbol; elliptic integrals; least squares.
Joseph Fourier	1768–1830	France	Analysis, PDE	Fourier series and transform; heat equation.
Sophie Germain	1776–1831	France	Number theory, elasticity	Work on Fermat's Last Theorem and vibrating plates; used the pseudonym M. LeBlanc.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Carl Friedrich Gauss	1777–1855	Brunswick (now Germany)	Number theory, geometry, statistics	*Disquisitiones Arithmeticae*; quadratic reciprocity; Gaussian distribution; least squares; curvature.
Siméon Denis Poisson	1781–1840	France	Analysis, probability, physics	Poisson distribution and equation; mathematical physics and potential theory.
Bernard Bolzano	1781–1848	Bohemia (now Czechia)	Analysis, logic	Early rigorous analysis; intermediate value theorem; Bolzano– Weierstrass theorem.
Augustin-Louis Cauchy	1789–1857	France	Analysis	Rigorous limits and convergence; Cauchy sequences; integral theorem; determinant theory.
August Ferdinand Möbius	1790–1868	Saxony (now Germany)	Geometry, number theory	Möbius strip; Möbius transformations and inversion function.
Charles Babbage	1791–1871	England	Computation, numerical mathematics	Difference Engine and Analytical Engine.
Nikolai Lobachevsky	1792–1856	Russia	Geometry	Independent creation of hyperbolic, non-Euclidean geometry.
George Green	1793–1841	England	Analysis, mathematical physics	Green's theorem, Green's identities, and Green functions.
Niels Henrik Abel	1802–1829	Norway	Algebra, analysis	Impossibility of a radical formula for the general quintic; elliptic functions; Abel Prize namesake.
János Bolyai	1802–1860	Transylvania (now Romania); Hungarian	Geometry	Independent creation of hyperbolic geometry.
Carl Gustav Jacobi	1804–1851	Prussia (now Germany)	Analysis, mechanics, number theory	Jacobian determinant; elliptic functions; Hamilton–Jacobi theory.
Peter Gustav Lejeune Dirichlet	1805–1859	France (now Germany); German	Number theory, analysis	Primes in arithmetic progressions; Dirichlet characters; modern function concept.
William Rowan Hamilton	1805–1865	Ireland	Algebra, mechanics	Quaternions; Hamiltonian mechanics; Hamilton–Jacobi equation.
Augustus De Morgan	1806–1871	British India (birthplace or historical origin) → United Kingdom (later residence/work)	Logic, algebra	De Morgan's laws; formal logic; mathematical induction terminology.
Joseph Liouville	1809–1882	France	Analysis, number theory	First proof that transcendental numbers exist; Liouville's theorem.
Hermann Grassmann	1809–1877	Prussia (now Poland); German	Algebra, geometry	*Ausdehnungslehre*; vector spaces and exterior algebra.
Ernst Kummer	1810–1893	Prussia (now Poland); German	Number theory, algebra	Ideal numbers; regular primes; major progress on Fermat's Last Theorem.
Évariste Galois	1811–1832	France	Algebra	Galois theory; solvability by radicals through groups.
James Joseph Sylvester	1814–1897	England	Algebra, number theory	Matrix and invariant theory; coined the term "matrix"; Sylvester's law of inertia.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Karl Weierstrass	1815–1897	Prussia (now Germany)	Analysis	ε–δ rigor; uniform convergence; nowhere-differentiable function.
George Boole	1815–1864	England (birthplace or historical origin) $\rightarrow$ Ireland (later residence/work)	Logic, algebra	Boolean algebra; algebraic treatment of logic.
Ada Lovelace	1815–1852	England	Computation	Notes on Babbage's Analytical Engine; an algorithm for Bernoulli numbers often described as the first published computer program.
George Gabriel Stokes	1819–1903	Ireland (birthplace or historical origin) $\rightarrow$ United Kingdom (later residence/work)	Analysis, mathematical physics	Stokes' theorem; fluid equations; optics.
Pafnuty Chebyshev	1821–1894	Russia	Number theory, approximation, probability	Chebyshev polynomials and inequality; work on prime distribution.
Arthur Cayley	1821–1895	England	Algebra, geometry	Matrix algebra; Cayley's theorem; Cayley–Hamilton theorem.
Charles Hermite	1822–1901	France	Analysis, number theory	Proved e transcendental in 1873; Hermite polynomials and normal forms.
Leopold Kronecker	1823–1891	Prussia (now Poland); German	Algebra, number theory	Kronecker product and delta; arithmetic approach to algebra.
Bernhard Riemann	1826–1866	Kingdom of Hanover (now Germany)	Analysis, geometry, number theory	Riemann surfaces and integral; Riemannian geometry; zeta function and hypothesis.
Richard Dedekind	1831–1916	Brunswick (now Germany)	Algebra, foundations, number theory	Dedekind cuts; ideals; rigorous construction of the real numbers.
Charles Lutwidge Dodgson (Lewis Carroll)	1832–1898	England	Logic, geometry, recreational mathematics	Oxford mathematics lecturer; wrote *Alice's Adventures in Wonderland* under the pen name Lewis Carroll; Dodgson condensation.
Camille Jordan	1838–1922	France	Algebra, analysis	Jordan normal form; Jordan curve theorem; permutation groups.
Sophus Lie	1842–1899	Norway	Geometry, differential equations	Lie groups and Lie algebras; continuous symmetries.
Georg Cantor	1845–1918	Russian Empire (birthplace or historical origin) $\rightarrow$ Germany (later residence/work)	Set theory	Founder of set theory; diagonal argument; cardinalities; Cantor set.
Wilhelm Killing	1847–1923	Prussia (now Germany)	Lie theory, geometry	Classification program for simple Lie algebras; Killing form.
Gottlob Frege	1848–1925	Germany	Logic, foundations	Predicate logic; *Begriffsschrift*; logicist foundations of arithmetic.
Felix Klein	1849–1925	Prussia (now Germany)	Geometry, group theory	Erlangen program; Klein bottle; Klein four-group.
Ferdinand Georg Frobenius	1849–1917	Prussia (now Germany)	Algebra, analysis	Frobenius theorem and endomorphism; character theory of finite groups.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Sofia Kovalevskaya	1850–1891	Russia (birthplace or historical origin) → Sweden (later residence/work)	Analysis, PDE	Cauchy–Kovalevskaya theorem; rigid-body dynamics.
Ferdinand von Lindemann	1852–1939	Germany	Number theory, analysis	Proved π transcendental in 1882, establishing the impossibility of squaring the circle.
Henri Poincaré	1854–1912	France	Topology, dynamics, geometry	Poincaré conjecture; qualitative dynamics; foundational topology.
Andrey Markov	1856–1922	Russia	Probability, number theory	Markov chains; Markov inequality; stochastic dependence.
Giuseppe Peano	1858–1932	Italy	Logic, foundations	Peano axioms; symbolic logic; space-filling Peano curve.
David Hilbert	1862–1943	Prussia (Königsberg; now Russia); German	Foundations, algebra, geometry	Twenty-three problems; Hilbert spaces and basis theorem; axiomatic geometry.
John Charles Fields	1863–1932	Canada	Complex analysis	Canadian mathematician who proposed and endowed the Fields Medal.
Hermann Minkowski	1864–1909	Russian Empire (now Lithuania) (birthplace or historical origin) → Germany (later residence/work)	Geometry, number theory	Geometry of numbers; Minkowski spacetime; convex body theorem.
Jacques Hadamard	8 Dec 1865–17 Oct 1963	France	Analysis, number theory	Prime number theorem; Cauchy–Hadamard theorem; Hadamard matrices.
Charles-Jean de la Vallée Poussin	1866–1962	Belgium	Analysis, number theory	Independent proof of the prime number theorem in 1896.
Felix Hausdorff	1868–1942	Germany	Topology, set theory	Hausdorff spaces; Hausdorff measure and dimension.
Élie Cartan	1869–1951	France	Differential geometry, Lie theory	Lie groups and algebras; differential forms; moving frames.
Ernst Zermelo	1871–1953	Germany	Set theory, foundations	Axiomatization of set theory; well-ordering theorem; Zermelo–Fraenkel set theory.
Bertrand Russell	1872–1970	Wales, United Kingdom	Logic, foundations	Russell's paradox; *Principia Mathematica* with Whitehead.
Alfred Young	1873–1940	England	Algebra, representation theory	Young diagrams and Young tableaux for symmetric groups.
Henri Lebesgue	1875–1941	France	Analysis, measure theory	Lebesgue measure and integral; dominated convergence theorem.
G. H. Hardy	1877–1947	England	Analysis, number theory	Hardy–Weinberg principle; Hardy–Littlewood circle method; collaboration with Ramanujan.
Maurice Fréchet	1878–1973	France	Analysis, topology	Metric spaces; Fréchet derivative; abstract functional spaces.
Frigyes Riesz	1880–1956	Austria-Hungary (now Hungary)	Functional analysis	Riesz representation theorem; L^p spaces and operator theory.
L. E. J. Brouwer	1881–1966	Netherlands	Topology, foundations	Brouwer fixed-point theorem; invariance of domain; intuitionism.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Wacław Sierpiński	1882–1969	Russian Empire (now Poland); Polish	Set theory, topology	Sierpiński triangle and carpet; continuum and descriptive set theory.
Emmy Noether	1882–1935	Germany (birthplace or historical origin) → United States (later residence/work)	Algebra, mathematical physics	Noetherian rings and modules; symmetry–conservation theorem.
J. E. Littlewood	1885–1977	England	Analysis, number theory	Hardy–Littlewood circle method and conjectures; analytic number theory.
Hermann Weyl	1885–1955	Germany (birthplace or historical origin) → Switzerland and United States (later residence/work)	Geometry, representation theory, physics	Weyl groups and character formula; gauge ideas; foundations.
Hugo Steinhaus	1887–1972	Austria-Hungary (now Ukraine); Polish	Analysis, probability	Lwów school; Steinhaus theorem; co-founder of *Studia Mathematica*.
George Pólya	1887–1985	Austria-Hungary (now Hungary) (birthplace or historical origin) → United States (later residence/work)	Combinatorics, probability	Pólya enumeration theorem; random walks; *How to Solve It*.
Srinivasa Ramanujan	1887–1920	British India	Number theory, analysis	Partitions, modular forms, mock theta functions; 1729.
Richard Courant	1888–1972	Prussia (now Poland) (birthplace or historical origin) → Germany and United States (later residence/work)	Analysis, PDE	Courant Institute namesake; finite elements; Courant–Friedrichs–Lewy condition.
Ludwig Wittgenstein	1889–1951	Austria (birthplace or historical origin) → United Kingdom (later residence/work)	Logic, philosophy	*Tractatus*; language games; major influence on logical positivism and philosophy of language.
Stefan Banach	1892–1945	Austria-Hungary (now Poland); Polish	Functional analysis	Banach spaces; contraction mapping theorem; Lwów school and *The Scottish Book*.
Norbert Wiener	1894–1964	United States	Analysis, probability	Wiener process and measure; harmonic analysis; founder of cybernetics.
Emil Artin	1898–1962	Austria-Hungary (now Czechia) (birthplace or historical origin) → Germany and United States (later residence/work)	Algebra, number theory	Artin reciprocity; braid groups; modern algebraic number theory.
Oscar Zariski	1899–1986	Russian Empire (now Belarus) (birthplace or historical origin) → Italy and United States (later residence/work)	Algebraic geometry	Zariski topology; foundations of modern algebraic geometry.
Juliusz Schauder	1899–1943	Austria-Hungary (now Ukraine); Polish	Functional analysis, PDE	Schauder fixed-point theorem and Schauder estimates.
Antoni Zygmund	1900–1992	Russian Empire (now Poland) (birthplace or historical origin) → United States (later residence/work)	Harmonic analysis	*Trigonometric Series*; Chicago school of harmonic analysis.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Alfred Tarski	1901–1983	Russian Empire (now Poland) (birthplace or historical origin) → United States (later residence/work)	Logic, foundations	Semantic definition of truth; Tarski's theorem; Banach–Tarski paradox.
Andrey Kolmogorov	1903–1987	Russia / USSR	Probability, dynamics	Axioms of probability; Kolmogorov complexity; turbulence and KAM theory.
Alonzo Church	1903–1995	United States	Logic, computation	Lambda calculus; Church's theorem; Church–Turing thesis.
John von Neumann	1903–1957	Austria-Hungary (now Hungary) (birthplace or historical origin) → United States (later residence/work)	Analysis, computation, game theory	Operator algebras; quantum mechanics; minimax theorem; stored-program architecture.
Stanisław Mazur	1905–1981	Austria-Hungary (now Ukraine); Polish	Functional analysis	Lwów school; Banach–Mazur theorem; Scottish Book basis problem and its live-goose prize.
Kurt Gödel	1906–1978	Austria-Hungary (now Czechia) (birthplace or historical origin) → United States (later residence/work)	Logic, foundations	Completeness and incompleteness theorems; consistency of choice and CH with ZF.
André Weil	1906–1998	France (birthplace or historical origin) → United States (later residence/work)	Number theory, algebraic geometry	Weil conjectures; adeles; founding member of Bourbaki.
Hans Fitting	1906–1938	Germany	Group theory, algebra	Fitting subgroup, lemma, decomposition, and ideals.
Lars Ahlfors	1907–1996	Finland (birthplace or historical origin) → United States (later residence/work)	Complex analysis	Riemann surfaces; Ahlfors theory; one of the first two Fields Medalists.
Claude Chevalley	1909–1984	South Africa; French (birthplace or historical origin) → United States (later residence/work)	Algebra, number theory	Chevalley groups; class field theory; founding member of Bourbaki.
Stanisław Ulam	1909–1984	Austria-Hungary (now Ukraine); Polish (birthplace or historical origin) → United States (later residence/work)	Set theory, probability	Monte Carlo method; Ulam spiral; Scottish Book and Manhattan Project.
Saunders Mac Lane	1909–2005	United States	Algebra, topology	Co-founder of category theory; Eilenberg–Mac Lane spaces.
Shiing-Shen Chern	1911–2004	China (birthplace or historical origin) → United States (later residence/work)	Differential geometry, topology	Chern classes; Chern–Gauss–Bonnet theorem; Chern Medal namesake.
Alan Turing	1912–1954	England	Logic, computation	Turing machine; computability; codebreaking; Turing test.
Paul Erdős	1913–1996	Austria-Hungary (now Hungary); Hungarian	Number theory, combinatorics	Probabilistic method; Erdős number; prolific collaboration.
Israel Gelfand	1913–2009	Russian Empire (now Ukraine) (birthplace or historical origin) → USSR and United States (later residence/work)	Functional analysis, representation theory	Gelfand representation; representation theory; influential seminar.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Samuel Eilenberg	1913–1998	Russian Empire (now Poland) (birthplace or historical origin) → United States (later residence/work)	Algebraic topology	Co-founder of category theory and homological algebra; Eilenberg–Mac Lane spaces.
Mark Kac	1914–1984	Russian Empire (now Ukraine); Polish (birthplace or historical origin) → United States (later residence/work)	Probability, mathematical physics	Feynman–Kac formula; “Can one hear the shape of a drum?”
George Dantzig	1914–2005	United States	Optimization, statistics	Linear programming; simplex method; the mistaken-homework anecdote.
Laurent Schwartz	1915–2002	France	Analysis	Theory of distributions; Fields Medal 1950.
Claude Shannon	1916–2001	United States	Information theory	Entropy, channel capacity, and the mathematical theory of communication.
Atle Selberg	1917–2007	Norway (birthplace or historical origin) → United States (later residence/work)	Analytic number theory	Selberg sieve and trace formula; elementary proof of the prime number theorem with Erdős.
Benoit Mandelbrot	1924–2010	Poland (birthplace or historical origin) → France and United States (later residence/work)	Geometry, probability	Fractal geometry; Mandelbrot set; popularized the term “fractal.”
Louis Nirenberg	1925–2020	Canada (birthplace or historical origin) → United States (later residence/work)	Analysis, PDE	Fundamental estimates for elliptic PDE; Abel Prize 2015 with John Nash.
Peter D. Lax	1926–2025	Hungary (birthplace or historical origin) → United States (later residence/work)	Analysis, PDE	Lax pair and Lax equivalence theorem; numerical analysis; Abel Prize 2005.
Jean-Pierre Serre	1926–	France	Algebraic topology, geometry, number theory	Youngest Fields Medalist; first Abel laureate; Serre duality and spectral sequences.
Alexander Grothendieck	1928–2014	Germany; stateless, later French	Algebraic geometry	Schemes, topoi, étale cohomology; transformed modern algebraic geometry.
John Nash	1928–2015	United States	Geometry, game theory, PDE	Nash equilibrium; Nash embedding theorem; Abel Prize 2015.
Michael Atiyah	1929–2019	England; British	Geometry, topology	Atiyah–Singer index theorem; K-theory; Fields and Abel prizes.
Goro Shimura	1930–2019	Japan (birthplace or historical origin) → United States (later residence/work)	Number theory, automorphic forms	Taniyama–Shimura modularity conjecture; Shimura varieties; work underlying the modular approach to Fermat’s Last Theorem.
Stephen Smale	1930–	United States	Topology, dynamics	Higher-dimensional Poincaré conjecture; Smale horseshoe; Smale’s problems.
Lars Hörmander	1931–2012	Sweden	Analysis, PDE	Linear partial differential operators; microlocal analysis; Fields Medal 1962.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
John Milnor	1931–	United States	Topology, geometry, dynamics	Exotic 7-spheres; Milnor fibration; Fields, Wolf, and Abel prizes.
Paul Cohen	1934–2007	United States	Logic, set theory	Forcing; independence of the continuum hypothesis and axiom of choice.
Yakov Sinai	1935–	USSR (now Russia) (birthplace or historical origin) → United States (later residence/work)	Dynamical systems, probability	Sinai billiards; Kolmogorov–Sinai entropy; Abel Prize 2014.
Robert Langlands	1936–	Canada (birthplace or historical origin) → United States (later residence/work)	Number theory, representation theory	Langlands program; reciprocity and functoriality; Abel Prize 2018.
Vladimir Arnold	1937–2010	USSR (now Ukraine); Russian	Dynamical systems, geometry	KAM theory; Arnold conjecture; singularity theory.
John Horton Conway	1937–2020	England (birthplace or historical origin) → United States (later residence/work)	Algebra, combinatorics	Surreal numbers; cellular automaton *Life*; Conway groups and knot notation.
Donald Knuth	1938–	United States	Algorithms, combinatorics	*The Art of Computer Programming*; TeX; analysis of algorithms.
Alan Baker	1939–2018	England	Number theory	Linear forms in logarithms; transcendence theory; Fields Medal 1970.
Dennis Sullivan	1941–	United States	Topology, dynamical systems	Surgery theory, rational homotopy theory, and conformal dynamics; Abel Prize 2022.
Karen Uhlenbeck	1942–	United States	Geometric analysis	Gauge theory and harmonic maps; first woman Abel laureate.
Mikhail Gromov	1943–	USSR (now Russia) (birthplace or historical origin) → France (later residence/work)	Geometry, topology	Gromov–Hausdorff distance; geometric group theory; Abel Prize 2009.
Per Enflo	1944–	Sweden	Functional analysis	Solved Mazur's Scottish Book basis problem and received the promised live goose.
Pierre Deligne	1944–	Belgium	Algebraic geometry	Proof of the Weil conjectures; Fields, Wolf, and Abel prizes.
Grigory Margulis	1946–	USSR (now Russia) (birthplace or historical origin) → United States (later residence/work)	Lie groups, dynamics	Superrigidity, arithmeticity, and homogeneous dynamics; Fields and Abel prizes.
William Thurston	1946–2012	United States	Topology, geometry	Geometrization conjecture; eight geometries; low-dimensional topology.
Masaki Kashiwara	1947–	Japan	Algebraic analysis, representation theory	D-modules, microlocal analysis, and crystal bases; Abel Prize 2025.
Luis Caffarelli	1948–	Argentina (birthplace or historical origin) → United States (later residence/work)	Nonlinear PDE	Regularity theory, free-boundary problems, and the Monge–Ampère equation; Abel Prize 2023.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
László Lovász	1948–	Hungary (birthplace or historical origin) → United States and Hungary (later residence/work)	Combinatorics, theoretical computer science	Lovász local lemma, graph limits, and algorithms; Abel Prize 2021.
Shing-Tung Yau	1949–	China (birthplace or historical origin) → Hong Kong and United States (later residence/work)	Geometric analysis	Calabi conjecture; Ricci-flat metrics; Fields Medal 1982.
Edward Witten	1951–	United States	Mathematical physics	First physicist to receive the Fields Medal; quantum field theory, topology, and string theory.
Michel Talagrand	1952–	France	Probability, functional analysis	Concentration inequalities, stochastic processes, and spin glasses; Abel Prize 2024.
Adi Shamir	1952–	Israel	Cryptography, theoretical computer science	Co-inventor of RSA and Shamir secret sharing; Turing Award and Wolf Prize 2024.
Andrew Wiles	1953–	England (birthplace or historical origin) → United States and United Kingdom (later residence/work)	Number theory	Fermat's Last Theorem via modularity; special IMU plaque and Abel Prize.
Gerd Faltings	1954–	West Germany (now Germany)	Arithmetic geometry	Mordell conjecture; Fields Medal 1986; Abel Prize 2026.
Ingrid Daubechies	1954–	Belgium (birthplace or historical origin) → United States (later residence/work)	Analysis, applied mathematics	Orthogonal wavelets; Daubechies wavelets; Wolf Prize 2023.
Efim Zelmanov	1955–	USSR (now Russia) (birthplace or historical origin) → United States (later residence/work)	Algebra, group theory	Restricted Burnside problem; Fields Medal 1994.
Yitang Zhang	1955–	China (birthplace or historical origin) → United States (later residence/work)	Analytic number theory	Proved the first bounded-gap theorem for infinitely many pairs of primes, initiating rapid progress on prime gaps.
Avi Wigderson	1956–	Israel (birthplace or historical origin) → United States (later residence/work)	Theoretical computer science	Complexity theory, randomness, and pseudorandomness; Abel Prize 2021 and Turing Award.
Noga Alon	1956–	Israel (birthplace or historical origin) → United States (later residence/work)	Combinatorics, graph theory	Probabilistic method, algebraic combinatorics, and theoretical computer science; Wolf Prize 2024.
Simon Donaldson	1957–	England	Geometry, topology	Gauge theory and smooth 4-manifolds; Fields Medal 1986.
Maxim Kontsevich	1964–	USSR (now Russia) (birthplace or historical origin) → France (later residence/work)	Geometry, mathematical physics	Deformation quantization; mirror symmetry; Fields Medal 1998.
Grigori Perelman	1966–	USSR (now Russia)	Geometry, topology	Poincaré and geometrization conjectures via Ricci flow; declined the Fields Medal and Millennium Prize.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Elon Lindenstrauss	1970–	Israel	Ergodic theory, number theory	Measure rigidity and applications to number theory; Fields Medal 2010.
Cédric Villani	1973–	France	PDE, optimal transport	Boltzmann equation and Landau damping; Fields Medal 2010; *Théorème vivant*.
Manjul Bhargava	1974–	Canada; Canadian–American	Number theory	Higher composition laws; arithmetic statistics; Fields Medal 2014.
Terence Tao	1975–	Australia (birthplace or historical origin) → United States (later residence/work)	Analysis, PDE, combinatorics	Green–Tao theorem; harmonic analysis; Fields Medal 2006.
Martin Hairer	1975–	Switzerland; Austrian (birthplace or historical origin) → United Kingdom (later residence/work)	Probability, stochastic PDE	Theory of regularity structures; Fields Medal 2014.
Maryam Mirzakhani	1977–2017	Iran (birthplace or historical origin) → United States (later residence/work)	Geometry, dynamics	Moduli spaces and Riemann surfaces; first woman Fields Medalist.
Caucher Birkar	1978–	Iran; Kurdish (birthplace or historical origin) → United Kingdom (later residence/work)	Algebraic geometry	Birational geometry and boundedness of Fano varieties; Fields Medal 2018.
Akshay Venkatesh	1981–	India (birthplace or historical origin) → Australia and United States (later residence/work)	Number theory, dynamics	Number theory and homogeneous dynamics; Fields Medal 2018.
June Huh	1983–	United States (birthplace or historical origin) → South Korea and United States (later residence/work)	Combinatorics, algebraic geometry	Hodge theory for matroids; log-concavity conjectures; Fields Medal 2022.
Alessio Figalli	1984–	Italy (birthplace or historical origin) → Switzerland (later residence/work)	Analysis, optimal transport	Optimal transport and PDE; Fields Medal 2018.
Maryna Viazovska	1984–	Ukraine (birthplace or historical origin) → Switzerland (later residence/work)	Number theory, discrete geometry	Optimal sphere packing in dimensions 8 and 24; Fields Medal 2022.
Hugo Duminil-Copin	1985–	France (birthplace or historical origin) → Switzerland (later residence/work)	Probability, mathematical physics	Phase transitions and percolation; Fields Medal 2022.
James Maynard	1987–	England	Number theory	Small gaps between primes and Diophantine approximation; Fields Medal 2022.
Peter Scholze	1987–	East Germany (now Germany)	Arithmetic geometry	Perfectoid spaces and p-adic geometry; Fields Medal 2018.

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Mathematician	Life	Birthplace / nationality / later residence	Main fields	Quiz anchors
Jacob Tsimerman	1988–	USSR (now Russia) (birthplace or historical origin) → Israel and Canada (later residence/work)	Arithmetic geometry	O-minimality in arithmetic and complex algebraic geometry; André–Oort and related conjectures; Fields Medal 2026.
Yu Deng	1989–	China (birthplace or historical origin) → United States (later residence/work)	PDE, kinetic theory	Boltzmann equation from hard-sphere dynamics, wave kinetic equations, and probabilistic nonlinear Schrödinger equations; Fields Medal 2026.
John Pardon	1989–	United States	Symplectic geometry, topology	Virtual fundamental cycles, Fukaya categories, group actions on 3-manifolds, and knot theory; Fields Medal 2026.
Hong Wang	1991–	China (birthplace or historical origin) → France and United States (later residence/work)	Harmonic analysis, geometric measure theory	Fourier restriction, local smoothing, and the three-dimensional Kakeya conjecture; third woman Fields Medalist, 2026.
Botong Wang	birth year not publicly listed	China (undergraduate education) → United States (doctoral study and later work)	Algebraic geometry, topology	Topology of algebraic varieties and interactions with combinatorics and applied mathematics.

Mathematical Institutions

The institutions below include learned societies, international unions, higher-education institutions, and research institutes. A research institute may host scholars and seminars without awarding degrees.

Institution	Founded	Location	Role and distinctive facts
Collège de France	1530	Paris, France	Public research chairs and lectures; admits listeners without enrollment and does not grant degrees.
École normale supérieure (ENS)	1794	Paris, France	Selective higher-education and research institution; unlike IAS, IHES, and SLMath, it has degree-granting programs.
London Mathematical Society (LMS)	1865	London, United Kingdom	Learned society that promotes mathematics through journals, meetings, grants, and prizes.
Société Mathématique de France (SMF)	1872	Paris, France	French learned society for mathematical research, publications, conferences, and professional exchange.
American Mathematical Society (AMS)	1888	Providence, United States	Began as the New York Mathematical Society and adopted its present name in 1894; publishes *Mathematical Reviews* and MathSciNet.

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Institution	Founded	Location	Role and distinctive facts
Deutsche Mathematiker-Vereinigung (DMV)	1890	Germany	German mathematical society; supports research, communication, education, and public engagement.
Institut Mittag-Leffler	1916	Djursholm, Sweden	International research center of the Royal Swedish Academy of Sciences; associated with *Acta Mathematica*.
International Mathematical Union (IMU)	1920; reconstituted 1951	Secretariat in Berlin, Germany	International union that supports global cooperation, organizes the ICM, and awards the Fields Medal and other IMU prizes.
Institut Henri Poincaré (IHP)	1928	Paris, France	International center for mathematics and theoretical physics; thematic programs, seminars, library, and outreach.
Institute for Advanced Study (IAS)	1930	Princeton, United States	Independent institute for fundamental research; home to a School of Mathematics; has no degree programs.
Mathematisches Forschungsinstitut Oberwolfach (MFO)	1944	Oberwolfach, Germany	International workshop and research center famous for small, intensive meetings and the Oberwolfach Research Institute for Mathematics.
Tata Institute of Fundamental Research (TIFR)	1945	Mumbai, India	National center for fundamental research in mathematics and the sciences; central to the development of modern mathematics in India.
Institut des Hautes Études Scientifiques (IHES)	1958	Bures-sur-Yvette, France	Private advanced research institute in mathematics and theoretical physics; hosts a small permanent faculty and many visitors; does not grant degrees.
Research Institute for Mathematical Sciences (RIMS)	1963	Kyoto, Japan	Kyoto University institute for mathematical research and collaboration; publishes *Publications of RIMS*.
Max Planck Institute for Mathematics (MPIM)	1980	Bonn, Germany	Max Planck Society institute for pure mathematics; long-term and visitor research programs.
Simons Laufer Mathematical Sciences Institute (SLMath)	1982	Berkeley, United States	Founded as MSRI and renamed in 2022; collaborative research programs, workshops, and outreach; no degree programs.
Fields Institute for Research in Mathematical Sciences	1992	Toronto, Canada	Collaborative research institute named after John Charles Fields; distinct from the Fields Medal.
Isaac Newton Institute for Mathematical Sciences	1992	Cambridge, United Kingdom	United Kingdom national and international visitor institute for long thematic programs in the mathematical sciences.
Korea Institute for Advanced Study (KIAS)	1996	Seoul, South Korea	Government-founded institute for basic research in mathematics, physics, and computational sciences; no formal curriculum or degree programs.

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Institution	Founded	Location	Role and distinctive facts
Clay Mathematics Institute (CMI)	1998	Denver, United States	Private foundation supporting research, fellowships, and publications; announced the seven Millennium Prize Problems in 2000.
Institute for Pure and Applied Mathematics (IPAM)	1999	Los Angeles, United States	UCLA-based institute connecting mathematics with science, engineering, technology, and society through long programs and workshops.
Banff International Research Station (BIRS)	2003	Banff, Canada	International mathematical research station for focused workshops, collaborative teams, and training programs.
National Institute for Mathematical Sciences (NIMS)	2005	Daejeon, South Korea	Government research institute for mathematical research, applications, computation, and public engagement.
IBS Center for Geometry and Physics (IBS-CGP)	2012	POSTECH, Pohang, South Korea	Institute for Basic Science center founded for geometry, mathematical physics, symplectic topology, and international collaboration.

Chapter 14

Mathematical History, Works, and Anecdotes

This chapter collects three complementary kinds of historical recall: chronological landmarks, mathematical works whose titles and authors are themselves worth recognizing, and well-known anecdotes. The three sections are kept distinct so that an event, a work, and a story do not become interchangeable kinds of facts.

Mathematical History

The following chronology records changes that altered mathematical language, method, or subject matter. It is intentionally written as short recall statements rather than as a continuous historical essay.

- **c. 1800–1600 BCE.** Babylonian tablets already contained sophisticated arithmetic, quadratic procedures, and examples of Pythagorean triples.
- **c. 1650 BCE.** The Rhind Mathematical Papyrus, copied by the scribe Ahmes from an older source, recorded Egyptian arithmetic and geometric problems.
- **c. 300 BCE.** Euclid organized Greek geometry and number theory axiomatically in the thirteen books of *Elements*.
- **3rd century BCE.** Archimedes used the method of exhaustion to obtain rigorous area and volume results and proved
$$\frac{223}{71} < \pi < \frac{22}{7}.$$
- **263.** Liu Hui completed his influential commentary on *The Nine Chapters on the Mathematical Art* and refined polygon methods for approximating π.
- **5th century.** Zu Chongzhi obtained $3.1415926 < \pi < 3.1415927$ and the approximation $355/113$.
- **628.** Brahmagupta's *Brahmasphutasiddhanta* gave systematic arithmetic rules involving zero and negative numbers.
- **c. 825.** Al-Khwarizmi's treatise on *al-jabr* gave algebra its name; the Latin form of his name produced the word *algorithm*.
- **1202.** Fibonacci's *Liber Abaci* promoted Hindu–Arabic numerals and place-value arithmetic in Latin Europe.

- **14th–16th centuries.** Madhava and the Kerala school developed infinite series for trigonometric functions and π, centuries before the European calculus.
- **1545.** Cardano published *Ars Magna*, containing methods for general cubic and quartic equations, based in part on work of del Ferro, Tartaglia, and Ferrari.
- **1557.** Robert Recorde introduced the equality sign $=$ in *The Whetstone of Witte.*
- **1614.** John Napier published logarithms, transforming difficult multiplication and astronomical computation into addition.
- **1631.** Thomas Harriot's work, published posthumously, introduced the inequality signs $<$ and $>$.
- **1637.** Descartes' *La Géométrie* joined algebra and geometry through coordinates and equations.
- **1654.** The Fermat–Pascal correspondence on games of chance helped establish mathematical probability.
- **1655.** John Wallis introduced the infinity symbol ∞ in *De sectionibus conicis.*
- **1665–1666.** During the plague years, Newton developed major parts of his calculus of fluxions, series methods, mechanics, and gravitation.
- **1675.** Leibniz first used the elongated-S integral sign $\int$ in a manuscript; his differential notation $\mathrm{d}x$ and $\mathrm{d}y$ became standard.
- **1684.** Leibniz published the first paper on differential calculus. Newton and Leibniz are now credited with independent developments of calculus.
- **1687.** Newton published the *Philosophiæ Naturalis Principia Mathematica*, placing mechanics and gravitation in a mathematical framework.
- **1706.** William Jones used the symbol π for the ratio of circumference to diameter; Euler later popularized it.
- **1713 and 1738.** Nicolaus Bernoulli proposed the St. Petersburg paradox, and Daniel Bernoulli analyzed it using expected utility.
- **1715.** Brook Taylor published the general expansion now called Taylor's theorem.
- **1730s.** Euler adopted e for the base of natural logarithms and $f(x)$ for a function; these notations became standard through his work.
- **1734–1735.** Euler solved the Basel problem,

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6},$$

- **1736.** Euler's Königsberg bridges paper founded graph theory and helped initiate topology.
- **1742.** Christian Goldbach wrote to Euler about the conjecture now usually stated as: every even integer greater than 2 is a sum of two primes.
- **1755.** Euler used capital sigma Σ for summation in *Institutiones calculi differentialis.*
- **1763.** Bayes' essay on inverse probability was published posthumously by Richard Price.

- **1777.** Euler used i for $\sqrt{-1}$ in a manuscript published in 1794; Gauss later helped make the notation standard.
- **1795.** John Playfair published a widely used reformulation of Euclid's parallel postulate: through a point outside a line, exactly one parallel line can be drawn.
- **1799.** Gauss's doctoral dissertation supplied the first substantially complete proof of the fundamental theorem of algebra.
- **1801.** Gauss published *Disquisitiones Arithmeticae*, reorganizing number theory; in the same year his least-squares calculations helped recover the orbit of Ceres.
- **1821.** Cauchy's *Cours d'Analyse* made limits, convergence, and continuity central to rigorous analysis.
- **1824.** Abel proved that the general quintic equation cannot be solved by radicals.
- **1829–1832.** Lobachevsky and Bolyai published independent constructions of hyperbolic geometry. Gauss had developed related ideas without publishing a systematic account; Riemann's 1854 lecture later broadened geometry through manifolds and curvature.
- **1830s–1846.** Galois connected polynomial equations with permutation groups; Liouville published Galois' manuscripts posthumously in 1846.
- **1843.** Hamilton discovered the quaternions, with
$$i^2 = j^2 = k^2 = ijk = -1.$$
- **1844.** Liouville proved that transcendental numbers exist, providing the first explicit examples.
- **1847.** Boole published *The Mathematical Analysis of Logic*, treating logic by algebraic operations.
- **1854.** Riemann's habilitation lecture introduced a broad theory of manifolds and curvature, transforming geometry.
- **1858.** August Ferdinand Möbius and Johann Benedict Listing independently described the one-sided surface now called the Möbius strip.
- **1872.** Dedekind published Dedekind cuts and Cantor published a construction of the real numbers, helping complete the arithmetization of analysis.
- **1873.** Charles Hermite proved that e is transcendental.
- **1874.** Cantor proved that the real algebraic numbers are countable and that transcendental numbers are uncountable.
- **1882.** Ferdinand von Lindemann proved that π is transcendental. Consequently, squaring the circle with straightedge and compass is impossible.
- **1889.** Peano published axioms for the natural numbers in a symbolic logical language.
- **1895.** Poincaré published *Analysis Situs*, a foundational work of algebraic topology.
- **1896.** Hadamard and de la Vallée Poussin independently proved the prime number theorem,
$$\pi(x) \sim \frac{x}{\ln x}.$$

- **1899.** Hilbert's *Foundations of Geometry* gave a modern axiomatic treatment of Euclidean geometry.
- **1900.** At the second ICM in Paris, Hilbert presented ten of the twenty-three problems in his published list; the list guided much of twentieth-century mathematics.
- **1901.** Russell discovered the paradox of the set of all sets that are not members of themselves, exposing a crisis in naive set theory.
- **1904–1908.** Zermelo used the axiom of choice to prove the well-ordering theorem and then proposed an axiomatization of set theory, later refined into ZF and ZFC.
- **1910–1913.** Whitehead and Russell published the three volumes of *Principia Mathematica*.
- **1918.** Emmy Noether proved that continuous symmetries of an action correspond to conservation laws.
- **1922.** Banach's work established complete normed spaces as central objects of functional analysis; his contraction principle became a standard existence theorem.
- **1930.** The Institute for Advanced Study was founded in Princeton as an independent research institution without degree programs.
- **1931.** Gödel proved his incompleteness theorems: every sufficiently strong consistent effectively axiomatized formal system is incomplete and cannot prove its own consistency.
- **1933.** Kolmogorov axiomatized probability using measure theory.
- **1935.** The Bourbaki group began publishing under the fictional name Nicolas Bourbaki; mathematicians of the Lwów school began recording problems in *The Scottish Book*.
- **1936.** Turing formalized computation through the Turing machine and proved the undecidability of the halting problem. The first Fields Medals were awarded in Oslo.
- **1940 and 1963.** Gödel proved that CH is consistent with ZF if ZF is consistent; Cohen introduced forcing and proved that CH is independent of ZFC, subject to consistency.
- **1947.** George Dantzig introduced the simplex method for linear programming.
- **1948.** Shannon founded information theory by defining entropy and channel capacity mathematically.
- **1950.** Nash introduced the equilibrium concept now called Nash equilibrium.
- **1956.** Milnor discovered exotic smooth structures on the 7-sphere.
- **1958.** The Institut des Hautes Études Scientifiques was founded near Paris as a research institute without degree programs.
- **1963.** The Atiyah–Singer index theorem connected the analytic index of an elliptic operator with topological data.
- **1970.** Matiyasevich completed earlier work of Davis, Putnam, and Robinson, proving that Hilbert's tenth problem has no algorithmic solution.
- **1976.** Appel and Haken proved the Four Color Theorem with extensive computer assistance, creating a landmark debate about computer-assisted proof.

- **1982.** The Mathematical Sciences Research Institute was founded in Berkeley; it was renamed the Simons Laufer Mathematical Sciences Institute in 2022.
- **1983.** Faltings proved the Mordell conjecture: a smooth projective curve of genus greater than 1 over a number field has only finitely many rational points.
- **1985.** Louis de Branges proved the Bieberbach conjecture on coefficients of univalent functions.
- **1994–1995.** Wiles, with the crucial repair developed with Richard Taylor, proved Fermat's Last Theorem through the modularity of semistable elliptic curves.
- **1998 and 2014.** Hales announced a computer-assisted proof of the Kepler conjecture; the Flyspeck project later completed a formal verification.
- **2002–2003.** Perelman posted papers completing Hamilton's Ricci-flow program and proving the Poincaré and geometrization conjectures.
- **2004.** Green and Tao proved that the prime numbers contain arbitrarily long arithmetic progressions.
- **2013.** Yitang Zhang proved the first finite bound on gaps between infinitely many pairs of primes; Maynard, Tao, and the Polymath project rapidly improved bounded-gap methods.
- **2014.** ICM Seoul was held in South Korea, and Maryam Mirzakhani became the first woman to receive the Fields Medal.
- **2016.** Maryna Viazovska proved the optimality of the E_8 lattice sphere packing in dimension 8; joint work soon settled dimension 24 using the Leech lattice.
- **2022.** June Huh received the Fields Medal for work joining combinatorics, algebraic geometry, and Hodge theory; Maryna Viazovska became the second woman Fields Medalist.
- **2025.** Hong Wang and Joshua Zahl proved the three-dimensional Kakeya conjecture: a subset of $\mathbb{R}^3$ that contains a unit line segment in every direction has full Hausdorff and Minkowski dimension 3.

Fact 14.1. Extensions of the Number System
The following historical associations organize several standard number systems without implying that each system was created by one person in a single step.

- **Zero and negative numbers:** Brahmagupta's *Brahmasphutasiddhanta* of 628 gave systematic arithmetic rules involving both.
- **Complex numbers:** Rafael Bombelli's *L'Algebra* of 1572 gave a systematic treatment of arithmetic with square roots of negative numbers.
- **Quaternions:** William Rowan Hamilton discovered them in 1843, with

$$i^2 = j^2 = k^2 = ijk = -1.$$

- **p-adic numbers:** Kurt Hensel introduced the p-adic viewpoint at the end of the nineteenth century, producing completions of $\mathbb{Q}$ different from $\mathbb{R}$.
- **Hyperreal numbers:** Edwin Hewitt introduced the term *hyper-real* in 1948; Abraham Robinson later developed nonstandard analysis using infinitesimal and infinitely large numbers.

- **Surreal numbers:** John Conway constructed the number system from combinatorial games around 1970; Donald Knuth introduced the name *surreal numbers*.

Fact 14.2. Königsberg and Kaliningrad
Königsberg, home of the Seven Bridges problem and birthplace of Hilbert, became Kaliningrad after World War II. Kaliningrad is now part of Russia.

Mathematical Works

The works below are ordered chronologically. They are included because the title, author, mathematical content, or historical role is itself a useful Science Quiz association. Ancient dates and dates of long compilations are necessarily approximate; for a work first circulated in one form and published later, both dates are recorded when that distinction is useful.

Date	Work	Author / association	Main content	Historical significance
c. 1650 BCE	*Rhind Mathematical Papyrus*	Ahmes, scribe; copied from an older Egyptian source	Arithmetic, Egyptian fractions, equations, progressions, areas, and volumes.	One of the principal surviving sources for ancient Egyptian mathematics.
c. 300 BCE	*Elements*	Euclid	Geometry, proportion, number theory, and the Euclidean algorithm in thirteen books.	The archetype of the axiomatic method and one of the most influential mathematical works in history.
c. 3rd century BCE	*On the Sphere and Cylinder*	Archimedes	Surface area and volume of the sphere and cylinder by methods anticipating integral geometry.	Contains Archimedes' celebrated sphere–cylinder results and exemplifies the Greek method of exhaustion.
c. 200 BCE	*Conics*	Apollonius of Perga	Systematic theory of the ellipse, parabola, and hyperbola.	Established much of the classical language and theory of conic sections.
c. 200 BCE–50 CE; commentary 263	*The Nine Chapters on the Mathematical Art*	Anonymous Chinese compilation; major commentary by Liu Hui	Algorithms for arithmetic, fractions, areas, volumes, linear systems, and practical problems.	Foundational Chinese mathematical classic; Liu Hui's commentary supplied explanations and refined polygonal approximations of π.
c. 250	*Arithmetica*	Diophantus of Alexandria	Problems on polynomial and indeterminate equations with rational solutions.	A major source for Diophantine analysis; Fermat's famous Last Theorem note was written in a later edition of this work.
c. 3rd–5th century	*Sunzi Suanjing* (*Sunzi Mathematical Classic*)	Traditionally associated with Sunzi	Arithmetic algorithms and remainder problems.	Contains the classical remainder problem that became a standard example of the Chinese Remainder Theorem.
499	*Aryabhatiya*	Aryabhata	Arithmetic, algebra, trigonometry, astronomical computation, and approximations to π.	A landmark of classical Indian mathematics and astronomy; influential in the development of sine tables and computational methods.

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Date	Work	Author / association	Main content	Historical significance
628	*Brahmasphutasiddhanta*	Brahmagupta	Arithmetic with zero and negative numbers, algebra, quadratic equations, geometry, and astronomy.	Gave systematic rules involving zero and signed numbers and includes Brahmagupta's formula for cyclic quadrilaterals.
c. 820	*Al-Kitab al-mukhtasar fi hisab al-jabr wa'l-muqabala*	Muhammad ibn Musa al-Khwarizmi	Systematic procedures for linear and quadratic equations.	The word *algebra* derives from *al-jabr* in the title; al-Khwarizmi's name is also the source of *algorithm*.
c. 1070	*Treatise on Demonstration of Problems of Algebra*	Omar Khayyam	Classification and geometric solution of cubic equations using conic sections.	A major medieval synthesis of algebra and geometry before general symbolic solutions of cubics were known.
1150	*Siddhanta Shiromani*	Bhaskara II	Arithmetic, algebra, planetary mathematics, and sphere mathematics; includes *Lilavati* and *Bijaganita*.	One of the major works of medieval Indian mathematics, especially noted for algebraic and indeterminate-equation techniques.
1202	*Liber Abaci*	Leonardo of Pisa (Fibonacci)	Hindu–Arabic numerals, commercial arithmetic, algebraic problems, and the rabbit problem leading to the Fibonacci sequence.	Helped popularize Hindu–Arabic positional arithmetic in medieval Europe.
1303	*The Precious Mirror of the Four Elements*	Zhu Shijie	Polynomial equations, elimination, finite sums, and a triangular array of binomial coefficients.	A high point of classical Chinese algebra, including methods for equations of high degree.
1494	*Summa de arithmetica, geometria, proportioni et proportionalita*	Luca Pacioli	Comprehensive printed survey of arithmetic, algebra, geometry, proportion, and commercial calculation.	One of the earliest major printed mathematical encyclopedias of Renaissance Europe.
1545	*Ars Magna*	Girolamo Cardano	General methods for cubic equations and Ferrari's solution of quartic equations.	A landmark in Renaissance algebra and an important step toward accepting complex quantities in algebraic calculation.
1572	*L'Algebra*	Rafael Bombelli	Algebraic rules for complex numbers and applications to cubic equations.	Gave the first systematic practical treatment of arithmetic with square roots of negative numbers.
1585	*De Thiende*	Simon Stevin	Decimal fractions and their arithmetic.	Helped establish decimal fractions as a practical computational language in Europe.
1591	*In artem analyticem isagoge*	Francois Viète	Symbolic algebra using letters for known and unknown quantities.	A decisive step toward modern symbolic algebra and systematic manipulation of general equations.
1614	*Mirifici Logarithmorum Canonis Descriptio*	John Napier	Logarithms and numerical tables designed to simplify multiplication and division.	Introduced logarithms to a wide audience and transformed numerical computation in astronomy and science.

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Date	Work	Author / association	Main content	Historical significance
1637	*La Géométrie*	René Descartes	Algebraic treatment of geometric curves and coordinate methods.	A founding work of analytic geometry, linking algebraic equations with geometric loci.
1654; published 1665	*Traité du triangle arithmétique*	Blaise Pascal	Binomial coefficients and identities organized through the arithmetical triangle.	One of the classic early systematic treatments of Pascal's triangle and combinatorial identities.
1684	*Nova Methodus pro Maximis et Minimis*	Gottfried Wilhelm Leibniz	Differential calculus, tangents, extrema, and rules of differentiation.	The first published account of Leibniz's differential calculus and the source of notation that strongly influenced modern calculus.
1687	*Philosophiæ Naturalis Principia Mathematica*	Isaac Newton	Laws of motion, universal gravitation, celestial mechanics, and geometric forms of infinitesimal reasoning.	One of the foundational works of mathematical physics and a defining achievement of the Scientific Revolution.
1713	*Ars Conjectandi*	Jacob Bernoulli	Combinatorics, probability, Bernoulli numbers, and the law of large numbers.	A foundational treatise in probability theory and one of the earliest systematic works on the subject.
1718	*The Doctrine of Chances*	Abraham de Moivre	Probability calculations for games of chance, independence, and approximations to the binomial distribution.	A classic early probability text; later editions contain work closely related to the normal approximation.
1748	*Introductio in analysin infinitorum*	Leonhard Euler	Functions, infinite series, exponential and logarithmic functions, trigonometry, and analytic methods.	A foundational text in modern analysis that helped establish the function-centered language of mathematics.
1788	*Mécanique analytique*	Joseph-Louis Lagrange	Mechanics formulated through algebra, differential equations, and variational principles.	Transformed mechanics into a systematic branch of mathematical analysis and introduced the Lagrangian viewpoint.
1801	*Disquisitiones Arithmeticae*	Carl Friedrich Gauss	Congruences, quadratic residues, binary quadratic forms, cyclotomy, and construction of the regular 17-gon.	A foundational work of modern number theory; it systematized congruence notation and quadratic reciprocity.
1821	*Cours d'Analyse*	Augustin-Louis Cauchy	Limits, continuity, derivatives, infinite series, and rigorous foundations of analysis.	A central nineteenth-century step toward the modern rigorous formulation of calculus and analysis.
1822	*Théorie analytique de la chaleur*	Joseph Fourier	Heat equation and representation of functions by trigonometric series.	Established Fourier series as a fundamental analytic tool and strongly influenced PDE, analysis, and mathematical physics.
1832	*Appendix scientiam spatii absolute veram exhibens*	János Bolyai	A systematic development of geometry independent of Euclid's parallel postulate.	One of the foundational publications of non-Euclidean geometry.

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Date	Work	Author / association	Main content	Historical significance
written 1831; published 1846	*Mémoire sur les conditions de résolubilité des équations par radicaux*	Èvariste Galois	Permutations of roots and criteria for solvability of polynomial equations by radicals.	Laid the foundations of Galois theory and connected group structure with solvability of equations.
1844	*Die lineale Ausdehnungslehre*	Hermann Grassmann	An algebra of directed and higher-dimensional geometric quantities.	A precursor of modern vector spaces, multilinear algebra, and exterior algebra.
1847	*The Mathematical Analysis of Logic*	George Boole	Algebraic formalization of logical operations.	Helped create mathematical logic and the algebraic tradition that led to Boolean algebra and digital logic.
1854; published 1868	*On the Hypotheses Which Lie at the Foundations of Geometry*	Bernhard Riemann	Manifolds, metrics, curvature, and the possibility of non-Euclidean geometries in higher dimensions.	A foundational source for Riemannian geometry and the geometric language later used in general relativity.
1859	*On the Number of Primes Less Than a Given Magnitude*	Bernhard Riemann	The zeta function, complex analytic continuation, and the distribution of prime numbers.	Introduced the Riemann Hypothesis and permanently linked complex analysis with prime-number theory.
1872	*Stetigkeit und irrationale Zahlen*	Richard Dedekind	A construction of the real numbers by Dedekind cuts and a precise treatment of continuity.	One of the classical arithmetizations of analysis and a rigorous foundation for the real number system.
1872	*Erlangen Program*	Felix Klein	Classification of geometries by transformation groups and invariants.	Reframed geometry through group actions and unified Euclidean, projective, and non-Euclidean viewpoints.
1874	*On a Property of the Collection of All Real Algebraic Numbers*	Georg Cantor	Countability of algebraic numbers and uncountability of the real numbers.	Commonly regarded as the first paper of Cantor's set theory and the beginning of the theory of different infinite cardinalities.
1879	*Begriffsschrift*	Gottlob Frege	A formal language for propositional and predicate logic with quantification.	A foundational work of modern mathematical logic and formal systems.
1888	*Was sind und was sollen die Zahlen?*	Richard Dedekind	Natural numbers, mappings, chains, and an abstract logical construction of arithmetic.	A foundational treatment of the natural numbers closely related to the later Peano axioms.
1889	*Arithmetices principia, nova methodo exposita*	Giuseppe Peano	Symbolic logic and axioms for the natural numbers.	Gave the famous Peano-style axiomatization and helped standardize a symbolic language for foundations.
1895	*Analysis Situs*	Henri Poincaré	Qualitative invariants of spaces, homology-type ideas, and the fundamental group.	A foundational paper of algebraic topology and the origin of much modern topological language.

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Date	Work	Author / association	Main content	Historical significance
1899	*Grundlagen der Geometrie*	David Hilbert	Axioms of geometry, independence, consistency, and incidence, order, congruence, parallels, and continuity.	A landmark in the axiomatic method and the modern foundations of geometry.
1900	*Mathematical Problems*	David Hilbert	Twenty-three problems proposed as major directions for future mathematical research.	Hilbert's Paris ICM address became one of the most influential research agendas in twentieth-century mathematics.
1902	*Intégrale, longueur, aire*	Henri Lebesgue	Measure, measurable functions, and the Lebesgue integral.	Established the measure-theoretic integration framework that underlies modern real analysis and probability.
1908	*Untersuchungen über die Grundlagen der Mengenlehre I*	Ernst Zermelo	An axiomatic system for set theory designed to avoid the paradoxes of naive set theory.	A direct ancestor of Zermelo–Fraenkel set theory, the standard foundational framework for modern mathematics.
1910–1913	*Principia Mathematica*	Alfred North Whitehead; Bertrand Russell	Symbolic logic and a logicist derivation of substantial parts of mathematics.	A landmark of mathematical logic and foundations; its program formed part of the background to Gödel's incompleteness results.
1913	*Die Idee der Riemannschen Fläche*	Hermann Weyl	Riemann surfaces treated through topology, analysis, and geometric structure.	A highly influential synthesis that helped shape the modern concept of a Riemann surface.
1914	*Grundzüge der Mengenlehre*	Felix Hausdorff	Set theory, topological spaces, neighborhoods, separation, and dimension-related ideas.	One of the foundational books of general topology; the Hausdorff separation condition bears the author's name.
1918	*Invariante Variationsprobleme*	Emmy Noether	Relations between continuous symmetries and conservation laws in variational systems.	Contains Noether's theorems, fundamental to modern mathematical physics and the calculus of variations.
1930–1931	*Moderne Algebra*	B. L. van der Waerden	Groups, rings, fields, ideals, and the structural algebra developed by Noether, Artin, and others.	A defining textbook of abstract algebra that helped standardize the structural viewpoint of twentieth-century algebra.
1931	*Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I*	Kurt Gödel	Arithmetization of syntax and the first and second incompleteness phenomena for formal systems.	Established fundamental limits on completeness and self-verification in sufficiently strong formal axiomatic systems.
1932	*Théorie des opérations linéaires*	Stefan Banach	Normed linear spaces, linear operators, functional analysis, and results now associated with Banach spaces.	A foundational monograph of functional analysis and the source of much of its modern structural language.
1933	*Grundbegriffe der Wahrscheinlichkeitsrechnung*	Andrey Kolmogorov	Probability formulated axiomatically using measure theory.	Established the standard modern axioms of probability on a probability space.

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Date	Work	Author / association	Main content	Historical significance
1935–1941	*The Scottish Book*	Mathematicians of the Lwów school; initiated around Stefan Banach's circle	Unsolved problems, conjectures, and prizes recorded at the Scottish Café.	A unique mathematical notebook associated with Banach, Steinhaus, Ulam, Mazur, and others; Problem 153 famously carried a live-goose prize.
1936	*On Computable Numbers, with an Application to the Entscheidungsproblem*	Alan Turing	Turing machines, computable numbers, and undecidability of the Entscheidungsproblem.	A foundational paper of computability theory and theoretical computer science.
1938	*An Introduction to the Theory of Numbers*	G. H. Hardy; E. M. Wright	Classical elementary and analytic number theory.	One of the most famous enduring number-theory textbooks and a standard historical reference for classical results.
1939 onward	*Éléments de mathématique*	Nicolas Bourbaki	A systematic axiomatic treatment of major areas of modern mathematics.	The collective project strongly influenced twentieth-century terminology, abstraction, and textbook style; Bourbaki is a pseudonymous group, not one mathematician.
1940	*A Mathematician's Apology*	G. H. Hardy	A reflective essay on pure mathematics, creativity, beauty, and the mathematician's life.	One of the best-known literary works about mathematical culture and the values of pure mathematics.
1944	*Theory of Games and Economic Behavior*	John von Neumann; Oskar Morgenstern	Strategic games, utility, minimax ideas, and mathematical models of economic behavior.	Established game theory as a systematic mathematical discipline with major applications beyond mathematics.
1945	*How to Solve It*	George Pólya	Heuristics for problem solving: understand the problem, devise a plan, carry it out, and look back.	One of the most influential books on mathematical problem solving and mathematical education.
1948	*A Mathematical Theory of Communication*	Claude Shannon	Entropy, information, coding, channel capacity, and quantitative communication theory.	Founded information theory and supplied the mathematical framework underlying modern digital communication.
1950–1951	*Equilibrium Points in N-Person Games*; *Non-Cooperative Games*	John Nash	Existence and theory of equilibrium in finite non-cooperative games.	Introduced the Nash equilibrium, a central solution concept in game theory and mathematical economics.
1955	*Faisceaux algébriques cohérents*	Jean-Pierre Serre	Coherent sheaves and cohomological methods in algebraic geometry.	The celebrated "FAC" paper helped establish sheaf cohomology as a fundamental tool of modern algebraic geometry.
1960–1967	*Éléments de géométrie algébrique*	Alexander Grothendieck; Jean Dieudonné	Schemes, morphisms, cohomological foundations, and a systematic language for algebraic geometry.	The "EGA" series transformed the foundations and language of modern algebraic geometry.

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Date	Work	Author / association	Main content	Historical significance
1963	*Solvability of Groups of Odd Order*	Walter Feit; John G. Thompson	Proof that every finite group of odd order is solvable.	A landmark 255-page group-theory paper and a major precursor to the classification of finite simple groups.
1968	*The Index of Elliptic Operators I*	Michael Atiyah; Isadore Singer	Index theory for elliptic differential operators connecting analysis, topology, and geometry.	Part of the foundational development of the Atiyah–Singer Index Theorem, one of the central results of twentieth-century mathematics.
1976	*On Numbers and Games*	John Horton Conway	Combinatorial game theory and a vast ordered field containing the surreal numbers.	Introduced a unified mathematical treatment of games and Conway's construction of the surreal numbers.
1977	*Every Planar Map Is Four Colorable*	Kenneth Appel; Wolfgang Haken; John Koch	Reduction to finitely many configurations and computer checking in the proof of the Four Color Theorem.	One of the first celebrated computer-assisted proofs, changing discussion about computation and mathematical proof.
1982	*The Fractal Geometry of Nature*	Benoit Mandelbrot	Fractals, self-similarity, scaling, and irregular geometric structures.	Popularized fractal geometry and made the term *fractal* central in mathematics and science.
1995	*Modular Elliptic Curves and Fermat's Last Theorem*	Andrew Wiles	Modularity of semistable elliptic curves using Galois representations and modular forms.	Together with the Taylor–Wiles companion paper, supplied the proof of Fermat's Last Theorem.
1995	*Ring-Theoretic Properties of Certain Hecke Algebras*	Richard Taylor; Andrew Wiles	The commutative-algebra and Hecke-algebra input needed to complete Wiles's modularity argument.	The companion paper that repaired the gap in the original argument and completed the published proof of Fermat's Last Theorem.
1998	*Proofs from THE BOOK*	Martin Aigner; Günter M. Ziegler	Elegant proofs in number theory, geometry, analysis, combinatorics, and graph theory.	Inspired by Paul Erdős' metaphor of "The Book" containing the most beautiful proofs; many topics were suggested by Erdős.
2002	*The Entropy Formula for the Ricci Flow and Its Geometric Applications*	Grigori Perelman	Entropy and monotonicity for Ricci flow, noncollapsing, and geometric applications.	The first of Perelman's three preprints advancing Hamilton's Ricci-flow program toward geometrization.
2003	*Ricci Flow with Surgery on Three-Manifolds*	Grigori Perelman	Construction and analysis of Ricci flow with surgery on 3-manifolds.	The technical continuation that supplied the surgery mechanism central to the proof of geometrization.
2003	*Finite Extinction Time for the Solutions to the Ricci Flow on Certain Three-Manifolds*	Grigori Perelman	Finite-time extinction for relevant Ricci flows with surgery.	Completed the sequence of preprints from which the Poincaré conjecture and Thurston geometrization theorem were established.

Continued on the next page.

Date	Work	Author / association	Main content	Historical significance
2012	*Théorème vivant*	Cédric Villani	A narrative of research with Clément Mouhot on nonlinear Landau damping.	A rare modern research memoir whose title and story became part of Villani's public mathematical identity after his 2010 Fields Medal.

Mathematical Anecdotes

Some famous stories are documented, while others survive as traditional legends. A story whose details are uncertain is explicitly identified below; the uncertainty is part of the fact to remember.

Fact 14.3. Frequently Named Mathematical Paradoxes
In quiz settings, a paradox is usually identified by its defining tension rather than by a full resolution.

- **Zeno's Achilles paradox:** infinitely many remaining subintervals appear to prevent a faster runner from overtaking a slower one; convergent series resolve the apparent contradiction.
- **Aristotle's wheel paradox:** concentric circles of different circumferences appear to roll through the same distance; the smaller circle does not undergo ordinary no-slip rolling against the line.
- **St. Petersburg paradox:** a game can have infinite expected monetary payoff while attracting only a modest entrance fee; expected utility explains the discrepancy.
- **Russell's or the barber paradox:** unrestricted self-reference creates a contradiction, motivating axiomatic set theory.
- **Banach–Tarski paradox:** assuming the axiom of choice, a solid ball in $\mathbb{R}^3$ can be partitioned into finitely many nonmeasurable pieces and reassembled into two balls congruent to the original.
- **Birthday paradox:** among 23 people, the probability that at least two share a birthday exceeds 1/2, under the usual uniform independent model.
- **Coastline paradox:** the measured length of an irregular boundary increases as the measuring scale becomes finer; fractal dimension describes this scale dependence.
- **Newcomb's paradox:** dominance reasoning and expected utility reasoning recommend different choices when a highly accurate predictor has already filled two boxes.
- **Parrondo's paradox:** alternating or mixing two losing stochastic games can produce a winning process.
- **c. fifth century BCE: Pythagorean irrationality legend.** A famous but historically uncertain legend says that Hippasus was drowned after revealing the irrationality of $\sqrt{2}$. The discovery itself shattered the belief that every magnitude is a ratio of integers.
- **c. 300 BCE: No royal road.** Proclus reports that when Ptolemy asked for an easier route to geometry, Euclid replied that there is no royal road to geometry.
- **third century BCE: Archimedes and the crown.** Vitruvius tells the traditional story that Archimedes recognized how displacement could test a crown's purity and ran out crying "Eureka!" The precise details are not independently verified.

- **212 BCE: Archimedes' circles.** Ancient accounts say that Archimedes was killed during the Roman capture of Syracuse while absorbed in a geometric diagram. The familiar wording "Do not disturb my circles" is traditional.

- **75 BCE: Archimedes' tomb.** Cicero reported rediscovering the tomb of Archimedes by recognizing the carved sphere and cylinder that commemorated Archimedes' favorite result.

- **1539–1545: Cardano and Tartaglia.** Tartaglia disclosed his cubic method to Cardano under a promise of secrecy. Cardano later published it in *Ars Magna* after learning that del Ferro had found the method earlier, beginning a celebrated priority dispute.

- **c. 1637: Fermat's margin.** Fermat wrote that he had a marvelous proof of his last theorem but that the margin was too small to contain it. No valid general proof by Fermat is known.

- **c. 1666: Newton's apple.** Newton later told William Stukeley that seeing an apple fall prompted thoughts about gravity. The story does not require that an apple struck Newton's head.

- **late seventeenth–early eighteenth centuries: The calculus priority dispute.** Newton and Leibniz developed calculus independently, but a bitter priority dispute divided British and continental mathematics for decades.

- **1766 onward: Euler's eyesight.** Euler continued producing mathematics at an extraordinary rate after losing almost all of his sight, dictating long calculations from memory.

- **c. 1780s: Gauss and the arithmetic sum.** A traditional schoolroom story says that young Gauss immediately summed $1 + 2 + \cdots + 100$ by pairing terms to obtain 5050. The details vary among later accounts.

- **1790s: Sophie Germain's pseudonym.** Because women were excluded from the École Polytechnique, Germain studied its lecture notes and submitted work under the name "M. LeBlanc." Lagrange became a supporter after learning her identity.

- **1832: Galois' last night.** Galois wrote urgent mathematical notes before the duel that killed him at age 20. He did not create all of Galois theory in one night; the notes summarized and revised ideas developed earlier.

- **1843: Hamilton's bridge.** On October 16, 1843, Hamilton carved the quaternion relations
$$i^2 = j^2 = k^2 = ijk = -1$$
into Brougham Bridge in Dublin after the key idea came to him during a walk.

- **1895: No Nobel Prize in Mathematics.** The popular story that Nobel excluded mathematics because of a romantic rivalry is unsupported. Nobel's will simply named physics, chemistry, physiology or medicine, literature, and peace.

- **1900: Hilbert's problems.** Hilbert's Paris address did not orally present all twenty-three problems; ten were delivered in the lecture, while the complete list appeared in print.

- **1915: Hilbert and Noether.** When objections were raised to Noether teaching at Göttingen because she was a woman, Hilbert reportedly replied that the university was not a bathhouse.

- **1918: The taxi-cab number.** Hardy told Ramanujan that taxi number 1729 seemed dull. Ramanujan immediately answered that it is the smallest number expressible as a sum of two positive cubes in two ways:
$$1729 = 1^3 + 12^3 = 9^3 + 10^3.$$
- **1935 onward: Nicolas Bourbaki.** Bourbaki is not one mathematician but the collective pseudonym of a group that sought a unified, axiomatic presentation of modern mathematics.
- **1935–1941: *The Scottish Book.*** Mathematicians of the Lwów school wrote problems in a notebook kept at the Scottish Café. Prizes ranged from coffee or beer to a live goose.
- **1936 and 1972: Mazur's goose.** Stanisław Mazur offered a live goose for Problem 153 in 1936. Per Enflo solved the basis problem in 1972, and Mazur presented the promised goose in a televised ceremony.
- **1939: Dantzig's homework.** George Dantzig arrived late to Jerzy Neyman's statistics class, copied two unsolved problems from the board believing they were homework, and later submitted solutions to both.
- **1947: Gödel's citizenship hearing.** Before his 1947 US citizenship examination, Gödel found what he believed was a logical route to dictatorship in the Constitution. Einstein and Oskar Morgenstern accompanied him and helped prevent the hearing from becoming a constitutional debate.
- **1947: Hilbert's hotel.** The infinite-hotel thought experiment, popularized by George Gamow, illustrates that a countably infinite full hotel can still accommodate more guests by moving each guest to another numbered room.
- **twentieth century: Erdős' language.** Erdős called children "epsilons" and imagined an ideal book in which God keeps the most elegant proofs. The Erdős number measures collaboration distance from him.
- **1954: Turing's apple.** Turing died from cyanide poisoning in 1954, and a bitten apple was found nearby. The popular claim that the apple was deliberately poisoned, or that it inspired the Apple logo, is not established.
- **1956: Milnor's exotic spheres.** Milnor discovered that a topological 7-sphere can carry differentiable structures not equivalent to the standard one. The group of oriented homotopy 7-spheres has order 28.
- **1976: The computer proof controversy.** The 1976 proof of the Four Color Theorem required computer checking of many cases, forcing mathematicians to debate what it means for a proof to be humanly inspectable.
- **1993–1994: Wiles' secret work.** Wiles worked privately for years on Fermat's Last Theorem. After announcing a proof in 1993, a gap was found; Wiles and Richard Taylor repaired it in 1994.
- **2006 and 2010: Perelman's refusals.** Perelman declined the 2006 Fields Medal and the Clay Millennium Prize for the Poincaré conjecture.
- **2010s: Villani's public image.** Cédric Villani became recognizable for a cravat and spider brooch. His book *Théorème vivant* recounts work with Clément Mouhot on nonlinear Landau damping.
- **2018: Birkar's stolen medal.** Caucher Birkar's Fields Medal was stolen shortly after the 2018 ceremony in Rio de Janeiro; the IMU later presented him with a replacement.

Chapter 15

Famous Problems and Conjectures

Famous problems are usually not tested by asking for complete proofs. They are tested through the following information: the standard statement, the person or group attached to the problem, whether it is solved, and the main idea or keyword attached to the solution. The goal of this chapter is to make those associations immediate.

Millennium Prize Problems

The Clay Mathematics Institute announced the seven Millennium Prize Problems in 2000. Each problem carries a prize of one million US dollars. Among the seven, only the Poincaré conjecture has been solved.

Problem	Standard mathematical form	Status	Quiz anchors
Poincaré Conjecture	Every closed simply connected 3-manifold is homeomorphic to S^3.	Solved.	Poincaré; Perelman; Ricci flow.
Riemann Hypothesis	Every nontrivial zero of $\zeta(s)$ has real part 1/2.	Open.	Riemann; zeta function; prime numbers.
P versus NP	Is $P = NP$?	Open.	Cook, Karp; computational complexity.
Birch and Swinnerton-Dyer Conjecture	For an elliptic curve E, $\operatorname{rk} E(\mathbb{Q})$ equals the order of vanishing of $L(E, s)$ at $s = 1$.	Open.	Elliptic curves; L-functions.
Hodge Conjecture	Rational Hodge classes on smooth projective complex varieties should come from algebraic cycles.	Open.	Hodge theory; algebraic geometry.
Navier–Stokes Existence and Smoothness	Smooth initial data for the 3-dimensional incompressible Navier–Stokes equations should have global smooth solutions, or one must find a breakdown.	Open.	Fluid mechanics; PDE; turbulence.
Yang–Mills Existence and Mass Gap	Construct quantum Yang–Mills theory on $\mathbb{R}^4$ for compact simple gauge groups and prove a positive mass gap.	Open.	Gauge theory; quantum field theory; mass gap.

Fact 15.1. Millennium Prize status check

Among the seven Millennium Prize Problems, the Poincaré conjecture is the only one that has been solved. The remaining six are the Riemann Hypothesis, P versus NP, the Birch and Swinnerton-Dyer conjecture, the Hodge conjecture, Navier–Stokes existence and smoothness, and Yang–Mills existence and mass gap.

Theorem 15.2. Poincaré Conjecture
Every closed simply connected 3*-manifold is homeomorphic to the* 3*-sphere* S^3*.*

Remark. Henri Poincaré posed the problem in the early twentieth century. Grigori Perelman proved it using Ricci flow and ideas from Hamilton's program. The principal association to retain is

$$\text{Poincaré conjecture} \quad \longleftrightarrow \quad \text{Perelman} \quad \longleftrightarrow \quad \text{Ricci flow.}$$

Hypothesis 15.3. Riemann Hypothesis
Let

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

be the Riemann zeta function, initially defined for $\mathrm{Re}(s) > 1$ *and continued meromorphically. The Riemann Hypothesis states that every nontrivial zero of* $\zeta(s)$ *has real part* $1/2$*.*

Idea. The zeta function is connected to prime numbers by Euler's product

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

Thus the Riemann Hypothesis is a statement about the hidden regularity of the distribution of prime numbers.

Open Problem 15.4. P versus NP
Determine whether

$$P = NP.$$

Here P is the class of decision problems solvable in polynomial time, and NP is the class of decision problems whose proposed solutions can be verified in polynomial time.

Remark. The usual slogan is: can every solution that can be checked quickly also be found quickly? The anchors are Cook, Karp, NP-completeness, and theoretical computer science.

Conjecture 15.5. Birch and Swinnerton-Dyer Conjecture
For an elliptic curve E *over* $\mathbb{Q}$*,* $\operatorname{rk} E(\mathbb{Q})$ *equals the order of vanishing of its* L*-function* $L(E, s)$ *at* $s = 1$*.*

Remark. The problem connects rational points on elliptic curves with analytic behavior of L-functions. The phrase "rk equals order of vanishing" is the main quiz memory.

Conjecture 15.6. Hodge Conjecture
On a smooth projective complex variety, every rational cohomology class of type (p, p) *should be a rational linear combination of classes of algebraic cycles.*

Remark. This is one of the deepest bridges between topology and algebraic geometry. The principal association to retain is

$$\text{Hodge classes} \quad \longleftrightarrow \quad \text{algebraic cycles.}$$

Open Problem 15.7. Navier–Stokes Existence and Smoothness
For the 3-dimensional incompressible Navier–Stokes equations, determine whether smooth initial data always lead to global smooth solutions, or whether finite-time singularities can occur.

Remark. This is a PDE problem motivated by fluid mechanics. The Quiz anchors are incompressible fluid, global regularity, singularity, and turbulence.

Open Problem 15.8. Yang–Mills Existence and Mass Gap
Construct a mathematically rigorous quantum Yang–Mills theory on $\mathbb{R}^4$ for every compact simple gauge group and prove that it has a positive mass gap.

Remark. The problem comes from quantum field theory. The relevant Quiz anchors are gauge theory, quantum field theory, and mass gap.

Hilbert's Problems

David Hilbert published a list of 23 problems after his address at the 1900 ICM in Paris. He presented ten of them orally. The list became a program for twentieth-century mathematics, but its items are not all yes–no questions; several are broad research programs whose status cannot be reduced to a single word.

No.	Subject	Modern status and principal names
1	Continuum hypothesis and well-ordering	CH is independent of ZFC, assuming consistency: Gödel proved relative consistency and Cohen proved independence using forcing.
2	Consistency of arithmetic	Gödel's incompleteness theorems rule out the strongest form of Hilbert's program; Gentzen later gave a relative consistency proof using transfinite induction.
3	Scissors congruence of polyhedra	Solved negatively by Max Dehn through the Dehn invariant: equal volume does not imply equidecomposability.
4	Geometries with straight-line geodesics	A broad program rather than one fixed proposition; substantially developed through metric and convex geometry.
5	Continuous groups and Lie groups	Solved affirmatively by Gleason and by Montgomery–Zippin: locally Euclidean topological groups are Lie groups.
6	Axiomatization of physics	An open-ended program involving probability, mechanics, and physical theories; no single final solution.
7	Transcendence of a^b	Solved by the Gelfond–Schneider theorem when a and b are algebraic, $a \neq 0, 1$, and b is irrational.
8	Prime-number problems	Includes the Riemann Hypothesis and Goldbach-type questions; central parts remain open.
9	General reciprocity laws	Largely answered by Artin reciprocity and class field theory.
10	Algorithm for Diophantine equations	Solved negatively by the Davis–Putnam–Robinson–Matiyasevich theorem: no such general algorithm exists over $\mathbb{Z}$.
11	Quadratic forms over algebraic number fields	Essentially solved through Hasse's local–global theory and later arithmetic developments.
12	Explicit class field theory	Solved for important special cases, but the general *Kronecker Jugendtraum* remains open.
13	Seventh-degree equations and functions	The continuous-function interpretation was solved by Kolmogorov and Arnold; stricter algebraic variants remain distinct.

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No.	Subject	Modern status and principal names
14	Finite generation of certain algebras	Solved negatively in general by Nagata's counterexample.
15	Rigorous foundation of Schubert calculus	Fulfilled through modern intersection theory and algebraic geometry.
16	Real algebraic curves and limit cycles	Both parts contain major open questions; the limit-cycle problem is open even in low degrees.
17	Nonnegative rational functions as sums of squares	Solved affirmatively by Emil Artin using sums of squares of rational functions.
18	Space groups, tilings, and sphere packing	Major parts were solved by Bieberbach and, for the three-dimensional Kepler conjecture, by Hales; it is a multi-part problem.
19	Regularity of variational solutions	Solved through work of De Giorgi, Nash, Moser, and others.
20	General boundary-value problems	A broad program developed through modern functional analysis and PDE; not one universally closed proposition.
21	Riemann–Hilbert problem	The unrestricted original existence claim is false by Bolibrukh's counterexample; important variants are solved.
22	Uniformization by automorphic functions	Solved by Koebe and Poincaré through the uniformization theorem.
23	Further development of the calculus of variations	A research program that stimulated extensive modern theory rather than a single proposition with one completion date.

Fact 15.9. Hilbert problem anchors
Hilbert's first problem concerns the continuum hypothesis; the second concerns consistency; the eighth includes the Riemann Hypothesis; the tenth concerns an algorithm for Diophantine equations; and the sixteenth includes the number and arrangement of limit cycles.

Landau's Problems

At the 1912 ICM in Cambridge, Edmund Landau identified four elementary prime-number problems that he regarded as exceptionally difficult. All four remain open.

Problem	Statement	Status and anchor
Goldbach Conjecture	Every even integer greater than 2 is a sum of two primes.	Open; additive number theory.
Twin Prime Conjecture	There are infinitely many primes p for which $p+2$ is prime.	Open; bounded prime gaps are known.
Legendre Conjecture	For every positive integer n, there is a prime between n^2 and $(n+1)^2$.	Open; primes in short intervals.
Primes of the form n^2+1	There are infinitely many positive integers n for which n^2+1 is prime.	Open; polynomial prime values.

Other Famous Problems and Conjectures

The following problems are not all open problems. Some are solved theorems, but they remain famous because of their history, statements, or solution methods. They are collected together rather than divided into mathematical fields.

Problem	Standard form	People and status	Quiz anchors
Basel Problem	Evaluate $\sum_{n=1}^{\infty} 1/n^2$.	Solved by Euler: $\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$.	Zeta value $\zeta(2)$; Euler.
Fermat's Last Theorem	$x^n + y^n = z^n$ has no positive integer solutions for $n \geq 3$.	Fermat; proved by Wiles with Taylor–Wiles.	Elliptic curves and modular forms.
Four Color Theorem	Every planar map can be colored with at most four colors.	Appel–Haken; later Robertson–Sanders–Seymour–Thomas.	First famous computer-assisted proof.
Kepler Conjecture	The maximum packing density of congruent spheres in $\mathbb{R}^3$ is $\pi/(3\sqrt{2})$ and is attained by the face-centered cubic packing.	Kepler; proved by Hales.	Sphere packing; Flyspeck project.
Thurston Geometrization Conjecture	Closed orientable 3-manifolds decompose into pieces modeled on eight geometries.	Thurston; completed through Perelman's work.	Eight Thurston geometries.
Continuum Hypothesis	There is no cardinality strictly between $\lvert\mathbb{N}\rvert$ and $\lvert\mathbb{R}\rvert$.	Cantor; independent of ZFC by Gödel and Cohen.	Forcing; independence.
Collatz Conjecture	Iterating $n \mapsto n/2$ for even n and $n \mapsto 3n+1$ for odd n eventually reaches 1.	Collatz; open.	Elementary iteration; nonlinear dynamics.
abc Conjecture	If $a + b = c$ and a, b, c are coprime, then c is controlled by the radical of abc, up to ε.	Masser–Oesterlé; open in the standard sense.	Radical, Diophantine equations.

Theorem 15.10. Basel Problem
Euler evaluated the reciprocal-square series:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \zeta(2) = \frac{\pi^2}{6}.$$

Remark. The problem was first posed by Pietro Mengoli in 1644 and became associated with the Bernoulli family in Basel. Euler's solution was a landmark because an infinite sum over integers produced a value involving π.

Theorem 15.11. Fermat's Last Theorem
For every integer $n \geq 3$, there do not exist positive integers x, y, z s.t.

$$x^n + y^n = z^n.$$

Remark. Fermat wrote that he had a marvelous proof, but no proof appeared in his margin. Andrew Wiles proved the theorem through the modularity of semistable elliptic curves. The principal association to retain is

$$\text{Fermat's Last Theorem} \quad \longleftrightarrow \quad \text{Wiles} \quad \longleftrightarrow \quad \text{elliptic curves and modular forms.}$$

Theorem 15.12. Four Color Theorem
Every planar map can be colored with at most four colors so that any two adjacent regions have different colors.

Remark. The theorem was proved by Kenneth Appel and Wolfgang Haken in 1976 with extensive computer assistance. This made it one of the most famous early examples of a computer-assisted proof.

Theorem 15.13. Kepler Conjecture
The maximum density of a packing of congruent spheres in $\mathbb{R}^3$ is equal to

$$\frac{\pi}{3\sqrt{2}}.$$

This density is attained by the face-centered cubic packing and also by other optimal packings, including the hexagonal close packing.

Remark. Thomas Hales proved the Kepler conjecture using a large computer-assisted argument. The Flyspeck project later produced a formal verification of the proof. In higher-dimensional sphere packing, Maryna Viazovska proved the optimality of the E_8 lattice in dimension 8.

Theorem 15.14. Thurston Geometrization Theorem
Every closed orientable 3-manifold can be decomposed into pieces that admit one of the eight Thurston geometries:

$$S^3,\ E^3,\ H^3,\ S^2 \times \mathbb{R},\ H^2 \times \mathbb{R},\ \widetilde{\mathrm{SL}}_2(\mathbb{R}),\ \mathrm{Nil},\ \mathrm{Sol}.$$

Remark. The Poincaré conjecture is a special case of the geometrization program. The 2018 Science Quiz asked for the number of Thurston geometries; the answer is 8.

Hypothesis 15.15. Continuum Hypothesis
There is no set whose cardinality is strictly between the cardinality of the natural numbers and the cardinality of the real numbers. Equivalently,

$$2^{\aleph_0} = \aleph_1.$$

Remark. Kurt Gödel proved that CH cannot be disproved from ZFC if ZFC is consistent. Paul Cohen proved that CH cannot be proved from ZFC if ZFC is consistent. Cohen introduced forcing and received the Fields Medal in 1966.

Open Problem 15.16. Goldbach Conjecture
Every even integer greater than 2 is a sum of two prime numbers.

Remark. Goldbach is one of the easiest famous open problems to state. It belongs to additive number theory. The weak Goldbach conjecture, concerning sums of three odd primes, was proved by Harald Helfgott.

Open Problem 15.17. Twin Prime Conjecture
There are infinitely many primes p s.t.

$$p+2$$

is also prime.

Remark. The modern quiz anchors are bounded gaps between primes, Yitang Zhang, James Maynard, Terence Tao, and the Polymath project. These results do not prove the twin prime conjecture, but they prove that infinitely many prime pairs occur within some fixed bounded distance.

Open Problem 15.18. Collatz Conjecture
Start with a positive integer n. If n is even, replace it by $n/2$. If n is odd, replace it by $3n+1$. The Collatz conjecture states that repeated iteration always eventually reaches 1.

Remark. This problem is famous because the rule is elementary but the global behavior is still unknown. It is also called the $3n+1$ problem.

Conjecture 15.19. *abc* **Conjecture**
For every $\varepsilon > 0$, there exists a constant K_ε s.t., whenever a, b, c are pairwise coprime positive integers satisfying $a + b = c$, we have

$$c < K_\varepsilon \operatorname{rad}(abc)^{1+\varepsilon},$$

where $\operatorname{rad}(n)$ *is the product of the distinct prime divisors of n.*

Remark. The conjecture is powerful because it predicts that the additive relation $a + b = c$ strongly restricts the repeated prime factors of abc. The keyword is "radical." Shinichi Mochizuki announced a claimed proof in 2012, and the papers were later published, but no broadly accepted proof has been established; the conjecture is therefore recorded here as open.

Fact 15.20. June Huh and combinatorial log-concavity
June Huh's work connects combinatorics with algebraic geometry and Hodge theory. The central associations are:

- Read's conjecture $\leftrightarrow$ chromatic polynomials.
- Brylawski's and Okounkov's conjectures $\leftrightarrow$ matroid characteristic polynomials and log-concavity.
- Rota's conjecture $\leftrightarrow$ log-concavity for matroids.
- Mason's conjecture $\leftrightarrow$ independent sets of matroids.
- Hodge theory $\leftrightarrow$ the geometric method behind these combinatorial log-concavity results.

In a quiz, if June Huh appears with a list of conjectures, the odd one out is often a famous conjecture from a different field, such as the Poincaré conjecture.

Conjecture 15.21. Hadwiger Conjecture
Every finite graph G with chromatic number $\chi(G) = k$ contains the complete graph K_k as a minor.

Remark. The conjecture is known for $k \leq 6$ and remains open in general. The case $k = 5$ is equivalent to the Four Color Theorem.

Conjecture 15.22. Hadamard Conjecture
For every positive integer n, there exists a $4n \times 4n$ matrix $\mathbf{H}$ with entries in $\{-1, 1\}$ s.t.

$$\mathbf{H}\mathbf{H}^{\mathsf{T}} = 4n\mathbf{I}_{4n}.$$

Remark. A matrix satisfying this orthogonality condition is a Hadamard matrix. The necessary divisibility condition is known, but existence for every positive multiple of 4 remains open.

Conjecture 15.23. Union-Closed Sets Conjecture
Let $\mathcal{F}$ be a finite nonempty family of finite sets s.t.

$$A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}.$$

Then some element belongs to at least half of the sets in $\mathcal{F}$.

Remark. The conjecture was proposed by Péter Frankl in 1979. It is also called Frankl's union-closed sets conjecture and remains open.

Conjecture 15.24. Lonely Runner Conjecture
Suppose that $n \geq 2$ runners move at distinct constant speeds around a unit circle. For each runner, there is a time at which that runner is at circular distance at least

$$\frac{1}{n}$$

from every other runner.

Remark. After subtracting the chosen runner's speed from every speed, the problem becomes one of simultaneous Diophantine approximation. It remains open in general.

Open Problem 15.25. Odd Perfect Number Problem
Determine whether there exists an odd positive integer n satisfying

$$\sigma(n) = 2n,$$

where $\sigma(n)$ is the sum of the positive divisors of n.

Remark. Euclid and Euler characterized the even perfect numbers through Mersenne primes. No odd perfect number is known, and nonexistence has not been proved.

Conjecture 15.26. Legendre Conjecture
For every positive integer n, there is a prime p satisfying

$$n^2 < p < (n+1)^2.$$

Remark. This is stronger than what follows from the prime number theorem and remains open.

Conjecture 15.27. Erdős–Straus Conjecture
For every integer $n \geq 2$, there exist positive integers x, y, z s.t.

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

Remark. The conjecture concerns Egyptian-fraction decompositions and is verified computationally for enormous ranges, but no general proof is known.

Conjecture 15.28. Beal Conjecture
If positive integers satisfy

$$A^x + B^y = C^z, \qquad x, y, z > 2,$$

then A, B, and C have a common prime factor.

Remark. Fermat's Last Theorem is the equal-exponent case with pairwise coprime bases. The Beal conjecture remains open.

Conjecture 15.29. Schanuel Conjecture
If $z_1, \ldots, z_n \in \mathbb{C}$ are linearly independent over $\mathbb{Q}$, then

$$\operatorname{trdeg}_{\mathbb{Q}} \mathbb{Q}(z_1, \ldots, z_n, e^{z_1}, \ldots, e^{z_n}) \geq n.$$

Remark. Schanuel's conjecture is a central organizing conjecture in transcendental number theory. It would imply many famous results and conjectures about algebraic independence.

Open Problem 15.30. Infinitely Many Mersenne Primes
Determine whether there are infinitely many primes of the form

$$2^p - 1,$$

where p is prime.

Remark. Every even perfect number has the Euclid–Euler form

$$2^{p-1}(2^p - 1)$$

with $2^p - 1$ prime. Thus the infinitude of even perfect numbers is equivalent to the infinitude of Mersenne primes.

Open Problem 15.31. Perfect Cuboid Problem
Determine whether there exists a rectangular box with integer edge lengths a, b, c for which all three face diagonals and the space diagonal are integers:

$$a^2 + b^2, \quad b^2 + c^2, \quad c^2 + a^2, \quad a^2 + b^2 + c^2$$

are all perfect squares.

Remark. Boxes with integer edges and integer face diagonals are known as Euler bricks. Requiring the space diagonal to be integral gives the still-open perfect cuboid problem.

Theorem 15.32. Catalan–Mihăilescu Theorem
The only solution in integers $x, y, a, b > 1$ of

$$x^a - y^b = 1$$

is

$$3^2 - 2^3 = 1.$$

Remark. Catalan posed the conjecture in 1844, and Preda Mihăilescu proved it in 2002. Thus 8 and 9 are the only consecutive positive perfect powers greater than 1.

Open Problem 15.33. Smooth Four-Dimensional Poincaré Conjecture
Determine whether every smooth closed 4-manifold that is homotopy equivalent to S^4 is diffeomorphic to S^4.

Remark. The topological 4-dimensional Poincaré conjecture was proved by Freedman. The smooth 4-dimensional case remains open, while the other dimensions are settled.

Theorem 15.34. Kakeya Conjecture in Three Dimensions
If a set $K \subseteq \mathbb{R}^3$ contains a unit line segment in every direction, then

$$\dim_{\mathrm{H}} K = 3,$$

and its Minkowski dimension is also 3.

Remark. Besicovitch showed that Kakeya sets may have Lebesgue measure 0. Hong Wang and Joshua Zahl proved the full dimension conjecture in $\mathbb{R}^3$ in 2025. The analogous conjecture remains open in dimensions $n \geq 4$.

Open Problem 15.35. Inscribed Square Problem
Determine whether every Jordan curve in the plane contains four points that are the vertices of a square.

Remark. Also called the Toeplitz square-peg problem, it is known for many regular classes of curves but remains open for arbitrary Jordan curves.

Fact 15.36. Kissing Numbers
The kissing number τ_n is the maximum number of nonoverlapping unit spheres that can touch one unit sphere in $\mathbb{R}^n$. Famous exact values include

$$\tau_2 = 6, \qquad \tau_3 = 12, \qquad \tau_4 = 24, \qquad \tau_8 = 240, \qquad \tau_{24} = 196560.$$

Exact values are unknown in most dimensions.

Open Problem 15.37. Hilbert–Smith Conjecture
Every locally compact topological group that acts faithfully and continuously on a connected manifold should be a Lie group.

Remark. The problem asks whether manifolds admit genuinely non-Lie continuous symmetry groups. It remains open in full generality.

Conjecture 15.38. Jacobian Conjecture
Let

$$F = (F_1, \dots, F_n) : \mathbb{C}^n \to \mathbb{C}^n$$

be a polynomial map. If

$$\det\left(\frac{\partial F_i}{\partial x_j}\right)$$

is a nonzero constant, then F has a polynomial inverse.

Remark. Keller posed the conjecture in 1939. It is open for every $n \geq 2$, even though many reductions are known.

Open Problem 15.39. Inverse Galois Problem
Determine whether every finite group G occurs as

$$\mathrm{Gal}(K/\mathbb{Q})$$

for some finite Galois extension K of $\mathbb{Q}$.

Remark. The problem is solved for many classes of groups but remains open in general.

Open Problem 15.40. Hilbert's Tenth Problem over $\mathbb{Q}$
Determine whether there is an algorithm that decides, for every polynomial

$$f \in \mathbb{Z}[x_1, \dots, x_n],$$

whether the equation $f = 0$ has a solution in $\mathbb{Q}^n$.

Remark. Matiyasevich's theorem gives a negative answer over $\mathbb{Z}$. The analogous decision problem over $\mathbb{Q}$ remains open.

Theorem 15.41. Classification of Finite Simple Groups
Every finite simple group is one of the following:

- *a cyclic group of prime order;*
- *an alternating group A_n with $n \geq 5$;*
- *a finite simple group of Lie type;*
- *one of the 26 sporadic simple groups.*

Remark. The proof is distributed across thousands of pages by many authors. The largest sporadic group is the Monster group.

Theorem 15.42. Restricted Burnside Problem
For fixed positive integers d and n, there are only finitely many finite d-generated groups of exponent n, up to isomorphism.

Remark. Efim Zelmanov solved the restricted Burnside problem using the theory of Lie algebras and received the 1994 Fields Medal. The unrestricted Burnside problem has a negative answer in general.

Fact 15.43. Langlands Program
The Langlands program predicts deep correspondences between arithmetic objects such as Galois representations and analytic objects such as automorphic representations. It is not one conjecture but a network of conjectures linking number theory, representation theory, and geometry.

Open Problem 15.44. Three-Dimensional Euler Singularity Problem
For the incompressible Euler equations in $\mathbb{R}^3$, determine whether smooth finite-energy initial data can develop a singularity in finite time.

Remark. The Euler equations model an ideal fluid without viscosity. Their global regularity problem is distinct from the Navier–Stokes Millennium problem.

Conjecture 15.45. Falconer Distance Conjecture
If a compact set $E \subseteq \mathbb{R}^n$ satisfies

$$\dim_{\mathrm{H}} E > \frac{n}{2},$$

then its distance set

$$\Delta(E) = \{\|\mathbf{x} - \mathbf{y}\| : \mathbf{x}, \mathbf{y} \in E\}$$

has positive Lebesgue measure.

Remark. The conjecture connects harmonic analysis, geometric measure theory, and discrete distance problems. It remains open in general.

Open Problem 15.46. Hilbert's Sixteenth Problem
The second part asks for a uniform upper bound, in terms of the degree n, on the number and arrangement of limit cycles of a planar polynomial vector field of degree n.

Remark. Even the quadratic case is not completely understood. The first part of Hilbert's sixteenth problem concerns the topology of real algebraic curves.

Theorem 15.47. Bieberbach Conjecture
If

$$f(z) = z + \sum_{n=2}^{\infty} a_n z^n$$

is analytic and injective on the unit disk, then

$$|a_n| \leq n \qquad (n \geq 2).$$

Remark. Ludwig Bieberbach conjectured the bound in 1916, and Louis de Branges proved it in 1985.

Theorem 15.48. Kadison–Singer Problem
Every pure state on the diagonal maximal abelian subalgebra of bounded operators on ℓ^2 has a unique extension to a state on all bounded operators.

Remark. Kadison and Singer posed the problem in 1959. Marcus, Spielman, and Srivastava solved it in 2013 through an equivalent finite-dimensional partition problem and the method of interlacing polynomials.

Part III

Mathematics in 2026

Chapter 16

Prizes and Congresses in 2026

This part records the mathematical events most relevant to the 2026 Science Quiz season. Its time window begins on January 1, 2025, and runs through the preparation cutoff of August 6, 2026. Later developments may be added before the competition. Unlike Part II, which collects durable facts that should remain useful in future years, this part emphasizes why a person, prize, theorem, conjecture, institution, or competition became especially important during 2025 or 2026.

The guiding association is

$$\begin{array}{lclcl} \text{event} & \longleftrightarrow & \text{person or institution} & & \\ & \longleftrightarrow & \text{mathematical achievement} & \longleftrightarrow & \text{year and status.} \end{array}$$

Fact 16.1. Status language in this part
Recent mathematics must be described with care.

- *Published* means that the work appeared in a scholarly publication.
- *Accepted* means that a journal announced acceptance, even if the final issue had not yet appeared.
- *Preprint* means that a manuscript was publicly posted but had not completed ordinary journal publication.
- *Formally verified* means that a proof assistant checked a formal proof term; this is different from ordinary peer review.
- *Scheduled* describes an officially announced future event or appointment.

Numerical evidence, an announced proof, a preprint, a formally verified argument, and a published theorem are therefore not interchangeable.

Strategy. Past Science Quiz questions frequently combine a recent event with an older mathematical fact. For each entry, remember the recent year and the shortest unmistakable mathematical anchor. Dates, locations, institutions, prize combinations, unusual records, and changes of affiliation are especially useful for multiple-choice elimination.

The chapter on mathematical prizes and congresses in Part II explains the Fields Medal, Abel Prize, and Wolf Prize. The present chapter records the season-specific results announced in 2025 and 2026.

Year	Prize or congress	Laureate(s) or location	Principal Quiz anchors
2025	Abel Prize	Masaki Kashiwara	Algebraic analysis, representation theory, D-modules, crystal bases.
2025	Breakthrough Prize in Mathematics	Dennis Gaitsgory	Geometric Langlands program in characteristic zero.
2025	Shaw Prize in Mathematical Sciences	Kenji Fukaya	Fukaya category, symplectic topology, mirror symmetry, gauge theory.
2025	ICTP–IMU Ramanujan Prize	Claudio Muñoz	Nonlinear dispersive PDE, soliton stability, multi-soliton dynamics.
2025	Basic Science Lifetime Awards	Shigefumi Mori; George Lusztig; Robert Tarjan	Minimal model program; representation theory; algorithms and data structures.
2026	Abel Prize	Gerd Faltings	Arithmetic geometry, Mordell conjecture, Lang conjectures, Diophantine geometry.
2026	International Congress of Mathematicians	Philadelphia, United States, July 23–30	Fields Medals and the principal IMU prizes; IMU General Assembly in New York City.
2026	Fields Medals	Yu Deng; John Pardon; Jacob Tsimerman; Hong Wang	PDE and kinetic theory; symplectic geometry; arithmetic geometry and o-minimality; harmonic analysis and Kakeya.
2026	IMU Abacus Medal	Shayan Oveis Gharan	Algorithms, combinatorial optimization, approximation algorithms.
2026	Carl Friedrich Gauss Prize	Yurii Nesterov	Convex optimization and accelerated first-order methods.
2026	Chern Medal Award	Graeme Segal	Topology, geometry, and mathematical physics.
2026	Leelavati Award	Hannah Fry	Public communication and popular understanding of mathematics.
2026	Breakthrough Prize in Mathematics	Frank Merle	Nonlinear evolution equations, singularity formation, stability, solitons.
2026	Shaw Prize in Mathematical Sciences	Emmanuel Candès; Camillo De Lellis	Compressed sensing, signal processing, statistics; geometric measure theory, singularities, and fluid dynamics.
2026	Clay Research Awards	Three research groups	Furstenberg and Kakeya sets; Telescope Conjecture counterexamples; Boltzmann equation from hard-sphere dynamics.
2026	Rolf Schock Prize in Mathematics	Tobias Holck Colding	Minimal surfaces and geometric flows.
2026	Maryam Mirzakhani Prize in Mathematics	Roman Bezrukavnikov	Geometric representation theory and modular representation theory.

Continued on the next page.

Year	Prize or congress	Laureate(s) or location	Principal Quiz anchors
2026	Michael and Sheila Held Prize	Irit Dinur; Subhash Khot; Guy Kindler; Dor Minzer; Muli Safra	The 2-to-2 Games Theorem, approximation hardness, Unique Games.

Fact 16.2. ICM 2026
The 2026 International Congress of Mathematicians was held in Philadelphia from July 23 to July 30. The twentieth IMU General Assembly met in New York City immediately before the Congress, on July 20–21. The opening ceremony included the Fields Medals, IMU Abacus Medal, Gauss Prize, and Chern Medal Award; the Leelavati Award was associated with the closing of the Congress.

2026 Fields Medalist	Main area	Season-specific Quiz anchors
Yu Deng	Partial differential equations and kinetic theory	Rigorous derivation of kinetic equations from many-particle dynamics; hard spheres, the Boltzmann equation, wave kinetic equations, and a concrete part of Hilbert's sixth-problem program.
John Pardon	Symplectic geometry and topology	Holomorphic curves, virtual fundamental cycles, Fukaya categories, symplectic topology, knot theory, and 3-manifolds.
Jacob Tsimerman	Arithmetic and complex algebraic geometry	O-minimality, the André–Oort conjecture for Siegel modular varieties, period maps, and the Griffiths conjecture.
Hong Wang	Harmonic analysis and geometric measure theory	Restriction theory, local smoothing, Furstenberg sets, and the three-dimensional Kakeya conjecture. She became the third woman to receive a Fields Medal.

Fact 16.3. The 2025 and 2026 Abel Prizes
The two Abel laureates in this preparation window are distinguished by sharply different anchors.

$$\begin{aligned} \text{Masaki Kashiwara} \quad &\longleftrightarrow \quad D\text{-modules and crystal bases,} \\ \text{Gerd Faltings} \quad &\longleftrightarrow \quad \text{Mordell conjecture and arithmetic geometry.} \end{aligned}$$

Faltings had already received the Fields Medal in 1986. Thus he joined the group of mathematicians who have received both the Fields Medal and the Abel Prize.

Fact 16.4. 2026 Clay Research Awards
The three recognized clusters are worth remembering as complete associations.

- Tuomas Orponen, Pablo Shmerkin, Hong Wang, and Joshua Zahl: Furstenberg sets in the plane and Kakeya sets in three-dimensional space.
- Robert Burklund, Jeremy Hahn, Ishan Levy, and Tomer Schlank: counterexamples to the Ravenel Telescope Conjecture in chromatic homotopy theory.
- Yu Deng and Zaher Hani: the long-time derivation of the Boltzmann equation from hard-sphere dynamics; Xiao Ma was a coauthor of the mathematical work.

Remark. Only officially confirmed prize results are recorded. In particular, no unannounced 2026 Wolf Prize in Mathematics laureate is inferred or supplied merely to complete the traditional prize triad.

Chapter 17

Mathematical Competitions in 2026

Competition questions differ from prize questions. The most useful chain is

$$\text{competition} \quad \longleftrightarrow \quad \text{host and year} \quad \longleftrightarrow \quad \text{team result} \quad \longleftrightarrow \quad \text{individual record.}$$

Competition	Host and scale	Republic of Korea	Notable records
IMO 2025	Sunshine Coast, Australia; 110 countries and 630 contestants	Third place with 203 points; four gold and two silver medals	Gold cutoff 35; five perfect scores; six problems over two contest days.
IMO 2026	Shanghai, China; 117 teams and 666 contestants	Sixth place with 167 points; three gold, two silver, and one honorable mention	Seven perfect scores; Hyeonjun Lee of Korea obtained 42/42.
APMO 2025	Thirty-seventh Asian Pacific Mathematical Olympiad	Kyungjun Park scored 35/35 and received gold; the next nine Korean results ranged from silver through honorable mention	Five problems, each worth seven points; official results.
APMO 2026	Thirty-eighth Asian Pacific Mathematical Olympiad	Seohyung Jo, Hyeonjun Lee, and Hyunjun Jang each scored 35/35	Results were still marked preliminary at the preparation cutoff; identical scores need not receive identical national award labels.

Fact 17.1. Republic of Korea at IMO 2025
The six Korean contestants and their scores were

$$37,\ 36,\ 36,\ 35,\ 31,\ 28.$$

Kyungjun Park, Wooju Ham, Hyeonjun Lee, and Hyewon Yoon received gold; Hyunjun Jang and Hyungjune Cho received silver. The team total was

$$37 + 36 + 36 + 35 + 31 + 28 = 203.$$

Fact 17.2. Republic of Korea at IMO 2026
Korea placed sixth with 167 points. Its aggregate problem scores were

$$42,\ 20,\ 17,\ 40,\ 37,\ 11.$$

Hyeonjun Lee was one of seven contestants worldwide to obtain a perfect score of 42.

Fact 17.3. The 2026 IMO medal design
The reverse of the 2026 IMO medal used Zhao Shuang's diagram for the Pythagorean theorem, a classical diagram associated with his commentary on the *Zhoubi Suanjing*. The medal had diameter 9 cm. This provides a direct connection between a current competition, the Pythagorean theorem, and the history of Chinese mathematics.

Remark. Human competition results belong in this chapter. Performance by AI systems on the same or similar problems is discussed separately, because an official contestant result, an external evaluation, a natural-language proof, and a machine-checked formal proof are different objects.

Chapter 18

Major Mathematical Developments in 2026

This chapter records developments that were prominent during 2025 and 2026. Complete proofs are generally far beyond the scope of Science Quiz. The intended memory unit is

standard problem or theorem + people + status + main method or field.

Development	Principal names	Status in this season	Quiz anchors
Three-dimensional Kakeya conjecture	Hong Wang; Joshua Zahl	2025 proof preprint; recognized by a 2026 Clay Research Award	Full Hausdorff and Minkowski dimension 3; harmonic analysis and geometric measure theory.
Hilbert's tenth problem over rings of integers	Levent Alpöge; Manjul Bhargava; Wei Ho; Ari Shnidman; independently Peter Koymans and Carlo Pagano	One proof accepted and published; an independent preprint	Diophantine equations, undecidability, number fields, elliptic curves.
Boltzmann equation from hard spheres	Yu Deng; Zaher Hani; Xiao Ma	Long-time derivation; recognized by a 2026 Clay Research Award	Kinetic theory, many-particle dynamics, Hilbert's sixth problem.
Modularity of ordinary abelian surfaces	George Boxer; Frank Calegari; Toby Gee; Vincent Pilloni	2025 preprint	Abelian surfaces, Siegel modular forms, Galois representations, Langlands philosophy.
Mizohata–Takeuchi conjecture counterexample	Hannah Mira Cairo	2025 preprint	Harmonic analysis, Fourier restriction, counterexample, logarithmic loss.
Compact Bonnet pairs	Alexander Bobenko; Tim Hoffmann; Andrew Sageman-Furnas	Published in 2025	Noncongruent immersed tori with the same metric and mean curvature.

Continued on the next page.

Development	Principal names	Status in this season	Quiz anchors
Unknotting number is not additive	Mark Brittenham; Susan Hermiller	Accepted in 2026	Knot theory, connected sum, explicit counterexamples.
Kissing-number lower bounds in dimensions 17–21	Henry Cohn; Anqi Li	Preprint widely publicized in 2025	Sphere packing, spherical codes, nonlattice configurations.
Furstenberg set conjecture	Tuomas Orponen; Pablo Shmerkin and collaborators	Major planar result; 2026 Clay recognition	Geometric measure theory, sets containing line segments in many directions.
Ravenel Telescope Conjecture	Robert Burklund; Jeremy Hahn; Ishan Levy; Tomer Schlank	Counterexamples recognized in 2026	Chromatic homotopy theory and stable homotopy theory.
Exacting and ultraexacting cardinals	Juan P. Aguilera; Joan Bagaria; Philipp Lücke	2025 preprint	Large cardinals, rank-into-rank embeddings, foundations of set theory.

Theorem 18.1. Three-Dimensional Kakeya Conjecture
A Kakeya set in $\mathbb{R}^3$ contains a unit line segment in every direction. The 2025 result of Hong Wang and Joshua Zahl establishes that every such set has Hausdorff dimension and Minkowski dimension equal to 3*.*

Idea. The theorem does not say that every Kakeya set has positive three-dimensional volume. It says that no Kakeya set can have dimension smaller than the ambient dimension. The main quiz association is

$$\text{Kakeya in } \mathbb{R}^3 \quad \longleftrightarrow \quad \text{Hong Wang and Joshua Zahl.}$$

Both the Hausdorff dimension and the Minkowski dimension are equal to 3.

Fact 18.2. Hilbert's tenth problem over number fields
Hilbert's tenth problem asks for an algorithm deciding whether an integer polynomial equation has an integer solution. Matiyasevich, building on Davis, Putnam, and Robinson, proved that no such general algorithm exists over $\mathbb{Z}$.

A paper by Levent Alpöge, Manjul Bhargava, Wei Ho, and Ari Shnidman was accepted and published during the 2025–2026 season and established the analogous negative answer for the ring of integers of every number field. Peter Koymans and Carlo Pagano posted an independent and more general undecidability result for infinite rings finitely generated over $\mathbb{Z}$.

Remark. The phrase "Hilbert's tenth problem over every number field" refers to the corresponding ring of algebraic integers. It is not the same as the still distinct question of Hilbert's tenth problem over the rational numbers $\mathbb{Q}$.

Fact 18.3. Boltzmann equation and Hilbert's sixth problem
Yu Deng, Zaher Hani, and Xiao Ma developed a rigorous long-time derivation of the Boltzmann equation from deterministic hard-sphere dynamics. This is a major realization of the kinetic-theory part of Hilbert's sixth-problem program.

Remark. Hilbert's sixth problem is a broad request for an axiomatic treatment of physics. The hard-sphere-to-Boltzmann result is therefore a major solution of a concrete component, not a declaration that every aspect of Hilbert's sixth problem has been completed.

Fact 18.4. Modularity beyond elliptic curves
George Boxer, Frank Calegari, Toby Gee, and Vincent Pilloni proved modularity for an important

class of ordinary abelian surfaces over $\mathbb{Q}$, under explicit local and residual hypotheses. In particular, their work proves modularity for a positive proportion of abelian surfaces in a natural ordering.

Idea. An elliptic curve is one-dimensional as an abelian variety, whereas an abelian surface has dimension 2. The association to retain is

$$\text{elliptic curves and modular forms} \quad \longrightarrow \quad \text{abelian surfaces and Siegel modular forms.}$$

This is part of the broader Langlands philosophy connecting arithmetic objects with automorphic forms and representations.

Fact 18.5. Hannah Cairo's counterexample
In 2025, Hannah Mira Cairo constructed a counterexample to the Mizohata–Takeuchi conjecture in Fourier restriction theory. Her result showed that the conjectured estimate fails in its original form and identified a logarithmic loss that cannot simply be removed.

Fact 18.6. Compact Bonnet pairs
A Bonnet pair consists of noncongruent surfaces with the same induced metric and the same mean-curvature function. In 2025, Alexander Bobenko, Tim Hoffmann, and Andrew Sageman-Furnas published the first compact examples: a pair of immersed tori.

Fact 18.7. Unknotting number is not additive
Let $u(K)$ denote the unknotting number of a knot K. The long-standing additivity expectation was

$$u(K_1 \# K_2) = u(K_1) + u(K_2).$$

Brittenham and Hermiller constructed examples for which

$$u(K_1 \# K_2) < u(K_1) + u(K_2),$$

and their paper was accepted in 2026. Thus a connected sum may require fewer crossing changes than separately unknotting the two summands.

Fact 18.8. Improved kissing configurations
Henry Cohn and Anqi Li improved the best known lower bounds for kissing numbers in dimensions 17 through 21. Their constructions used spherical codes that need not come from lattices, reinforcing the lesson that the most symmetric construction need not be the best known one.

Fact 18.9. New large-cardinal terminology
Exacting cardinals and ultraexacting cardinals were introduced in a 2025 preprint by Aguilera, Bagaria, and Lücke. They belong to the study of very strong rank-into-rank principles. The names are season-specific anchors; their relative consistency strength, rather than an elementary numerical value, is the mathematical issue.

Strategy. For a newly announced result, do not memorize only the headline "problem solved." Memorize whether the event was a proof, a counterexample, an improved bound, a construction, or a new framework. These distinctions are common sources of plausible but incorrect answer choices.

Chapter 19

Artificial Intelligence and Mathematics in 2026

The years 2025 and 2026 marked a transition from isolated demonstrations to sustained use of AI in olympiad reasoning, algorithm discovery, formal proof, and mathematical research. The correct comparison is not merely "human versus machine." One must identify the type of mathematical output and the manner in which it was checked.

Type of claim	What was produced	What the label does and does not mean
Gold-medal-level score	A score at or above the human gold-medal cutoff on a specified problem set	It is a benchmark statement, not automatically a formal proof or a new research theorem.
Natural-language proof	A human-readable written argument	It may be judged by expert graders, but it is not mechanically checked merely because a model produced it.
Formal proof	A proof term accepted by a proof assistant such as Lean	The kernel checks the formal derivation; the formal statement and its relation to the intended theorem must still be correct.
AI-assisted discovery	A construction, algorithm, conjecture, counterexample candidate, or proof idea	Human mathematical validation, literature search, attribution, and publication status remain essential.
Numerical candidate	Computational evidence suggesting a phenomenon or singularity	It is not a theorem until an appropriate rigorous argument is supplied.

Fact 19.1. Gold-medal-level performance at IMO 2025
Google DeepMind reported that an advanced version of Gemini with Deep Think solved five of the six IMO 2025 problems and received 35 out of 42 points under an official evaluation process. Since the human gold cutoff was also 35, the result met the gold-medal standard. The system worked from the natural-language problem statements and produced natural-language proofs.

Remark. The phrase "gold-medal-level" is exact but limited: an AI system was not an official contestant and did not receive an IMO medal. The statement compares its evaluated score with the human cutoff.

Fact 19.2. Two AI systems at IMO 2025
OpenAI separately reported that a general-purpose reasoning model also scored 35 out of 42 on the IMO 2025 problem set. Thus both OpenAI and Google DeepMind announced gold-medal-

level natural-language performance in 2025, using different systems. Neither system was an official contestant, and neither received an IMO medal.

Fact 19.3. AlphaProof and formal mathematics
A paper on AlphaProof appeared in *Nature* in 2025. AlphaProof combines reinforcement learning with the Lean proof assistant and produces machine-checkable formal proofs. Its well-known benchmark used the 2024 IMO problem set and reached silver-medal-level performance when combined with a geometry system.

Fact 19.4. AlphaEvolve
Google DeepMind introduced AlphaEvolve in 2025 as an evolutionary coding agent for discovering and improving algorithms. Its mathematical significance lies in searching large spaces of executable constructions and evaluating candidates automatically, rather than merely generating prose solutions.

Fact 19.5. ICPC-level algorithmic performance
In 2025, Gemini Deep Think was also evaluated on the ICPC World Finals problem set and achieved a gold-medal-level result. This is primarily an achievement in algorithm design and competitive programming, whereas the IMO emphasizes proof-based olympiad mathematics.

Fact 19.6. First Proof
In February 2026, OpenAI released proof attempts by an internal model for all ten problems in the research-level First Proof challenge. After expert feedback, OpenAI assessed at least five attempts—Problems 4, 5, 6, 9, and 10—as having a high chance of being correct, while other attempts remained under review. Its initially favored attempt for Problem 2 was later acknowledged to be incorrect. The episode is a useful reminder that a persuasive research proof attempt still requires independent mathematical checking.

Fact 19.7. AI disproof of the Erdős unit-distance conjecture
Let $u(n)$ be the largest possible number of pairs at unit distance among n points in the plane. Erdős had conjectured that

$$u(n) = n^{1+o(1)}.$$

In May 2026, OpenAI released a proof generated by an internal general-purpose model that disproved this conjecture. The construction gives, for infinitely many n and some fixed $\delta > 0$,

$$u(n) \geq n^{1+\delta}.$$

The argument uses class-field towers and other ideas from algebraic number theory, and external mathematicians checked the released proof. The result disproves the conjectured near-linear upper bound; it does not determine the exact asymptotic order of $u(n)$.

Fact 19.8. Ten AI-generated research advances
On August 1, 2026, OpenAI released manuscripts for ten results generated by an internal version of its Astra model. Humans prepared the manuscripts with the same model, after which the model formalized each argument as a Lean certificate. The announced results were:

1. new upper bounds for high-dimensional sphere packing down to the Cohn–Elkies threshold;
2. exponentially improved bounds for binary and spherical codes;
3. a construction establishing the existence of non-sofic groups;
4. a disproof of Connes's rigidity conjecture;
5. new arithmetic-circuit lower bounds for the permanent, including an arithmetic-formula lower bound of order $n^4/\log n$;

6. an exponential parallel-repetition theorem for general two-player quantum games;
7. polynomial-factor hardness of approximation for the closest vector problem;
8. a resolution of Ehrhart's volume conjecture;
9. a superexponential lower bound for multicolor triangle Ramsey numbers, resolving Erdős Problem 183;
10. results on compactness and degeneracy in extremal graph theory, resolving Erdős Problems 146 and 180.

These were freshly released manuscripts with formal certificates at the preparation cutoff. Formal verification of the stated Lean theorems and long-term independent assessment by the mathematical community are distinct stages.

Fact 19.9. Leiden Declaration on AI and Mathematics
The Leiden Declaration, released in 2026 and endorsed by the IMU, articulated principles for responsible AI-assisted mathematics. The central requirements are transparent disclosure of significant AI use, accurate attribution, openness about methods and evidence, and continued human responsibility for mathematical correctness and scholarship.

Strategy. A Science Quiz question may deliberately mix the following terms:

IMO score, formal verification, peer review, AI-assisted discovery, new theorem.

Treat each as a separate claim. A result may satisfy more than one, but none implies all the others.

Chapter 20

Mathematicians and Institutions in 2026

Recent-person questions often use a move, retirement, death, or unusual career event as the first clue, followed by an older theorem or prize. The following tables record the principal institutional changes and memorial events of the season.

Mathematician	2025–2026 event	Institution	Quiz anchors
Hong Wang	Became a permanent professor in 2025	Institut des Hautes Ètudes Scientifiques, jointly with New York University	Harmonic analysis, Kakeya, 2026 Fields Medal.
Jacob Tsimerman	Announced in July 2026 that he would begin a position at OpenAI	OpenAI; on leave from the University of Toronto	Arithmetic geometry, o-minimality, 2026 Fields Medal, AI safety.
Tim Roughgarden	Joined the faculty on July 1, 2026	Institute for Advanced Study	Algorithms, algorithmic game theory, mechanism design.
James Maynard	Appointed Regius Professor; scheduled to begin October 1, 2026	University of Oxford	Prime gaps, sieve methods, 2022 Fields Medal; successor to Andrew Wiles.
Ryan Chen; Alex Cohen; Anna Skorobogatova	Began Clay Research Fellowships in 2025	Clay Mathematics Institute appointments hosted at research institutions	Clay Research Fellowships and early-career research appointments.
Oliver Edtmair; Qiuyu Ren	Began Clay Research Fellowships in 2026	Clay Mathematics Institute appointments hosted at research institutions	Clay Research Fellowships and early-career research appointments.

Fact 20.1. Jacob Tsimerman and OpenAI
After receiving a 2026 Fields Medal, Jacob Tsimerman announced at ICM 2026 that he would begin a position at OpenAI. He went on leave from the University of Toronto to work on AI safety, while remaining a professor and faculty member there. The accurate season-specific association is therefore a Fields Medalist moving temporarily from ordinary university duties into AI-safety research, not an unqualified claim that he quit mathematics or resigned from the university.

Mathematician	Death	Principal associations
Peter D. Lax	May 16, 2025, aged 99	Partial differential equations, conservation laws, shock waves, numerical analysis, Lax equivalence theorem, Wolf Prize 1987, Abel Prize 2005.
Heisuke Hironaka	March 18, 2026, aged 94	Resolution of singularities in characteristic zero, algebraic geometry, Fields Medal 1970.
Michael O. Rabin	April 14, 2026	Finite automata, probabilistic algorithms, Rabin–Karp string search, Turing Award 1976 shared with Dana Scott.

Fact 20.2. Peter Lax
Peter Lax spent his academic career at New York University's Courant Institute. His work connected partial differential equations, numerical methods, fluid dynamics, shock waves, scattering theory, and integrable systems. The Lax equivalence theorem states, in its standard numerical analysis setting, that a consistent finite-difference approximation of a well-posed linear initial-value problem is convergent if and only if it is stable.

Fact 20.3. Heisuke Hironaka
Hironaka received the 1970 Fields Medal for proving resolution of singularities for algebraic varieties over fields of characteristic zero. The season-specific clue is his death in 2026; the durable mathematical clue is

$$\text{Hironaka} \quad \longleftrightarrow \quad \text{resolution of singularities.}$$

Fact 20.4. From Wiles to Maynard at Oxford
Andrew Wiles was the inaugural holder of Oxford's modern Regius Professorship of Mathematics. James Maynard was appointed as his successor, with the new appointment scheduled to begin on October 1, 2026. This creates a memorable association between Fermat's Last Theorem and modern work on prime gaps.

Chapter 21

Mathematics in Korea in 2026

This chapter records events whose specifically Korean institutional or biographical connection makes them useful for the POSTECH–KAIST Science War. International results already discussed above are not duplicated merely because a Korean mathematician or contestant was involved.

Year	Person or event	Korean connection	Principal Quiz anchors
2025	Sug Woo Shin received the Samsung Ho-Am Prize in Physics and Mathematics	Distinguished professor at KIAS; professor at the University of California, Berkeley	Number theory, automorphic forms, the Langlands program.
2025	Jae Kyoung Kim delivered a plenary lecture at the SIAM Annual Meeting	Chief investigator at the IBS Biomedical Mathematics Group	First Korean plenary speaker at a SIAM Annual Meeting; mathematical and computational biology.
2025	Nam-Gyu Kang received the Korean Mathematical Society DI Prize	Professor at KIAS	Probability, complex analysis, conformal field theory.
2025	Inhyeok Choi received the Sangsan Prize and joined KIAS	KIAS fellow and professor	Geometric group theory, random walks, hyperbolic groups.
2026	Sung-Jin Oh received the Samsung Ho-Am Prize in Physics and Mathematics	Professor at KIAS and associate professor at the University of California, Berkeley	Nonlinear hyperbolic PDE, general relativity, instability inside black holes.
2026	Kyeongsu Choi received the Science Prize at the POSCO TJ Park Prize	Professor at KIAS	Mean-curvature flow, Gauss-curvature flow, geometric PDE.
2026	IMU Executive Committee met in Seoul	Hosted by KIAS on March 13–14	International Mathematical Union and Korean mathematical institutions.
2025–2026	Jineon Baek and the moving sofa problem	POSTECH alumnus; June E Huh Fellow at KIAS	Gerver's sofa, optimal area approximately 2.21953166, preprint status.

Open Problem 21.1. Moving Sofa Problem
Determine the planar shape of maximum area that can be moved around a right-angled corner in a corridor of unit width.

Fact 21.2. Jineon Baek's claimed resolution
Jineon Baek posted the preprint *Optimality of Gerver's Sofa*, claiming that Gerver's construction is optimal. The proposed optimal area is approximately

$$2.21953166.$$

During the 2025–2026 preparation season, the work remained a preprint and had not completed ordinary peer-reviewed journal publication. The correct wording is therefore "a preprint claiming a proof," not an unqualified statement that journal publication had been completed.

Fact 21.3. Two consecutive Samsung Ho-Am mathematics laureates
The Physics and Mathematics laureates in the two-year window are

$$\begin{aligned} &2025: \quad \text{Sug Woo Shin} \quad \longleftrightarrow \quad \text{Langlands program,} \\ &2026: \quad \text{Sung-Jin Oh} \quad \longleftrightarrow \quad \text{nonlinear hyperbolic PDE and black holes.} \end{aligned}$$

The prize category combines physics and mathematics, so the laureate's own discipline must be identified from the citation rather than from the category name alone.

Fact 21.4. Korean competition results belong to the competition chapter
Korea's third-place finish at IMO 2025, sixth-place finish at IMO 2026, and Hyeonjun Lee's perfect 2026 score are recorded under Mathematical Competitions. They are not repeated here, preserving one primary home for each event.

Strategy. Korea-related questions are often easiest when three levels are kept separate:

$$\text{nationality or origin,} \qquad \text{current institution,} \qquad \text{institution connected to the event.}$$

A Korean or Korean-American mathematician may work abroad, hold a joint appointment, or receive a Korean prize for research performed in an international setting.

Part IV

Past Problems

Problem 2010.1. Determine whether there exist positive integers x and y satisfying

$$x^2 + y^2 = 963.$$

Solution. No such positive integers exist. By the quadratic-residue criterion modulo 4 from Mathematical Concepts, for any integer a, we have

$$(a^2 \equiv 0 \pmod 4) \vee (a^2 \equiv 1 \pmod 4).$$

Hence the sum of two integer squares can be congruent only to

$$0, \quad 1, \quad \text{or} \quad 2 \pmod 4.$$

However,

$$963 \equiv 3 \pmod 4.$$

Thus 963 cannot be written as the sum of two integer squares.

Therefore, the answer is $\boxed{\text{no}}$.

□

Problem 2010.2. Identify the mathematician who became famous for refusing the Fields Medal after solving the Poincaré conjecture.

Solution. Grigori Perelman proved the Poincaré conjecture using Ricci flow. He was awarded the Fields Medal in 2006, but declined it.

Therefore, the answer is $\boxed{\text{Grigori Perelman}}$.

□

Problem 2010.3. Determine the country of origin of John Charles Fields, after whom the Fields Medal is named.

Solution. John Charles Fields was a Canadian mathematician. The Fields Medal is named after him.

Therefore, the answer is $\boxed{\text{Canada}}$.

□

Problem 2010.4. Determine the maximum possible number of distinct eigenvalues of a 5×5 matrix.

Solution. The eigenvalues of a 5×5 matrix $\mathbf{A}$ are the roots of its characteristic polynomial

$$p_{\mathbf{A}}(\lambda) = \det(\lambda \mathbf{I}_5 - \mathbf{A}).$$

This polynomial has degree 5. Therefore it can have at most 5 distinct roots, and hence $\mathbf{A}$ can have at most 5 distinct eigenvalues.

Therefore, the answer is $\boxed{5}$.

□

Problem 2010.5. In 1742, a Prussian mathematician wrote to Euler and proposed a conjecture. Euler later stated it in the following form:

Every even integer greater than 2 can be expressed as the sum of two primes.

This is an old unsolved problem in number theory. It has been verified for very large numbers, but it has not been proved in general. Identify this conjecture.

Solution. The statement that every even integer greater than 2 can be expressed as the sum of two primes is known as Goldbach's conjecture.

Therefore, the answer is $\boxed{\text{Goldbach's conjecture}}$.

□

Problem 2010.6. Brownian motion is modeled by a stochastic process named after the American mathematician who developed its mathematical theory and later founded cybernetics. For this process, increments over disjoint time intervals are independent, and an increment over an interval of length t is Gaussian with mean 0 and variance t. Identify the mathematician.

Solution. The process described is the Wiener process, the standard mathematical model of Brownian motion. It is named after Norbert Wiener.

Hence the mathematician is $\boxed{\text{Norbert Wiener}}$.

□

Problem 2010.7. This set is a closed uncountable set and is also perfect. Although it is uncountable, it has Lebesgue measure 0. It is self-similar and is often used in the construction of fractals. It is obtained from the closed interval $[0, 1]$ by repeatedly removing the open middle third, then doing the same to the remaining intervals. It is named after the mathematician widely known as the founder of set theory. Identify this set.

Solution. The set obtained by repeatedly removing the open middle third from $[0, 1]$ is the Cantor set. It is closed, perfect, uncountable, self-similar, and has Lebesgue measure 0.

Therefore, the answer is $\boxed{\text{Cantor set}}$.

□

Problem 2010.8. Determine the year in which Terence Tao received the Fields Medal.

Solution. Terence Tao received the Fields Medal at the 2006 International Congress of Mathematicians.

Therefore, the answer is $\boxed{2006}$.

□

Problem 2011.1. Identify the international mathematical congress, held in South Korea in 2014, at which the Fields Medal is awarded.

Solution. The Fields Medal is awarded at the International Congress of Mathematicians, commonly abbreviated as ICM. The 2014 ICM was held in Seoul, South Korea.

Therefore, the answer is $\boxed{\text{International Congress of Mathematicians, or ICM}}$.

□

Problem 2011.2. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. If

$$\mathbf{AB} - \mathbf{I}_n = \mathbf{0},$$

where $\mathbf{I}_n$ is the $n \times n$ identity matrix. Determine

$$\mathbf{BA} - \mathbf{I}_n.$$

Solution. The condition

$$\mathbf{AB} - \mathbf{I}_n = \mathbf{0}$$

means that

$$\mathbf{AB} = \mathbf{I}_n.$$

Since $\mathbf{A}$ and $\mathbf{B}$ are square matrices, this implies that $\mathbf{B}$ is the inverse of $\mathbf{A}$. Hence

$$\mathbf{BA} = \mathbf{I}_n.$$

Therefore,

$$\mathbf{BA} - \mathbf{I}_n = \mathbf{I}_n - \mathbf{I}_n = \mathbf{0}.$$

Therefore, the answer is $\boxed{\mathbf{0}}$.

□

Problem 2011.3. Differentiate $(1+x)^{10}$ three times, substitute $x = 0$, and determine the resulting value.

Solution. We compute the third derivative:

$$\frac{\mathrm{d}^3}{\mathrm{d}x^3}(1+x)^{10} = 10 \cdot 9 \cdot 8(1+x)^7.$$

Substituting $x = 0$, we get

$$10 \cdot 9 \cdot 8(1+0)^7 = 720.$$

Therefore, the answer is $\boxed{720}$.

□

Problem 2011.4. Give at least two real-valued functions defined on $\mathbb{R}$ that remain unchanged after being differentiated twice. In other words, give at least two functions $f : \mathbb{R} \to \mathbb{R}$ satisfying

$$f'' = f.$$

Solution. Two simple examples are

$$f(x) = e^x \qquad \text{and} \qquad g(x) = e^{-x}.$$

Indeed,

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}e^x = e^x$$

and

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}e^{-x} = e^{-x}.$$

Thus both functions remain unchanged after being differentiated twice.

Therefore, two valid examples are $\boxed{e^x \text{ and } e^{-x}}$.

□

Problem 2011.5. Let $\mathbf{A}$ be a 5×5 matrix s.t. every entry in the ith row is equal to i. Find the sum of the five eigenvalues of $\mathbf{A}$, counted with multiplicity.

Solution. By the trace–eigenvalue identity in Mathematical Concepts, the sum of the eigenvalues of a square matrix, counted with algebraic multiplicity, is equal to $\operatorname{tr}(\mathbf{A})$.

In this matrix, the diagonal entries are

$$1, \quad 2, \quad 3, \quad 4, \quad 5.$$

Therefore,

$$\operatorname{tr}(\mathbf{A}) = 1 + 2 + 3 + 4 + 5 = 15.$$

Hence the sum of the five eigenvalues of $\mathbf{A}$, counted with multiplicity, is $\boxed{15}$.

$\square$

Problem 2011.6. A train takes 22 seconds to completely cross an 82 m bridge. When its speed is doubled, it takes 50 seconds to completely cross a 706 m bridge. Determine the length of the train.

Solution. Let L be the length of the train, and let v be its original speed in meters per second.

To completely cross a bridge, the train must travel the length of the bridge plus its own length. Hence

$$\frac{L+82}{v} = 22.$$

When the speed is doubled, the train takes 50 seconds to completely cross a 706 m bridge, so

$$\frac{L+706}{2v} = 50.$$

Equivalently,

$$L + 82 = 22v, \qquad L + 706 = 100v.$$

Subtracting the first equation from the second gives

$$624 = 78v,$$

so

$$v = 8.$$

Therefore,

$$L + 82 = 22 \cdot 8 = 176,$$

and hence

$$L = 94.$$

Thus the length of the train is $\boxed{94\,\text{m}}$.

$\square$

Problem 2011.7. Form an integer using the digits

$$1, 2, 3, 4, 6, 7, 8, 9.$$

Each of these digits must be used at least once, and digits may be repeated. Determine the smallest such integer divisible by all of

$$1, 2, 3, 4, 6, 7, 8, 9.$$

Solution. The number must be divisible by

$$\operatorname{lcm}(1, 2, 3, 4, 6, 7, 8, 9) = 504.$$

Thus it must be divisible by 7, 8, and 9.

If each digit is used exactly once, then the sum of the digits is

$$1 + 2 + 3 + 4 + 6 + 7 + 8 + 9 = 40,$$

which is not divisible by 9. Hence no 8-digit number works.

If we use exactly one extra digit d, then the digit sum becomes $40 + d$. For divisibility by 9, we need

$$40 + d \equiv 0 \pmod 9,$$

so

$$d \equiv 5 \pmod 9.$$

But the digit 5 is not allowed. Hence no 9-digit number works.

Therefore, the answer must have at least 10 digits. For a 10-digit number, we need to add two extra digits. Their sum must be congruent to 5 modulo 9. Among the allowed digits, the possible extra pairs are

$$(1,4), \qquad (2,3), \qquad (6,8), \qquad (7,7).$$

The lexicographically smallest admissible multiset is

$$1, 1, 2, 3, 4, 4, 6, 7, 8, 9.$$

With these digits, the smallest possible beginning is

$$112344.$$

It remains to arrange the last four digits $6, 7, 8, 9$ so that the whole number is divisible by 504.

Since the digit sum is already divisible by 9, it remains to check divisibility by 7 and 8. The last three digits must be divisible by 8. Among the permutations of three of the digits $6, 7, 8, 9$, the possible last three digits divisible by 8 are

$$768, \qquad 896, \qquad 968, \qquad 976.$$

Hence the only candidates with prefix 112344 are

$$1123449768, \qquad 1123447896, \qquad 1123447968, \qquad 1123448976.$$

Checking divisibility by 7, only

$$1123449768$$

is divisible by 7. Indeed,

$$1123449768 = 504 \cdot 2229067.$$

Therefore it is divisible by all of

$$1, 2, 3, 4, 6, 7, 8, 9.$$

Thus the smallest possible integer is $\boxed{1123449768}$.

□

Problem 2011.8. Determine the greatest integer less than or equal to

$$-\frac{1}{\ln\left(\frac{365}{366}\right)}.$$

Hint. Use the Taylor approximation of $\ln(1 - x)$.

Solution. We write

$$\frac{365}{366} = 1 - \frac{1}{366}.$$

Using

$$\ln(1 - x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots,$$

with $x = \dfrac{1}{366}$, we get

$$-\ln\left(\frac{365}{366}\right) = \frac{1}{366} + \frac{1}{2\cdot 366^2} + \frac{1}{3\cdot 366^3} + \cdots.$$

Hence

$$\frac{1}{366} < -\ln\left(\frac{365}{366}\right) < \frac{1}{365}.$$

Taking reciprocals gives

$$365 < -\frac{1}{\ln\left(\frac{365}{366}\right)} < 366.$$

Therefore, the greatest integer less than or equal to the given number is $\boxed{365}$.

□

Problem 2011.9. Evaluate the integral

$$\int_0^{\pi/2} \frac{1}{1+\tan^{\sqrt{2}} t}\, dt.$$

Hint. Use the substitution $x = \dfrac{\pi}{2} - t$.

Solution. Let

$$I = \int_0^{\pi/2} \frac{1}{1+\tan^{\sqrt{2}} t}\, dt.$$

Use the substitution

$$x = \frac{\pi}{2} - t.$$

Then

$$\tan t = \tan\left(\frac{\pi}{2} - x\right) = \cot x = \frac{1}{\tan x}.$$

Hence

$$I = \int_0^{\pi/2} \frac{1}{1+\cot^{\sqrt{2}} x}\, dx = \int_0^{\pi/2} \frac{\tan^{\sqrt{2}} x}{1+\tan^{\sqrt{2}} x}\, dx.$$

Adding this expression to the original expression for I, we obtain

$$2I = \int_0^{\pi/2} \left(\frac{1}{1+\tan^{\sqrt{2}} x} + \frac{\tan^{\sqrt{2}} x}{1+\tan^{\sqrt{2}} x}\right) dx = \int_0^{\pi/2} 1\, dx = \frac{\pi}{2}.$$

Therefore,

$$I = \frac{\pi}{4}.$$

Therefore, the answer is $\boxed{\dfrac{\pi}{4}}$.

□

Problem 2011.10. This French mathematician was born on December 8, 1865, and died on October 17, 1963, living to the age of 97. He is well known for his proof of the prime number theorem, which states that the number of primes less than or equal to x is asymptotically approximated by $\dfrac{x}{\ln x}$ for large x. He was also personally connected to the Dreyfus affair and became a strong supporter of Zionism.

Identify this mathematician.

Hint. The following clues are useful:

(1) The Cauchy–__________ theorem gives the radius of convergence of a power series.
(2) A __________ matrix is a square matrix whose entries are all $+1$ or -1, and such matrices are also used in error-correcting codes.

Solution. The mathematician described in the problem is Jacques Hadamard.

The first hint refers to the Cauchy–Hadamard theorem, which gives the radius of convergence of a power series. The second hint refers to a Hadamard matrix, a square matrix whose entries are all $+1$ or -1 and whose rows are mutually orthogonal.

Hadamard is also famous for proving the prime number theorem in 1896, independently of Charles Jean de la Vallée Poussin. Therefore, the answer is $\boxed{\text{Jacques Hadamard}}$.

□

Problem 2012.1. This British mathematical genius died of cyanide poisoning shortly before his 42nd birthday. A famous account says that he died after eating an apple laced with cyanide. Identify this mathematician.

Solution. The mathematician described in the problem is Alan Turing.

Turing was a British mathematician, logician, and computer scientist. He made fundamental contributions to computability theory through the concept of the Turing machine, and he also played a crucial role in codebreaking during World War II.

Turing died in 1954 from cyanide poisoning, shortly before his 42nd birthday. The story involving an apple laced with cyanide is widely known, although some details of the event remain historically uncertain.

Therefore, the answer is $\boxed{\text{Alan Turing}}$.

□

Problem 2012.2. Evaluate the infinite series

$$1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots .$$

Solution. The given series is

$$\sum_{n=1}^{\infty} \frac{1}{n^2}.$$

This is the famous Basel problem. Euler proved that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Therefore, the value of the series is $\boxed{\dfrac{\pi^2}{6}}$.

□

Problem 2012.3. Identify the mathematician who first proved the law of quadratic reciprocity and discovered more than seven proofs of it.

Solution. The law of quadratic reciprocity is one of the central theorems in elementary number theory. It describes a deep symmetry between the solvability of

$$x^2 \equiv p \pmod{q} \qquad \text{and} \qquad x^2 \equiv q \pmod{p}$$

for odd primes p and q.

Carl Friedrich Gauss gave the first complete proof of the law of quadratic reciprocity. He regarded it as one of the jewels of number theory and found several different proofs of the theorem.

Therefore, the answer is $\boxed{\text{Carl Friedrich Gauss}}$.

□

Problem 2012.4. Let $\mathbf{A}$ be a 10×10 matrix s.t. every entry in the kth row is equal to k. Find the sum of the eigenvalues of $\mathbf{A}$, counted with multiplicity.

Solution. By the trace–eigenvalue identity in Mathematical Concepts, the sum of the eigenvalues of a square matrix, counted with algebraic multiplicity, is equal to $\operatorname{tr}(\mathbf{A})$.

In this matrix, the diagonal entries are

$$1, \quad 2, \quad 3, \quad \ldots, \quad 10.$$

Therefore,

$$\operatorname{tr}(\mathbf{A}) = 1 + 2 + 3 + \cdots + 10 = \frac{10 \cdot 11}{2} = 55.$$

Hence the sum of the eigenvalues of $\mathbf{A}$, counted with multiplicity, is $\boxed{55}$.

□

Problem 2012.5. This philosopher was born as the youngest son of a wealthy Austrian steel magnate who supported many artists. He showed talent in several fields, especially mathematics and philosophy. His work had a major influence on logical positivism and on the philosophy of language, and it also influenced the humanities, social sciences, and the arts. His later philosophy is especially known for emphasizing ordinary language use.

Identify this person.

Solution. The person described in the problem is Ludwig Wittgenstein.

Wittgenstein was born into a wealthy Austrian family and studied logic, mathematics, and philosophy. His early work, especially the *Tractatus Logico-Philosophicus*, strongly influenced logical positivism and the Vienna Circle.

Later, in works such as *Philosophical Investigations*, Wittgenstein emphasized the importance of ordinary language and argued that the meaning of a word is closely connected to its use in language.

Therefore, the answer is $\boxed{\text{Ludwig Wittgenstein}}$.

□

Problem 2013.1. A complex number is called *algebraic* if it is a root of a nonzero polynomial with integer coefficients; otherwise, it is called *transcendental*. Joseph Liouville first proved the existence of transcendental numbers in 1844. Hilbert's seventh problem, one of the 23 problems presented at the 1900 International Congress of Mathematicians in Paris, also concerns transcendental numbers. It was solved independently by Aleksandr Gelfond and Theodor Schneider in the 1930s, and Alan Baker received the 1970 Fields Medal in part for major advances in this area.

Which mathematician proved in 1873 that e, the base of the natural logarithm, is transcendental? Ferdinand von Lindemann later generalized this mathematician's method and proved in 1882 that π is transcendental.

Solution. The mathematician was Charles Hermite. In 1873, Hermite proved that

$$e$$

is transcendental. Lindemann subsequently developed related ideas to prove the transcendence of π in 1882.

Therefore, the answer is $\boxed{\text{Charles Hermite}}$.

□

Problem 2013.2. Beginning around 1935, mathematicians associated with the Lwów school gathered in cafés near the university and recorded problems in a notebook. The names appearing in it include Banach, Mazur, Ulam, Kac, Zygmund, Schauder, Erdős, Dugundji, Steinhaus, and Ahlfors. Some problems carried prizes. In particular, on November 6, 1936, Stanisław Mazur offered a live goose for Problem 153, later associated with the basis problem. Per Enflo solved the problem in 1972, and Mazur presented him with the promised goose in a televised ceremony.

Determine the title by which this famous notebook is known.

Solution. The notebook is known as *The Scottish Book*, named after the Scottish Café in Lwów, where members of the Lwów school of mathematics frequently met and discussed problems.

Therefore, the answer is $\boxed{\textit{The Scottish Book}}$.

□

Problem 2013.3. Using exactly three occurrences of the digit 9, together with standard arithmetic operations, parentheses, powers, and square roots, construct each of the numbers

$$1, \qquad 6, \qquad 8.$$

Solution. One set of constructions is

$$\left(\frac{9}{9}\right)^9 = 1, \qquad \frac{9+9}{\sqrt{9}} = 6, \qquad 9 - \frac{9}{9} = 8.$$

Each expression uses exactly three occurrences of the digit 9. Alternative constructions include

$$9^{9-9} = 1, \qquad \sqrt{9 \cdot 9} - \sqrt{9} = 6, \qquad 9 - \frac{\sqrt{9}}{\sqrt{9}} = 8.$$

□

Problem 2014.1. This twentieth-century German mathematician is well known for a theorem stating that symmetries of Hamiltonian dynamical systems correspond to conservation laws. Identify this mathematician.

Solution. The theorem described in the problem is Noether's theorem. It states, roughly speaking, that continuous symmetries of a physical system correspond to conserved quantities. For example, time-translation symmetry corresponds to conservation of energy, and spatial-translation symmetry corresponds to conservation of momentum.

The mathematician associated with this theorem is Emmy Noether, one of the most important algebraists of the twentieth century.

Therefore, the answer is $\boxed{\text{Emmy Noether}}$.

□

Problem 2014.2. In the famous Chinese mathematical classic *The Nine Chapters on the Mathematical Art*, often compared with Euclid's *Elements* as a foundational mathematical classic, the value of π appears as 3. Later, Liu Hui, who wrote a commentary on *The Nine Chapters*, obtained the approximation 3.14 by refining the method of inscribed polygons. Identify the ancient Chinese mathematician who further refined this method and obtained the approximations

$$\frac{355}{113} \quad \text{and} \quad \frac{22}{7}$$

for π.

Solution. The mathematician described in the problem is Zu Chongzhi.

Liu Hui improved the approximation of π by using polygons with many sides. Zu Chongzhi later obtained the very accurate bounds

$$3.1415926 < \pi < 3.1415927,$$

and gave two famous rational approximations:

$$\frac{22}{7} \qquad \text{and} \qquad \frac{355}{113}.$$

The approximation $\frac{355}{113}$ is especially accurate and is traditionally known as the Milü, or "accurate ratio."

Therefore, the answer is $\boxed{\text{Zu Chongzhi}}$.

□

Problem 2014.3. Consider the 2014×2014 matrix whose diagonal entries are all 0 and whose off-diagonal entries are all 1. Find all eigenvalues of this matrix.

Solution. Let $\mathbf{A}$ be the given matrix, and let $\mathbf{J}_{2014}$ be the 2014×2014 matrix whose entries are all 1. Then

$$\mathbf{A} = \mathbf{J}_{2014} - \mathbf{I}_{2014}.$$

The matrix $\mathbf{J}_{2014}$ has eigenvalue 2014 for the vector

$$\mathbf{1} = (1, 1, \ldots, 1)^{\mathsf{T}},$$

since

$$\mathbf{J}_{2014}\mathbf{1} = 2014\mathbf{1}.$$

Therefore,

$$\mathbf{A}\mathbf{1} = (\mathbf{J}_{2014} - \mathbf{I}_{2014})\mathbf{1} = 2014\mathbf{1} - \mathbf{1} = 2013\mathbf{1}.$$

Hence 2013 is an eigenvalue of $\mathbf{A}$.

Now take any vector $\mathbf{v} = (v_1, \ldots, v_{2014})^{\mathsf{T}}$ s.t.

$$v_1 + \cdots + v_{2014} = 0.$$

Then $\mathbf{J}_{2014}\mathbf{v} = \mathbf{0}$, so

$$\mathbf{A}\mathbf{v} = (\mathbf{J}_{2014} - \mathbf{I}_{2014})\mathbf{v} = -\mathbf{v}.$$

Thus -1 is an eigenvalue. The subspace of vectors whose coordinates sum to 0 has dimension 2013, so the multiplicity of -1 is 2013.

Therefore, the eigenvalues are

$$\boxed{2013 \text{ with multiplicity } 1, \qquad -1 \text{ with multiplicity } 2013}.$$

□

Problem 2014.4. Let $m \geq 3$ and $n \geq 3$. Suppose that

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \subseteq \mathbb{R}^m, \qquad \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\} \subseteq \mathbb{R}^n.$$

Assume that

$$V = \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$$

is a 2-dimensional subspace of $\mathbb{R}^m$, and that

$$W = \operatorname{span}(\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3)$$

is a 3-dimensional subspace of $\mathbb{R}^n$. Determine $\operatorname{rk}(\mathbf{A})$ for the $m \times n$ matrix

$$\mathbf{A} = \mathbf{v}_1\mathbf{w}_1^\mathsf{T} + \mathbf{v}_2\mathbf{w}_2^\mathsf{T} + \mathbf{v}_3\mathbf{w}_3^\mathsf{T}.$$

Solution. Let

$$\mathbf{P} = \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{pmatrix}, \qquad \mathbf{Q} = \begin{pmatrix} \mathbf{w}_1 & \mathbf{w}_2 & \mathbf{w}_3 \end{pmatrix}.$$

Then

$$\mathbf{A} = \mathbf{P}\mathbf{Q}^\mathsf{T}.$$

Since $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ span a 2-dimensional subspace, we have

$$\operatorname{rk}(\mathbf{P}) = 2.$$

Since $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ span a 3-dimensional subspace, they are linearly independent, and hence

$$\operatorname{rk}(\mathbf{Q}) = 3.$$

Therefore, the linear map $\mathbf{Q}^\mathsf{T} : \mathbb{R}^n \to \mathbb{R}^3$ is surjective.

It follows that the image of $\mathbf{A} = \mathbf{P}\mathbf{Q}^\mathsf{T}$ is

$$\operatorname{im}(\mathbf{A}) = \mathbf{P}(\operatorname{im}(\mathbf{Q}^\mathsf{T})) = \mathbf{P}(\mathbb{R}^3) = \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3).$$

This space has dimension 2. Hence

$$\operatorname{rk}(\mathbf{A}) = 2.$$

Therefore, the answer is $\boxed{2}$.

□

Problem 2015.1. Read the following passage and fill in the blank with the title of the mathematical classic, either in English or in Korean.

> For 2000 years, __________ was not only the core of mathematical education but also at the heart of Western culture. The most glowing tributes to __________ do not, in fact, come from mathematicians, but from philosophers, politicians, and others.
>
> Here are some examples:
>
> (1) He studied and nearly mastered the six books of __________ since he was a member of Congress. He regrets his want of education, and does what he can to supply the want.
>
> Abraham Lincoln, *Short Autobiography*
>
> (2) He studied __________ until he could demonstrate with ease all the propositions in the six books.
>
> *Herndon's Life of Lincoln*
>
> (3) At the age of eleven, I began __________. This was one of the great events of my life, as dazzling as first love. I had not imagined there was anything so delicious in the world.
>
> Bertrand Russell, *Autobiography*, Vol. 1

Solution. The passage is referring to Euclid's *Elements*, one of the most influential works in the history of mathematics.

For centuries, Euclid's *Elements* served as a standard model of rigorous mathematical reasoning and was central to mathematical education. The quotations from Abraham Lincoln and Bertrand Russell emphasize how deeply the book influenced not only mathematicians, but also philosophers, politicians, and intellectual culture more broadly.

Therefore, the answer is $\boxed{\textit{The Elements}, \text{ or Euclid's } \textit{Elements}}$.

□

Problem 2015.2. Let $\mathbf{A}$ be an invertible square matrix. Suppose that, for a suitable permutation matrix $\mathbf{P}$, Gaussian elimination gives an LU-decomposition

$$\mathbf{PA} = \mathbf{LU}.$$

It may also happen that another permutation matrix $\widetilde{\mathbf{P}}$ gives another LU-decomposition

$$\widetilde{\mathbf{P}}\mathbf{A} = \widetilde{\mathbf{L}}\widetilde{\mathbf{U}}.$$

Show that the sets of absolute values of the pivots can be different. In other words, show by example that

$$\{|u_{kk}| \mid 1 \leq k \leq n\} \neq \{|\widetilde{u}_{kk}| \mid 1 \leq k \leq n\}.$$

Hint. Look for a counterexample in the 2×2 case.

Solution. Consider the invertible matrix

$$\mathbf{A} = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{3} & 1 \end{pmatrix}.$$

With no row exchange, we have the LU-decomposition

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ \frac{1}{3} & 1 \end{pmatrix} \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{5}{6} \end{pmatrix}.$$

Hence the absolute values of the pivots are

$$\left\{1, \frac{5}{6}\right\}.$$

Now let $\widetilde{\mathbf{P}}$ be the permutation matrix that swaps the two rows. Then

$$\widetilde{\mathbf{P}}\mathbf{A} = \begin{pmatrix} \frac{1}{3} & 1 \\ 1 & \frac{1}{2} \end{pmatrix}.$$

This matrix has the LU-decomposition

$$\widetilde{\mathbf{P}}\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{3} & 1 \\ 0 & -\frac{5}{2} \end{pmatrix}.$$

Hence the absolute values of the pivots are

$$\left\{\frac{1}{3}, \frac{5}{2}\right\}.$$

Therefore,

$$\left\{1, \frac{5}{6}\right\} \neq \left\{\frac{1}{3}, \frac{5}{2}\right\}.$$

This shows that the set of absolute values of the pivots can depend on the chosen row permutation. □

Problem 2015.3. In $\mathbb{R}^2$, the so-called Mercedes-Benz system is defined by

$$\{\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2\}, \qquad \mathbf{v}_k = \begin{pmatrix} \cos\left(\frac{2k\pi}{3}\right) \\ \sin\left(\frac{2k\pi}{3}\right) \end{pmatrix} \qquad (k = 0, 1, 2).$$

Find vectors $\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2 \in \mathbb{R}^3$ s.t. they have the same length, are mutually orthogonal, have positive z-coordinates, and satisfy

$$\mathbf{v}_k = P_{xy}(\mathbf{w}_k) \qquad (k = 0, 1, 2),$$

where P_{xy} denotes the projection onto the xy-plane. Here $\mathbb{R}^2$ is regarded as the subspace of $\mathbb{R}^3$ with z-coordinate equal to 0.

Solution. Since $P_{xy}(\mathbf{w}_k) = \mathbf{v}_k$, each $\mathbf{w}_k$ must have the form

$$\mathbf{w}_k = \begin{pmatrix} \cos\left(\frac{2k\pi}{3}\right) \\ \sin\left(\frac{2k\pi}{3}\right) \\ c \end{pmatrix}$$

for some positive constant c.

For $i \neq j$, the angle between $\mathbf{v}_i$ and $\mathbf{v}_j$ is $120°$, so

$$\mathbf{v}_i \cdot \mathbf{v}_j = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}.$$

Therefore,

$$\mathbf{w}_i \cdot \mathbf{w}_j = \mathbf{v}_i \cdot \mathbf{v}_j + c^2 = -\frac{1}{2} + c^2.$$

To make the vectors mutually orthogonal, we need

$$-\frac{1}{2} + c^2 = 0,$$

so

$$c = \frac{1}{\sqrt{2}}.$$

Hence one possible choice is

$$\boxed{\mathbf{w}_k = \begin{pmatrix} \cos\left(\frac{2k\pi}{3}\right) \\ \sin\left(\frac{2k\pi}{3}\right) \\ \frac{1}{\sqrt{2}} \end{pmatrix} \qquad (k = 0, 1, 2)}.$$

These vectors all have squared length

$$1 + \frac{1}{2} = \frac{3}{2},$$

have positive z-coordinates, project to the given vectors $\mathbf{v}_k$, and are mutually orthogonal. □

Problem 2015.4. This French mathematician was born in 1973 and received the Fields Medal in 2010 at the age of 36. On his website, he described himself as “Mathématicien, Professeur de l'Université de Lyon, Directeur de l'Institut Henri Poincaré." He is also famous for his stylish clothing.

Together with Clément Mouhot, he completed a theorem that played a central role in his Fields Medal-winning work. He later wrote a book about the process of developing this theorem, and the title of the book is also one of his nicknames.

Write the name of this mathematician, using the correct French spelling, and the title of the book.

Solution. The mathematician described in the problem is Cédric Villani.

Villani received the Fields Medal in 2010 for his work on nonlinear Landau damping and the Boltzmann equation. He is also widely recognized for his distinctive style, often including a cravat and a spider brooch.

The book mentioned in the problem is *Théorème vivant*, which may be translated as *Living Theorem*. It describes the process through which Villani and Clément Mouhot developed the theorem that contributed to Villani's Fields Medal-winning work.

Therefore, the answer is $\boxed{\text{Cédric Villani, } \textit{Théorème vivant}}$.

□

Problem 2015.5. On the front side of the Fields Medal, there is a profile of Archimedes together with the phrase by Marcus Manilius:

"TRANSIRE SUUM PECTUS MUNDOQUE POTIRI."

On the back side, there is another inscription together with the two solid figures that were engraved on Archimedes' tomb.

Identify the two solid figures and determine their volume ratio.

Solution. The two solid figures are a sphere and its circumscribed cylinder.

Suppose the sphere has radius R. Then the volume of the sphere is

$$V_S = \frac{4}{3}\pi R^3.$$

The circumscribed cylinder has radius R and height $2R$, so its volume is

$$V_C = \pi R^2 \cdot 2R = 2\pi R^3.$$

Therefore,

$$V_S : V_C = \frac{4}{3}\pi R^3 : 2\pi R^3 = 2 : 3.$$

Thus the two figures are a sphere and its circumscribed cylinder, and the volume ratio is $\boxed{2 : 3}$.

□

Problem 2015.6. Let $f : \mathbb{R} \to \mathbb{R}$ be a periodic function with period 2π. For $x \in (0, 2\pi]$, define

$$f(x) = \lceil \max(3, |\pi - x|) - 3 \rceil,$$

and extend f periodically, so that

$$f(x + 2\pi) = f(x)$$

for all $x \in \mathbb{R}$. Determine whether the following limit converges:

$$\lim_{n\to\infty} f(n).$$

Hint. Use $3.14 < \pi < 3.145$.

Solution. For $x \in (0, 2\pi]$, the function f takes only the values 0 and 1. More precisely,

$$f(x) = 1$$

when

$$0 < x < \pi - 3 \qquad \text{or} \qquad \pi + 3 < x \le 2\pi,$$

and

$$f(x) = 0$$

when

$$\pi - 3 \le x \le \pi + 3.$$

Thus f is a 2π-periodic function which is 1 near multiples of 2π, and 0 on the large middle interval.

Since $1/(2\pi)$ is irrational, the fractional parts

$$\left\{ \frac{n}{2\pi} \right\}, \qquad n = 1, 2, 3, \ldots,$$

are dense in $[0, 1]$. Equivalently, the representatives of the integers modulo 2π are dense in $(0, 2\pi]$. Hence there are infinitely many positive integers n whose representatives modulo 2π lie in

$$(0, \pi - 3) \cup (\pi + 3, 2\pi],$$

and for these integers we have

$$f(n) = 1.$$

There are also infinitely many positive integers n whose residues modulo 2π lie in

$$(\pi - 3, \pi + 3),$$

and for these integers we have

$$f(n) = 0.$$

The sequence $f(n)$ has a subsequence equal to 1 and another subsequence equal to 0, so it does not converge as $n \to \infty$.

Therefore, the answer is $\boxed{\text{the limit does not exist}}$.

□

Problem 2015.7. This person is believed to have lived approximately from 624 BCE to 546 BCE. He is often regarded as one of the earliest figures worthy of being called a mathematician. He introduced the method of proof to geometric facts that had been known empirically through land surveying, helping to lay the foundations of geometry that were later systematized in Euclid's *Elements*.

The theorems attributed to him include:

(1) Vertical angles are equal.
(2) The base angles of an isosceles triangle are equal.
(3) Triangle congruence criteria such as side-angle-side and angle-side-angle.
(4) An angle inscribed in a semicircle is a right angle.

Identify this person.

Solution. The person described in the problem is Thales of Miletus.

Thales is traditionally regarded as one of the earliest Greek mathematicians. He is especially important because he is associated not merely with observing geometric facts, but with proving them. Several elementary geometric results are attributed to him, including the theorem that an angle inscribed in a semicircle is a right angle.

Therefore, the answer is $\boxed{\text{Thales of Miletus}}$.

□

Problem 2015.8. Determine which of the following mathematicians was born earliest.

(1) Henri Poincaré
(2) Karl Weierstrass
(3) Julius Wilhelm Richard Dedekind
(4) Andrei Andreyevich Markov
(5) Georg Cantor

Solution. We compare their years of birth:

Karl Weierstrass	1815,
Julius Wilhelm Richard Dedekind	1831,
Georg Cantor	1845,
Henri Poincaré	1854,
Andrei Andreyevich Markov	1856.

Among these mathematicians, the one born earliest is Karl Weierstrass.

Therefore, the answer is $\boxed{\text{Karl Weierstrass}}$.

□

Problem 2015.9. The first International Congress of Mathematicians at which the Fields Medal was awarded was held in 1936. Determine the country in which it was held.

Solution. The Fields Medal was first awarded at the 1936 International Congress of Mathematicians. The 1936 ICM was held in Oslo, Norway.

Therefore, the answer is $\boxed{\text{Norway}}$.

□

Problem 2015.10. Find the sum of the distinct prime numbers appearing in the prime factorization of 15!.

Solution. The prime numbers appearing in the prime factorization of 15! are exactly the primes less than or equal to 15:

$$2, \quad 3, \quad 5, \quad 7, \quad 11, \quad 13.$$

Therefore, their sum is

$$2 + 3 + 5 + 7 + 11 + 13 = 41.$$

Therefore, the answer is $\boxed{41}$.

□

Problem 2015.11. In three-dimensional space, suppose that the lengths of the orthogonal projections of a line segment ℓ onto the yz-plane, the zx-plane, and the xy-plane are 5, 5, and 6, respectively. Determine the length of ℓ.

Solution. Let the displacement vector of the line segment be

$$\mathbf{v} = (a, b, c).$$

Then the length of ℓ is

$$\|\mathbf{v}\| = \sqrt{a^2 + b^2 + c^2}.$$

The projection of $\mathbf{v}$ onto the yz-plane has length

$$\sqrt{b^2 + c^2} = 5.$$

Similarly, the projections onto the zx-plane and the xy-plane give

$$\sqrt{c^2 + a^2} = 5, \qquad \sqrt{a^2 + b^2} = 6.$$

Squaring and adding these three equations, we get

$$(b^2 + c^2) + (c^2 + a^2) + (a^2 + b^2) = 5^2 + 5^2 + 6^2.$$

Hence

$$2(a^2 + b^2 + c^2) = 25 + 25 + 36 = 86,$$

so

$$a^2 + b^2 + c^2 = 43.$$

Therefore, the length of ℓ is

$$\|\mathbf{v}\| = \sqrt{43}.$$

Therefore, the answer is $\boxed{\sqrt{43}}$.

□

Problem 2015.12. In three-dimensional space, consider the matrix obtained by rotating space by 30° around the line passing through the origin and the point $(1, 1, 2)$. Determine the number of real eigenvalues of this matrix.

Solution. A rotation in $\mathbb{R}^3$ around an axis fixes every vector lying on the rotation axis. Therefore, the direction vector

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

is an eigenvector with eigenvalue 1.

On the plane perpendicular to the rotation axis, the matrix acts as a 30° rotation. A nontrivial rotation in a two-dimensional plane has no real eigenvalues unless the rotation angle is 0° or 180°. Since the angle here is 30°, the two remaining eigenvalues are non-real complex conjugates.

Hence the only real eigenvalue is 1. Therefore, the number of real eigenvalues is $\boxed{1}$.

□

Problem 2015.13. A right circular cone has base radius 3 and height 6. Determine the maximum possible volume of a cylinder inscribed in this cone.

Solution. Let x be the radius of the inscribed cylinder, and let h be its height. By similarity of triangles in the cone, the radius of the horizontal cross-section decreases linearly from 3 to 0 as the height increases from 0 to 6. Hence

$$x = 3 - \frac{h}{2}.$$

Equivalently,

$$h = 6 - 2x.$$

Therefore, the volume of the cylinder is

$$V(x) = \pi x^2 h = \pi x^2(6 - 2x) = 6\pi x^2 - 2\pi x^3,$$

where $0 \le x \le 3$.

Differentiating, we get

$$V'(x) = 12\pi x - 6\pi x^2 = 6\pi x(2 - x).$$

Thus the critical point in the interval is

$$x = 2.$$

At this point,

$$h = 6 - 2 \cdot 2 = 2,$$

so the maximum volume is

$$V(2) = \pi \cdot 2^2 \cdot 2 = 8\pi.$$

Therefore, the maximum possible volume is $\boxed{8\pi}$.

□

Problem 2015.14. Suppose that a polynomial equation of degree 7 with real coefficients has three non-real complex roots whose absolute values are all different. Determine the number of real roots of the equation.

Solution. For a polynomial with real coefficients, every non-real complex root appears together with its complex conjugate.

Suppose the three given non-real roots are

$$\alpha_1, \quad \alpha_2, \quad \alpha_3,$$

and their absolute values are all different. Then their complex conjugates

$$\overline{\alpha_1}, \quad \overline{\alpha_2}, \quad \overline{\alpha_3}$$

are also roots. Since

$$|\alpha_i| = |\overline{\alpha_i}|,$$

and the three given roots have distinct absolute values, these conjugate roots give three additional non-real roots.

Thus the polynomial has at least 6 non-real roots. Since the degree is 7, there is exactly one remaining root, and it must be real.

Therefore, the equation has $\boxed{1}$ real root.

□

Problem 2015.15. This mathematician was born in 1882 and died in 1935. When she was being considered for a professorship at the University of Göttingen, there was strong opposition. In response, Hilbert famously argued, "I do not see that the sex of the candidate is an argument against her admission as a Privatdozent. After all, we are a university, not a bathhouse."

She made remarkable contributions to several areas of mathematics, especially modern algebra, and much of twentieth-century abstract algebra was deeply influenced by her work. Identify this mathematician.

Solution. The mathematician described in the problem is Emmy Noether.

Noether made fundamental contributions to modern algebra, especially through her work on rings, ideals, and modules. She is also famous for Noether's theorem, which connects symmetries and conservation laws in physics.

The biographical details in the problem, including her connection with Göttingen and Hilbert's defense of her academic appointment, point to Emmy Noether.

Therefore, the answer is $\boxed{\text{Emmy Noether}}$.

$\square$

Problem 2015.16. Determine the number of distinct terms in the expansion of

$$(x_1 + x_2 + \cdots + x_7)^4.$$

Solution. Each term in the expansion has the form

$$x_1^{k_1} x_2^{k_2} \cdots x_7^{k_7},$$

where

$$k_1 + k_2 + \cdots + k_7 = 4$$

and each k_i is a nonnegative integer.

Therefore, the number of distinct terms is the number of nonnegative integer solutions to

$$k_1 + k_2 + \cdots + k_7 = 4.$$

By the stars and bars formula, this number is

$$\binom{4+7-1}{7-1} = \binom{10}{6} = 210.$$

Therefore, the answer is $\boxed{210}$.

$\square$

Problem 2015.17. Let

$$\mathbf{v} = \mathbf{w} = (1, 2, 3, 4)$$

be row vectors. Consider the 4×4 matrix obtained as the product

$$\mathbf{v}^\mathsf{T}\mathbf{w}.$$

Find the sum and the product of the eigenvalues of this matrix.

Solution. The matrix is

$$\mathbf{v}^\mathsf{T}\mathbf{w} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \end{pmatrix}.$$

By the trace–eigenvalue identity in Mathematical Concepts, the sum of the eigenvalues of a square matrix, counted with algebraic multiplicity, is equal to its trace. Therefore,

$$\sum_{i=1}^{4} \lambda_i = \operatorname{tr}(\mathbf{v}^\mathsf{T}\mathbf{w}) = 1^2 + 2^2 + 3^2 + 4^2 = 30.$$

The product of the eigenvalues is equal to the determinant. Since $\mathbf{v}^\mathsf{T}\mathbf{w}$ is an outer product of two vectors, we have $\operatorname{rk}(\mathbf{v}^\mathsf{T}\mathbf{w}) = 1$. In particular, as a 4×4 matrix, it is not invertible, so

$$\det(\mathbf{v}^\mathsf{T}\mathbf{w}) = 0.$$

Hence the product of the eigenvalues is

$$\prod_{i=1}^{4} \lambda_i = 0.$$

Therefore, the sum and product of the eigenvalues are $\boxed{30 \text{ and } 0}$, respectively.

□

Problem 2015.18. Evaluate the infinite series

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!}.$$

Hint. Use

$$\frac{e^x - 1}{x} = \sum_{n=0}^{\infty} \frac{x^n}{(n+1)!},$$

then differentiate and substitute $x = 1$.

Solution. We use a simple telescoping identity:

$$\frac{n}{(n+1)!} = \frac{1}{n!} - \frac{1}{(n+1)!}.$$

Therefore,

$$\sum_{n=1}^{N} \frac{n}{(n+1)!} = \sum_{n=1}^{N} \left(\frac{1}{n!} - \frac{1}{(n+1)!}\right) = 1 - \frac{1}{(N+1)!}.$$

Taking $N \to \infty$, we obtain

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!} = 1.$$

Therefore, the answer is $\boxed{1}$.

□

Problem 2017.1. Determine all possible positive integers that can occur as the number of faces of a regular polyhedron, that is, a Platonic solid.

Solution. There are exactly five Platonic solids:

Solid	Number of faces
Tetrahedron	4
Cube	6
Octahedron	8
Dodecahedron	12
Icosahedron	20

Therefore, the possible numbers of faces of a regular polyhedron are

$$\boxed{4,\ 6,\ 8,\ 12,\ 20}.$$

□

Problem 2017.2. Let (F_n) be the Fibonacci sequence defined by

$$F_0 = 0, \qquad F_1 = 1, \qquad F_{n+1} = F_n + F_{n-1} \quad (n \geq 1).$$

Determine

$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n}.$$

Solution. By the linear-recurrence method developed in Mathematical Concepts, Binet's formula is

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \qquad \psi = \frac{1-\sqrt{5}}{2}.$$

Since $|\psi/\varphi| < 1$,

$$\frac{F_{n+1}}{F_n} = \varphi \, \frac{1-(\psi/\varphi)^{n+1}}{1-(\psi/\varphi)^n} \longrightarrow \varphi.$$

Therefore,

$$\boxed{\frac{1+\sqrt{5}}{2}}.$$

$\square$

Problem 2017.3. For a positive integer n, define the $n \times n$ matrix $\mathbf{C}$ by

$$c_{ij} = \binom{i+j-2}{i-1} \qquad (1 \le i, j \le n).$$

Now define another $n \times n$ matrix $\mathbf{B}$ by

$$b_{ij} = \begin{cases} c_{ij}, & (i,j) \ne (n,n), \\ c_{ij} + 2016, & (i,j) = (n,n). \end{cases}$$

Determine $\det(\mathbf{B})$.

Solution. We first compute the determinant of $\mathbf{C}$. By Vandermonde's identity,

$$\binom{i+j-2}{i-1} = \sum_{k=1}^{\min(i,j)} \binom{i-1}{k-1}\binom{j-1}{k-1}.$$

Hence

$$\mathbf{C} = \mathbf{L}\mathbf{L}^{\mathsf{T}},$$

where

$$l_{ik} = \begin{cases} \binom{i-1}{k-1}, & k \le i, \\ 0, & k > i. \end{cases}$$

The matrix $\mathbf{L}$ is lower triangular and all of its diagonal entries are 1. Therefore,

$$\det(\mathbf{L}) = 1,$$

and so

$$\det(\mathbf{C}) = \det(\mathbf{L}\mathbf{L}^{\mathsf{T}}) = (\det(\mathbf{L}))^2 = 1.$$

The matrix $\mathbf{B}$ differs from $\mathbf{C}$ only in the (n,n)-entry. Since the determinant is linear in each entry, we have

$$\det(\mathbf{B}) = \det(\mathbf{C}) + 2016 \cdot M_{nn},$$

where M_{nn} is the (n,n)-minor of $\mathbf{C}$.

This minor is the determinant of the upper-left $(n-1)\times(n-1)$ submatrix of $\mathbf{C}$, which has the same form as $\mathbf{C}$ with size $n-1$. Therefore,

$$M_{nn} = 1.$$

Hence

$$\det(\mathbf{B}) = 1 + 2016 \cdot 1 = 2017.$$

Therefore, the answer is $\boxed{2017}$.

□

Problem 2017.4. In 2017, this prize was awarded to the French mathematician Yves Meyer. The Norwegian Academy of Science and Letters announced on March 21 that Meyer would receive the prize, worth six million Norwegian kroner, in recognition of his decisive contributions to the development of wavelet theory, which is used in areas ranging from gravitational wave detection to movie file compression.

Identify this prize, which is often called the Nobel Prize of mathematics.

Solution. The prize described in the problem is the Abel Prize.

The Abel Prize is awarded by the Norwegian Academy of Science and Letters and is often described as one of the most prestigious prizes in mathematics. In 2017, it was awarded to Yves Meyer for his fundamental contributions to the mathematical theory of wavelets.

Therefore, the answer is $\boxed{\text{Abel Prize}}$.

□

Problem 2017.5. Express the integral

$$\int_0^1 x^m(1-x)^n \, \mathrm{d}x$$

in terms of a binomial coefficient closely related to the positive integers m and n.

Solution. This integral is a beta integral. For positive integers m and n, we have

$$\int_0^1 x^m(1-x)^n \, \mathrm{d}x = \frac{m!n!}{(m+n+1)!}.$$

This is the factorial form of the answer.

To express it using a binomial coefficient, recall that

$$\binom{m+n}{m} = \frac{(m+n)!}{m!n!}.$$

Hence

$$\frac{m!n!}{(m+n+1)!} = \frac{1}{m+n+1}\cdot\frac{m!n!}{(m+n)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}.$$

Therefore,

$$\int_0^1 x^m(1-x)^n \, \mathrm{d}x = \frac{m!n!}{(m+n+1)!} = \frac{1}{(m+n+1)\binom{m+n}{m}}.$$

Therefore, the answer is $\boxed{\dfrac{m!n!}{(m+n+1)!} = \dfrac{1}{(m+n+1)\binom{m+n}{m}}}$.

□

Problem 2018.1. The following people are this year's Fields Medalists. Excluding choice (5), identify the second-oldest medalist among them.

(1) Caucher Birkar
(2) Alessio Figalli
(3) Peter Scholze
(4) Akshay Venkatesh
(5) The person who stole Birkar's medal

Solution. The 2018 Fields Medalists were

(1)	Caucher Birkar	born in 1978,
(2)	Alessio Figalli	born in 1984,
(3)	Peter Scholze	born in 1987,
(4)	Akshay Venkatesh	born in 1981.

Among these four, the second oldest is Akshay Venkatesh. Therefore, the answer is

$$\boxed{(4)}.$$

Choice (5) is a joke referring to the incident in which Caucher Birkar's Fields Medal was stolen shortly after the 2018 Fields Medal ceremony. Birkar later received a replacement medal.

□

Problem 2018.2. According to William Thurston's geometrization conjecture, a 3-dimensional manifold can be cut along spheres and tori so that each resulting piece has one of n possible geometric structures. Determine n.

Solution. The 3-dimensional geometric structures appearing in Thurston's geometrization conjecture are the eight Thurston geometries:

$$S^3, \quad \mathrm{E}^3, \quad \mathbb{H}^3, \quad S^2 \times \mathbb{R}, \quad \mathbb{H}^2 \times \mathbb{R}, \quad \widetilde{\mathrm{SL}}_2(\mathbb{R}), \quad \mathrm{Nil}, \quad \mathrm{Sol}.$$

Hence $n = 8$, so the answer is

$$\boxed{8}.$$

□

Problem 2018.3. Find the sum of the following infinite series:

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \cdots .$$

Solution. The given series is one half of the alternating harmonic series:

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{1}{2}\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\right).$$

Recall the Taylor expansion

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots .$$

Substituting $x = 1$, we obtain

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots .$$

Therefore,

$$\frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = \frac{\ln 2}{2}.$$

Therefore, the answer is $\boxed{\dfrac{\ln 2}{2}}$.

$\square$

Problem 2018.4. According to the information used in the original 2018 Science Quiz, determine which of the following five Fields Medalists did not have a Korean doctoral student at that time.

(1) William Thurston
(2) Maxim Kontsevich
(3) Grigory Margulis
(4) Terence Tao
(5) Efim Zelmanov

Solution. The original answer key identifies the exceptional mathematician among the five listed Fields Medalists as

$$\boxed{\text{(2) Maxim Kontsevich}}.$$

$\square$

Problem 2019.1. Consider the 2019×2019 matrix whose diagonal entries are all 0 and whose off-diagonal entries are all 1. Find all eigenvalues of this matrix.

Solution. Let $\mathbf{A}$ be the given matrix, and let $\mathbf{J}_{2019}$ be the 2019×2019 matrix whose entries are all 1. Then

$$\mathbf{A} = \mathbf{J}_{2019} - \mathbf{I}_{2019}.$$

First, consider the vector

$$\mathbf{1} = (1, 1, \ldots, 1)^{\mathsf{T}}.$$

Since

$$\mathbf{J}_{2019}\mathbf{1} = 2019\mathbf{1},$$

we get

$$\mathbf{A}\mathbf{1} = (\mathbf{J}_{2019} - \mathbf{I}_{2019})\mathbf{1} = 2019\mathbf{1} - \mathbf{1} = 2018\mathbf{1}.$$

Hence 2018 is an eigenvalue.

Now take any vector $\mathbf{v} = (v_1, \ldots, v_{2019})^{\mathsf{T}}$ satisfying

$$v_1 + \cdots + v_{2019} = 0.$$

Then

$$\mathbf{J}_{2019}\mathbf{v} = \mathbf{0},$$

and therefore

$$\mathbf{A}\mathbf{v} = (\mathbf{J}_{2019} - \mathbf{I}_{2019})\mathbf{v} = -\mathbf{v}.$$

Thus -1 is an eigenvalue. The subspace of vectors whose coordinates sum to 0 has dimension 2018, so the multiplicity of -1 is 2018.

Therefore, the eigenvalues are

$$\boxed{2018 \text{ with multiplicity } 1, \qquad -1 \text{ with multiplicity } 2018}.$$

$\square$

Problem 2019.2. Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{8}{1 + 4x^2}\, \mathrm{d}x.$$

Solution. Let
$$u = 2x.$$
Then
$$\mathrm{d}x = \frac{1}{2}\,\mathrm{d}u.$$
Therefore,
$$\int_{-\infty}^{\infty} \frac{8}{1+4x^2}\,\mathrm{d}x = \int_{-\infty}^{\infty} \frac{8}{1+u^2}\cdot\frac{1}{2}\,\mathrm{d}u = 4\int_{-\infty}^{\infty} \frac{1}{1+u^2}\,\mathrm{d}u.$$
Since
$$\int_{-\infty}^{\infty} \frac{1}{1+u^2}\,\mathrm{d}u = \pi,$$
we obtain
$$\int_{-\infty}^{\infty} \frac{8}{1+4x^2}\,\mathrm{d}x = 4\pi.$$

Therefore, the answer is $\boxed{4\pi}$.

□

Problem 2019.3. Compute the value of $\pi + e$ up to five decimal places.

Solution. Using the approximations
$$\pi \approx 3.14159265, \qquad e \approx 2.71828183,$$
we get
$$\pi + e \approx 3.14159265 + 2.71828183 = 5.85987448.$$
Therefore, up to five decimal places,
$$\pi + e \approx 5.85987.$$

Therefore, the answer is $\boxed{5.85987}$.

□

Problem 2020.1. The International Congress of Mathematicians, abbreviated as ICM, is a congress for mathematicians from around the world. It is organized every four years by the International Mathematical Union and may be thought of as a kind of Olympic Games for mathematicians. The 2014 ICM was held in South Korea, and the 2018 ICM was held in Brazil. As of 2019, determine the country in which the 2022 ICM was scheduled to be held.

Solution. ICM stands for the International Congress of Mathematicians. It is the main international congress in mathematics and is organized every four years by the International Mathematical Union.

The 2014 ICM was held in Seoul, South Korea, and the 2018 ICM was held in Rio de Janeiro, Brazil. At the reference date specified in the problem, the 2022 ICM was scheduled to be held in Saint Petersburg, Russia. Therefore, the answer is $\boxed{\text{Russia}}$.

The congress was later changed to a virtual event, with the 2022 awards ceremony held in Helsinki, Finland. The answer above therefore records the planned host country at the date specified in the original problem.

□

Problem 2020.2. A fair six-sided die has the numbers $-2, 0, 2, 3, 4, 5$ written on its faces. The die is rolled 10 times, and X is defined to be the product of the 10 numbers obtained. Determine $\mathbb{E}[X]$.

Solution. Let Y be the number obtained from one roll of the die. Since all six faces are equally likely,

$$\mathbb{E}[Y] = \frac{-2+0+2+3+4+5}{6} = 2.$$

Now let $Y_1, Y_2, \ldots, Y_{10}$ be the numbers obtained from the 10 rolls. Then

$$X = Y_1 Y_2 \cdots Y_{10}.$$

Since the rolls are independent, the random variables $Y_1, Y_2, \ldots, Y_{10}$ are independent. Therefore,

$$\mathbb{E}[X] = \mathbb{E}[Y_1 Y_2 \cdots Y_{10}] = \mathbb{E}[Y_1]\mathbb{E}[Y_2] \cdots \mathbb{E}[Y_{10}].$$

All the Y_i have the same distribution as Y, so

$$\mathbb{E}[X] = (\mathbb{E}[Y])^{10} = 2^{10} = 1024.$$

Therefore, the answer is $\boxed{1024}$.

□

Problem 2020.3. At the time of the original 2020 Science Quiz, which of the following Fields Medalists was identified as not serving as a professor?

(1) Cédric Villani
(2) Elon Lindenstrauss
(3) Peter Scholze
(4) Martin Hairer
(5) Terence Tao

Solution. The original answer key identifies Cédric Villani. At the time relevant to the question, he was serving in French politics; the item is therefore interpreted according to its original temporal context rather than as a statement about the mathematicians' present positions.

Therefore, the answer is $\boxed{\text{(1) Cédric Villani}}$.

□

Problem 2020.4. In a three-dimensional gravitational field, when an object moves from a point $\mathbf{a}$ to another point $\mathbf{b}$, the change in its potential energy is independent of the path taken from $\mathbf{a}$ to $\mathbf{b}$. Determine which of the following mathematical facts is least directly related to this physical phenomenon.

(1) On a simply connected domain, a continuously differentiable curl-free vector field is the gradient of a potential function.
(2) Stokes' theorem
(3) The Laplacian of $\dfrac{1}{\|\mathbf{x}\|}$ is a delta distribution, up to a constant factor.
(4) The curl of a gradient is identically zero.

Solution. Path independence means that the gravitational force field is conservative. If $\mathbf{F} = -\nabla U$, then

$$\Delta U = -\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot \mathrm{d}\mathbf{r}$$

depends only on the endpoints. The conservative-field criterion from Mathematical Concepts connects this statement with choice (1), while

$$\nabla \times \nabla U = \mathbf{0}$$

gives choice (4). Stokes' theorem relates zero curl to vanishing circulation around closed curves, so choice (2) is also directly tied to path independence.

The distributional identity

$$\Delta\left(\frac{1}{\|\mathbf{x}\|}\right) = -4\pi\delta_0$$

is fundamental in Newtonian gravitation and Poisson's equation, but it is not the direct reason that line integrals of a conservative field are path-independent. Therefore, the intended answer is

$$\boxed{(3)}.$$

□

Problem 2020.5. Determine which of the following mathematics-related academic institutions is a degree-granting school.

(1) Institut des Hautes Études Scientifiques, IHES
(2) École normale supérieure
(3) Mathematical Sciences Research Institute, MSRI
(4) Institute for Advanced Study, IAS
(5) Collège de France

Solution. École normale supérieure is one of France's most prestigious higher education institutions. It operates educational programs and is naturally regarded as a degree-granting school.

By contrast, IHES, MSRI, and IAS are research institutes rather than degree-granting schools. Collège de France is also a famous academic institution that offers public lectures, but it does not grant degrees.

Therefore, the answer is $\boxed{\text{(2) École normale supérieure}}$.

□

Problem 2021.1. Determine which of A and B is larger, where

$$A = 1 + \frac{1}{1 + \frac{1}{1 - \frac{1}{1+4}}}, \qquad B = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1+3}}}.$$

Solution. First, we compute A:

$$A = 1 + \frac{1}{1 + \frac{1}{1-\frac{1}{5}}} = 1 + \frac{1}{1 + \frac{1}{\frac{4}{5}}} = 1 + \frac{1}{1 + \frac{5}{4}} = 1 + \frac{1}{\frac{9}{4}} = 1 + \frac{4}{9} = \frac{13}{9}.$$

Next, we compute B:

$$B = 1 + \frac{1}{1 + \frac{1}{1+\frac{1}{4}}} = 1 + \frac{1}{1 + \frac{1}{\frac{5}{4}}} = 1 + \frac{1}{1 + \frac{4}{5}} = 1 + \frac{1}{\frac{9}{5}} = 1 + \frac{5}{9} = \frac{14}{9}.$$

Since

$$\frac{13}{9} < \frac{14}{9},$$

we have $A < B$. Therefore, the answer is $\boxed{B}$.

□

Problem 2021.2. Let

$$A(n) = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^{n+1}.$$

Determine the integer closest to $A(7)$.

Solution. By Binet's formula,

$$F_8 = \frac{\varphi^8 - \psi^8}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \qquad \psi = \frac{1-\sqrt{5}}{2}.$$

Hence

$$A(7) = \frac{\varphi^8}{\sqrt{5}} = F_8 + \frac{\psi^8}{\sqrt{5}}.$$

Since $|\psi| < 1$,

$$0 < \frac{|\psi|^8}{\sqrt{5}} < \frac{1}{2}.$$

Thus the nearest integer is $F_8 = 21$, and therefore

$$\boxed{21}.$$

□

Problem 2021.3. Choose all groups among the following that are not finitely generated:

(1) The fundamental group of an open unit disk with finitely many punctures.
(2) $\mathrm{SL}_2(\mathbb{Z})$, the group of 2×2 integer matrices with determinant 1.
(3) $\mathbb{C}^\times$, the group of nonzero complex numbers under multiplication.
(4) The Monster group, the largest sporadic finite simple group.

Solution. First, consider an open unit disk with finitely many punctures. If the removed points are $p_1, \ldots, p_n$, then the fundamental group is isomorphic to the free group on n generators:

$$\pi_1 \left(D \setminus \{p_1, \ldots, p_n\} \right) \cong F_n.$$

Hence the group in choice (1) is finitely generated.

The group $\mathrm{SL}_2(\mathbb{Z})$ is also finitely generated. For example, it is generated by the two matrices

$$\mathbf{S} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \mathbf{T} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Thus choice (2) is finitely generated.

On the other hand, $\mathbb{C}^\times$ is an uncountable abelian group. Every finitely generated group is countable, so $\mathbb{C}^\times$ cannot be finitely generated. Therefore, choice (3) is not finitely generated.

Finally, the Monster group is a finite group. Every finite group is finitely generated, since one may simply take all of its elements as generators. Hence choice (4) is finitely generated.

Therefore, the only group in the list that is not finitely generated is $\boxed{(3)\ \mathbb{C}^\times}$.

□

Problem 2021.4. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}.$$

Find all eigenvalues of $\mathbf{A}$ together with their multiplicities. Also determine whether $\mathbf{A}$ is diagonalizable.

Solution. We can write $\mathbf{A}$ as

$$\mathbf{A} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}.$$

Thus $\mathrm{rk}(\mathbf{A}) = 1$.

First, let

$$\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Then

$$\mathbf{A}\mathbf{v} = \begin{pmatrix} 6 \\ 6 \\ 6 \end{pmatrix} = 6\mathbf{v}.$$

Hence 6 is an eigenvalue of $\mathbf{A}$.

Since $\mathrm{rk}(\mathbf{A}) = 1$, its nullity is 2. Therefore, the eigenspace corresponding to the eigenvalue 0 has dimension 2. Indeed,

$$\mathbf{A} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 2y + 3z \\ x + 2y + 3z \\ x + 2y + 3z \end{pmatrix},$$

so

$$E_0 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y + 3z = 0 \right\}.$$

Thus the geometric multiplicity of 0 is 2.

Now

$$\mathrm{tr}(\mathbf{A}) = 1 + 2 + 3 = 6.$$

Since the sum of the eigenvalues, counted with algebraic multiplicity, is equal to $\mathrm{tr}(\mathbf{A})$, the eigenvalues are

$$6, \quad 0, \quad 0.$$

Hence the algebraic multiplicities are

$$6 : 1, \qquad 0 : 2.$$

Finally, the eigenspace for 6 has dimension 1, and the eigenspace for 0 has dimension 2. The total dimension of the eigenspaces is therefore

$$1 + 2 = 3.$$

Hence there exists a basis of $\mathbb{R}^3$ consisting of eigenvectors of $\mathbf{A}$. Therefore, $\mathbf{A}$ is diagonalizable.

Thus the eigenvalues are $\boxed{6, 0, 0}$, with algebraic multiplicities 1 and 2, respectively, and $\boxed{\mathbf{A} \text{ is diagonalizable}}$.

□

Problem 2022.1. Determine the largest of the following numbers.

(1) e^{π}

(2) π^{e}

(3) 7π
(4) $\binom{7}{2}$
(5) $4!$

Solution. We compare the values directly:

$$e^\pi \approx 23.1407, \qquad \pi^e \approx 22.4592, \qquad 7\pi \approx 21.9911.$$

Also,

$$\binom{7}{2} = \frac{7 \cdot 6}{2} = 21, \qquad 4! = 24.$$

Therefore, the largest number among the choices is $\boxed{(5)\ 4!}$.

□

Problem 2022.2. Three people store a jointly owned treasure in a safe and want to lock it. They want no single person to be able to open the safe alone, but any two or more people should always be able to open it together. This can be done by putting n locks on the safe, making m copies of the key for each lock, and distributing the keys appropriately among the three people. Find n and m so that the number of locks n is minimized.

Solution. Let the three people be A, B, and C.

Since any two people should be able to open the safe, the key to each lock must be given to at least two people. If a certain lock had its key given to only one person, then the other two people would not be able to open that lock together.

Thus, for each lock, exactly one of the three people should not receive the key. On the other hand, no single person should be able to open the safe alone. Therefore, each person must be missing the key to at least one lock.

Hence we need at least three locks: one lock whose key is not given to A, one lock whose key is not given to B, and one lock whose key is not given to C. This lower bound is achievable by distributing the keys as follows:

	A	B	C
L_A	X	O	O
L_B	O	X	O
L_C	O	O	X

That is, the key to L_A is given to B and C, the key to L_B is given to A and C, and the key to L_C is given to A and B.

Then each person is unable to open one of the locks, so no one can open the safe alone. However, any two people together can open all three locks. Therefore, the minimum is achieved by $\boxed{n = 3,\ m = 2}$.

□

Problem 2022.3. Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{3}{(x-2)^2 + 7}\, dx.$$

Solution. Let

$$u = x - 2.$$

Then the limits of integration remain $-\infty$ and ∞, so

$$\int_{-\infty}^{\infty} \frac{3}{(x-2)^2 + 7}\, dx = 3 \int_{-\infty}^{\infty} \frac{1}{u^2 + 7}\, du.$$

We use the standard formula
$$\int_{-\infty}^{\infty} \frac{1}{u^2+a^2}\,\mathrm{d}u = \frac{\pi}{a}.$$
Here $a = \sqrt{7}$, and therefore
$$3\int_{-\infty}^{\infty} \frac{1}{u^2+7}\,\mathrm{d}u = 3 \cdot \frac{\pi}{\sqrt{7}} = \frac{3\pi}{\sqrt{7}}.$$
Therefore, the answer is $\boxed{\dfrac{3\pi}{\sqrt{7}}}$.

□

Problem 2022.4. A couple has two children.

(a) Given that at least one child is a daughter, determine the probability that both children are daughters.

(b) Given that a specified child is a daughter named Youngmi, determine the probability that the other child is also a daughter.

Solution. Let G denote a girl and B denote a boy. If we distinguish the two children by birth order, then the possible gender combinations are
$$GG, \quad GB, \quad BG, \quad BB.$$
For part (a), the condition that one child is a daughter means that at least one child is a girl. Thus the possible cases are
$$GG, \quad GB, \quad BG.$$
Among these three cases, only GG has the property that the other child is also a daughter. Therefore,
$$\mathbb{P}(\text{both are daughters} \mid \text{at least one is a daughter}) = \frac{1}{3}.$$
For part (b), the information that one child is a daughter named Youngmi is more specific. In the intended interpretation of the problem, this information identifies one particular child. Then the gender of the other child is still equally likely to be G or B. Hence
$$\mathbb{P}(\text{the other child is a daughter}) = \frac{1}{2}.$$
Therefore, the answers are $\boxed{\text{(a) } \frac{1}{3}, \text{ (b) } \frac{1}{2}}$.

□

Problem 2022.5. Determine which of the following conjectures was not solved by Professor June Huh.

(1) Read's conjecture

(2) Brylawski's conjecture

(3) Poincaré conjecture

(4) Rota's conjecture

(5) Okounkov's conjecture

Solution. Professor June Huh is known for connecting combinatorics with algebraic geometry and using these ideas to solve several major conjectures in combinatorics. His work is related to conjectures such as Read's conjecture, Brylawski's conjecture, Rota's conjecture, and Okounkov's conjecture.

On the other hand, the Poincaré conjecture is a famous problem in 3-dimensional topology. It was proved by Grigori Perelman using Ricci flow, not by June Huh.

Therefore, the conjecture in the list that was not solved by June Huh is

$$\boxed{\text{(3) Poincaré conjecture}}.$$

□

Problem 2022.6. This mathematician, often called the father of linear programming, is famous for solving difficult problems while he was a doctoral student. One day, while studying at UC Berkeley, he arrived late to a class. The statistics professor Jerzy Neyman had written two famous unsolved problems on the blackboard, but the student thought they were homework problems. He took them home, solved them, and submitted complete solutions a few days later, thinking only that he had turned in the homework late. Identify this mathematician.

Solution. The mathematician described in the problem is George Dantzig.

George Dantzig is known as one of the founders of linear programming and as the creator of the simplex method. For this reason, he is often called the father of linear programming.

The anecdote in the problem is also a famous story about Dantzig. While he was a doctoral student at UC Berkeley, he arrived late to Jerzy Neyman's statistics class and copied down two problems written on the blackboard, believing that they were homework problems. He later submitted solutions, not realizing that the problems were actually open problems in statistics.

Therefore, the answer is $\boxed{\text{George Dantzig}}$.

□

Problem 2022.7. Two fair coins are each tossed three times, with all six tosses independent. Let X and Y be the numbers of heads obtained from the two coins, respectively. Determine $\mathbb{P}(X = Y)$.

Solution. We have

$$X, Y \sim \mathrm{Bin}\left(3, \frac{1}{2}\right),$$

and X and Y are independent. Therefore,

$$\begin{aligned}
\mathbb{P}(X = Y) &= \sum_{k=0}^{3} \mathbb{P}(X = k)\mathbb{P}(Y = k) \\
&= \frac{1}{64} \sum_{k=0}^{3} \binom{3}{k}^2 \\
&= \frac{1 + 9 + 9 + 1}{64} = \boxed{\frac{5}{16}}.
\end{aligned}$$

□

Problem 2023.1. Determine the year in which the Fields Medal, received by Professor June Huh, was first awarded.

Solution. The Fields Medal is one of the most prestigious awards in mathematics, and it is awarded by the International Mathematical Union. Professor June Huh received the Fields Medal in 2022.

The Fields Medal was first awarded in 1936. The first two recipients were Lars Ahlfors and Jesse Douglas.

Therefore, the answer is $\boxed{1936}$.

$\square$

Problem 2023.2. Consider the 6×6 square grid with vertices $(0,0)$, $(6,0)$, $(0,6)$, and $(6,6)$. A path starts at $(0,0)$ and ends at $(6,6)$, moving only one unit to the right or one unit upward at each step. Determine the number of such paths that never go above the line $x = y$. Equivalently, determine the number of such paths that stay in the region

$$\{(x,y) \in \mathbb{R}^2 \mid y \leq x\}.$$

Solution. To go from $(0,0)$ to $(6,6)$, a path must consist of 6 right steps and 6 upward steps. Without the condition $y \leq x$, the total number of such paths is

$$\binom{12}{6}.$$

We now count the paths that do go above the line $y = x$. By the reflection principle stated in Mathematical Concepts, the number of paths from $(0,0)$ to $(6,6)$ that at some point satisfy $y > x$ is

$$\binom{12}{5}.$$

Therefore, the number of paths that always stay in the region $y \leq x$ is

$$\binom{12}{6} - \binom{12}{5}.$$

Computing this gives

$$\binom{12}{6} - \binom{12}{5} = 924 - 792 = 132.$$

Therefore, the answer is $\boxed{132}$.

$\square$

Problem 2023.3. A point on the surface of the Earth has the following property: starting from that point, one moves 5 km south, then 5 km east, and then 5 km north, and returns to the original point. Determine the number of such points on the surface of the Earth.

Assume that the surface of the Earth is a perfect two-dimensional sphere, and that the North Pole and the South Pole are antipodal points. At the North Pole, the directions north, east, and west are not defined, and at the South Pole, the directions south, east, and west are not defined. Starting points for which one of these undefined directions occurs during the movement are excluded.

Solution. The North Pole is one such point. If we start at the North Pole, move 5 km south, then move 5 km east along a latitude circle, and finally move 5 km north, we return to the North Pole.

There are also many such points near the South Pole. Let Q be the point reached after moving 5 km south from the starting point. Suppose that the latitude circle passing through Q has circumference

$$\frac{5}{n}\,\text{km}$$

for some positive integer n. Then moving 5 km east from Q goes exactly n full times around that latitude circle and returns to Q. After that, moving 5 km north returns to the original starting point.

Near the South Pole, the circumferences of latitude circles approach 0. Hence, for every positive integer n, there is a latitude circle near the South Pole whose circumference is exactly $\frac{5}{n}$ km. For each such latitude circle, every point lying 5 km north of it gives a valid starting point.

Therefore, there are not just one but infinitely many such points on the Earth's surface. In fact, this construction gives entire latitude circles of valid starting points, so there are uncountably many such points. In particular, the number of such points is

$$\boxed{\text{infinitely many}}.$$

□

Problem 2023.4. Let $f : (-1, 1) \to \mathbb{R}$ be a continuous function. For each $\lambda > 0$, define

$$f_\lambda(x) := \lambda f\left(\frac{x}{\lambda}\right),$$

where f_λ is defined for $x \in (-\lambda, \lambda)$.

Suppose that for every sequence $\lambda_i \to \infty$, the functions f_{λ_i} converge locally uniformly to 0 on $\mathbb{R}$. Determine whether the following statement is true or false:

$$(f \text{ is differentiable at } 0) \wedge (f'(0) = 0).$$

Solution. We prove that the statement is true.

First, evaluate at $x = 0$. For every sequence $\lambda_i \to \infty$, we have

$$f_{\lambda_i}(0) = \lambda_i f(0).$$

Since f_{λ_i} converges locally uniformly to 0, in particular $f_{\lambda_i}(0) \to 0$ at the fixed point $x = 0$. Hence

$$\lambda_i f(0) \to 0.$$

This is possible for all sequences $\lambda_i \to \infty$ only if

$$f(0) = 0.$$

Now we show that $f'(0) = 0$. Let $h_i \to 0$ be any sequence with $h_i \neq 0$. For all sufficiently large i, we have $h_i \in (-1, 1)$. Define

$$\lambda_i = \frac{1}{|h_i|}.$$

Then $\lambda_i \to \infty$, and

$$h_i = \frac{\operatorname{sgn}(h_i)}{\lambda_i}.$$

By the assumption, $f_{\lambda_i} \to 0$ locally uniformly on $\mathbb{R}$. In particular, this convergence is uniform on the compact set $\{-1, 1\}$. Therefore,

$$f_{\lambda_i}(\operatorname{sgn}(h_i)) \to 0.$$

But

$$f_{\lambda_i}(\operatorname{sgn}(h_i)) = \lambda_i f\left(\frac{\operatorname{sgn}(h_i)}{\lambda_i}\right) = \lambda_i f(h_i).$$

Hence
$$\lambda_i f(h_i) \to 0.$$
Since $f(0) = 0$, we get
$$\left|\frac{f(h_i) - f(0)}{h_i}\right| = \left|\frac{f(h_i)}{h_i}\right| = \lambda_i |f(h_i)| \to 0.$$
Thus
$$\frac{f(h_i) - f(0)}{h_i} \to 0$$
for every sequence $h_i \to 0$ with $h_i \neq 0$. Therefore, f is differentiable at 0 and
$$f'(0) = 0.$$

Therefore, the answer is $\boxed{\text{True}}$.

□

Problem 2024.1. Determine the number of occurrences of the digit 0 when all natural numbers from 100 to 999 are written consecutively.

Solution. Every natural number from 100 to 999 is a three-digit number.

In a three-digit number, the hundreds digit cannot be 0. Therefore, the digit 0 can appear only in the tens digit or the units digit.

First, count the numbers whose tens digit is 0. The hundreds digit can be one of $1, 2, \ldots, 9$, and the units digit can be one of $0, 1, \ldots, 9$. Hence there are
$$9 \cdot 10 = 90$$
such numbers.

Next, count the numbers whose units digit is 0. The hundreds digit can be one of $1, 2, \ldots, 9$, and the tens digit can be one of $0, 1, \ldots, 9$. Hence there are again
$$9 \cdot 10 = 90$$
such numbers.

Therefore, the total number of appearances of the digit 0 is
$$90 + 90 = 180.$$
Therefore, the answer is $\boxed{180}$.

□

Problem 2024.2. Determine which of the following people was born latest.

(1) John von Neumann
(2) Alan Turing
(3) John Nash
(4) Pierre de Fermat

Solution. We compare their years of birth:

(1) John von Neumann: born in 1903,
(2) Alan Turing: born in 1912,
(3) John Nash: born in 1928,
(4) Pierre de Fermat: born in 1607.

Among these four, the person born the latest is John Nash.

Therefore, the answer is $\boxed{\text{(3) John Nash}}$.

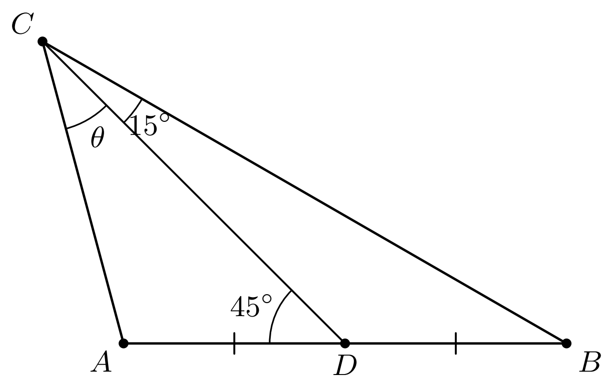
□

Problem 2024.3. In triangle ABC, the point D is the midpoint of $\overline{AB}$. Suppose that

$$\angle BCD = 15^\circ \qquad \text{and} \qquad \angle ADC = 45^\circ.$$

Determine

$$\theta = \angle ACD.$$



Solution. Since A, D, B are collinear,

$$\angle BDC = 180^\circ - \angle ADC = 135^\circ.$$

Therefore, in $\triangle BCD$,

$$\angle CBD = 180^\circ - 135^\circ - 15^\circ = 30^\circ.$$

By the Sine Rule,

$$\frac{DB}{CD} = \frac{\sin 15^\circ}{\sin 30^\circ} = 2\sin 15^\circ.$$

Let $\theta = \angle ACD$. In $\triangle ACD$,

$$\angle CAD = 180^\circ - 45^\circ - \theta = 135^\circ - \theta.$$

Again by the Sine Rule,

$$\frac{AD}{CD} = \frac{\sin\theta}{\sin(135^\circ - \theta)}.$$

Since $AD = DB$, we obtain

$$\frac{\sin\theta}{\sin(135^\circ - \theta)} = 2\sin 15^\circ.$$

Expanding the denominator gives

$$\sin\theta = \sqrt{2}\sin 15^\circ(\sin\theta + \cos\theta).$$

Since

$$\sqrt{2}\sin 15^\circ = \frac{\sqrt{3}-1}{2},$$

it follows that

$$\tan\theta = \frac{1}{\sqrt{3}}.$$

As θ is an interior angle of the triangle, we conclude that

$$\boxed{\theta = 30^\circ}.$$

$\square$

Problem 2024.4. Let C be a convex subset of $\mathbb{R}^3$. Here convex means that for any two points $\mathbf{p}, \mathbf{q} \in C$, the line segment joining $\mathbf{p}$ and $\mathbf{q}$ is also contained in C.

Suppose that the following two facts are known:

(1) The intersection of C with the xy-plane is

$$C \cap \{z = 0\} = \{(x, y, 0) \mid x^2 + 4y^2 \leq 1\}.$$

(2) The set C contains the z-axis.

Determine the volume of C in the region $-1 \leq z \leq 1$, or determine that the given information is insufficient.

(1) This cannot be determined.
(2) π
(3) 2π
(4) 4π

Solution. Let

$$E = \{(x, y) \in \mathbb{R}^2 \mid x^2 + 4y^2 \leq 1\}.$$

By condition (1),

$$C \cap \{z = 0\} = E \times \{0\}.$$

The ellipse E has semiaxes 1 and $\frac{1}{2}$, so its area is

$$\operatorname{area}(E) = \pi \cdot 1 \cdot \frac{1}{2} = \frac{\pi}{2}.$$

For each height z, define the horizontal cross-section

$$C_z = \{(x, y) \in \mathbb{R}^2 \mid (x, y, z) \in C\}.$$

We claim that C_z has the same area as E for every z.

First, we show that $C_z \subseteq E$. Suppose that $(x, y, z) \in C$ and

$$x^2 + 4y^2 > 1.$$

Since C contains the z-axis, the point $(0, 0, t)$ belongs to C for every $t \in \mathbb{R}$. By convexity, the line segment joining (x, y, z) and $(0, 0, t)$ is contained in C.

For any $\alpha \in (0, 1)$, we may choose

$$t = -\frac{\alpha z}{1 - \alpha}$$

so that this segment intersects the xy-plane at the point

$$(\alpha x, \alpha y, 0).$$

Since α can be chosen sufficiently close to 1, we may arrange that

$$(\alpha x)^2 + 4(\alpha y)^2 > 1.$$

This would give a point of $C \cap \{z = 0\}$ lying outside $E \times \{0\}$, contradicting condition (1). Therefore,

$$C_z \subseteq E$$

for every z.

Conversely, we show that the interior of E is contained in every C_z. Take (x, y) with

$$x^2 + 4y^2 < 1.$$

Choose $\alpha \in (0, 1)$ sufficiently close to 1 so that

$$\left(\frac{x}{\alpha}\right)^2 + 4\left(\frac{y}{\alpha}\right)^2 \leq 1.$$

Then

$$\left(\frac{x}{\alpha}, \frac{y}{\alpha}, 0\right) \in C.$$

Also, since C contains the z-axis,

$$\left(0, 0, \frac{z}{1-\alpha}\right) \in C.$$

By convexity, their convex combination also belongs to C, and

$$\alpha\left(\frac{x}{\alpha}, \frac{y}{\alpha}, 0\right) + (1-\alpha)\left(0, 0, \frac{z}{1-\alpha}\right) = (x, y, z).$$

Hence $(x, y) \in C_z$.

Thus, for every z,

$$\operatorname{int}(E) \subseteq C_z \subseteq E.$$

Since the boundary of the ellipse has area 0, we have

$$\operatorname{area}(C_z) = \operatorname{area}(E) = \frac{\pi}{2}$$

for every z. Therefore, the volume of C in the region $-1 \leq z \leq 1$ is

$$\int_{-1}^{1} \operatorname{area}(C_z)\,\mathrm{d}z = \int_{-1}^{1} \frac{\pi}{2}\,\mathrm{d}z = \pi.$$

Therefore, the answer is $\boxed{(2)\ \pi}$.

□

Problem 2025.1. A fair coin is tossed repeatedly until either two heads in a row or two tails in a row appear. Determine the expected number of tosses before the process stops.

Solution. On the first toss, it is impossible to have two consecutive equal outcomes. After the first toss, the process stops exactly when the next toss gives the same result as the previous toss.

Since the coin is fair, from the second toss onward we have

$$\mathbb{P}(\text{the new toss matches the previous toss}) = \frac{1}{2}.$$

Therefore, after the first toss, the number of additional tosses needed until the process stops follows a geometric distribution with success probability $\frac{1}{2}$. Its expected value is

$$\frac{1}{1/2} = 2.$$

Since the first toss is always made, the expected total number of tosses is

$$1 + 2 = 3.$$

Therefore, the answer is $\boxed{3}$.

□

Problem 2025.2. The nineteenth-century mathematicians Cauchy and Weierstrass established rigorous definitions of limits. Identify the first mathematician to prove that the sum of the reciprocals of the squares of all positive integers is

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

(1) David Hilbert
(2) Georg Cantor
(3) Bernhard Riemann
(4) Leonhard Euler

Solution. The problem of finding the exact value of the infinite series

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

is known as the Basel problem.

This problem had been known since the seventeenth century, and it was Leonhard Euler who first found and proved the famous identity

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Therefore, the answer is $\boxed{\text{(4) Leonhard Euler}}$.

□

Problem 2025.3. Find the equation of the smallest sphere tangent to all four planes

$$x = 0, \qquad y = 0, \qquad z = 0, \qquad x + y + z = 3.$$

Solution. If a sphere of radius $r > 0$ is tangent to the three coordinate planes, its center has the form

$$\mathbf{c} = (\sigma_1 r, \sigma_2 r, \sigma_3 r), \qquad \sigma_1, \sigma_2, \sigma_3 \in \{-1, 1\}.$$

Put $k = \sigma_1 + \sigma_2 + \sigma_3$, so

$$k \in \{-3, -1, 1, 3\}.$$

Tangency to $x + y + z = 3$ requires

$$\frac{|kr - 3|}{\sqrt{3}} = r.$$

For $k = 3$, the positive solutions are

$$r = \frac{3}{3 + \sqrt{3}} \qquad \text{and} \qquad r = \frac{3}{3 - \sqrt{3}}.$$

For $k = 1$, the only positive solution is

$$r = \frac{3}{1+\sqrt{3}},$$

and for $k = -1$, the only positive solution is

$$r = \frac{3}{\sqrt{3}-1}.$$

The case $k = -3$ has no positive solution. Therefore the smallest radius is

$$r = \frac{3}{3+\sqrt{3}} = \frac{3-\sqrt{3}}{2},$$

attained when $\sigma_1 = \sigma_2 = \sigma_3 = 1$. Hence the center is

$$\mathbf{c} = \left(\frac{3-\sqrt{3}}{2}, \frac{3-\sqrt{3}}{2}, \frac{3-\sqrt{3}}{2}\right),$$

and the required sphere is

$$\boxed{\left(x - \frac{3-\sqrt{3}}{2}\right)^2 + \left(y - \frac{3-\sqrt{3}}{2}\right)^2 + \left(z - \frac{3-\sqrt{3}}{2}\right)^2 = \left(\frac{3-\sqrt{3}}{2}\right)^2}.$$

□

Problem 2025.4. There are 200 cards numbered from 1 to 200, all initially facing up. For each integer from 1 to 200, we call out that integer and flip every card whose number is a multiple of it. Determine the number of cards facing up after all 200 calls.

(1) 142
(2) 153
(3) 164
(4) 175
(5) 186

Solution. A card numbered n is flipped once for each positive divisor of n. Hence it is flipped $\tau(n)$ times, where $\tau(n)$ is the number of positive divisors of n.

A positive integer has an odd number of positive divisors if and only if it is a perfect square. Therefore, exactly the cards numbered

$$1^2, \quad 2^2, \quad 3^2, \quad \dots, \quad 14^2$$

are flipped an odd number of times, since

$$14^2 = 196 \leq 200 < 225 = 15^2.$$

These 14 cards end up facing down. The remaining

$$200 - 14 = 186$$

cards are facing up.

Therefore, the answer is $\boxed{(5)\ 186}$.

□

Problem 2025.5. Determine which of 11^9 and 9^{11} is larger.

Solution. Taking logarithms, it is enough to compare

$$9\ln 11 \qquad \text{and} \qquad 11\ln 9.$$

Equivalently, compare

$$\frac{\ln 11}{11} \qquad \text{and} \qquad \frac{\ln 9}{9}.$$

The function $f(x) = \ln x/x$ is decreasing for $x > e$. Since $9 < 11$ and both are greater than e, we have

$$\frac{\ln 9}{9} > \frac{\ln 11}{11}.$$

Hence

$$11\ln 9 > 9\ln 11,$$

so

$$9^{11} > 11^9.$$

Therefore, the larger number is $\boxed{9^{11}}$.

□

Problem 2025.6. Roll a fair six-sided die repeatedly until a 6 appears. Determine the expected sum of all rolls, including the final 6.

Solution. Let E be the desired expected sum. On the first roll, if the result is 6, the process stops and contributes 6. If the result is one of $1, 2, 3, 4, 5$, then the process returns to the same situation after contributing that roll.

Therefore,

$$E = \frac{1}{6}\cdot 6 + \frac{1}{6}\sum_{k=1}^{5}(k+E).$$

Hence

$$E = 1 + \frac{15+5E}{6} = \frac{7}{2} + \frac{5}{6}E.$$

Thus

$$\frac{1}{6}E = \frac{7}{2}, \qquad E = 21.$$

Therefore, the answer is $\boxed{21}$.

□

Problem 2025.7. Determine the probability that at least two people in a group of 10 share the same birthday. Assume a 365-day year and that each day is equally likely.

Solution. It is easier to compute the complementary probability that all 10 birthdays are distinct. The first person may have any birthday, the second must avoid that birthday, the third must avoid the first two birthdays, and so on. Hence

$$\mathbb{P}(\text{all birthdays are distinct}) = \frac{365}{365}\cdot\frac{364}{365}\cdot\frac{363}{365}\cdots\frac{356}{365}.$$

Therefore,

$$\mathbb{P}(\text{at least two people share a birthday}) = 1 - \frac{365}{365}\cdot\frac{364}{365}\cdot\frac{363}{365}\cdots\frac{356}{365}.$$

Therefore, the answer is

$$\boxed{1 - \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{356}{365}}.$$

□

Problem 2025.8. Find the largest of the following numbers:

(1) e^{π}
(2) π^{e}
(3) $\sqrt{500}$
(4) 7π

Solution. Using the standard approximations

$$e \approx 2.718, \qquad \pi \approx 3.142,$$

we get

$$e^{\pi} \approx 23.14, \qquad \pi^{e} \approx 22.46, \qquad \sqrt{500} \approx 22.36, \qquad 7\pi \approx 21.99.$$

Therefore, the answer is $\boxed{(1)\ e^{\pi}}$.

□

Problem 2025.9. There are three doors. One hides a car and the other two hide goats. After the contestant chooses a door, a host who knows the car's location must open an unchosen door hiding a goat; when two such doors are available, the host follows a rule independent of the car's location. The contestant may then keep the original choice or switch to the only other unopened door. Determine which choice gives the higher probability of winning the car.

(1) The probabilities are the same.
(2) It is better not to switch.
(3) It cannot be determined.
(4) It is better to switch.

Solution. The original choice contains the car with probability 1/3 and is wrong with probability 2/3. Under the stated host protocol, switching wins exactly when the original choice was wrong. Hence staying wins with probability 1/3, whereas switching wins with probability 2/3.

Therefore, the answer is

$$\boxed{\text{(4) It is better to switch}}.$$

□

Problem 2025.10. Determine which of the following Fields Medalists majored in history rather than mathematics as an undergraduate.

(1) June Huh
(2) Caucher Birkar
(3) Edward Witten
(4) Simon Donaldson
(5) William Thurston

Solution. Edward Witten is famous as a mathematical physicist and was the first physicist to receive the Fields Medal. As an undergraduate, he majored in history rather than mathematics.

Therefore, the answer is $\boxed{\text{(3) Edward Witten}}$.

□

Problem 2025.11. Determine which of the following mathematicians was not alive during the twentieth century.

(1) Sophus Lie
(2) Wilhelm Killing
(3) Alfred Young
(4) Hans Fitting
(5) Botong Wang

Solution. Sophus Lie lived from 1842 to 1899, so he died before the twentieth century began. The other listed mathematicians were alive at some point during the twentieth century.

Therefore, the answer is $\boxed{\text{(1) Sophus Lie}}$.

□

Problem 2025.12. Königsberg was the capital of East Prussia and a major center of European scholarship for many years. It was also the hometown of Immanuel Kant, David Hilbert, and Jürgen Moser. The famous Seven Bridges of Königsberg problem, in which Leonhard Euler proved that no solution exists, marked the beginning of both topology and graph theory. Determine the present-day country containing the former city of Königsberg.

(1) Germany
(2) Poland
(3) Lithuania
(4) Russia
(5) Korea

Solution. The historical city Königsberg is now Kaliningrad. Kaliningrad is a city in Russia.

Therefore, the answer is $\boxed{\text{(4) Russia}}$.

□

Problem 2025.13. A fair coin is flipped 50 times, and each outcome is recorded as either heads (H) or tails (T). A run is a consecutive sequence of the same outcome. For example, if the result is $HHTTTH$, there are 3 runs: HH, TTT, and H. Determine the expected number of runs.

(1) 24.5
(2) 25
(3) 25.5
(4) 26

Solution. The first toss always starts one run. For each position from the second toss onward, a new run begins exactly when the current toss differs from the previous toss.

For a fair coin,

$$\mathbb{P}(\text{current toss differs from previous toss}) = \frac{1}{2}.$$

For $2 \leq i \leq 50$, let

$$I_i = \mathbb{1}_{\{\text{a new run starts at position } i\}}.$$

Then

$$R = 1 + \sum_{i=2}^{50} I_i.$$

By linearity of expectation,

$$\mathbb{E}[R] = 1 + \sum_{i=2}^{50} \mathbb{E}[I_i] = 1 + 49 \cdot \frac{1}{2} = 25.5.$$

Therefore, the answer is $\boxed{(3)\ 25.5}$.

□

Problem 2025.14. Determine which of the following results appeared first historically.

(1) Bayes' Theorem
(2) Intermediate Value Theorem
(3) Cauchy–Schwarz Inequality
(4) Fundamental Theorem of Calculus

Solution. Interpreting the question by the earliest recognizable form of each result, the inverse relationship between differentiation and integration was established in seventeenth-century work associated with Barrow, Newton, and Leibniz. Bayes' theorem was published in the eighteenth century, while rigorous forms of the Intermediate Value Theorem and the Cauchy–Schwarz inequality appeared in the nineteenth century.

Therefore, the intended answer is

$$\boxed{\text{(4) Fundamental Theorem of Calculus}}.$$

□

Problem 2025.15. At the time of the original 2025 Science Quiz, determine which of the following amounts was closest to the Abel Prize award.

(1) None
(2) About \$1,000,000
(3) About \$1,500,000
(4) About \$2,000,000

Solution. In 2025, the Abel Prize amounted to NOK 7.5 million. Under the exchange rate relevant to the original question, this was closest to one million US dollars among the listed choices.

Therefore, the answer is $\boxed{\text{(2) About \$1,000,000}}$.

□

Problem 2025.16. Determine which of the following mathematicians has received both the Fields Medal and the Abel Prize.

(1) John Milnor
(2) Terence Tao
(3) June Huh
(4) No one

Solution. John Milnor received the Fields Medal in 1962 and the Abel Prize in 2011. Therefore he has received both prizes.

Therefore, the answer is $\boxed{\text{(1) John Milnor}}$.

□

Problem 2025.17. A fair six-sided die is rolled until all 6 faces have appeared. Determine the probability that no odd-numbered face appears before all even-numbered faces have appeared.

(1) 20%
(2) 15%
(3) 10%
(4) 5%

Solution. Ignore repeated rolls and look only at the order in which new faces appear. The condition is that the first three distinct faces to appear are exactly the even faces $2, 4, 6$ in some order.

The probability that the first new face is even is $3/6$. Given that one even face has appeared, the probability that the next new face is also even is $2/5$. Given that two even faces have appeared, the probability that the next new face is the remaining even face is $1/4$. Therefore the desired probability is

$$\frac{3}{6} \cdot \frac{2}{5} \cdot \frac{1}{4} = \frac{1}{20} = 5\%.$$

Therefore, the answer is $\boxed{(4)\ 5\%}$.

□

Problem 2025.18. In the Tower of Hanoi puzzle, 5 disks are stacked on peg A from largest to smallest. The objective is to move all disks to peg C using auxiliary peg B, subject to the following rules:

(1) Only one disk may be moved at a time.
(2) A larger disk may not be placed on a smaller disk.
(3) Peg B may be used freely.

Determine the minimum number of moves required.

(1) 25
(2) 31
(3) 32
(4) 63
(5) 120

Solution. Let $T(n)$ be the minimum number of moves needed for n disks. To move n disks, first move the top $n-1$ disks to the auxiliary peg, then move the largest disk once, and finally move the $n-1$ disks onto the largest disk. Hence

$$T(n) = 2T(n-1) + 1, \qquad T(1) = 1.$$

Solving this recurrence gives

$$T(n) = 2^n - 1.$$

Therefore,

$$T(5) = 2^5 - 1 = 31.$$

Therefore, the answer is $\boxed{(2)\ 31}$.

□

Problem 2025.19. Determine the last two digits of 7^{123}, expressed as a two-digit number.

(1) 37
(2) 41
(3) 43
(4) 49

Solution. Work modulo 100. Since

$$7^4 = 2401 \equiv 1 \pmod{100},$$

and

$$123 = 4 \cdot 30 + 3,$$

we have

$$7^{123} = (7^4)^{30} 7^3 \equiv 1^{30} \cdot 343 \equiv 43 \pmod{100}.$$

Therefore, the answer is $\boxed{(3)\ 43}$.

$\square$

Part V

Expected Problems

Standard Mathematical Problems

Problem 1. (1988 IMO, Vieta Jumping). Let a and b be positive integers s.t. $ab+1$ divides a^2+b^2. Show that

$$\frac{a^2+b^2}{ab+1}$$

is the square of an integer.

Solution. Let

$$k = \frac{a^2+b^2}{ab+1} \in \mathbb{N}.$$

Suppose, for contradiction, that k is not a perfect square. Among all positive integer solutions of

$$a^2+b^2 = k(ab+1),$$

choose one for which $a+b$ is minimal, and assume without loss of generality that $a \geq b$.

If $a=b$, then

$$k = \frac{2a^2}{a^2+1} < 2,$$

so $k=1$, a contradiction. Hence $a>b$. Regard the equation as a quadratic equation in a:

$$x^2 - kbx + b^2 - k = 0.$$

One root is a, so the other root is

$$c = kb - a \in \mathbb{Z}.$$

We claim that $c>0$. From

$$a(a-kb) = k - b^2,$$

if $a=kb$, then $k=b^2$, contrary to the assumption. If $a>kb$, then the left-hand side is at least $a \geq kb \geq k$, whereas the right-hand side is less than k, which is impossible. Therefore $a<kb$, and hence $c>0$.

Since k is a positive integer that is not a perfect square, we have $k \geq 2$. Let

$$f(x) = x^2 - kbx + b^2 - k.$$

Since

$$f(b) = (2-k)b^2 - k < 0$$

and the two roots are c and a with $a>b$, it follows that

$$0 < c < b.$$

By Viète's formulas, c is the other root, and therefore

$$c^2 + b^2 = k(cb+1).$$

Thus (c,b) is another positive integer solution with

$$c+b < a+b,$$

contradicting the minimality of $a+b$. Hence k must be a perfect square.

□

Problem 2. (Sophomore's Dream). Prove the identity

$$\int_0^1 x^{-x}\,\mathrm{d}x = \sum_{n=1}^{\infty} \frac{1}{n^n}.$$

Solution. For $0 < x \le 1$, we have $-\ln x \ge 0$, and hence

$$x^{-x} = e^{-x\ln x} = \sum_{n=0}^{\infty} \frac{x^n(-\ln x)^n}{n!}.$$

Every term is nonnegative, so Tonelli's theorem from Mathematical Concepts permits term-by-term integration:

$$\int_0^1 x^{-x}\,\mathrm{d}x = \sum_{n=0}^{\infty} \frac{1}{n!}\int_0^1 x^n(-\ln x)^n\,\mathrm{d}x.$$

In the nth integral, let $t = -(n+1)\ln x$. Then

$$\int_0^1 x^n(-\ln x)^n\,\mathrm{d}x = \frac{1}{(n+1)^{n+1}}\int_0^{\infty} t^n e^{-t}\,\mathrm{d}t = \frac{n!}{(n+1)^{n+1}}.$$

Therefore

$$\int_0^1 x^{-x}\,\mathrm{d}x = \sum_{n=0}^{\infty} \frac{1}{(n+1)^{n+1}} = \boxed{\sum_{n=1}^{\infty} \frac{1}{n^n}}.$$

Numerically, the common value is approximately

$$1.291285997\ldots.$$

$\square$

Problem 3. (1982 IMO). A function

$$f : \mathbb{Z}_{>0} \to \mathbb{Z}_{\ge 0}$$

satisfies

$$f(m+n) - f(m) - f(n) \in \{0, 1\}$$

for all positive integers m and n. Moreover,

$$f(2) = 0, \qquad f(3) > 0, \qquad f(9999) = 3333.$$

Determine $f(1982)$.

Solution. Repeatedly applying the given condition shows that, for all positive integers r and t,

$$rf(t) \le f(rt) \le rf(t) + r - 1.$$

Apply this estimate to

$$N = 9999 \cdot 1982$$

in two different ways. Using $t = 1982$ and then $t = 9999$ gives

$$9999f(1982) \le f(N) \le 9999f(1982) + 9998$$

and

$$1982 \cdot 3333 \le f(N) \le 1982 \cdot 3333 + 1981.$$

Consequently,

$$\frac{1982 \cdot 3333 - 9998}{9999} \le f(1982) \le \frac{1982 \cdot 3333 + 1981}{9999}.$$

The left endpoint lies strictly between 659 and 660, whereas the right endpoint lies strictly between 660 and 661. Since $f(1982)$ is an integer, it follows that

$$\boxed{f(1982) = 660}.$$

□

Problem 4. (Happy Ending Problem). Five points are given in the plane, no three of which are collinear. Prove that four of them are the vertices of a convex quadrilateral.

Solution. Consider the convex hull of the five points. If the convex hull has at least four vertices, any four of its vertices, taken in cyclic order, form a convex quadrilateral.

It remains to consider the case in which the convex hull is a triangle with vertices A, B, C. Let the remaining two points be P and Q; both lie in the interior of $\triangle ABC$. The line PQ contains none of A, B, C, so two of the three vertices lie in the same open half-plane determined by PQ. Relabel them as A and B.

The triangle $\triangle ABQ$ meets the line PQ only at Q, because A and B lie strictly on the same side of that line. Hence $P \notin \triangle ABQ$. Similarly, $Q \notin \triangle ABP$. Since A and B are vertices of the convex hull of all five points, neither lies in the triangle determined by the other three points. Consequently, all four points A, B, P, Q are vertices of their convex hull, which is a convex quadrilateral.

□

Problem 5. (Sylvester–Gallai Theorem). Let S be a finite set of points in the plane, not all collinear. Prove that there exists a line containing exactly two points of S.

Solution. Call a line determined by points of S *ordinary* if it contains exactly two points of S. Consider all pairs (P, ℓ) s.t. $P \in S$, the line ℓ is determined by at least two points of S, and $P \notin \ell$. Since S is finite and not collinear, the set of positive distances arising in this way is finite and nonempty. Choose (P, ℓ) so that

$$h = \operatorname{dist}(P, \ell) > 0$$

is as small as possible, and let H be the foot of the perpendicular from P to ℓ.

Suppose, for contradiction, that ℓ contains at least three points of S. We can choose distinct points $A, B \in S \cap \ell$ so that either $A = H$, or A and B lie on the same side of H with A closer to H. In either case,

$$AB < PB.$$

Indeed, if $A = H$, then $AB = HB < PB$; otherwise, $AB = HB - HA < HB < PB$.

Computing the area of $\triangle PAB$ with bases AB and PB gives

$$\frac{1}{2} AB\, h = [PAB] = \frac{1}{2} PB \operatorname{dist}(A, PB).$$

Hence

$$\operatorname{dist}(A, PB) = \frac{AB}{PB} h < h.$$

However, the line PB is determined by two points of S, and $A \notin PB$. This contradicts the minimality of h.

Therefore, ℓ contains exactly two points of S, so ℓ is the required ordinary line.

□

Problem 6. (Moser's Circle Problem). Let n be a positive integer. Place n points on a circle and join every pair of points by a chord. Assume that no three chords meet at a common point in the interior of the circle. Determine, with proof, the maximum number of regions into which the chords divide the circle.

Solution. There are

$$\binom{n}{2}$$

chords. Draw them one at a time. If a new chord contains r existing interior intersection points, those points divide it into $r+1$ segments, and each segment splits one existing region into two. Therefore the new chord creates $r+1$ new regions.

It follows that the final number of regions is

$$1 + \#\{\text{chords}\} + \#\{\text{interior intersections}\}.$$

Every interior intersection is determined by four boundary points: among the six chords joining those four points, exactly the two diagonals of their cyclic quadrilateral intersect in the interior. Conversely, every choice of four boundary points produces one such intersection. The assumption that no three chords are concurrent ensures that all these intersections are distinct. Hence there are

$$\binom{n}{4}$$

interior intersections.

Thus the maximum number of regions is

$$\boxed{1 + \binom{n}{2} + \binom{n}{4}}.$$

In particular, the first values are

$$1,\ 2,\ 4,\ 8,\ 16,\ 31, \ldots,$$

so the apparent pattern 2^{n-1} fails when $n = 6$.

□

Problem 7. (1976 IMO). Determine, with proof, the largest number that can be written as the product of positive integers whose sum is 1976. The number of factors is not prescribed.

Solution. A maximizing product contains no factor 1, since replacing a pair $1, a$ by $a+1$ preserves the sum and increases the product. It also contains no factor $k \geq 5$, because replacing k by 3 and $k-3$ preserves the sum and increases the product:

$$3(k-3) > k.$$

A factor 4 may be replaced by $2, 2$ without changing the product. Thus it suffices to use only 2 and 3.

Three factors equal to 2 should be replaced by two factors equal to 3, since

$$2+2+2 = 3+3 \qquad \text{and} \qquad 2^3 < 3^2.$$

Hence at most two factors equal to 2 occur. Since

$$1976 = 3 \cdot 658 + 2,$$

the only optimal remainder pattern is one factor equal to 2 and 658 factors equal to 3.

Therefore, the largest possible product is

$$\boxed{2 \cdot 3^{658}}.$$

□

Problem 8. (2021 Putnam). A grasshopper starts at the origin in the coordinate plane and makes a sequence of hops. Each hop has length 5, and after every hop the grasshopper is at a point whose coordinates are both integers. Determine the smallest number of hops needed to reach

$$(2021, 2021).$$

Solution. If a hop has displacement vector (p, q), then

$$p^2 + q^2 = 25.$$

Thus the possible absolute coordinate pairs are $(5, 0)$ and $(4, 3)$, in either order. In particular, a single hop can increase the sum of the two coordinates by at most

$$4 + 3 = 7.$$

Since

$$2021 + 2021 = 4042 = 7 \cdot 577 + 3,$$

at least 578 hops are necessary.

This bound is attained because

$$(2021, 2021) = 288(3, 4) + 288(4, 3) + (5, 0) + (0, 5).$$

The right-hand side is a sum of

$$288 + 288 + 1 + 1 = 578$$

legal hop vectors. Therefore, the minimum number of hops is

$$\boxed{578}.$$

□

Problem 9. (Buffon's Needle). Infinitely many parallel lines are drawn in the plane, with distance d between consecutive lines. A needle of length ℓ, where

$$0 < \ell \le d,$$

is dropped at random. Assume that its midpoint is uniformly distributed modulo one strip and, independently, that its direction is uniformly distributed. Determine the probability that the needle intersects one of the parallel lines.

Solution. Let X be the distance from the midpoint of the needle to the nearest parallel line, and let Θ be the acute angle between the needle and the lines. Then

$$0 \le X \le \frac{d}{2}, \qquad 0 \le \Theta \le \frac{\pi}{2},$$

and X and Θ are independent and uniform on these intervals.

For a fixed angle θ, the needle intersects a line exactly when its perpendicular half-length reaches the nearest line, that is, when

$$X \le \frac{\ell}{2} \sin\theta.$$

Since $\ell \le d$, the right-hand side never exceeds $d/2$. Therefore,

$$\begin{aligned}
\mathbb{P}(\text{intersection}) &= \frac{2}{\pi} \int_0^{\pi/2} \mathbb{P}\left(X \le \frac{\ell}{2} \sin\theta\right) \mathrm{d}\theta \\
&= \frac{2}{\pi} \int_0^{\pi/2} \frac{\ell \sin\theta}{d} \mathrm{d}\theta \\
&= \boxed{\frac{2\ell}{\pi d}}.
\end{aligned}$$

□

Problem 10. (2011 IMO, Windmill Problem). Let S be a finite set of at least two points in the plane, and assume that no three points of S are collinear. A windmill is the following process. Start with a line ℓ through a point $P \in S$. Rotate ℓ clockwise about the pivot P until it first meets another point $Q \in S$. The point Q then becomes the new pivot, and the line continues to rotate clockwise. The process is repeated indefinitely.

Prove that one can choose the initial point P and the initial line ℓ so that every point of S is used as a pivot infinitely many times.

Solution. Orient the rotating line and call its two open sides the left and right sides. When the pivot changes from T to U, retain the orientation of the line. Immediately after the change, the old pivot T lies on the same side on which U lay immediately before the change. Hence the ordered numbers of points on the two sides are invariant throughout the process, except at the instants when the line contains two points of S.

First suppose that

$$|S| = 2m + 1.$$

For every $T \in S$, there is an oriented line through T with exactly m points of S on each side. To see this, rotate a line through T by an angle π. The number of points on its left changes by one at each encounter with another point of S, and its initial and final values sum to $2m$; therefore the value m occurs.

Choose such a balanced line through any point as the initial state. The side counts remain (m, m). Fix $T \in S$ and choose a balanced line r through T. Among all lines parallel to r, the line r is the unique one that contains a point of S and has m points on each side: translating it in either direction changes the balance at the first point crossed. Consequently, when the windmill becomes parallel to r, it must coincide with r, so T is a pivot. Thus every point is visited during each half-turn of the windmill line.

Now suppose that

$$|S| = 2m.$$

For every $T \in S$, the same half-turn argument gives an oriented line through T with $m-1$ points on its left and m points on its right. Start the windmill from one such line. Its ordered side counts remain $(m-1, m)$. For any prescribed T, choose an oriented line r through T with these counts. It is the unique oriented translate in its direction with these counts, so the windmill must pass through T when it reaches that oriented direction. Hence every point is visited during each full turn.

Only finitely many line directions are determined by pairs of points of S. Therefore the positive angular increments between successive pivot changes have a positive minimum. Since the process continues indefinitely, the total angle of rotation tends to infinity. The preceding half-turn or full-turn argument then applies repeatedly, and every point of S is used as a pivot infinitely many times.

□

Problem 11. (Art Gallery Problem). A museum has the shape of a simple polygon with n vertices. A guard must stand at a vertex of the polygon and can see a point whenever the entire line segment joining the guard to that point lies in the polygon. Prove that the entire museum can always be guarded by at most

$$\left\lfloor \frac{n}{3} \right\rfloor$$

guards.

Solution. Triangulate the polygon using nonintersecting diagonals between its vertices. The graph formed by the polygon sides and the triangulation diagonals admits a proper coloring with three colors.

To prove this, use induction on n. The assertion is immediate for a triangle. Every triangulated polygon has an ear: a triangle in the triangulation with two sides on the boundary. Remove the ear vertex and color the remaining triangulated polygon by induction. The two neighbors of the removed vertex are adjacent and therefore have different colors, so the removed vertex can be assigned the third color. This completes the induction.

Every triangle of the triangulation has one vertex of each color. Of the three color classes, one contains at most

$$\left\lfloor \frac{n}{3} \right\rfloor$$

vertices. Place guards at all vertices in that smallest color class. Every triangle contains one of these guards, and a guard at a vertex of a triangle sees every point of that triangle because a triangle is convex. Since the triangles cover the polygon, these guards see the entire museum.

□

Problem 12. (Bertrand's Ballot Problem). In an election, candidate A receives p votes and candidate B receives q votes, where

$$p > q.$$

The $p + q$ ballots are counted in a uniformly random order. Determine the probability that, after every ballot is counted, the cumulative number of votes for A is strictly greater than the cumulative number of votes for B.

Solution. Represent a vote for A by a horizontal step and a vote for B by a vertical step. A counting order is then a lattice path from $(0, 0)$ to (p, q), and the required condition is that the path remain strictly below the diagonal $y = x$ after leaving the origin.

The first ballot must be for A. Remove this first horizontal step and translate the remaining path one unit to the left. We obtain a path from $(0, 0)$ to $(p - 1, q)$ that never enters the region $y > x$. By the reflection principle, the number of such paths is

$$\binom{p+q-1}{q} - \binom{p+q-1}{q-1}.$$

Since all

$$\binom{p+q}{q}$$

ballot orders are equally likely, the desired probability is

$$\frac{\binom{p+q-1}{q} - \binom{p+q-1}{q-1}}{\binom{p+q}{q}} = \frac{p-q}{p+q}.$$

Therefore the probability is

$$\boxed{\frac{p-q}{p+q}}.$$

□

Problem 13. (1964 IMO). Seventeen people correspond with one another. Every pair discusses exactly one of three topics in their correspondence. Prove that there are three people s.t. every pair among them discusses the same topic.

Solution. Choose one person P. Among the 16 correspondences involving P, at least

$$\left\lceil \frac{16}{3} \right\rceil = 6$$

concern the same topic; call it topic 1, and let the six other people be $P_1, \ldots, P_6$.

If some pair P_i, P_j also discusses topic 1, then P, P_i, P_j form the required triple. Suppose instead that no correspondence among $P_1, \ldots, P_6$ concerns topic 1. Their mutual correspondences therefore use only topics 2 and 3.

It remains to use the elementary fact $R(3,3) \leq 6$. Indeed, choose one of six vertices. At least three of its five incident edges have the same color. If one of the three edges joining their other endpoints has that color, there is a monochromatic triangle; otherwise those three edges form a triangle of the other color. Thus among $P_1, \ldots, P_6$ there is a triangle all of whose edges have topic 2 or all of whose edges have topic 3.

□

Problem 14. (Sperner's Lemma). A triangle ABC is triangulated into finitely many smaller triangles. Label every vertex by one of $1, 2, 3$ so that A, B, C receive labels $1, 2, 3$, respectively; vertices on AB receive only 1 or 2; vertices on BC receive only 2 or 3; and vertices on CA receive only 3 or 1. Interior vertices may receive any label.

Prove that the number of small triangles whose three vertex labels are $1, 2, 3$ is odd. In particular, at least one such triangle exists.

Solution. Call an edge a 12-edge if its endpoints have labels 1 and 2. Along the subdivided side AB, the sequence of labels begins with 1 and ends with 2. Hence the number of changes between 1 and 2 on AB is odd, so AB contains an odd number of 12-edges. The other two sides contain none.

Count incidences between small triangles and their 12-edges modulo 2. Every interior 12-edge is incident with two small triangles and contributes 0 modulo 2, whereas every boundary 12-edge contributes 1. Thus the total incidence count is odd.

A small triangle with labels $1, 2, 3$ has exactly one 12-edge. A small triangle using only labels 1 and 2 has either zero or two such edges, and every other small triangle has none. Therefore the parity of the incidence count is exactly the parity of the number of fully labeled triangles. That number is consequently odd.

□

Problem 15. (Morley's Trisector Theorem). In a triangle ABC, trisect each interior angle. Let P be the intersection of the two trisectors adjacent to BC, let Q be the intersection of the two trisectors adjacent to CA, and let R be the intersection of the two trisectors adjacent to AB. Prove that $\triangle PQR$ is equilateral.

Solution. Write

$$\angle A = 3\alpha, \qquad \angle B = 3\beta, \qquad \angle C = 3\gamma.$$

Then

$$\alpha + \beta + \gamma = \frac{\pi}{3}.$$

Let $\mathcal{R}$ be the circumradius of $\triangle ABC$.

In $\triangle BCP$, the angles at B and C are β and γ. The sine rule and the identity

$$\sin(3t) = 4 \sin t \sin\left(\frac{\pi}{3} + t\right) \sin\left(\frac{\pi}{3} - t\right)$$

give

$$BP = \frac{BC \sin\gamma}{\sin(\beta + \gamma)} = 8\mathcal{R} \sin\alpha \sin\gamma \sin\left(\frac{\pi}{3} + \alpha\right).$$

Similarly, using $AB = 2\mathcal{R}\sin(3\gamma)$ in $\triangle ABR$,

$$BR = 8\mathcal{R}\sin\alpha\sin\gamma\sin\left(\frac{\pi}{3}+\gamma\right).$$

Since $\angle PBR = \beta$, the law of cosines yields

$$\begin{aligned} PR^2 = (8\mathcal{R}\sin\alpha\sin\gamma)^2 \Bigg(&\sin^2\left(\frac{\pi}{3}+\alpha\right)+\sin^2\left(\frac{\pi}{3}+\gamma\right) \\ &- 2\sin\left(\frac{\pi}{3}+\alpha\right)\sin\left(\frac{\pi}{3}+\gamma\right)\cos\beta\Bigg). \end{aligned}$$

Because

$$\left(\frac{\pi}{3}+\alpha\right)+\left(\frac{\pi}{3}+\gamma\right)+\beta = \pi,$$

the expression in parentheses is $\sin^2\beta$. Hence

$$PR = 8\mathcal{R}\sin\alpha\sin\beta\sin\gamma.$$

Cyclically applying the same calculation gives the identical value for PQ and QR. Therefore $\triangle PQR$ is equilateral.

$\square$

Problem 16. (Steiner Tree for a Square). Determine the minimum possible total length of a connected road network containing all four vertices of a square of side length a. Roads may meet at additional points in the interior of the square.

Solution. The standard structure theorem for Euclidean Steiner minimal trees says that a shortest network is a tree, every nonterminal branching point has degree 3, and its three incident edges meet at angles of 120°. The possible square topologies have zero, one, or two Steiner points. With zero Steiner points, the shortest network is a minimum spanning tree of length $3a$. With one Steiner point, one first joins three square vertices by their three-terminal Steiner tree and then attaches the remaining vertex by an edge of length at least a. The three-terminal tree for a right isosceles triangle of legs a has length

$$a\sqrt{2+\sqrt{3}},$$

so this case is also longer than the construction below. Thus it remains to examine the full topology with two Steiner points, each joined to two adjacent vertices and to the other Steiner point.

Place the square at

$$(0,-a/2), \quad (0,a/2), \quad (a,-a/2), \quad (a,a/2).$$

The two Steiner points lie on the horizontal symmetry axis. Write them as

$$S_1 = (d,0), \qquad S_2 = (a-d,0).$$

The angle between the two edges from S_1 to the left-hand vertices is 120°. Taking their dot product gives

$$\frac{d^2-a^2/4}{d^2+a^2/4} = -\frac{1}{2},$$

so

$$d = \frac{a}{2\sqrt{3}}.$$

Each of the four terminal edges has length

$$\sqrt{d^2 + \frac{a^2}{4}} = \frac{a}{\sqrt{3}},$$

while

$$S_1S_2 = a - 2d = a - \frac{a}{\sqrt{3}}.$$

Thus the total length is

$$4\frac{a}{\sqrt{3}} + a - \frac{a}{\sqrt{3}} = \boxed{a(1+\sqrt{3})}.$$

Since

$$a(1+\sqrt{3}) < a\left(1 + \sqrt{2+\sqrt{3}}\right) < 3a,$$

the two-Steiner construction is the global minimum.

□

Problem 17. (1975 IMO). Let A be the sum of the decimal digits of 4444^{4444}, let B be the sum of the decimal digits of A, and let C be the sum of the decimal digits of B. Determine C.

Solution. Since $4444 < 10^4$,

$$4444^{4444} < 10^{17776}.$$

Hence

$$A \leq 9 \cdot 17776 = 159984.$$

Thus A has at most six digits, so $B \leq 54$. Among the integers from 1 to 54, the largest possible digit sum is 13, attained at 49. Therefore

$$1 \leq C \leq 13.$$

Repeated digit sums preserve a number modulo 9. Since

$$4444 \equiv 7 \pmod 9, \qquad 7^3 \equiv 1 \pmod 9, \qquad 4444 \equiv 1 \pmod 3,$$

we have

$$C \equiv 4444^{4444} \equiv 7 \pmod 9.$$

The only integer between 1 and 13 congruent to 7 modulo 9 is

$$\boxed{C = 7}.$$

□

Problem 18. (Golomb's Tromino Theorem). Remove an arbitrary unit square from a $2^n \times 2^n$ board. Prove that the remaining board can be tiled by L-shaped trominoes, each consisting of three unit squares.

Solution. We use induction on n. For $n = 1$, the board is 2×2, and the three remaining squares form one L-tromino.

Suppose the statement holds for $n-1$. Divide a $2^n \times 2^n$ board into four $2^{n-1} \times 2^{n-1}$ quadrants. The missing square lies in one quadrant. Place one L-tromino at the center so that it covers one central square in each of the other three quadrants.

Each quadrant is now a $2^{n-1} \times 2^{n-1}$ board with exactly one square missing: the original missing square in one quadrant and a central square in each of the other three. By the induction hypothesis, every quadrant can be tiled. Together with the central tromino, these tilings cover the whole deficient board.

□

Problem 19. A rectangle is tiled by finitely many smaller rectangles whose sides are parallel to the sides of the large rectangle. Each small rectangle has at least one side of integer length. Prove that the large rectangle also has at least one side of integer length.

Solution. Place the large rectangle at

$$[0, a] \times [0, b].$$

For any measurable region E, define

$$I(E) = \iint_E \sin(2\pi x) \sin(2\pi y) \, \mathrm{d}x \, \mathrm{d}y.$$

This quantity is additive over a finite dissection.

Consider a tile

$$[u, u + w] \times [v, v + h].$$

If w is an integer, then

$$\int_u^{u+w} \sin(2\pi x) \, \mathrm{d}x = 0;$$

if h is an integer, the corresponding integral in y is zero. Therefore every tile has invariant 0, and additivity gives

$$I([0, a] \times [0, b]) = 0.$$

On the other hand,

$$I([0, a] \times [0, b]) = \frac{(1 - \cos(2\pi a))(1 - \cos(2\pi b))}{4\pi^2}.$$

Both factors in the numerator are nonnegative. Their product is zero only when

$$\cos(2\pi a) = 1 \qquad \text{or} \qquad \cos(2\pi b) = 1,$$

which means that a or b is an integer.

□

Problem 20. (Hall's Marriage Theorem). A standard deck of 52 cards is divided into 13 piles of four cards. Prove that one can choose exactly one card from each pile so that the chosen cards have the thirteen distinct ranks

$$A, 2, 3, \ldots, 10, J, Q, K.$$

Solution. Construct a bipartite graph whose left vertices are the 13 piles and whose right vertices are the 13 ranks. Join a pile to every rank that occurs in it. We verify Hall's condition.

Take any k piles. Together they contain $4k$ cards. If the union of their neighboring ranks contained at most $k - 1$ ranks, those ranks could account for at most

$$4(k - 1)$$

cards in the entire deck, since each rank occurs in four suits. This is impossible. Thus every k piles collectively contain at least k distinct ranks.

Hall's Marriage Theorem therefore gives a matching from all piles to distinct ranks. Choosing the matched card from each pile produces the required selection.

□

Problem 21. (Zeckendorf's Theorem). Define

$$F_1 = 1, \qquad F_2 = 2, \qquad F_{n+2} = F_{n+1} + F_n.$$

Prove that every positive integer can be written uniquely as a sum of nonconsecutive terms of this Fibonacci sequence.

Solution. For existence, apply the greedy algorithm. Given $N > 0$, choose the largest $F_k \le N$. Since $N < F_{k+1} = F_k + F_{k-1}$, the remainder satisfies

$$0 \le N - F_k < F_{k-1}.$$

Thus the next Fibonacci number selected by the same procedure has index at most $k - 2$. Repeating this process terminates and produces a sum of nonconsecutive Fibonacci numbers.

For uniqueness, use the identity

$$F_{k-1} + F_{k-3} + F_{k-5} + \cdots < F_k.$$

In fact, the sum on the left is $F_k - 1$. Suppose two valid representations differ, and let F_k be the largest term appearing in exactly one of them. The representation containing F_k exceeds the sum of every possible collection of nonconsecutive smaller terms, whereas the other representation uses only such smaller terms. They therefore cannot have the same value, a contradiction.

□

Problem 22. (1986 Putnam). Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices of the same size over a field, and suppose that

$$\mathbf{A} + \mathbf{B} = \mathbf{AB}.$$

Prove that

$$\mathbf{AB} = \mathbf{BA}.$$

Solution. Rearranging the hypothesis gives

$$(\mathbf{A} - \mathbf{I})(\mathbf{B} - \mathbf{I}) = \mathbf{AB} - \mathbf{A} - \mathbf{B} + \mathbf{I} = \mathbf{I}.$$

For square matrices, a one-sided inverse is a two-sided inverse. Hence

$$(\mathbf{B} - \mathbf{I})(\mathbf{A} - \mathbf{I}) = \mathbf{I}.$$

Expanding this identity gives

$$\mathbf{BA} - \mathbf{A} - \mathbf{B} = \mathbf{0}.$$

Comparing with the original relation yields

$$\mathbf{BA} = \mathbf{A} + \mathbf{B} = \mathbf{AB}.$$

□

Problem 23. (Erdős–Szekeres Monotone Subsequence Theorem). Let

$$a_1, a_2, \ldots, a_{n^2+1}$$

be distinct real numbers. Prove that the sequence contains an increasing subsequence of length $n+1$ or a decreasing subsequence of length $n+1$.

Solution. For each index i, let p_i be the length of the longest increasing subsequence ending at a_i, and let q_i be the length of the longest decreasing subsequence ending at a_i.

Suppose that neither required subsequence exists. Then

$$1 \le p_i \le n, \qquad 1 \le q_i \le n$$

for every i. There are only n^2 possible ordered pairs (p_i, q_i), so two indices $i < j$ have

$$(p_i, q_i) = (p_j, q_j).$$

Since the terms are distinct, either $a_i < a_j$ or $a_i > a_j$. In the first case, an increasing subsequence ending at a_i can be extended by a_j, so $p_j > p_i$. In the second case, $q_j > q_i$. Both alternatives contradict equality of the ordered pairs.

□

Problem 24. (2002 Putnam). Given any five distinct points on a sphere, prove that some four of them lie in a closed hemisphere.

Solution. Choose two of the points that are not antipodal, and draw a great circle through them. The two chosen points lie on the boundary of each of the two closed hemispheres determined by that great circle.

Each of the remaining three points belongs to at least one of the two hemispheres. By the pigeonhole principle, one hemisphere contains at least two of those three points. Together with the two boundary points, it contains at least four of the five points.

□

Problem 25. (Prouhet–Thue–Morse Partition). For a positive integer m, partition

$$\{0, 1, \ldots, 2^m - 1\}$$

into sets A and B s.t.

$$\sum_{a \in A} a^r = \sum_{b \in B} b^r$$

for every $r = 0, 1, \ldots, m-1$.

Solution. Put an integer j in A when the number $s_2(j)$ of 1's in its binary expansion is even, and put it in B when $s_2(j)$ is odd. Consider

$$P(x) = \sum_{j=0}^{2^m-1} (-1)^{s_2(j)} x^j = \prod_{k=0}^{m-1} (1 - x^{2^k}).$$

Every factor on the right vanishes at $x = 1$, so P has a zero of order m there.

Apply the operator

$$x \frac{\mathrm{d}}{\mathrm{d}x}$$

exactly $r < m$ times and then set $x = 1$. Since the zero has order m, the result is 0. On the other hand,

$$\left(x \frac{\mathrm{d}}{\mathrm{d}x} \right)^r P(x) \bigg|_{x=1} = \sum_{j=0}^{2^m-1} (-1)^{s_2(j)} j^r.$$

Separating the even and odd binary parities gives the required equal power sums.

□

Problem 26. (Cayley's Formula). For an integer $n \geq 2$, determine the number of trees with vertex set

$$\{1, 2, \ldots, n\}.$$

Solution. Associate a Prüfer sequence with a labeled tree as follows. Repeatedly delete the leaf with the smallest label and record its unique neighbor. Stop when two vertices remain. This produces a sequence of length $n - 2$, every term of which lies in $\{1, \ldots, n\}$.

Conversely, start with a sequence

$$(p_1, p_2, \ldots, p_{n-2}).$$

At each step, choose the smallest remaining label that does not occur in the current sequence, join it to the first term of the sequence, and then delete both that label and the first sequence term. After all terms have been removed, join the two labels that remain. This reconstructs a unique tree and reverses the encoding process.

Thus labeled trees are in bijection with length-$(n-2)$ sequences over an alphabet of n labels. Their number is therefore

$$\boxed{n^{n-2}}.$$

□

Problem 27. (Tournament King Theorem). In a tournament, every pair of vertices is joined by exactly one directed edge. Prove that there is a vertex A s.t. every other vertex B is either an out-neighbor of A or an out-neighbor of some out-neighbor of A.

Solution. Choose a vertex A of maximum outdegree. Let B be any vertex that defeats A. If no vertex defeated by A defeats B, then B defeats A and every out-neighbor of A. Consequently,

$$\deg^+(B) \geq \deg^+(A) + 1,$$

contradicting the maximality of the outdegree of A.

Therefore, for every B that defeats A, there is a vertex C s.t. A defeats C and C defeats B. Hence A is a king.

□

Problem 28. (Gale–Shapley Stable Marriage Problem). Two sets X and Y each contain n people. Every person has a strict preference ordering of all members of the other set. Prove that there is a perfect matching with no blocking pair, and give an algorithm that finds one.

Solution. Use the deferred-acceptance algorithm. While some person $x \in X$ is unmatched and has not proposed to everyone, let x propose to the most preferred member of Y who has not yet rejected x. Each $y \in Y$ keeps the most preferred proposal received so far and rejects every other proposer. A tentative engagement may therefore be replaced by a more preferred one.

Each ordered pair (x, y) is used in at most one proposal, so the algorithm terminates after at most n^2 proposals. At termination the matching is perfect: if some x were unmatched, then x would have proposed to every member of Y, so every member of Y would be holding a proposal; since the two sets have equal size, no member of X could then be unmatched.

Suppose that (x, y) is a blocking pair in the final matching. Since x prefers y to the assigned partner, x must have proposed to y earlier. The person y either rejected x immediately or later replaced x by someone preferred to x. Since tentative partners of y only improve, the final partner is preferred to x, contradicting that (x, y) blocks the matching.

□

Problem 29. (Buffon's Noodle). Parallel lines at distance D apart are drawn in the plane. A rectifiable plane curve of total length L is placed with uniformly random translation modulo the strips and uniformly random orientation. If N is the number of intersections of the curve with the parallel lines, determine $\mathbb{E}[N]$.

Solution. First consider a line segment of length ℓ making an acute angle θ with the parallel lines. Its perpendicular span is

$$\ell \sin\theta.$$

Averaging over a uniformly random offset modulo D, the expected number of parallel lines crossed by this segment is

$$\frac{\ell \sin\theta}{D}.$$

Averaging over a uniformly random direction gives

$$\frac{2}{\pi}\int_0^{\pi/2} \frac{\ell\sin\theta}{D}\,\mathrm{d}\theta = \frac{2\ell}{\pi D}.$$

For a polygonal curve, add the intersection counts of its segments. Linearity of expectation gives the same formula with ℓ replaced by the total length. A general rectifiable curve follows by polygonal approximation, which is the corresponding special case of the Cauchy–Crofton formula. Therefore,

$$\boxed{\mathbb{E}[N] = \frac{2L}{\pi D}}.$$

□

Problem 30. (1987 IMO). Prove that there is no function

$$f : \mathbb{N}_0 \to \mathbb{N}_0$$

s.t.

$$f(f(n)) = n + 1987$$

for every $n \in \mathbb{N}_0$.

Solution. Applying f to the given identity in two different ways gives

$$f(n + 1987) = f(f(f(n))) = f(n) + 1987.$$

Thus the residue of $f(n)$ modulo 1987 depends only on the residue of n modulo 1987.

For each $r \in \{0, 1, \ldots, 1986\}$, write

$$f(r) = 1987q_r + s_r, \qquad 0 \le s_r < 1987.$$

Then

$$r + 1987 = f(f(r)) = 1987q_r + f(s_r).$$

Reducing modulo 1987 shows that

$$s_{s_r} = r.$$

Hence $r \mapsto s_r$ is an involution on a set of 1987 elements.

This involution has no fixed point. Indeed, if $s_r = r$, then the same displayed identity gives

$$1987(q_r + q_r) = 1987,$$

so $2q_r = 1$, which is impossible. An involution without fixed points partitions its underlying set into pairs, but a set of odd cardinality cannot be partitioned into pairs. Therefore no such function exists.

$\square$

Problem 31. (1994 IMO). Let

$$A = \{a_1, a_2, \ldots, a_m\} \subseteq \{1, 2, \ldots, n\}$$

be a set of distinct positive integers. Suppose that whenever $x, y \in A$ and $x + y \leq n$, one also has $x + y \in A$. Prove that

$$\frac{a_1 + a_2 + \cdots + a_m}{m} \geq \frac{n+1}{2}.$$

Solution. Let d be the least element of A. Partition A according to residue classes modulo d. Consider one nonempty class, and let b be its least element. Since $d, b \in A$, the closure condition implies that

$$b, \quad b + d, \quad b + 2d, \quad \ldots$$

all belong to A for as long as they do not exceed n. Therefore this residue class of A is a complete arithmetic progression

$$b, b + d, \ldots, c,$$

where c is the largest integer not exceeding n that is congruent to b modulo d.

We have $b \geq d$ and $c > n - d$, so

$$b + c > n.$$

Since $b + c$ is an integer,

$$b + c \geq n + 1.$$

The average of the progression is therefore at least $(n + 1)/2$. Every residue class has average at least this value, so their weighted average, which is the average of all elements of A, also satisfies

$$\boxed{\frac{a_1 + a_2 + \cdots + a_m}{m} \geq \frac{n+1}{2}}.$$

$\square$

Problem 32. (1974 IMO). Three cards bear distinct positive integers

$$0 < p < q < r.$$

In each round the cards are shuffled and dealt, one to each of three players A, B, C. Each player receives as many counters as the number on the card dealt to that player, and all counters are retained. At least two rounds are played.

At the end, the players have respectively

$$20, \qquad 10, \qquad 9$$

counters, and in the final round player B received the card bearing r. Determine who received the card bearing q in the first round.

Solution. Suppose that t rounds were played. Adding the final totals gives

$$t(p+q+r) = 20 + 10 + 9 = 39.$$

Since $t \geq 2$ and player C received at least one counter per round, $t \leq 9$. Hence

$$t = 3, \qquad p+q+r = 13.$$

Player A received 20 counters in three rounds, so $r \geq 7$. In the last round player B received r, so the two earlier cards dealt to B have total $10 - r$. Checking the possibilities with $p < q < r$ and $p + q + r = 13$ gives the unique choice

$$(p, q, r) = (1, 4, 8).$$

Indeed, for $r = 7, 9$, or 10, the required sum $10 - r$ cannot be formed by two cards, while for $r = 8$ it forces both earlier cards of B to be $p = 1$.

In each of the first two rounds, players A and C therefore received q and r. To reach a total of 20, player A must have received

$$r, r, q,$$

over the three rounds. Consequently player C received

$$q, q, p,$$

and in particular received q in the first round. Thus the required player is

$$\boxed{C}.$$

□

Problem 33. (2004 Putnam). A basketball player's cumulative free-throw percentage was below 80% at one point in a season and above 80% at a later point. Prove that at some intermediate point the cumulative percentage was exactly 80%.

Solution. After n attempts, let m be the number of successful free throws and define

$$D = 5m - 4n.$$

The cumulative percentage is below, equal to, or above 80% exactly when D is negative, zero, or positive, respectively.

After a successful free throw, D increases by

$$5 - 4 = 1,$$

whereas after a missed free throw, D decreases by 4. Thus the only way for the integer D to pass from a negative value to a positive value is through 0: an upward step has size only 1. Therefore at some intermediate point

$$5m - 4n = 0,$$

and the cumulative percentage was exactly $4/5 = 80\%$.

□

Problem 34. (2005 Putnam). Find a nonzero polynomial $P(x, y)$ in two variables s.t.

$$P(\lfloor a \rfloor, \lfloor 2a \rfloor) = 0$$

for every real number a.

Solution. Write

$$a = \lfloor a \rfloor + \{a\}, \qquad 0 \le \{a\} < 1.$$

Then

$$\lfloor 2a \rfloor - 2\lfloor a \rfloor = \lfloor 2\{a\} \rfloor \in \{0, 1\}.$$

Hence one may take

$$\boxed{P(x, y) = (y - 2x)(y - 2x - 1)}.$$

This polynomial is nonzero and vanishes for both possible values of $y - 2x$.

□

Problem 35. (1996 Putnam). Let $[n] = \{1, 2, \ldots, n\}$. A subset $X \subseteq [n]$ is called *selfish* if

$$|X| \in X.$$

A selfish set is minimal if none of its proper subsets is selfish. Determine the number of minimal selfish subsets of $[n]$.

Solution. Let X be a minimal selfish set and put $|X| = k$. Then $k \in X$. We claim that X contains no integer $j < k$. If it did, one could choose a j-element subset $Y \subsetneq X$ containing j, and then

$$|Y| = j \in Y,$$

so Y would be selfish, contradicting minimality.

Conversely, any k-element set that contains k and whose other elements are all greater than k is minimal selfish. Thus, for a fixed k, there are

$$\binom{n-k}{k-1}$$

choices. Summing over all possible k gives

$$\sum_{k \ge 1} \binom{n-k}{k-1} = \sum_{j \ge 0} \binom{n-j-1}{j} = F_n,$$

where $F_1 = F_2 = 1$ and $F_n = F_{n-1} + F_{n-2}$. Therefore the number of minimal selfish sets is

$$\boxed{F_n}.$$

□

Problem 36. (2002 AIME I). Let S be a set of distinct positive integers s.t.

$$1 \in S, \qquad \max S = 2002.$$

Suppose that, for every $x \in S$, the arithmetic mean of the elements of $S \setminus \{x\}$ is an integer. Determine the maximum possible value of $|S|$.

Solution. Let

$$|S| = k, \qquad T = \sum_{x \in S} x.$$

The hypothesis says that

$$T - x \equiv 0 \pmod{k-1}$$

for every $x \in S$. Hence all elements of S are congruent modulo $k-1$. Since $1, 2002 \in S$,

$$k - 1 \mid 2001.$$

Put $d = k-1$. There must be at least $d+1$ distinct positive integers not exceeding 2002 that are congruent to 1 modulo d. The number of such integers is $2001/d + 1$, so

$$d + 1 \leq \frac{2001}{d} + 1, \qquad \text{hence} \qquad d^2 \leq 2001.$$

Since

$$2001 = 3 \cdot 23 \cdot 29,$$

the largest divisor of 2001 not exceeding $\sqrt{2001}$ is 29. Therefore $k \leq 30$.

Equality is attainable. Take

$$S = \{1 + 29j \mid 0 \leq j \leq 28\} \cup \{2002\}.$$

This set has 30 elements, all congruent to 1 modulo 29. Its total is also congruent to 1 modulo 29, so deleting any one element leaves a sum divisible by 29. Thus the maximum is

$$\boxed{30}.$$

□

Problem 37. (1961 UTokyo Entrance Examination). In a triangle ABC, points $L \in \overline{BC}$, $M \in \overline{CA}$, and $N \in \overline{AB}$ satisfy

$$\frac{LC}{BL} = \frac{MA}{CM} = \frac{NB}{AN} = 2.$$

Let

$$P = AL \cap CN, \qquad Q = AL \cap BM, \qquad R = BM \cap CN.$$

Determine

$$\frac{[PQR]}{[ABC]}.$$

Solution. Area ratios are preserved by affine transformations, so take

$$A = (0,0), \qquad B = (1,0), \qquad C = (0,1).$$

The division conditions give

$$L = \left(\frac{2}{3}, \frac{1}{3}\right), \qquad M = \left(0, \frac{2}{3}\right), \qquad N = \left(\frac{1}{3}, 0\right).$$

Solving the three pairs of line equations yields

$$P = \left(\frac{2}{7}, \frac{1}{7}\right), \quad Q = \left(\frac{4}{7}, \frac{2}{7}\right), \quad R = \left(\frac{1}{7}, \frac{4}{7}\right).$$

Therefore

$$[PQR] = \frac{1}{2}\left|\det\begin{pmatrix} 2/7 & 1/7 \\ -1/7 & 3/7 \end{pmatrix}\right| = \frac{1}{14}.$$

Since $[ABC] = 1/2$,

$$\boxed{\frac{[PQR]}{[ABC]} = \frac{1}{7}}.$$

□

Problem 38. (1995 KyotoU Entrance Examination). For a positive integer n, let $f(n)$ be the remainder when n is divided by 7, and define

$$g(n) = 3f\left(\sum_{k=1}^{7} k^n\right).$$

Prove that

$$f(n^7) = f(n)$$

for every positive integer n. Then determine the maximum possible value of $g(n)$.

Solution. Fermat's little theorem gives

$$n^7 \equiv n \pmod 7$$

for every integer n, including multiples of 7. Hence

$$f(n^7) = f(n).$$

Since f takes values in $\{0, 1, \ldots, 6\}$, we have

$$g(n) \leq 18$$

for every positive integer n. This upper bound is attained at $n = 6$. Indeed, for $1 \leq k \leq 6$,

$$k^6 \equiv 1 \pmod 7,$$

while $7^6 \equiv 0 \pmod 7$. Therefore

$$f\left(\sum_{k=1}^{7} k^6\right) = 6,$$

and the maximum possible value is

$$\boxed{18}.$$

□

Problem 39. (1993 Tokyo Tech Entrance Examination). Let $P(x)$ be a polynomial of degree n. Suppose that

$$P(0), P(1), \ldots, P(n)$$

are all integers. Prove that $P(m)$ is an integer for every $m \in \mathbb{Z}$.

Solution. We use induction on the degree. The assertion is immediate for constant polynomials. Let

$$Q(x) = P(x+1) - P(x).$$

Then Q has degree at most $n-1$, and

$$Q(0), Q(1), \ldots, Q(n-1)$$

are integers. By the induction hypothesis,

$$Q(m) \in \mathbb{Z}$$

for every $m \in \mathbb{Z}$.

Since $P(0) \in \mathbb{Z}$, for $m > 0$ we have

$$P(m) = P(0) + \sum_{j=0}^{m-1} Q(j) \in \mathbb{Z}.$$

For $m < 0$,

$$P(m) = P(0) - \sum_{j=m}^{-1} Q(j) \in \mathbb{Z}.$$

Thus $P(m)$ is an integer for every integer m.

$\square$

Problem 40. (2009 Hitotsubashi Entrance Examination). Let $m, n \geq 2$ be integers satisfying

$$m^3 + 1 = n^3 + 10^3.$$

Determine m and n.

Solution. Rearranging and factoring gives

$$(m - n)(m^2 + mn + n^2) = 999.$$

Put $d = m - n > 0$. Then $d \mid 999$ and

$$3n^2 + 3dn + d^2 = \frac{999}{d}.$$

Since $n \geq 2$, the left-hand side exceeds d^2, so $d^3 < 999$. Hence the only possible positive divisors are

$$d = 1, 3, 9.$$

Substitution gives no integer $n \geq 2$ for $d = 1$, gives $n = 9$ for $d = 3$, and gives only $n = 1$ or a negative value for $d = 9$. Therefore

$$\boxed{(m, n) = (12, 9)}.$$

The common value is the taxicab number

$$12^3 + 1^3 = 9^3 + 10^3 = 1729.$$

$\square$

Problem 41. (2025 UTokyo Entrance Examination). For a positive integer a, define

$$f_a(x) = x^2 + x - a.$$

A square number means the square of a nonnegative integer. First prove that if n is a positive integer and $f_a(n)$ is a square number, then $n \leq a$.

Let N_a be the number of positive integers n for which $f_a(n)$ is a square number. Prove that

$$N_a = 1 \iff 4a + 1 \text{ is prime.}$$

Solution. Suppose that

$$n^2 + n - a = m^2$$

for some $m \in \mathbb{N}_0$. If $n \geq a+1$, then

$$n^2 < n^2 + n - a < (n+1)^2,$$

which is impossible for a square. Hence $n \leq a$.

Multiplying the equation by 4 and completing the square gives

$$(2n+1-2m)(2n+1+2m) = 4a+1.$$

Both factors are positive odd integers. Conversely, let

$$uv = 4a+1, \qquad 1 \leq u \leq v,$$

with u and v odd. Since $uv \equiv 1 \pmod 4$, the two factors are congruent modulo 4, and therefore

$$n = \frac{u+v-2}{4}, \qquad m = \frac{v-u}{4}$$

are nonnegative integers satisfying $f_a(n) = m^2$. This gives a bijection between positive factor pairs of $4a+1$ and admissible values of n.

The factorization

$$4a+1 = 1 \cdot (4a+1)$$

always gives the solution $n = a$. There is exactly one factor pair if and only if $4a+1$ is prime. Hence

$$\boxed{N_a = 1 \iff 4a+1 \text{ is prime}}.$$

□

Problem 42. (1977 IMO). Let

$$f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$$

satisfy

$$f(n+1) > f(f(n))$$

for every positive integer n. Prove that

$$f(n) = n$$

for every positive integer n.

Solution. We first prove, by induction on m, that

$$k \geq m \implies f(k) \geq m.$$

The assertion is immediate for $m = 1$. Suppose it holds for some m, and let $k \geq m+1$. Write $k = n+1$, so $n \geq m$. By the induction hypothesis,

$$f(n) \geq m,$$

and applying the same hypothesis once more gives

$$f(f(n)) \geq m.$$

Hence

$$f(k) = f(n+1) > f(f(n)) \geq m,$$

so $f(k) \geq m+1$. The induction is complete. Taking $k = m$ shows that

$$f(n) \geq n$$

for every n.

Applying this inequality with the input $f(n)$ gives

$$f(f(n)) \geq f(n).$$

Therefore

$$f(n+1) > f(f(n)) \geq f(n),$$

so f is strictly increasing.

If $f(n) > n$ for some n, then $f(n) \geq n+1$. Since f is increasing,

$$f(f(n)) \geq f(n+1),$$

contradicting the given inequality. Thus $f(n) \leq n$ for every n. Combined with $f(n) \geq n$, this yields

$$\boxed{f(n) = n}$$

for every positive integer n.

□

Problem 43. (1985 IMO). Let n and k be relatively prime positive integers with $k < n$. Each element of

$$M = \{1, 2, \ldots, n-1\}$$

is colored either blue or white. Suppose that

(1) i and $n-i$ have the same color for every $i \in M$;

(2) i and $|i-k|$ have the same color whenever $i \in M$ and $i \neq k$.

Prove that all elements of M have the same color.

Solution. For $1 \leq j \leq n-1$, let r_j be the least positive residue of jk modulo n. Since

$$\gcd(n, k) = 1,$$

the numbers

$$r_1, r_2, \ldots, r_{n-1}$$

are precisely the elements of M in some order.

We show that consecutive terms in this list have the same color. Let $1 \leq j \leq n-2$. The equality $r_j + k = n$ cannot occur, because it would imply

$$(j+1)k \equiv 0 \pmod{n}$$

with $1 \leq j+1 < n$.

If $r_j + k < n$, then

$$r_{j+1} = r_j + k.$$

Since $r_{j+1} \neq k$, the second coloring condition gives

$$r_{j+1} \sim |r_{j+1} - k| = r_j,$$

where $x \sim y$ means that x and y have the same color.

If $r_j + k > n$, then

$$r_{j+1} = r_j + k - n.$$

The first condition gives $r_j \sim n - r_j$, and the second gives

$$n - r_j \sim |n - r_j - k| = r_j + k - n = r_{j+1}.$$

Thus $r_j \sim r_{j+1}$ in either case. Consequently all the residues r_1, $ldots, r_{n-1}$, and hence all elements of M, have the same color.

□

Problem 44. (2001 IMO). Twenty-one girls and twenty-one boys took part in a mathematical competition. Each contestant solved at most six problems, and every girl–boy pair solved at least one problem in common. Prove that some problem was solved by at least three girls and at least three boys.

Solution. For each girl–boy pair, choose one problem that both contestants solved. Suppose, for contradiction, that no problem was solved by at least three girls and at least three boys.

Call a problem *girl-sparse* if it was solved by at most two girls. Fix a boy. The problems solved by this boy must collectively cover all 21 girls. Since he solved at most six problems, at least one of those problems was solved by at least three girls. Hence he solved at most five girl-sparse problems. Each such problem accounts for at most two of his 21 chosen girl–boy pairs. Therefore, for this boy, at most

$$5 \cdot 2 = 10$$

chosen pairs use a girl-sparse problem. Over all boys, there are at most

$$21 \cdot 10 = 210$$

such pairs.

By symmetry, at most 210 chosen pairs use a problem solved by at most two boys. Under the contrary assumption, every problem is girl-sparse or is solved by at most two boys. Thus all

$$21^2 = 441$$

chosen pairs would belong to the union of two sets containing at most 210 pairs each. This is impossible because

$$441 > 210 + 210.$$

Therefore some problem was solved by at least three girls and at least three boys.

□

Problem 45. (2012 Putnam). Let

$$d_1, d_2, \ldots, d_{12}$$

be real numbers in the open interval $(1, 12)$. Prove that there exist distinct indices i, j, k s.t. d_i, d_j, d_k are the side lengths of an acute triangle.

Solution. Reorder the numbers so that

$$d_1 \leq d_2 \leq \cdots \leq d_{12}.$$

Suppose that no three of them are the side lengths of an acute triangle. For every $r \geq 3$, the three consecutive numbers d_{r-2}, d_{r-1}, d_r then satisfy

$$d_r^2 \geq d_{r-1}^2 + d_{r-2}^2.$$

Indeed, the reverse strict inequality would imply both the triangle inequality and, by the converse of the Pythagorean theorem, that the triangle is acute.

Let $F_1 = F_2 = 1$ and $F_r = F_{r-1} + F_{r-2}$. Since

$$d_1^2 > 1 = F_1, \qquad d_2^2 > 1 = F_2,$$

induction gives

$$d_r^2 > F_r$$

for every r. In particular,

$$d_{12}^2 > F_{12} = 144,$$

so $d_{12} > 12$, contradicting $d_{12} \in (1, 12)$. Therefore the required three indices exist.

□

Problem 46. (2017 Putnam). Let S be the smallest set of positive integers satisfying the following conditions:

(1) $2 \in S$;

(2) if $n^2 \in S$, then $n \in S$;

(3) if $n \in S$, then $(n+5)^2 \in S$.

Determine all positive integers that do not belong to S.

Solution. Let

$$U = \{n \in \mathbb{Z}_{>0} \mid (n \neq 1) \wedge (5 \nmid n)\}.$$

The set U contains 2 and satisfies both closure conditions. Hence, by the minimality of S,

$$S \subseteq U.$$

We prove the reverse inclusion by showing that every set T satisfying the three conditions contains U. Combining the second and third conditions gives

$$n \in T \quad \Longrightarrow \quad n + 5 \in T.$$

Thus T is closed under adding any nonnegative multiple of 5.

Starting from $2 \in T$, the following chain is valid:

$$2 \longrightarrow 49 \longrightarrow 54^2 \longrightarrow 56^2 \longrightarrow 56 \longrightarrow 121 \longrightarrow 11.$$

Here the first two arrows use the third condition, the arrows from 54^2 to 56^2 and from 56 to 121 use closure under adding multiples of 5, and the square-root arrows use the second condition. From $11 \in T$, closure under adding multiples of 5 gives

$$16 \in T, \qquad 36 \in T.$$

Taking square roots gives $4, 6 \in T$. Since $4 \in T$, adding 5 gives $9 \in T$, and taking a square root gives $3 \in T$. Hence

$$2, 3, 4, 6 \in T.$$

Repeatedly adding 5 now shows that T contains every positive integer other than 1 and the positive multiples of 5. Thus $U \subseteq T$ for every admissible T, and in particular $U \subseteq S$.

Therefore the positive integers not in S are exactly

$$\boxed{\text{1 and the positive multiples of 5}}.$$

□

Problem 47. (Fagnano's Problem). Let ABC be an acute triangle. Choose points

$$P \in \overline{BC}, \qquad Q \in \overline{CA}, \qquad R \in \overline{AB}.$$

Determine the triangle PQR whose perimeter

$$PQ + QR + RP$$

is as small as possible.

Solution. Reflect P across the lines CA and AB to points P_1 and P_2, respectively. Since $Q \in CA$ and $R \in AB$,

$$PQ = P_1Q, \qquad PR = P_2R.$$

Therefore

$$PQ + QR + RP = P_1Q + QR + RP_2 \geq P_1P_2.$$

Moreover,

$$AP_1 = AP_2 = AP, \qquad \angle P_1AP_2 = 2\angle A,$$

so

$$P_1P_2 = 2AP \sin A.$$

This quantity is minimized when AP is the altitude from A to BC.

Let D, E, F be the feet of the altitudes from A, B, C, respectively. When $P = D$, the right-angle cyclic quadrilaterals determined by the altitude feet show that the reflections D_1, D_2 of D across CA and AB satisfy

$$D_1, E, F, D_2$$

collinear. Hence equality holds in the preceding triangle inequality for

$$(P, Q, R) = (D, E, F).$$

Thus the unique minimizing inscribed triangle is the orthic triangle of ABC.

□

Problem 48. (Fermat–Torricelli Problem). Given a triangle ABC, determine the point P in the plane for which

$$PA + PB + PC$$

is as small as possible.

Solution. First suppose that every angle of ABC is less than $120°$. The standard construction with outward equilateral triangles produces a unique interior point P satisfying

$$\angle APB = \angle BPC = \angle CPA = 120°.$$

Let $\mathbf{u}_A, \mathbf{u}_B, \mathbf{u}_C$ be the unit vectors from P toward A, B, C, respectively. Since these vectors are separated by angles of $120°$,

$$\mathbf{u}_A + \mathbf{u}_B + \mathbf{u}_C = \mathbf{0}.$$

For any point Q, let $\mathbf{q}$ be the vector from P to Q. The Cauchy–Schwarz inequality gives

$$QA \geq PA - \mathbf{u}_A \cdot \mathbf{q},$$

and similarly for B and C. Adding the three inequalities yields

$$QA + QB + QC \geq PA + PB + PC.$$

Thus this interior point is the minimizer.

Now suppose, without loss of generality, that

$$\angle BAC \geq 120°.$$

Let $\mathbf{u}$ and $\mathbf{v}$ be the unit vectors from A toward B and C, and let $\mathbf{q}$ be the vector from A to an arbitrary point Q. Then

$$QB \geq AB - \mathbf{u} \cdot \mathbf{q}, \qquad QC \geq AC - \mathbf{v} \cdot \mathbf{q}.$$

Since

$$\|\mathbf{u} + \mathbf{v}\| = 2 \cos \left(\frac{\angle BAC}{2} \right) \leq 1,$$

we also have

$$QA = \|\mathbf{q}\| \geq (\mathbf{u} + \mathbf{v}) \cdot \mathbf{q}.$$

Adding gives

$$QA + QB + QC \geq AB + AC,$$

with equality at $Q = A$. Therefore the minimizing point is

$$\boxed{\begin{cases} \text{the interior point making three } 120^\circ \text{ angles,} & \text{if all triangle angles are } < 120^\circ, \\ \text{the vertex whose angle is } \geq 120^\circ, & \text{otherwise.} \end{cases}}$$

□

Problem 49. (Napoleon's Theorem). On the three sides BC, CA, AB of an arbitrary triangle ABC, construct equilateral triangles externally. Let X, Y, Z be the centers of the equilateral triangles constructed on BC, CA, AB, respectively. Prove that XYZ is equilateral.

Solution. Identify the plane with $\mathbb{C}$, and let a, b, c be the complex coordinates of A, B, C, listed counterclockwise. Put

$$\rho = e^{-i\pi/3}.$$

The third vertices of the outward equilateral triangles on BC, CA, AB are

$$p = b + \rho(c - b), \qquad q = c + \rho(a - c), \qquad r = a + \rho(b - a),$$

respectively. Since the center of an equilateral triangle is its centroid, the complex coordinates x, y, z of X, Y, Z are

$$x = \frac{b + c + p}{3}, \qquad y = \frac{c + a + q}{3}, \qquad z = \frac{a + b + r}{3}.$$

Direct simplification gives

$$z - y = -\rho(y - x).$$

The complex number $-\rho$ has modulus 1 and argument $2\pi/3$. Hence the vector from Y to Z is obtained from the vector from X to Y by a rotation through 120°, without changing its length. Therefore

$$XY = YZ \qquad \text{and} \qquad \angle XYZ = 60^\circ,$$

so XYZ is equilateral.

□

Problem 50. (Butterfly Theorem). Let M be the midpoint of a chord PQ of a circle. Draw two other chords AB and CD through M, and define

$$X = AD \cap PQ, \qquad Y = BC \cap PQ.$$

Prove that M is the midpoint of XY.

Solution. Choose Cartesian coordinates with

$$M = (0,0)$$

and PQ as the x-axis. Since the perpendicular bisector of PQ passes through the center of the circle, its equation has the form

$$x^2 + (y-h)^2 = R^2.$$

Excluding only immediate degenerate cases, write the two chords through M as

$$AB: \ x = \alpha y, \qquad CD: \ x = \beta y, \qquad \alpha \neq \beta.$$

Let y_A, y_B, y_C, y_D be the y-coordinates of the corresponding points. Substitution into the circle equation and Viète's formulas give

$$\frac{y_A + y_B}{y_A y_B} = \frac{2h}{h^2 - R^2} = \frac{y_C + y_D}{y_C y_D}.$$

Write $X = (x_X, 0)$ and $Y = (x_Y, 0)$. Collinearity of A, X, D and of B, Y, C gives

$$x_X(y_A - y_D) = (\alpha - \beta) y_A y_D,$$

$$x_Y(y_B - y_C) = (\alpha - \beta) y_B y_C.$$

Hence

$$\frac{x_X + x_Y}{\alpha - \beta} = \frac{y_A y_B (y_C + y_D) - y_C y_D (y_A + y_B)}{(y_A - y_D)(y_B - y_C)} = 0.$$

Thus $x_X = -x_Y$, so X and Y are equidistant from the origin M. Therefore

$$\boxed{MX = MY}.$$

□

Problem 51. The number N of typhoons arriving in a given year is modeled by a Poisson distribution with parameter 1. Determine the probability that at least one typhoon arrives during the next year.

Solution. Since $N \sim \text{Poi}(1)$,

$$\mathbb{P}(N \geq 1) = 1 - \mathbb{P}(N = 0) = 1 - e^{-1}.$$

Therefore, the required probability is

$$\boxed{1 - \frac{1}{e}}.$$

□

Problem 52. Consider the functions

$$z^2, \qquad \sin z, \qquad \tan z, \qquad \text{Re}(z), \qquad e^z.$$

Determine:

(a) which functions are entire;

(b) which function is meromorphic on $\mathbb{C}$ but not entire;

(c) which function is not holomorphic on any nonempty open subset of $\mathbb{C}$.

Solution. The functions

$$z^2, \qquad \sin z, \qquad e^z$$

are entire. The function $\tan z$ is meromorphic but not entire because it has poles at

$$\frac{\pi}{2} + k\pi, \qquad k \in \mathbb{Z}.$$

Finally, $\mathrm{Re}(z)$ is not complex differentiable at any point: its difference quotient equals 1 along real increments and 0 along purely imaginary increments. Thus the answers are

$$\boxed{\begin{aligned}&\text{(a) } z^2,\ \sin z,\ e^z,\\ &\text{(b) } \tan z,\\ &\text{(c) } \mathrm{Re}(z).\end{aligned}}$$

□

Problem 53. In the usual topology on $\mathbb{R}$, determine whether each statement is true or false.

(1) There exists a perfect subset of $\mathbb{R}$ containing no nondegenerate interval.
(2) The set $\mathbb{Z}$ is open.
(3) The set $\mathbb{Z}$ is closed.
(4) There exists a countable subset of $\mathbb{R}$ that is neither open nor closed.

Solution. The Cantor set is perfect and contains no nondegenerate interval, so (1) is true. The set $\mathbb{Z}$ is not open, because no open interval centered at an integer lies inside $\mathbb{Z}$, but it is closed because $\mathbb{R} \setminus \mathbb{Z}$ is open. Hence (2) is false and (3) is true. Finally, $\mathbb{Q}$ is countable and is neither open nor closed in $\mathbb{R}$, so (4) is true. Therefore,

$$\boxed{(1)\ \mathrm{T}, \quad (2)\ \mathrm{F}, \quad (3)\ \mathrm{T}, \quad (4)\ \mathrm{T}}.$$

□

Problem 54. Which of the following integers can be the order of a nonabelian simple group?

(1) 60;
(2) 48;
(3) 30;
(4) 77.

Solution. The alternating group A_5 is simple and has order 60, so 60 is possible.

A group of order $77 = 7 \cdot 11$ has a unique Sylow 11-subgroup, since

$$n_{11} \equiv 1 \pmod{11}, \qquad n_{11} \mid 7,$$

so it is not simple. If a group of order 30 were simple, then it would have 6 Sylow 5-subgroups and 10 Sylow 3-subgroups. These would already contain

$$6(5-1) + 10(3-1) = 44$$

nonidentity elements, impossible in a group with only 29 nonidentity elements.

Finally, for a group of order 48, the number of Sylow 2-subgroups is 1 or 3. The first case gives a normal subgroup. In the second case, conjugation gives a homomorphism to S_3; it cannot be injective because $48 > 6$, so its kernel is a nontrivial proper normal subgroup. Hence the only possible order is

$$\boxed{60}.$$

□

Problem 55. Let
$$\mathbf{F}(x,y,z)=(z,x,y).$$
A plane through the origin intersects the sphere
$$x^2+y^2+z^2=4$$
in a great circle. Determine the plane for which the positively oriented circulation of $\mathbf{F}$ around this circle is as large as possible, and find the maximum circulation.

Solution. The curl is constant:
$$\nabla\times\mathbf{F}=(1,1,1).$$
Let $\mathbf{n}$ be the unit normal that determines the positive orientation of the circle. The circle bounds a disk of radius 2 and area 4π. By Stokes' Theorem,
$$\oint_{\partial S}\mathbf{F}\cdot\mathrm{d}\mathbf{r}=\iint_S(\nabla\times\mathbf{F})\cdot\mathbf{n}\,\mathrm{d}S=4\pi(1,1,1)\cdot\mathbf{n}.$$
Cauchy–Schwarz gives
$$(1,1,1)\cdot\mathbf{n}\le\sqrt{3},$$
with equality when
$$\mathbf{n}=\frac{1}{\sqrt{3}}(1,1,1).$$
Therefore the required plane and maximum circulation are
$$\boxed{x+y+z=0},\qquad\boxed{4\pi\sqrt{3}}.$$

□

Problem 56. (Cantor's Diagonal Argument). Let $\{0,1\}^{\mathbb{N}}$ be the set of all infinite binary sequences. Prove that no function
$$f:\mathbb{N}\to\{0,1\}^{\mathbb{N}}$$
is surjective.

Solution. Write
$$f(n)=(a_{n1},a_{n2},a_{n3},\ldots),\qquad a_{nk}\in\{0,1\}.$$
Define a binary sequence $b=(b_1,b_2,\ldots)$ by
$$b_n=1-a_{nn}.$$
For every $n\in\mathbb{N}$, the sequence b differs from $f(n)$ in its nth coordinate. Hence
$$b\ne f(n)\qquad\text{for every } n\in\mathbb{N}.$$
Therefore b is not in the image of f, so f is not surjective.

□

Problem 57. On $\mathbb{R}^2$, determine which of the following functions is not a metric. For $\mathbf{x}=(x_1,x_2)$ and $\mathbf{y}=(y_1,y_2)$, let

(1) $d(\mathbf{x},\mathbf{y})=\sqrt{(x_1-y_1)^2+(x_2-y_2)^2}$;

(2) $d(\mathbf{x},\mathbf{y})=\min\{|x_1-y_1|,|x_2-y_2|\}$;

(3) $d(\mathbf{x}, \mathbf{y}) = |x_1 - y_1| + |x_2 - y_2|$;
(4) $d(\mathbf{x}, \mathbf{y}) = 0$ if $\mathbf{x} = \mathbf{y}$ and $d(\mathbf{x}, \mathbf{y}) = 1$ otherwise.

Solution. Choice (2) is not a metric. For example,

$$\mathbf{x} = (0, 0), \qquad \mathbf{y} = (0, 1)$$

are distinct, but

$$d(\mathbf{x}, \mathbf{y}) = \min\{0, 1\} = 0.$$

Thus the identity-of-indiscernibles axiom fails. The other three functions are, respectively, the Euclidean, taxicab, and discrete metrics. Therefore, the answer is

$$\boxed{(2)}.$$

□

Direct Calculation Problems

Problem 1. Finite Sums and Products.

Evaluate each of the following finite sums or products exactly. State any restrictions on the integer parameters that are needed for the displayed expression to be defined.

1.1. (Vandermonde's Identity). For $n \in \mathbb{N}_0$, determine

$$\sum_{k=0}^{n} \binom{n}{k}^2.$$

Solution 1.1. This is the $r = s = n$ specialization of Vandermonde's identity from Mathematical Concepts. Its coefficient interpretation gives the same result immediately: since

$$\binom{n}{k} = \binom{n}{n-k},$$

the required sum is the coefficient of x^n in

$$(1 + x)^n (1 + x)^n = (1 + x)^{2n}.$$

Therefore,

$$\boxed{\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}}.$$

□

1.2. For $n \in \mathbb{N}_0$, determine

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{1}{k+1}.$$

Solution 1.2. Using

$$\frac{1}{k+1} = \int_0^1 x^k \, dx,$$

we obtain

$$\begin{aligned}\sum_{k=0}^{n}(-1)^k\binom{n}{k}\frac{1}{k+1} &= \int_0^1\sum_{k=0}^{n}\binom{n}{k}(-x)^k\,\mathrm{d}x\\ &= \int_0^1(1-x)^n\,\mathrm{d}x\\ &= \boxed{\frac{1}{n+1}}.\end{aligned}$$

□

1.3. (Sine Product Formula). For an integer $n \geq 2$, determine

$$\prod_{k=1}^{n-1}\sin\left(\frac{k\pi}{n}\right).$$

Solution 1.3. Let $\omega_n = e^{2\pi i/n}$. Factoring z^n-1 and then dividing by $z-1$ gives

$$\frac{z^n-1}{z-1} = \prod_{k=1}^{n-1}(z-\omega_n^k).$$

Letting $z \to 1$ and taking absolute values yields

$$n = \prod_{k=1}^{n-1}|1-\omega_n^k| = \prod_{k=1}^{n-1}2\sin\left(\frac{k\pi}{n}\right).$$

Hence

$$\boxed{\prod_{k=1}^{n-1}\sin\left(\frac{k\pi}{n}\right) = \frac{n}{2^{n-1}}}.$$

□

1.4. For an integer $n \geq 2$, determine

$$\prod_{k=2}^{n}\frac{k^3-1}{k^3+1}.$$

Solution 1.4. Factor each term as

$$\frac{k^3-1}{k^3+1} = \frac{k-1}{k+1}\frac{k^2+k+1}{k^2-k+1}.$$

The two products telescope separately:

$$\prod_{k=2}^{n}\frac{k-1}{k+1} = \frac{2}{n(n+1)},$$

and, since $k^2+k+1 = (k+1)^2-(k+1)+1$,

$$\prod_{k=2}^{n}\frac{k^2+k+1}{k^2-k+1} = \frac{n^2+n+1}{3}.$$

Therefore,

$$\boxed{\prod_{k=2}^{n}\frac{k^3-1}{k^3+1} = \frac{2(n^2+n+1)}{3n(n+1)}}.$$

□

1.5. For $m \in \mathbb{N}$ and $\sin x \neq 0$, determine

$$\prod_{k=0}^{m-1} \cos(2^k x).$$

Solution 1.5. Repeatedly apply

$$\sin(2t) = 2 \sin t \cos t.$$

This gives

$$\sin(2^m x) = 2^m \sin x \prod_{k=0}^{m-1} \cos(2^k x).$$

Since $\sin x \neq 0$,

$$\boxed{\prod_{k=0}^{m-1} \cos(2^k x) = \frac{\sin(2^m x)}{2^m \sin x}}.$$

□

1.6. For $n \in \mathbb{N}_0$, determine

$$\sum_{\substack{0 \le k \le n \\ 3 \mid k}} \binom{n}{k}.$$

Solution 1.6. Let

$$\omega = e^{2\pi i/3}.$$

Since

$$1 + \omega^k + \omega^{2k} = \begin{cases} 3, & 3 \mid k, \\ 0, & 3 \nmid k, \end{cases}$$

the roots-of-unity filter gives

$$\begin{aligned} 3S_n &= \sum_{k=0}^{n} \binom{n}{k} (1 + \omega^k + \omega^{2k}) \\ &= 2^n + (1+\omega)^n + (1+\omega^2)^n. \end{aligned}$$

Now

$$1 + \omega = e^{i\pi/3}, \qquad 1 + \omega^2 = e^{-i\pi/3}.$$

Therefore,

$$\boxed{S_n = \frac{2^n + 2\cos(n\pi/3)}{3}}.$$

□

1.7. (1997 CSAT). For a permutation

$$(a_1, a_2, a_3, a_4)$$

of $\{1, 2, 3, 4\}$, let s_k be the number of entries to the right of a_k that are smaller than a_k, for $k = 1, 2, 3$. Define

$$|(a_1, a_2, a_3, a_4)| = s_1 + s_2 + s_3.$$

Determine the sum of this quantity over all 24 permutations.

Solution 1.7. For each unordered pair of distinct elements, exactly half of the 24 permutations place the larger element before the smaller one. Thus each of the

$$\binom{4}{2} = 6$$

pairs contributes one inversion in exactly 12 permutations. Hence the total inversion number is

$$\boxed{6 \cdot 12 = 72}.$$

□

Problem 2. Recurrence Relations.

Determine the requested term, closed form, or residue for each of the following recursively defined sequences.

2.1. (2006 VTRMC). Let $F_0 = 0$, $F_1 = 1$, and

$$F_{n+2} = F_{n+1} + F_n.$$

Determine the last decimal digit of F_{2006}.

Solution 2.1. Modulo 2, the Fibonacci sequence has period 3:

$$0, 1, 1, 0, 1, 1, \ldots.$$

Hence

$$F_{2006} \equiv F_2 \equiv 1 \pmod{2}.$$

Also

$$F_{n+5} = 5F_{n+1} + 3F_n,$$

so modulo 5,

$$F_{n+5} \equiv 3F_n.$$

Iterating four times gives $F_{n+20} \equiv F_n \pmod{5}$. Therefore

$$F_{2006} \equiv F_6 = 8 \equiv 3 \pmod{5}.$$

The unique decimal digit that is odd and congruent to 3 modulo 5 is

$$\boxed{3}.$$

□

2.2. Let

$$a_0 = 2, \qquad a_1 = \sqrt{2}, \qquad a_{n+2} = \sqrt{2}\, a_{n+1} - a_n.$$

Determine a_{2026}.

Solution 2.2. The sequence

$$b_n = 2\cos\left(\frac{n\pi}{4}\right)$$

has the same initial values and satisfies

$$b_{n+2} = 2\cos\left(\frac{\pi}{4}\right) b_{n+1} - b_n = \sqrt{2}\, b_{n+1} - b_n.$$

Hence $a_n = b_n$. Since $2026 \equiv 2 \pmod 8$,

$$\boxed{a_{2026} = 2\cos\left(\frac{2026\pi}{4}\right) = 0}.$$

$\square$

2.3. Let $a_0 = 1$ and

$$a_{n+1} = \frac{1}{2}\left(a_n + \frac{2}{a_n}\right).$$

Determine a closed form for a_n.

Solution 2.3. Set

$$r_n = \frac{a_n - \sqrt{2}}{a_n + \sqrt{2}}.$$

Direct substitution into the recurrence gives

$$r_{n+1} = r_n^2.$$

Moreover,

$$r_0 = \frac{1 - \sqrt{2}}{1 + \sqrt{2}} = -(3 - 2\sqrt{2}).$$

Thus

$$r_n = \left(-(3 - 2\sqrt{2})\right)^{2^n}.$$

Solving the defining relation for a_n yields

$$\boxed{a_n = \sqrt{2}\,\frac{1 + r_n}{1 - r_n}, \qquad r_n = \left(-(3 - 2\sqrt{2})\right)^{2^n}}.$$

$\square$

2.4. Let

$$a_0 = 2\cos\left(\frac{2\pi}{7}\right), \qquad a_{n+1} = a_n^2 - 2.$$

Determine a_{2026}.

Solution 2.4. The double-angle identity gives

$$(2\cos\theta)^2 - 2 = 2\cos(2\theta).$$

Hence induction yields

$$a_n = 2\cos\left(\frac{2^{n+1}\pi}{7}\right).$$

Powers of 2 modulo 14 have period 3, and

$$2^{2027} \equiv 4 \pmod{14}.$$

Therefore,

$$\boxed{a_{2026} = 2\cos\left(\frac{4\pi}{7}\right)}.$$

$\square$

2.5. Let $a_0 = 0$ and

$$a_{n+1} = \frac{1 + a_n}{2 + a_n}.$$

Determine a closed form for a_n in terms of the Fibonacci sequence.

Solution 2.5. Let (F_n) be the Fibonacci sequence with $F_0 = 0$ and $F_1 = 1$. We claim that

$$a_n = \frac{F_{2n}}{F_{2n+1}}.$$

The formula holds for $n = 0$. If it holds for n, then

$$\begin{aligned} a_{n+1} &= \frac{F_{2n} + F_{2n+1}}{F_{2n} + 2F_{2n+1}} \\ &= \frac{F_{2n+2}}{F_{2n+3}}. \end{aligned}$$

Thus

$$\boxed{a_n = \frac{F_{2n}}{F_{2n+1}}}.$$

□

2.6. (2014 UTokyo Entrance Examination). A bag initially contains $a + 2$ white balls and one red ball, where a is a positive integer. After each draw, perform the following operation:

(1) if a white ball is drawn, replace the contents by a white balls and one red ball;

(2) if the red ball is drawn, remove it and leave the other balls unchanged.

Let p_n be the probability that the red ball is drawn on the nth draw. Determine p_n.

Solution 2.6. The first probability is

$$p_1 = \frac{1}{a + 3}.$$

If the red ball is drawn on the nth draw, it cannot be drawn next. If a white ball is drawn, the next state has $a + 1$ balls, exactly one of which is red. Therefore

$$p_{n+1} = \frac{1 - p_n}{a + 1}.$$

The fixed point of this recurrence is $1/(a + 2)$. Hence, with

$$q_n = p_n - \frac{1}{a + 2},$$

we have

$$q_{n+1} = -\frac{q_n}{a + 1}, \qquad q_1 = -\frac{1}{(a + 3)(a + 2)}.$$

Consequently,

$$\boxed{p_n = \frac{1}{a + 2} + \frac{(-1)^n}{(a + 3)(a + 2)(a + 1)^{n-1}}}$$

for $n \geq 1$. In particular,

$$p_2 = \frac{a + 2}{(a + 3)(a + 1)}.$$

□

Problem 3. Limits.

Evaluate each of the following limits. In parameterized problems, determine the exact value for every parameter regime stated in the subproblem.

3.1. (2012 KAIST Problem of the Week). Let $a_0 = 3$ and

$$a_n = a_{n-1} + \sqrt{a_{n-1}^2 + 3}, \qquad n \geq 1.$$

Determine

$$\lim_{n\to\infty} \frac{a_n}{2^n}.$$

Solution 3.1. Write

$$a_n = \sqrt{3}\cot\theta_n.$$

Since $a_0 = 3$, we may take $\theta_0 = \pi/6$. The half-angle identity

$$\cot\frac{\theta}{2} = \cot\theta + \csc\theta$$

transforms the recurrence into

$$a_n = \sqrt{3}\,(\cot\theta_{n-1} + \csc\theta_{n-1}) = \sqrt{3}\cot\frac{\theta_{n-1}}{2}.$$

Hence

$$\theta_n = \frac{\pi}{6\cdot 2^n} \qquad \text{and} \qquad a_n = \sqrt{3}\cot\left(\frac{\pi}{6\cdot 2^n}\right).$$

Using $t\cot t \to 1$ as $t \to 0$, we obtain

$$\boxed{\lim_{n\to\infty} \frac{a_n}{2^n} = \frac{6\sqrt{3}}{\pi}}.$$

□

3.2. (2013 KAIST Problem of the Week). For $a, b \in \mathbb{R}$, determine

$$\lim_{n\to\infty} n\left(1 - \frac{a}{n} - \frac{b\ln(n+1)}{n}\right)^n.$$

Solution 3.2. Put

$$c_n = a + b\ln(n+1).$$

Since $c_n/n \to 0$ and $c_n^2/n \to 0$, the logarithmic expansion gives

$$n\ln\left(1 - \frac{c_n}{n}\right) = -c_n + o(1).$$

Therefore,

$$n\left(1 - \frac{c_n}{n}\right)^n = e^{-a} n(n+1)^{-b}(1 + o(1)).$$

It follows that the limit is

$$\boxed{\begin{cases} \infty, & b < 1, \\ e^{-a}, & b = 1, \\ 0, & b > 1. \end{cases}}$$

□

3.3. (2014 KAIST Problem of the Week). For $x \geq 0$, let

$$f_n(x) = \frac{\prod_{k=1}^{n-1}(x+k)(x+k+1)}{(n!)^2}.$$

Determine

$$\lim_{n\to\infty} f_n(x).$$

Solution 3.3. The Gamma function gives

$$f_n(x) = \frac{\Gamma(n+x)\Gamma(n+x+1)}{\Gamma(x+1)\Gamma(x+2)\Gamma(n+1)^2}.$$

Using

$$\frac{\Gamma(n+\alpha)}{\Gamma(n+\beta)} \sim n^{\alpha-\beta},$$

we obtain

$$f_n(x) \sim \frac{n^{2x-1}}{\Gamma(x+1)\Gamma(x+2)}.$$

Consequently,

$$\boxed{\lim_{n\to\infty} f_n(x) = \begin{cases} 0, & 0 \leq x < \frac{1}{2}, \\ \frac{8}{3\pi}, & x = \frac{1}{2}, \\ \infty, & x > \frac{1}{2}. \end{cases}}$$

$\square$

3.4. (2016 KAIST Problem of the Week). For $a \geq 0$, determine

$$\lim_{n\to\infty} n \int_{-1}^{0} \left(x + \frac{x^2}{2} + e^{ax}\right)^n \mathrm{d}x.$$

Solution 3.4. Let

$$g(x) = x + \frac{x^2}{2} + e^{ax}.$$

On $[-1, 0]$, we have $-1/2 \leq g(x) \leq 1$, with equality $g(x) = 1$ only at $x = 0$. Moreover,

$$g(0) = 1 \qquad \text{and} \qquad g'(0) = 1 + a.$$

The contribution from every interval $[-1, -\delta]$ is exponentially small. On the remaining interval, set $x = -t/n$. For each fixed $t \geq 0$,

$$g\left(-\frac{t}{n}\right)^n \longrightarrow e^{-(1+a)t}.$$

A uniform exponential bound near the endpoint permits dominated convergence, and therefore

$$\begin{aligned} \lim_{n\to\infty} n \int_{-1}^{0} g(x)^n \,\mathrm{d}x &= \int_0^\infty e^{-(1+a)t} \,\mathrm{d}t \\ &= \boxed{\frac{1}{1+a}}. \end{aligned}$$

□

3.5. (2023 KAIST Problem of the Week). Determine

$$\lim_{n\to\infty}\left(\frac{\sum_{k=1}^{n+2} k^k}{\sum_{k=1}^{n+1} k^k}-\frac{\sum_{k=1}^{n+1} k^k}{\sum_{k=1}^{n} k^k}\right).$$

Solution 3.5. Let

$$S_n=\sum_{k=1}^{n} k^k.$$

The final two terms dominate the sum:

$$S_n=n^n\left(1+\frac{1}{en}+O\left(\frac{1}{n^2}\right)\right).$$

Hence

$$\begin{aligned}\frac{S_{n+1}}{S_n}&=(n+1)\left(1+\frac{1}{n}\right)^n\frac{1+\dfrac{1}{e(n+1)}+O(n^{-2})}{1+\dfrac{1}{en}+O(n^{-2})}\\&=en+\frac{e}{2}+O\left(\frac{1}{n}\right).\end{aligned}$$

Replacing n by $n+1$ and subtracting gives

$$\boxed{\lim_{n\to\infty}\left(\frac{S_{n+2}}{S_{n+1}}-\frac{S_{n+1}}{S_n}\right)=e}.$$

□

Problem 4. Derivatives.

Compute each requested derivative or derivative-dependent quantity exactly. For high-order derivatives, a closed form involving standard sequences or harmonic numbers is acceptable.

4.1. (2006 HMMT). A nonzero polynomial $f\in\mathbb{R}[x]$ satisfies

$$f(x)=f'(x)f''(x).$$

Determine the leading coefficient of f.

Solution 4.1. Suppose that the leading term of f is cx^n, where $c\neq 0$. The leading term of $f'f''$ is

$$c^2n^2(n-1)x^{2n-3}.$$

Comparing degrees gives

$$n=2n-3,$$

so $n=3$. Comparing leading coefficients then gives

$$c=18c^2.$$

Since $c\neq 0$,

$$\boxed{c=\frac{1}{18}}.$$

$\square$

4.2. (1993 VTRMC). For $x > 0$, define

$$f_1(x) = x, \qquad f_{n+1}(x) = x^{f_n(x)}.$$

Determine $f''_{2026}(1)$.

Solution 4.2. Since $f_n(1) = 1$ for every n, take logarithms:

$$\ln f_{n+1}(x) = f_n(x) \ln x.$$

Differentiating and evaluating at $x = 1$ shows inductively that

$$f'_n(1) = 1.$$

Put $g(x) = \ln f_{n+1}(x)$. Then

$$g'(1) = 1 \qquad \text{and} \qquad g''(1) = 2f'_n(1) - f_n(1) = 1.$$

Since $f_{n+1} = e^g$,

$$f''_{n+1}(1) = f_{n+1}(1)\big(g'(1)^2 + g''(1)\big) = 2.$$

Therefore,

$$\boxed{f''_{2026}(1) = 2}.$$

$\square$

4.3. Let

$$f(x) = \frac{1}{1 - x - x^2}.$$

Determine $f^{(2026)}(0)$ in terms of the Fibonacci sequence.

Solution 4.3. The ordinary generating function of the Fibonacci sequence is

$$\frac{1}{1 - x - x^2} = \sum_{n=0}^{\infty} F_{n+1} x^n,$$

where $F_0 = 0$ and $F_1 = 1$. Therefore the coefficient of x^{2026} is F_{2027}, and

$$\boxed{f^{(2026)}(0) = 2026!\, F_{2027}}.$$

$\square$

4.4. Determine

$$\frac{\mathrm{d}^{2026}}{\mathrm{d}x^{2026}} (e^x \sin x)\bigg|_{x=0}.$$

Solution 4.4. Since

$$e^x \sin x = \operatorname{Im}\big(e^{(1+i)x}\big),$$

the required derivative is

$$\operatorname{Im}\big((1+i)^{2026}\big).$$

Now

$$(1+i)^{2026} = 2^{1013} e^{i2026\pi/4} = 2^{1013} i.$$

Hence the value is

$$\boxed{2^{1013}}.$$

□

4.5. Let

$$H_n = \sum_{k=1}^{n} \frac{1}{k}.$$

Determine

$$\frac{\mathrm{d}^{100}}{\mathrm{d}x^{100}}(x^{100}\ln x)\bigg|_{x=1}.$$

Solution 4.5. For a real parameter α,

$$\frac{\mathrm{d}^{100}}{\mathrm{d}x^{100}}x^{\alpha} = \alpha(\alpha-1)\cdots(\alpha-99)x^{\alpha-100}.$$

Differentiate both sides with respect to α and then set $\alpha = 100$ and $x = 1$. Since

$$\frac{\mathrm{d}}{\mathrm{d}\alpha}\prod_{j=0}^{99}(\alpha-j)\Bigg|_{\alpha=100} = 100!\sum_{k=1}^{100}\frac{1}{k},$$

we obtain

$$\boxed{100!\,H_{100}}.$$

□

Problem 5. Series.

Determine whether each of the following series converges. For every convergent series, determine its exact sum. In parameterized problems, state all parameter values for which convergence occurs.

5.1. (2016 KAIST Problem of the Week). Let k be a positive integer, and define

$$a_n = \begin{cases} 1, & (k+1) \nmid n, \\ -k, & (k+1) \mid n. \end{cases}$$

Determine

$$\sum_{n=1}^{\infty} \frac{a_n}{n}.$$

Solution 5.1. At the end of the Nth complete block of length $k+1$, the partial sum is

$$\begin{aligned} \sum_{n=1}^{(k+1)N} \frac{a_n}{n} &= \sum_{n=1}^{(k+1)N} \frac{1}{n} - (k+1)\sum_{m=1}^{N} \frac{1}{(k+1)m} \\ &= H_{(k+1)N} - H_N. \end{aligned}$$

Since $H_N = \ln N + \gamma + o(1)$, these partial sums tend to $\ln(k+1)$. The terms inside the final incomplete block tend uniformly to zero, so the full series has the same limit. Therefore,

$$\boxed{\sum_{n=1}^{\infty} \frac{a_n}{n} = \ln(k+1)}.$$

□

5.2. (2017 KAIST Problem of the Week). Determine whether

$$\sum_{n=0}^{\infty} \frac{1}{4^n}\binom{2n}{n}$$

converges.

Solution 5.2. By the central binomial coefficient asymptotic,

$$\frac{1}{4^n}\binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}}.$$

Since the p-series $\sum n^{-1/2}$ diverges, the limit comparison test gives

$$\boxed{\sum_{n=0}^{\infty} \frac{1}{4^n}\binom{2n}{n} \text{ diverges}}.$$

□

5.3. (2014 KAIST Problem of the Week). Determine all positive integers ℓ for which

$$\sum_{n=1}^{\infty} \frac{n^3}{(n+1)(n+2)\cdots(n+\ell)}$$

converges, and compute its value whenever it converges.

Solution 5.3. The nth term is asymptotic to $n^{3-\ell}$, so the series converges exactly when

$$\ell \geq 5.$$

For such ℓ, the beta-integral identity gives

$$\frac{1}{(n+1)(n+2)\cdots(n+\ell)} = \frac{1}{(\ell-1)!}\int_0^1 x^n(1-x)^{\ell-1}\,\mathrm{d}x.$$

Since

$$\sum_{n=1}^{\infty} n^3x^n = \frac{x(1+4x+x^2)}{(1-x)^4},$$

term-by-term integration yields

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{n^3}{(n+1)\cdots(n+\ell)} &= \frac{1}{(\ell-1)!}\int_0^1 x(1+4x+x^2)(1-x)^{\ell-5}\,\mathrm{d}x \\ &= \boxed{\frac{\ell(\ell+5)}{(\ell-4)(\ell-3)(\ell-2)(\ell-1)(\ell-1)!}}.\end{aligned}$$

□

5.4. Determine

$$\sum_{n=1}^{\infty} \frac{1}{n^2\binom{2n}{n}}.$$

Solution 5.4. The Taylor expansion

$$2\arcsin^2\left(\frac{x}{2}\right) = \sum_{n=1}^{\infty} \frac{x^{2n}}{n^2\binom{2n}{n}}$$

holds for $|x| \le 2$. It may be verified by differentiating both sides and using the power series for $1/\sqrt{1-x^2/4}$.

Setting $x = 1$ gives

$$\sum_{n=1}^{\infty} \frac{1}{n^2\binom{2n}{n}} = 2\arcsin^2\left(\frac{1}{2}\right) = 2\left(\frac{\pi}{6}\right)^2 = \boxed{\frac{\pi^2}{18}}.$$

□

5.5. Evaluate

$$\sum_{n=1}^{\infty} \frac{n}{(n+1)!}$$

in two ways.

Solution 5.5. First,

$$\frac{n}{(n+1)!} = \frac{1}{n!} - \frac{1}{(n+1)!},$$

so the series telescopes:

$$\sum_{n=1}^{N} \frac{n}{(n+1)!} = 1 - \frac{1}{(N+1)!} \longrightarrow 1.$$

Alternatively, put $m = n+1$ and use the exponential series:

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{n}{(n+1)!} &= \sum_{m=2}^{\infty} \frac{m-1}{m!} \\ &= \sum_{m=2}^{\infty} \frac{1}{(m-1)!} - \sum_{m=2}^{\infty} \frac{1}{m!} \\ &= (e-1) - (e-2) = 1.\end{aligned}$$

Thus both methods give

$$\boxed{1}.$$

□

5.6. (1998 CSAT). Let

$$a_1 = 1, \qquad a_2 = 2, \qquad a_{n+2} = a_{n+1} + a_n.$$

Determine

$$\sum_{n=1}^{\infty} \frac{a_n}{a_{n+1}a_{n+2}}.$$

Solution 5.6. The recurrence gives

$$\frac{a_n}{a_{n+1}a_{n+2}} = \frac{a_{n+2} - a_{n+1}}{a_{n+1}a_{n+2}} = \frac{1}{a_{n+1}} - \frac{1}{a_{n+2}}.$$

Thus the series telescopes:

$$\sum_{n=1}^{N} \frac{a_n}{a_{n+1}a_{n+2}} = \frac{1}{a_2} - \frac{1}{a_{N+2}}.$$

Since $a_n \to \infty$,

$$\boxed{\sum_{n=1}^{\infty} \frac{a_n}{a_{n+1}a_{n+2}} = \frac{1}{2}}.$$

$\square$

Problem 6. Integrals.

Evaluate each of the following integrals. For indefinite integrals, the constant of integration may be omitted; for definite integrals, determine the exact value.

6.1. (2010 MIT Integration Bee). $\displaystyle\int_0^{\pi/2} \sin x \sin(2x) \sin(3x)\, \mathrm{d}x$

Solution 6.1. By the product-to-sum identities,

$$\sin x \sin(3x) = \frac{1}{2}(\cos(2x) - \cos(4x)),$$

and hence

$$\sin x \sin(2x) \sin(3x) = \frac{1}{4}(\sin(2x) + \sin(4x) - \sin(6x)).$$

Therefore,

$$\int_0^{\pi/2} \sin x \sin(2x) \sin(3x)\, \mathrm{d}x = \frac{1}{4}\left(1 - \frac{1}{3}\right) = \boxed{\frac{1}{6}}.$$

$\square$

6.2. (2010 MIT Integration Bee). $\displaystyle\int_0^1 \sin^2(\ln x)\, \mathrm{d}x$

Solution 6.2. Let $t = -\ln x$. Then $x = e^{-t}$ and $\mathrm{d}x = -e^{-t}\, \mathrm{d}t$, so

$$\int_0^1 \sin^2(\ln x)\, \mathrm{d}x = \int_0^\infty e^{-t} \sin^2 t\, \mathrm{d}t.$$

Since $\sin^2 t = (1 - \cos(2t))/2$,

$$\int_0^\infty e^{-t} \sin^2 t\, \mathrm{d}t = \frac{1}{2}\left(1 - \frac{1}{5}\right) = \boxed{\frac{2}{5}}.$$

$\square$

6.3. (2010 MIT Integration Bee). $\displaystyle\int_1^\infty \frac{\mathrm{d}x}{x\sqrt{x^4 - 1}}$

Solution 6.3. Let $u = x^2$. Then

$$\frac{\mathrm{d}x}{x} = \frac{\mathrm{d}u}{2u},$$

and therefore

$$\int_1^\infty \frac{\mathrm{d}x}{x\sqrt{x^4 - 1}} = \frac{1}{2}\int_1^\infty \frac{\mathrm{d}u}{u\sqrt{u^2 - 1}}.$$

With $u = \sec\theta$, this becomes

$$\frac{1}{2}\int_0^{\pi/2} \mathrm{d}\theta = \boxed{\frac{\pi}{4}}.$$

$\square$

6.4. (2010 MIT Integration Bee). $\displaystyle\int_0^1 \frac{\ln(1+x)}{1+x^2}\, \mathrm{d}x$

Solution 6.4. Let

$$I = \int_0^1 \frac{\ln(1+x)}{1+x^2}\, \mathrm{d}x.$$

Substituting $x = \tan t$ gives

$$I = \int_0^{\pi/4} \ln(1 + \tan t)\,\mathrm{d}t.$$

Under $t \mapsto \pi/4 - t$,

$$1 + \tan\left(\frac{\pi}{4} - t\right) = \frac{2}{1 + \tan t}.$$

Thus

$$I = \frac{\pi}{4}\ln 2 - I,$$

and consequently

$$\boxed{I = \frac{\pi}{8}\ln 2}.$$

□

6.5. (2010 MIT Integration Bee). $\displaystyle\int \frac{\cos x + x\sin x}{x(x + \cos x)}\,\mathrm{d}x$

Solution 6.5. We recognize the logarithmic derivative

$$\frac{\mathrm{d}}{\mathrm{d}x}(\ln|x| - \ln|x + \cos x|) = \frac{\cos x + x\sin x}{x(x + \cos x)}.$$

Hence

$$\boxed{\int \frac{\cos x + x\sin x}{x(x + \cos x)}\,\mathrm{d}x = \ln|x| - \ln|x + \cos x|}.$$

□

6.6. (2010 MIT Integration Bee). $\displaystyle\int_0^{\pi/2} \frac{\mathrm{d}x}{\sin x + \sec x}$

Solution 6.6. Write

$$I = \int_0^{\pi/2} \frac{\cos x}{1 + \sin x\cos x}\,\mathrm{d}x.$$

Replacing x by $\pi/2 - x$ also gives

$$I = \int_0^{\pi/2} \frac{\sin x}{1 + \sin x\cos x}\,\mathrm{d}x.$$

Adding the two expressions and setting $u = \sin x - \cos x$, we obtain

$$2I = \int_{-1}^{1} \frac{2}{3 - u^2}\,\mathrm{d}u.$$

Therefore,

$$\boxed{I = \frac{1}{\sqrt{3}}\ln(2 + \sqrt{3})}.$$

□

6.7. (2010 MIT Integration Bee). $\displaystyle\int_0^1 \sqrt{1 + x\sqrt{1 + x\sqrt{1 + \cdots}}}\,\mathrm{d}x$

Solution 6.7. For $x \in [0, 1]$, define the finite truncations by

$$R_1(x) = 1, \qquad R_{n+1}(x) = \sqrt{1 + xR_n(x)}.$$

Since $y \mapsto \sqrt{1+xy}$ is increasing for $x \in [0,1]$, induction gives

$$1 \le R_n(x) \le \varphi \qquad \text{and} \qquad R_n(x) \le R_{n+1}(x),$$

where $\varphi = (1+\sqrt{5})/2$ and $\varphi^2 = 1+\varphi$. Hence $R_n(x)$ converges pointwise to a limit $R(x)$. Passing to the limit in the recurrence gives

$$R(x)^2 = 1 + xR(x),$$

so positivity yields

$$R(x) = \frac{x+\sqrt{x^2+4}}{2}.$$

The uniform bound $0 \le R_n(x) \le \varphi$ permits integration by the dominated convergence theorem. Therefore,

$$\begin{aligned}\int_0^1 R(x)\,\mathrm{d}x &= \frac{1}{2}\int_0^1 x\,\mathrm{d}x + \frac{1}{2}\int_0^1 \sqrt{x^2+4}\,\mathrm{d}x \\ &= \boxed{\frac{1+\sqrt{5}}{4} + \ln\left(\frac{1+\sqrt{5}}{2}\right)}.\end{aligned}$$

□

6.8. (2010 MIT Integration Bee). $\displaystyle\int \left(\frac{1}{\ln x} - \frac{1}{(\ln x)^2}\right)\mathrm{d}x, \qquad x>0, \quad x \neq 1$

Solution 6.8. On any interval contained in $(0,\infty)\setminus\{1\}$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x}{\ln x}\right) = \frac{1}{\ln x} - \frac{1}{(\ln x)^2}.$$

Therefore,

$$\boxed{\int \left(\frac{1}{\ln x} - \frac{1}{(\ln x)^2}\right)\mathrm{d}x = \frac{x}{\ln x}}.$$

□

6.9. (2010 MIT Integration Bee). $\displaystyle\int_1^2 \sqrt{x-1}\sqrt{2-x}\,\mathrm{d}x$

Solution 6.9. Let $u = x-1$. Then

$$\int_1^2 \sqrt{x-1}\sqrt{2-x}\,\mathrm{d}x = \int_0^1 \sqrt{u(1-u)}\,\mathrm{d}u.$$

Setting $v = u - 1/2$ gives

$$\sqrt{u(1-u)} = \sqrt{\frac{1}{4} - v^2}.$$

Thus the integral is the area of a semicircle of radius $1/2$:

$$\boxed{\frac{\pi}{8}}.$$

□

6.10. (2011 MIT Integration Bee). $\displaystyle\int \cos(\ln x)\,\mathrm{d}x, \qquad x>0$

Solution 6.10. Let $t = \ln x$. Then $x = e^t$ and $\mathrm{d}x = e^t\,\mathrm{d}t$, so

$$\int \cos(\ln x)\,\mathrm{d}x = \int e^t \cos t\,\mathrm{d}t.$$

Therefore,

$$\boxed{\int \cos(\ln x)\,\mathrm{d}x = \frac{x}{2}(\cos(\ln x) + \sin(\ln x))}.$$

□

6.11. (2011 MIT Integration Bee). $\displaystyle\int_0^\pi \cos x \cos(3x)\cos(5x)\,\mathrm{d}x$

Solution 6.11. By repeated use of the product-to-sum identity,

$$\cos x \cos(3x)\cos(5x) = \frac{1}{4}(\cos x + \cos(3x) + \cos(7x) + \cos(9x)).$$

Each term has integral 0 on $[0, \pi]$. Hence

$$\boxed{\int_0^\pi \cos x \cos(3x)\cos(5x)\,\mathrm{d}x = 0}.$$

□

6.12. (2011 MIT Integration Bee). $\displaystyle\int \sin(101x)\sin^{99} x\,\mathrm{d}x$

Solution 6.12. Differentiating a suitable product gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{100}\sin(100x)\sin^{100} x\right) &= \sin^{99} x(\sin x\cos(100x) + \cos x \sin(100x)) \\ &= \sin(101x)\sin^{99} x.\end{aligned}$$

Therefore,

$$\boxed{\int \sin(101x)\sin^{99} x\,\mathrm{d}x = \frac{1}{100}\sin(100x)\sin^{100} x}.$$

□

6.13. (2011 MIT Integration Bee). $\displaystyle\int x e^{e^{x^2}+x^2}\,\mathrm{d}x$

Solution 6.13. Since

$$e^{e^{x^2}+x^2} = e^{e^{x^2}} e^{x^2},$$

let $u = e^{x^2}$. Then $\mathrm{d}u = 2xe^{x^2}\,\mathrm{d}x$, and

$$\int x e^{e^{x^2}+x^2}\,\mathrm{d}x = \frac{1}{2}\int e^u\,\mathrm{d}u.$$

Hence

$$\boxed{\int x e^{e^{x^2}+x^2}\,\mathrm{d}x = \frac{1}{2}e^{e^{x^2}}}.$$

□

6.14. (2011 MIT Integration Bee). $\displaystyle\int \sqrt{\frac{1-x}{1+x}}\,\mathrm{d}x, \qquad -1 < x < 1$

Solution 6.14. For $-1 < x < 1$,

$$\sqrt{\frac{1-x}{1+x}} = \frac{1-x}{\sqrt{1-x^2}}.$$

Therefore,

$$\int \sqrt{\frac{1-x}{1+x}}\,\mathrm{d}x = \int \frac{\mathrm{d}x}{\sqrt{1-x^2}} - \int \frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x$$
$$= \boxed{\arcsin x + \sqrt{1-x^2}}.$$

□

6.15. (Gabriel's Horn). Determine the volume and the area of the curved surface obtained by rotating

$$y = \frac{1}{x}, \qquad x \geq 1,$$

about the x-axis.

Solution 6.15. The volume is

$$V = \pi \int_1^\infty \frac{1}{x^2}\,\mathrm{d}x = \boxed{\pi}.$$

The curved surface area is

$$S = 2\pi \int_1^\infty \frac{1}{x}\sqrt{1+\frac{1}{x^4}}\,\mathrm{d}x.$$

Since the integrand is at least $1/x$, this integral dominates

$$2\pi \int_1^\infty \frac{1}{x}\,\mathrm{d}x,$$

which diverges. Hence

$$\boxed{S = \infty}.$$

□

6.16. (Napkin Ring Problem). A cylindrical hole is drilled through the center of a sphere. The remaining solid has height 6. Determine its volume without being given either the sphere radius or the hole radius.

Solution 6.16. Let the original sphere radius be R and the cylindrical hole radius be r. If the remaining solid has height 6, then its half-height is 3, so

$$R^2 - r^2 = 9.$$

At height $z \in [-3, 3]$, the cross-section of the remaining solid is an annulus of area

$$\pi\big((R^2 - z^2) - r^2\big) = \pi(9 - z^2).$$

Therefore,

$$V = \pi \int_{-3}^{3} (9 - z^2)\,\mathrm{d}z = \boxed{36\pi}.$$

□

6.17. (Steinmetz Solid). Two infinite circular cylinders of radius R have perpendicular axes that intersect. Determine the volume of their intersection.

Solution 6.17. Choose coordinates so that the cylinders are

$$x^2 + y^2 \leq R^2, \qquad x^2 + z^2 \leq R^2.$$

For fixed $x \in [-R, R]$, both y and z range independently over

$$\left[-\sqrt{R^2 - x^2}, \sqrt{R^2 - x^2}\right].$$

The cross-section is therefore a square of area

$$4(R^2 - x^2).$$

Hence

$$V = \int_{-R}^{R} 4(R^2 - x^2)\,\mathrm{d}x = \boxed{\frac{16}{3}R^3}.$$

□

6.18. (Dirichlet Integral). Determine the improper integral

$$\int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x.$$

Solution 6.18. For $a > 0$, define

$$F(a) = \int_0^\infty e^{-ax}\frac{\sin x}{x}\,\mathrm{d}x.$$

Differentiation under the integral sign gives

$$F'(a) = -\int_0^\infty e^{-ax}\sin x\,\mathrm{d}x = -\frac{1}{1+a^2}.$$

Also $F(a) \to 0$ as $a \to \infty$. Therefore,

$$F(a) = \int_a^\infty \frac{\mathrm{d}t}{1+t^2} = \frac{\pi}{2} - \arctan a.$$

Letting $a \to 0^+$ and applying Abel's limiting argument for the Dirichlet integral yields

$$\boxed{\int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}}.$$

□

6.19. (1944 Dalzell, On 22/7). Evaluate

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,\mathrm{d}x,$$

and use the result to prove that

$$\pi < \frac{22}{7}.$$

Solution 6.19. Polynomial division gives

$$\frac{x^4(1-x)^4}{1+x^2} = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}.$$

Therefore,

$$\begin{aligned} I &= \int_0^1 \left(x^6 - 4x^5 + 5x^4 - 4x^2 + 4\right)\mathrm{d}x - 4\int_0^1 \frac{\mathrm{d}x}{1+x^2} \\ &= \frac{22}{7} - \pi. \end{aligned}$$

Since the original integrand is positive on $(0,1)$,

$$0 < I = \frac{22}{7} - \pi,$$

and hence

$$\boxed{\pi < \frac{22}{7}}.$$

□

6.20. (Borwein Integrals). For $n \in \mathbb{N}_0$, define

$$I_n = \int_0^\infty \prod_{k=0}^{n} \frac{\sin(x/(2k+1))}{x/(2k+1)}\,\mathrm{d}x.$$

Determine how long the pattern $I_n = \pi/2$ persists, and show that the next case is smaller.

Solution 6.20. Put

$$a_k = \frac{1}{2k+1},$$

and let X_k be independent random variables uniformly distributed on $[-a_k, a_k]$. The characteristic function of X_k is

$$\frac{\sin(a_k t)}{a_k t}.$$

If $S_n = X_0 + \cdots + X_n$ has density f_n, Fourier inversion at the origin gives

$$I_n = \pi f_n(0).$$

Write $S_n = X_0 + Y_n$, where X_0 is uniform on $[-1, 1]$. If

$$\sum_{k=1}^{n} a_k \le 1,$$

then $|Y_n| \le 1$ always, and conditioning on Y_n gives

$$f_n(0) = \frac{1}{2}.$$

Thus $I_n = \pi/2$. Now

$$\sum_{k=1}^{6} \frac{1}{2k+1} = \frac{43024}{45045} < 1,$$

whereas

$$\sum_{k=1}^{7} \frac{1}{2k+1} = \frac{46027}{45045} > 1.$$

Hence

$$\boxed{I_0 = I_1 = \cdots = I_6 = \frac{\pi}{2}}.$$

For $n = 7$, the continuous random variable Y_7 has positive probability of satisfying $|Y_7| > 1$. Therefore

$$f_7(0) = \frac{1}{2}\mathbb{P}(|Y_7| \le 1) < \frac{1}{2},$$

and consequently

$$\boxed{I_7 < \frac{\pi}{2}}.$$

□

6.21. (1987 Putnam). Determine

$$\int_2^4 \frac{\ln(9-x)}{\ln(9-x)+\ln(x+3)}\,\mathrm{d}x.$$

Solution 6.21. Let

$$h(x)=\frac{\ln(9-x)}{\ln(9-x)+\ln(x+3)}.$$

The substitution $x\mapsto 6-x$ gives

$$h(6-x)=1-h(x).$$

Since the interval $[2,4]$ is symmetric about 3,

$$2\int_2^4 h(x)\,\mathrm{d}x=\int_2^4 (h(x)+h(6-x))\,\mathrm{d}x=2.$$

Therefore the integral is

$$\boxed{1}.$$

□

6.22. (1989 UTokyo Entrance Examination). Let

$$f(x)=\pi x^2\sin(\pi x^2).$$

Determine the volume obtained by revolving the region bounded by $y=f(x)$, the x-axis, $x=0$, and $x=1$ about the y-axis.

Solution 6.22. Using cylindrical shells, the volume is

$$V=2\pi\int_0^1 xf(x)\,\mathrm{d}x=2\pi^2\int_0^1 x^3\sin(\pi x^2)\,\mathrm{d}x.$$

Let $u=x^2$. Then

$$V=\pi^2\int_0^1 u\sin(\pi u)\,\mathrm{d}u.$$

Integration by parts gives

$$\int_0^1 u\sin(\pi u)\,\mathrm{d}u=\frac{1}{\pi}.$$

Hence

$$\boxed{V=\pi}.$$

□

6.23. (2014 CSAT). Let $f:\mathbb{R}\to\mathbb{R}$ be continuous and odd. Suppose that

$$f(x)=\frac{\pi}{2}\int_1^{x+1} f(t)\,\mathrm{d}t$$

for every real x, and that $f(1)=1$. Determine

$$\pi^2\int_0^1 xf(x+1)\,\mathrm{d}x.$$

Solution 6.23. Differentiating the integral equation gives

$$f'(x)=\frac{\pi}{2}f(x+1),$$

so

$$f(x+1) = \frac{2}{\pi} f'(x).$$

Therefore the required expression is

$$2\pi \int_0^1 x f'(x)\,\mathrm{d}x.$$

Integration by parts yields

$$2\pi \left([xf(x)]_0^1 - \int_0^1 f(x)\,\mathrm{d}x \right) = 2\pi - 2\pi \int_0^1 f(x)\,\mathrm{d}x.$$

Since f is odd and $f(1) = 1$, we have $f(-1) = -1$. Substituting $x = -1$ into the original equation gives

$$-1 = \frac{\pi}{2} \int_1^0 f(t)\,\mathrm{d}t = -\frac{\pi}{2} \int_0^1 f(t)\,\mathrm{d}t.$$

Thus

$$\int_0^1 f(t)\,\mathrm{d}t = \frac{2}{\pi},$$

and the required value is

$$\boxed{2\pi - 4}.$$

□

6.24. For a real parameter $p > 1$, evaluate

$$\int_0^\infty \frac{\mathrm{d}x}{1 + x^p}.$$

Solution 6.24. Set

$$t = \frac{x^p}{1 + x^p}.$$

Then

$$x^p = \frac{t}{1-t}$$

and the integral becomes

$$\frac{1}{p} \int_0^1 t^{1/p-1} (1-t)^{-1/p}\,\mathrm{d}t = \frac{1}{p} B\left(\frac{1}{p}, 1 - \frac{1}{p}\right).$$

By the beta–gamma identity and Euler's reflection formula,

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin(\pi z)},$$

so

$$\boxed{\int_0^\infty \frac{\mathrm{d}x}{1 + x^p} = \frac{\pi}{p} \csc\left(\frac{\pi}{p}\right)}.$$

□

Problem 7. Complex Numbers.

Evaluate each of the following expressions exactly. When roots of unity occur, use $\omega_n = e^{2\pi i/n}$ unless another root is specified.

7.1. Determine $(1+i)^{2026}$.

Solution 7.1. Using polar form,

$$1+i=\sqrt{2}\,e^{i\pi/4}.$$

Hence

$$(1+i)^{2026}=2^{1013}e^{i2026\pi/4}=\boxed{2^{1013}i}.$$

□

7.2. Let $z\in\mathbb{C}\setminus\{0\}$ satisfy

$$z+\frac{1}{z}=1.$$

Determine

$$z^{2026}+\frac{1}{z^{2026}}.$$

Solution 7.2. The equation is equivalent to

$$z^2-z+1=0.$$

Thus $z=e^{\pm i\pi/3}$, and therefore

$$z^m+z^{-m}=2\cos\left(\frac{m\pi}{3}\right).$$

Since $2026\equiv 4 \pmod 6$,

$$\boxed{z^{2026}+z^{-2026}=2\cos\left(\frac{4\pi}{3}\right)=-1}.$$

□

7.3. For an integer $n\geq 2$, determine

$$\sum_{k=0}^{n-1}(1+\omega_n^k)^n.$$

Solution 7.3. Expand by the binomial theorem and interchange the finite sums:

$$\sum_{k=0}^{n-1}(1+\omega_n^k)^n=\sum_{j=0}^{n}\binom{n}{j}\sum_{k=0}^{n-1}\omega_n^{kj}.$$

The inner sum is n when $n\mid j$ and 0 otherwise. For $0\leq j\leq n$, only $j=0$ and $j=n$ contribute. Hence

$$\boxed{\sum_{k=0}^{n-1}(1+\omega_n^k)^n=2n}.$$

□

7.4. (Quadratic Gauss Sum). Let $\omega=e^{2\pi i/5}$. Determine

$$\sum_{k=0}^{4}\omega^{k^2}.$$

Solution 7.4. The quadratic residues modulo 5 are

$$0,1,4,4,1.$$

Therefore,

$$\sum_{k=0}^{4} \omega^{k^2} = 1 + 2(\omega + \omega^{-1}) = 1 + 4\cos\left(\frac{2\pi}{5}\right).$$

Since

$$2\cos\left(\frac{2\pi}{5}\right) = \frac{\sqrt{5}-1}{2},$$

the sum is

$$\boxed{\sqrt{5}}.$$

□

7.5. For an integer $n \geq 2$, determine

$$\sum_{k=0}^{n-1} \frac{1}{2 - \omega_n^k}.$$

Solution 7.5. Let

$$p(z) = z^n - 1 = \prod_{k=0}^{n-1} (z - \omega_n^k).$$

Its logarithmic derivative is

$$\frac{p'(z)}{p(z)} = \sum_{k=0}^{n-1} \frac{1}{z - \omega_n^k}.$$

Setting $z = 2$ gives

$$\boxed{\sum_{k=0}^{n-1} \frac{1}{2 - \omega_n^k} = \frac{n2^{n-1}}{2^n - 1}}.$$

□

Problem 8. Polynomials and Roots.

Use the coefficients, factorization, or symmetric functions of the roots to compute each requested quantity exactly. Solving every polynomial explicitly is not required unless the problem asks for the roots.

8.1. (2006 HMMT). Determine the sum of the distinct real roots of

$$2x^6 - 3x^5 + 3x^4 + x^3 - 3x^2 + 3x - 1 = 0.$$

Solution 8.1. The polynomial factors as

$$2x^6 - 3x^5 + 3x^4 + x^3 - 3x^2 + 3x - 1 = (x^2 - x + 1)^2(2x^2 + x - 1).$$

The first factor has no real roots. The distinct real roots are therefore the roots of

$$2x^2 + x - 1 = (2x - 1)(x + 1),$$

namely $1/2$ and -1. Their sum is

$$\boxed{-\frac{1}{2}}.$$

□

8.2. (1997 VTRMC). Two roots r_1, r_2 of

$$x^4 - x^3 + ax^2 - 8x - 8 = 0$$

satisfy $r_1 r_2 = 2$. Determine a, r_1, and r_2.

Solution 8.2. Since $r_1 r_2 = 2$, their quadratic factor has the form

$$x^2 + px + 2.$$

The remaining factor must have constant term -4, so write

$$x^4 - x^3 + ax^2 - 8x - 8 = (x^2 + px + 2)(x^2 + qx - 4).$$

Comparing the coefficients of x^3 and x gives

$$p + q = -1, \qquad 2q - 4p = -8.$$

Hence $p = 1$ and $q = -2$. Comparing the coefficient of x^2 gives $a = -4$. Thus

$$r_1, r_2 = \frac{-1 \pm i\sqrt{7}}{2}.$$

Therefore,

$$\boxed{a = -4, \qquad \{r_1, r_2\} = \left\{ \frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2} \right\}}.$$

□

8.3. Let $\alpha_1, \ldots, \alpha_5$ be the roots of

$$x^5 - 5x + 3 = 0.$$

Determine

$$\sum_{j=1}^{5} \left(\alpha_j^4 + \alpha_j^5 \right).$$

Solution 8.3. Let

$$s_k = \sum_{j=1}^{5} \alpha_j^k.$$

Newton's identities for

$$x^5 + 0x^4 + 0x^3 + 0x^2 - 5x + 3$$

give

$$s_1 = s_2 = s_3 = 0,$$

$$s_4 + 4(-5) = 0, \qquad s_5 + 5(3) = 0.$$

Hence $s_4 = 20$ and $s_5 = -15$, so

$$\boxed{s_4 + s_5 = 5}.$$

□

8.4. Let $\alpha_1, \ldots, \alpha_4$ be the roots of

$$x^4 + x^3 + x^2 + x + 1 = 0.$$

Determine

$$\sum_{j=1}^{4} \frac{1}{(1-\alpha_j)^2}.$$

Solution 8.4. Let

$$p(x) = x^4 + x^3 + x^2 + x + 1 = \prod_{j=1}^{4}(x - \alpha_j).$$

Then

$$\frac{p'(x)}{p(x)} = \sum_{j=1}^{4} \frac{1}{x-\alpha_j}.$$

Differentiating gives

$$\sum_{j=1}^{4} \frac{1}{(x-\alpha_j)^2} = \left(\frac{p'(x)}{p(x)}\right)^2 - \frac{p''(x)}{p(x)}.$$

Since

$$p(1) = 5, \qquad p'(1) = 10, \qquad p''(1) = 20,$$

the required sum is

$$\boxed{\left(\frac{10}{5}\right)^2 - \frac{20}{5} = 0}.$$

□

8.5. Let $\alpha_1, \alpha_2, \alpha_3$ be the roots of

$$x^3 - x - 1 = 0.$$

Determine

$$\prod_{j=1}^{3}(1 + \alpha_j^2).$$

Solution 8.5. Let

$$p(x) = x^3 - x - 1 = \prod_{j=1}^{3}(x - \alpha_j).$$

For each root α_j,

$$(i - \alpha_j)(-i - \alpha_j) = 1 + \alpha_j^2.$$

Therefore,

$$\prod_{j=1}^{3}(1 + \alpha_j^2) = p(i)p(-i).$$

Since

$$p(i) = -1 - 2i, \qquad p(-i) = -1 + 2i,$$

we obtain

$$\boxed{p(i)p(-i) = 5}.$$

□

8.6. (1995 CSAT). Suppose that

$$\frac{1}{(x-1)(x-2)\cdots(x-10)} = \sum_{k=1}^{10} \frac{a_k}{x-k}$$

whenever the denominators are nonzero. Determine

$$a_1 + a_2 + \cdots + a_{10}.$$

Solution 8.6. Multiply the identity by x and let $x \to \infty$. The left-hand side tends to 0, while

$$\frac{x}{x-k} \to 1$$

for every k. Therefore

$$\boxed{a_1 + a_2 + \cdots + a_{10} = 0}.$$

□

Problem 9. Determinants.

Compute each of the following determinants in exact form.

9.1. (2016 KAIST Problem of the Week). Let $\mathbf{A} = (a_{ij})$ be the $n \times n$ matrix defined by

$$a_{ij} = \begin{cases} 3, & i = j, \\ (-1)^{|i-j|}, & i \neq j. \end{cases}$$

Determine $\det(\mathbf{A})$.

Solution 9.1. Let

$$\mathbf{S} = \operatorname{diag}((-1)^1, (-1)^2, \ldots, (-1)^n).$$

Then $\det(\mathbf{S})^2 = 1$, and

$$\mathbf{SAS} = 2\mathbf{I}_n + \mathbf{J}_n,$$

where $\mathbf{J}_n$ is the all-ones matrix. The eigenvalues of $2\mathbf{I}_n + \mathbf{J}_n$ are $n+2$ once and 2 with multiplicity $n-1$. Therefore,

$$\boxed{\det(\mathbf{A}) = 2^{n-1}(n+2)}.$$

□

9.2. (2016 KAIST Problem of the Week). Let T be a tree on the vertex set $\{1, 2, \ldots, n\}$, and let d_{ij} be the distance between vertices i and j. Define

$$\mathbf{D}_T(x) = \left(x^{d_{ij}}\right)_{1 \le i,j \le n}.$$

Determine $\det(\mathbf{D}_T(x))$.

Solution 9.2. We use induction on n. The formula is immediate for $n = 1$. For $n \geq 2$, choose a leaf n whose unique neighbor is p. For every $j \neq n$,

$$d_{nj} = 1 + d_{pj}.$$

Apply the row operation

$$R_n \longleftarrow R_n - xR_p.$$

The last row becomes zero except for its last entry $1 - x^2$. Next apply

$$C_n \longleftarrow C_n - xC_p.$$

The last column also becomes zero except for the same last entry, while the remaining principal block is the matrix associated with the tree obtained by deleting the leaf. Thus

$$\det(\mathbf{D}_T(x)) = (1 - x^2)\det(\mathbf{D}_{T-n}(x)).$$

Induction gives

$$\boxed{\det(\mathbf{D}_T(x)) = (1 - x^2)^{n-1}}.$$

□

9.3. (2015 KAIST Problem of the Week). For $n \in \mathbb{N}$ and $x \in \mathbb{R}$, let $\mathbf{T}_n(x) = (t_{ij})$ be the $n \times n$ tridiagonal matrix defined by

$$t_{ij} = \begin{cases} 1, & i = j, \\ x, & |i-j| = 1, \\ 0, & |i-j| \geq 2. \end{cases}$$

Determine $\det(\mathbf{T}_n(x))$.

Solution 9.3. Put

$$D_n(x) = \det(\mathbf{T}_n(x)), \qquad D_0(x) = 1.$$

Expansion along the final row gives

$$D_n(x) = D_{n-1}(x) - x^2 D_{n-2}(x), \qquad D_0(x) = D_1(x) = 1.$$

The polynomial

$$\sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \binom{n-k}{k} x^{2k}$$

has these initial values and satisfies the same recurrence by Pascal's identity. Therefore,

$$\boxed{\det(\mathbf{T}_n(x)) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \binom{n-k}{k} x^{2k}}.$$

Equivalently, if $x \neq \pm 1/2$ and

$$r_{\pm} = \frac{1 \pm \sqrt{1 - 4x^2}}{2},$$

then

$$D_n(x) = \frac{r_+^{n+1} - r_-^{n+1}}{r_+ - r_-};$$

for $x = \pm 1/2$, this becomes $D_n(x) = (n+1)/2^n$.

□

9.4. (2011 KAIST Problem of the Week). Let $\mathbf{M} = (m_{ij})$ be the $n \times n$ matrix defined by

$$m_{ij} = i(i+1)(i+2)\cdots(i+j-2),$$

where the empty product m_{i1} is 1. Determine $\det(\mathbf{M})$.

Solution 9.4. For $j \geq 1$, define the monic polynomial

$$p_{j-1}(t) = t(t+1)\cdots(t+j-2),$$

with $p_0(t) = 1$. Then

$$m_{ij} = p_{j-1}(i).$$

Replacing the monomial basis $1, t, t^2, \ldots, t^{n-1}$ by the monic basis $p_0, p_1, \ldots, p_{n-1}$ uses a unit upper-triangular change-of-basis matrix. Hence $\det(\mathbf{M})$ equals the Vandermonde determinant at $1, 2, \ldots, n$:

$$\det(\mathbf{M}) = \prod_{1 \le i < j \le n} (j - i)$$
$$= \boxed{\prod_{k=1}^{n-1} k!}.$$

□

9.5. (2012 KAIST Problem of the Week). List all nonempty subsets of $\{1, 2, \ldots, n\}$ as $S_1, S_2, \ldots, S_{2^n-1}$, and define the $(2^n - 1) \times (2^n - 1)$ matrix $\mathbf{A} = (a_{ij})$ by

$$a_{ij} = \begin{cases} 1, & S_i \cap S_j \neq \emptyset, \\ 0, & S_i \cap S_j = \emptyset. \end{cases}$$

Determine $|\det(\mathbf{A})|$.

Solution 9.5. Index rows and columns by the nonempty subsets of $[n]$. Define

$$c_{S,U} = \begin{cases} 1, & U \subseteq S, \\ 0, & U \nsubseteq S, \end{cases}$$

and let $\mathbf{C} = (c_{S,U})$. Also let $\mathbf{E}$ be diagonal with

$$e_{U,U} = (-1)^{|U|+1}.$$

Inclusion-exclusion gives

$$(\mathbf{C}\mathbf{E}\mathbf{C}^\mathsf{T})_{S,T} = \sum_{\emptyset \neq U \subseteq S \cap T} (-1)^{|U|+1} = \begin{cases} 1, & S \cap T \neq \emptyset, \\ 0, & S \cap T = \emptyset. \end{cases}$$

Thus

$$\mathbf{A} = \mathbf{C}\mathbf{E}\mathbf{C}^\mathsf{T}.$$

Order the subsets by increasing cardinality. Then $\mathbf{C}$ is triangular with every diagonal entry equal to 1, so $\det(\mathbf{C}) = 1$. Since every diagonal entry of $\mathbf{E}$ is ± 1,

$$\boxed{|\det(\mathbf{A})| = 1}.$$

□

Problem 10. Eigenvalues and Matrix Powers.

Determine the requested eigenvalues or matrix powers exactly. Use diagonalization, polynomial identities, or invariant subspaces whenever they make the computation shorter than direct multiplication.

10.1. (1997 VTRMC). Let

$$\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix}.$$

Determine the $(1, 1)$-entry of $\mathbf{A}^{100}$.

Solution 10.1. The eigenvalues of $\mathbf{A}$ are

$$\lambda_{\pm} = 1 \pm \sqrt{3}.$$

The matrix is diagonalizable, and the two spectral projections have equal $(1,1)$-entries $1/2$. Hence

$$(\mathbf{A}^{100})_{11} = \frac{\lambda_+^{100} + \lambda_-^{100}}{2}.$$

Therefore,

$$\boxed{(\mathbf{A}^{100})_{11} = \frac{(1+\sqrt{3})^{100} + (1-\sqrt{3})^{100}}{2}}.$$

□

10.2. (Fibonacci Q-Matrix). Let

$$\mathbf{Q} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Determine $\mathbf{Q}^n$ for $n \in \mathbb{N}$ in terms of the Fibonacci sequence.

Solution 10.2. Let $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$. We claim that

$$\mathbf{Q}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$$

for $n \geq 1$. The formula is true for $n = 1$, and multiplying the displayed matrix by $\mathbf{Q}$ advances every entry according to the Fibonacci recurrence. Thus

$$\boxed{\mathbf{Q}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}}.$$

□

10.3. Let $\mathbf{A}$ be a $d \times d$ matrix satisfying

$$\mathbf{A}^2 - 3\mathbf{A} + 2\mathbf{I}_d = \mathbf{0}.$$

Determine $\mathbf{A}^{2026}$ as a linear combination of $\mathbf{A}$ and $\mathbf{I}_d$.

Solution 10.3. Reduce the polynomial x^{2026} modulo

$$(x-1)(x-2) = x^2 - 3x + 2.$$

Its remainder has the form $r(x) = ux + v$ and must satisfy

$$r(1) = 1, \qquad r(2) = 2^{2026}.$$

Hence

$$u = 2^{2026} - 1, \qquad v = 2 - 2^{2026}.$$

Substituting $\mathbf{A}$ for x gives

$$\boxed{\mathbf{A}^{2026} = (2^{2026} - 1)\mathbf{A} + (2 - 2^{2026})\mathbf{I}_d}.$$

□

10.4. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Determine $\mathbf{A}^n$ for $n \in \mathbb{N}_0$.

Solution 10.4. Write

$$\mathbf{A} = \mathbf{I}_3 + \mathbf{N}, \qquad \mathbf{N} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Since $\mathbf{N}^3 = \mathbf{0}$, the binomial theorem gives

$$\mathbf{A}^n = \mathbf{I}_3 + n\mathbf{N} + \binom{n}{2}\mathbf{N}^2.$$

Therefore,

$$\boxed{\mathbf{A}^n = \begin{pmatrix} 1 & n & \binom{n}{2} \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}}.$$

$\square$

10.5. Determine the eigenvalues of

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}.$$

Solution 10.5. Write

$$\mathbf{A} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}.$$

This is a rank-one outer product. Its only possible nonzero eigenvalue is

$$\begin{pmatrix} 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 6.$$

Since the matrix has rank 1, the remaining two eigenvalues are 0. Therefore,

$$\boxed{6,\ 0,\ 0}.$$

$\square$

10.6. Determine the eigenvalues of

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 3 & 0 \\ -2 & 3 & 0 & 1 \\ 3 & 0 & 1 & -2 \\ 0 & 1 & -2 & 3 \end{pmatrix}.$$

Solution 10.6. Reorder the coordinates as $(1, 3, 2, 4)$. The resulting similar matrix has the block form

$$\begin{pmatrix} \mathbf{B} & -2\mathbf{I}_2 \\ -2\mathbf{I}_2 & \mathbf{C} \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix}, \qquad \mathbf{C} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}.$$

The vectors $(1,1)$ and $(1,-1)$ simultaneously diagonalize $\mathbf{B}$ and $\mathbf{C}$. On the corresponding two-dimensional invariant subspaces, the matrix reduces respectively to

$$\begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -2 & -2 \\ -2 & 2 \end{pmatrix}.$$

Their eigenvalues are $6, 2$ and $\pm 2\sqrt{2}$. Hence

$$\boxed{6,\ 2,\ 2\sqrt{2},\ -2\sqrt{2}}.$$

□

10.7. Let $\mathbf{A}$ be the 2022×2022 matrix whose diagonal entries are 0 and whose off-diagonal entries are 1. Determine all eigenvalues, including multiplicities.

Solution 10.7. We have

$$\mathbf{A} = \mathbf{J}_{2022} - \mathbf{I}_{2022}.$$

The all-ones vector is an eigenvector of $\mathbf{J}_{2022}$ with eigenvalue 2022, while every vector whose coordinates sum to zero has eigenvalue 0. Subtracting the identity shifts all eigenvalues by -1. Therefore,

$$\boxed{2021 \text{ with multiplicity } 1, \qquad -1 \text{ with multiplicity } 2021}.$$

□

10.8. Let $\mathbf{J}_n$ be the $n \times n$ all-ones matrix and set

$$\mathbf{A} = \mathbf{I}_n + \mathbf{J}_n.$$

Determine the eigenvalues of $\mathbf{A}$ and compute $\mathbf{A}^{2026}$.

Solution 10.8. The vector $\mathbf{1}$ is an eigenvector of $\mathbf{J}_n$ with eigenvalue n, while every vector orthogonal to $\mathbf{1}$ is an eigenvector with eigenvalue 0. Hence $\mathbf{A}$ has eigenvalues

$$n+1 \quad \text{with multiplicity } 1, \qquad 1 \quad \text{with multiplicity } n-1.$$

Since $\mathbf{J}_n^2 = n\mathbf{J}_n$, the projection onto $\operatorname{span}(\mathbf{1})$ is $\mathbf{J}_n/n$. Thus

$$\mathbf{A}^{2026} = \left(\mathbf{I}_n - \frac{1}{n}\mathbf{J}_n\right) + (n+1)^{2026}\frac{1}{n}\mathbf{J}_n.$$

Therefore,

$$\boxed{\mathbf{A}^{2026} = \mathbf{I}_n + \frac{(n+1)^{2026} - 1}{n}\mathbf{J}_n}.$$

□

10.9. Determine all eigenvalues of

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Solution 10.9. The matrix cyclically shifts the coordinates and satisfies

$$\mathbf{P}^4 = \mathbf{I}_4.$$

Hence every eigenvalue λ satisfies $\lambda^4 = 1$. The characteristic polynomial is

$$\lambda^4 - 1,$$

whose roots are

$$1, \quad -1, \quad i, \quad -i.$$

Therefore the eigenvalues are

$$\boxed{1,\ -1,\ i,\ -i}.$$

□

Memorization Problems

Problem 1. The Clay Mathematics Institute's Yang–Mills Existence and Mass Gap problem asks for a nontrivial quantum Yang–Mills theory on $\mathbb{R}^d$ with a positive mass gap. Determine d, and state whether this case has been solved.

Solution. The Millennium Prize Problem is posed on four-dimensional Euclidean space:

$$\boxed{d = 4}.$$

As of 2026, the required rigorous construction and proof of a positive mass gap in this dimension remain open.

□

Problem 2. John Milnor first exhibited exotic spheres in dimension n: smooth manifolds homeomorphic, but not diffeomorphic, to the standard sphere S^n. Later work showed that the group Θ_n of oriented diffeomorphism classes of homotopy n-spheres has exactly k elements in this dimension. Determine

$$n + k.$$

Solution. Milnor's first exotic spheres were seven-dimensional, and Kervaire–Milnor theory gives

$$\Theta_7 \cong \mathbb{Z}/28\mathbb{Z}.$$

Thus, with orientation taken into account, there are 28 classes in total: one standard class and 27 exotic classes. Therefore $n = 7$ and $k = 28$, so

$$\boxed{n + k = 35}.$$

□

Problem 3. This public-key cryptosystem was developed in 1977 by Ronald Rivest, Adi Shamir, and Leonard Adleman. It supports public-key encryption and digital signatures, and its standard key construction uses a modulus formed from two large primes. The three researchers received the 2002 ACM A. M. Turing Award for their contribution to practical public-key cryptography. Identify the cryptosystem.

Solution. The cryptosystem is

$$\boxed{\text{RSA}}.$$

Its name consists of the initials of Rivest, Shamir, and Adleman.

□

Problem 4. For a connected finite graph, Euler's criterion states that an Euler circuit exists exactly when every vertex has even degree, while an Euler trail with distinct endpoints exists exactly when precisely two vertices have a certain degree parity. What kind of vertices occur zero times in the first case and exactly twice in the second?

Solution. They are

$$\boxed{\text{vertices of odd degree}}.$$

□

Problem 5. This Norwegian mathematician lived from 1802 to 1829 and died at the age of 26. Despite poverty and illness, he made fundamental contributions to analysis and algebra. He is especially known for proving that the general quintic equation cannot be solved by radicals. Identify the mathematician.

Solution. The mathematician is

$$\boxed{\text{Niels Henrik Abel}}.$$

□

Problem 6. This closed surface has no boundary and is nonorientable, so it has no globally consistent distinction between two sides. Unlike a Möbius strip, it is closed. It cannot be embedded in three-dimensional Euclidean space, although it can be represented there by a self-intersecting immersion. Identify the surface.

Solution. The surface is the

$$\boxed{\text{Klein bottle}}.$$

□

Problem 7. This German mathematician was born near Königsberg in 1862, held a professorship at the University of Königsberg, and moved to the University of Göttingen in 1895. His work shaped algebra, number theory, analysis, and the axiomatic foundations of geometry, and his *Grundlagen der Geometrie* appeared in 1899. Identify the mathematician.

Solution. The mathematician is

$$\boxed{\text{David Hilbert}}.$$

□

Problem 8. In game theory, this is a strategy profile in which no player can improve their payoff by changing only their own strategy while all other players keep theirs fixed. It is named after the mathematician portrayed in the film *A Beautiful Mind*. What is this concept?

Solution. The concept is a

$$\boxed{\text{Nash equilibrium}}.$$

□

Problem 9. A complex number is called this if it is not a root of any nonzero polynomial with rational coefficients. The numbers π and e are standard examples. What is such a number called?

Solution. It is called a

$$\boxed{\text{transcendental number}}.$$

$\square$

Problem 10. A positive integer is called this if it equals the sum of its positive proper divisors. For example,

$$6 = 1 + 2 + 3, \qquad 28 = 1 + 2 + 4 + 7 + 14.$$

The Euclid–Euler theorem states that every even integer of this type has the form

$$2^{p-1}(2^p - 1),$$

where $2^p - 1$ is prime. What is such an integer called?

Solution. It is called a

$$\boxed{\text{perfect number}}.$$

$\square$

Problem 11. An Euler path traverses every edge of a graph exactly once. What is the standard name for a path that visits every vertex of a graph exactly once?

Solution. It is called a

$$\boxed{\text{Hamiltonian path}}.$$

$\square$

Problem 12. The obverse of the Fields Medal depicts a mathematician from antiquity, while the reverse depicts a sphere inscribed in a cylinder. Identify the mathematician and state the volume ratio of the sphere to the circumscribed cylinder.

Solution. The mathematician is Archimedes. If the sphere has radius r, then the cylinder has radius r and height $2r$, so

$$\frac{V_{\text{sphere}}}{V_{\text{cylinder}}} = \frac{\frac{4}{3}\pi r^3}{2\pi r^3} = \frac{2}{3}.$$

Thus the answer is

$$\boxed{\text{Archimedes and } 2 : 3}.$$

$\square$

Problem 13. Match the four 2018 Fields Medalists with the following principal mathematical anchors:

(1) p-adic geometry;
(2) boundedness of Fano varieties and the BAB conjecture;
(3) optimal transport;
(4) number theory and homogeneous dynamics.

Solution. The correct matching is

$$\boxed{\begin{aligned} \text{Peter Scholze} &\longleftrightarrow p\text{-adic geometry}, \\ \text{Caucher Birkar} &\longleftrightarrow \text{Fano varieties and BAB}, \\ \text{Alessio Figalli} &\longleftrightarrow \text{optimal transport}, \\ \text{Akshay Venkatesh} &\longleftrightarrow \text{number theory and homogeneous dynamics}. \end{aligned}}$$

$\square$

Problem 14. Maryna Viazovska proved optimal sphere-packing results in dimensions 8 and 24. What is the classical three-dimensional sphere-packing conjecture called, and who proved it?

Solution. The three-dimensional statement is the

$$\boxed{\text{Kepler conjecture}},$$

proved by

$$\boxed{\text{Thomas Hales}}.$$

The later Flyspeck project formally verified the proof.

□

Problem 15. Which of the following is not one of the combinatorial log-concavity conjectures associated with June Huh's work?

(1) Rota's conjecture;
(2) Mason's conjecture;
(3) Dowling–Wilson conjecture;
(4) Sullivan conjecture.

Solution. The first three belong to the circle of matroid and combinatorial log-concavity results associated with June Huh and his collaborators. The Sullivan conjecture belongs to algebraic topology. Therefore, the odd one out is

$$\boxed{\text{Sullivan conjecture}}.$$

□

Problem 16. Identify the Abel Prize laureate or laureates for each year:

$$2018, \quad 2019, \quad 2020, \quad 2021, \quad 2022.$$

Solution. The sequence is

$$\boxed{\begin{aligned} &2018: \text{ Robert Langlands,}\\ &2019: \text{ Karen Uhlenbeck,}\\ &2020: \text{ Hillel Furstenberg and Gregory Margulis,}\\ &2021: \text{ László Lovász and Avi Wigderson,}\\ &2022: \text{ Dennis Sullivan.}\end{aligned}}$$

□

Problem 17. Give the Fields Medalists conventionally associated with the following national firsts:

$$\text{Brazil}, \qquad \text{Japan}, \qquad \text{Iran}, \qquad \text{Vietnam}.$$

Solution. The standard associations are

$$\boxed{\begin{aligned} \text{Brazil} &\longleftrightarrow \text{Artur Avila,}\\ \text{Japan} &\longleftrightarrow \text{Kunihiko Kodaira,}\\ \text{Iran} &\longleftrightarrow \text{Maryam Mirzakhani,}\\ \text{Vietnam} &\longleftrightarrow \text{Ngô Bào Châu.}\end{aligned}}$$

□

Problem 18. The 2026 International Congress of Mathematicians was held in which city, and who received the four Fields Medals awarded there?

Solution. ICM 2026 was held in

$$\boxed{\text{Philadelphia, United States}}.$$

The four Fields Medalists were

$$\boxed{\text{Yu Deng, John Pardon, Jacob Tsimerman, and Hong Wang}}.$$

□

Problem 19. Which 2026 Fields Medalist is especially associated with harmonic analysis, Furstenberg sets, and the three-dimensional Kakeya problem? What historical distinction did she attain?

Solution. The mathematician is

$$\boxed{\text{Hong Wang}}.$$

She became the

$$\boxed{\text{third woman to receive a Fields Medal}}.$$

□

Problem 20. The 2026 Abel Prize was awarded for powerful tools in arithmetic geometry and for resolving long-standing Diophantine conjectures of Mordell and Lang. Identify the laureate.

Solution. The 2026 Abel Prize laureate is

$$\boxed{\text{Gerd Faltings}}.$$

He had previously received the Fields Medal in 1986.

□

Problem 21. Which of the following mathematicians was born earliest?

(1) Guillaume de l'Hôpital;
(2) Isaac Newton;
(3) Gottfried Wilhelm Leibniz;
(4) Colin Maclaurin.

Solution. Newton was born in 1643 under the modern Gregorian calendar convention, earlier than Leibniz in 1646, l'Hôpital in 1661, and Maclaurin in 1698. Therefore the answer is

$$\boxed{\text{Isaac Newton}}.$$

□

Problem 22. What is Hilbert's first problem in the list of twenty-three problems presented in 1900?

Solution. Hilbert's first problem asks for the resolution of the

$$\boxed{\text{continuum hypothesis}}.$$

In modern set theory, Gödel and Cohen showed that the continuum hypothesis is independent of ZFC, assuming ZFC is consistent.

□

Problem 23. Match each number system with the mathematician most directly associated with its introduction or early systematic development. The mathematicians are Hamilton, Hensel, Hewitt, and Conway.

(1) quaternions;
(2) p-adic numbers;
(3) hyperreal numbers;
(4) surreal numbers.

Solution. The standard associations are

$$\boxed{\begin{aligned} \text{quaternions} &\longleftrightarrow \text{Hamilton},\\ p\text{-adic numbers} &\longleftrightarrow \text{Hensel},\\ \text{hyperreal numbers} &\longleftrightarrow \text{Hewitt},\\ \text{surreal numbers} &\longleftrightarrow \text{Conway}.\end{aligned}}$$

Robinson later made hyperreal methods central to nonstandard analysis, and Knuth introduced the name *surreal numbers*.

□

Problem 24. Match each clue with the corresponding paradox.

(a) Among 23 people, a shared birthday has probability greater than 1/2.
(b) Alternating two losing games can create a winning game.
(c) A ball can be reassembled into two congruent copies using finitely many nonmeasurable pieces.
(d) Dominance reasoning and expected utility reasoning recommend different choices when a highly accurate predictor has already filled two boxes.

Solution. The matching is

$$\boxed{\begin{array}{l} \text{(a) birthday paradox,}\\ \text{(b) Parrondo's paradox,}\\ \text{(c) Banach–Tarski paradox,}\\ \text{(d) Newcomb's paradox.}\end{array}}$$

□

Problem 25. Euclid's fifth postulate is often replaced in modern elementary geometry by the statement that through a point outside a given line there is exactly one parallel line. What is this equivalent statement called?

Solution. It is

$$\boxed{\text{Playfair's axiom}}.$$

□

Problem 26. Match the three constant-curvature geometries with the number of parallels through a point outside a given line and with the angle sum of a triangle.

(a) Euclidean geometry;
(b) hyperbolic geometry;
(c) elliptic or spherical geometry.

Solution. The matching is

$$\boxed{\begin{array}{rl} \text{(a) Euclidean:} & \text{exactly one parallel, angle sum } = \pi, \\ \text{(b) hyperbolic:} & \text{infinitely many disjoint lines, angle sum } < \pi, \\ \text{(c) elliptic / spherical:} & \text{no disjoint geodesic line, angle sum } > \pi. \end{array}}$$

□

Problem 27. Match each foundational clue with the corresponding mathematician or pair of mathematicians.

(1) diagonal argument and the comparison of infinite cardinalities;
(2) the paradox of the set of all sets that do not contain themselves;
(3) the axiomatic system ZF, with ZFC obtained by adding the Axiom of Choice;
(4) the independence of the Continuum Hypothesis from ZFC, assuming consistency.

Solution. The correct associations are

$$\boxed{\begin{array}{l} \text{(1) Georg Cantor,} \\ \text{(2) Bertrand Russell,} \\ \text{(3) Ernst Zermelo and Abraham Fraenkel,} \\ \text{(4) Kurt Gödel and Paul Cohen.} \end{array}}$$

□

Problem 28. Arrange the following classes of commutative rings from strongest to weakest:

(1) unique factorization domain;
(2) field;
(3) integral domain;
(4) principal ideal domain;
(5) Euclidean domain.

Solution. The standard implication chain is

$$\boxed{\text{field} \Longrightarrow \text{Euclidean domain} \Longrightarrow \text{PID} \Longrightarrow \text{UFD} \Longrightarrow \text{integral domain}}.$$

□

Problem 29. Identify the Joseon mathematician and chief state councillor who wrote *Gusuryak*, exhibited orthogonal Latin squares of order 9, and devised Jisuguimundo, also called the Hexagonal Tortoise Problem.

Solution. The mathematician is

$$\boxed{\text{Choi Seok-jeong}}.$$

□

Problem 30. Identify the Oxford mathematics lecturer Charles Lutwidge Dodgson by the pen name under which he wrote *Alice's Adventures in Wonderland*.

Solution. His pen name was

$$\boxed{\text{Lewis Carroll}}.$$

□

Problem 31. Who received the first Abel Prize, awarded in 2003?

Solution. The first Abel laureate was

$$\boxed{\text{Jean-Pierre Serre}}.$$

□

Problem 32. Identify the mathematician associated with the modularity conjecture formerly called the Taniyama–Shimura conjecture, a central ingredient in the proof of Fermat's Last Theorem.

Solution. The mathematician is

$$\boxed{\text{Goro Shimura}}.$$

□

Problem 33. What collective pseudonym was adopted by the group of mainly French mathematicians who sought a unified axiomatic presentation of modern mathematics?

Solution. The collective pseudonym is

$$\boxed{\text{Nicolas Bourbaki}}.$$

□

Reasoning and Puzzle Problems

Problem 1. (Sum-and-Product Puzzle). Two integers x and y satisfy

$$1 < x < y, \qquad x + y \leq 100.$$

The sum $x + y$ is disclosed privately to S, and the product xy is disclosed privately to P. The preceding information is common knowledge. The following conversation then takes place:

(1) P: "I do not know the pair."
(2) S: "I knew that you did not know the pair."
(3) P: "Now I know the pair."
(4) S: "Now I also know the pair."

Determine (x, y).

Solution. Let

$$\Omega_0 = \{(x, y) \in \mathbb{N}^2 \mid (1 < x < y) \wedge (x + y \leq 100)\}.$$

For $\Omega \subseteq \Omega_0$ and $\omega = (x, y) \in \Omega$, define the information sets

$$\mathcal{I}_S^{\Omega}(\omega) = \{(u, v) \in \Omega \mid u + v = x + y\},$$

$$\mathcal{I}_P^{\Omega}(\omega) = \{(u,v) \in \Omega \mid uv = xy\}.$$

The four announcements successively restrict the state space as follows:

$$\Omega_1 = \{\omega \in \Omega_0 \mid |\mathcal{I}_P^{\Omega_0}(\omega)| > 1\},$$

$$\Omega_2 = \{\omega \in \Omega_1 \mid \mathcal{I}_S^{\Omega_0}(\omega) \subseteq \Omega_1\},$$

$$\Omega_3 = \{\omega \in \Omega_2 \mid |\mathcal{I}_P^{\Omega_2}(\omega)| = 1\},$$

and

$$\Omega_4 = \{\omega \in \Omega_3 \mid |\mathcal{I}_S^{\Omega_3}(\omega)| = 1\}.$$

The finite elimination is summarized by

stage	public information retained	number of states
0	$(1 < x < y) \wedge (x + y \leq 100)$	2352
1	P does not know the pair	1747
2	S knew that P did not know	145
3	P now knows the pair	86
4	S now knows the pair	1

At the second stage, the possible sums are exactly

$$11,\ 17,\ 23,\ 27,\ 29,\ 35,\ 37,\ 41,\ 47,\ 53.$$

These values follow by checking, for each admissible sum, whether every compatible factorization has at least two representations in Ω_0. After the third announcement, the only sum-information set of cardinality 1 is

$$\mathcal{I}_S^{\Omega_3}(4,13) = \{(4,13)\}.$$

Therefore,

$$\Omega_4 = \{(4,13)\},$$

and the unique pair is

$$\boxed{(x,y) = (4,13)}.$$

□

Problem 2. (Bertrand's Box Paradox). Three boxes contain coins as follows:

$$GG, \qquad GS, \qquad SS,$$

where G denotes a gold coin and S denotes a silver coin. A box is chosen uniformly at random, and then one of its two coins is chosen uniformly at random. The chosen coin is gold. Determine the conditional probability that the other coin in the same box is also gold.

Solution. Let B_{GG} be the event that the chosen box is GG, and let G be the event that the observed coin is gold. Then

$$\mathbb{P}(G \mid B_{GG}) = 1, \qquad \mathbb{P}(G \mid B_{GS}) = \frac{1}{2}, \qquad \mathbb{P}(G \mid B_{SS}) = 0.$$

Since the three boxes are initially equally likely, Bayes' theorem gives

$$\begin{aligned} \mathbb{P}(B_{GG} \mid G) &= \frac{\mathbb{P}(G \mid B_{GG})\mathbb{P}(B_{GG})}{\mathbb{P}(G \mid B_{GG})\mathbb{P}(B_{GG}) + \mathbb{P}(G \mid B_{GS})\mathbb{P}(B_{GS})} \\ &= \frac{1 \cdot (1/3)}{1 \cdot (1/3) + (1/2) \cdot (1/3)} = \boxed{\frac{2}{3}}. \end{aligned}$$

Equivalently, after observing gold, the selected coin is equally likely to be any of the three gold coins present among the boxes. Two of those three coins lie in the GG box.

□

Problem 3. (Penney's Game). The first player chooses a pattern of three coin tosses, such as HHT. After seeing this choice, the second player chooses a different pattern of length 3. A fair coin is then tossed repeatedly until one of the two patterns first appears as three consecutive outcomes; the player who chose that pattern wins.

Prove that the second player can always choose a pattern whose winning probability is greater than 1/2.

Solution. For two patterns X and Y of length 3, define the overlap score

$$C(X,Y) = \sum_{\substack{1\le k\le 3\\ \text{suffix}_k(X)=\text{prefix}_k(Y)}} 2^k.$$

A standard fair-betting argument gives the probability that Y occurs before X:

$$\mathbb{P}(Y \text{ before } X) = \frac{C(X,X)-C(X,Y)}{C(X,X)-C(X,Y)+C(Y,Y)-C(Y,X)}.$$

Indeed, start a new unit-stake gambler before each toss, with one team betting successively on X and the other on Y, doubling its capital after each correct fair bet. At the stopping time, the surviving capital of the team betting on Y is $C(X,Y)$ if X wins and $C(Y,Y)$ if Y wins; the analogous statement holds for the other team. Equality of the two expected net gains yields the displayed formula.

Write the first player's pattern as

$$X = x_1x_2x_3,$$

and let $\overline{x_2}$ denote the opposite face of x_2. The second player chooses

$$Y = \overline{x_2}x_1x_2.$$

By interchanging heads and tails if necessary, it is enough to check the following four cases:

X	Y	$\mathbb{P}(Y \text{ before } X)$
HHH	THH	7/8
HHT	THH	3/4
HTH	HHT	2/3
HTT	HHT	2/3

These values follow immediately from the overlap formula. Each is strictly greater than 1/2, so the second player always has the required choice.

□

Problem 4. (Josephus Problem). The people numbered

$$1, 2, \ldots, n$$

stand in a circle. Starting with person 1, count repeatedly around the circle and eliminate every second person. After each elimination, resume counting with the next remaining person. Determine the number of the last survivor.

Solution. Let $J(n)$ denote the number of the survivor. If $n = 2m$, the first circuit eliminates all even-numbered people. Relabeling the survivors

$$1, 3, \ldots, 2m-1$$

as $1, 2, \ldots, m$ gives

$$J(2m) = 2J(m) - 1.$$

A similar relabeling when $n = 2m+1$ gives

$$J(2m+1) = 2J(m) + 1.$$

Write

$$n = 2^r + \ell, \qquad 0 \le \ell < 2^r.$$

Starting from $J(2^r) = 1$, a strong induction using these recurrences yields

$$\boxed{J(n) = 2\ell + 1}.$$

In binary notation this has a particularly simple interpretation. If

$$n = (1b_{r-1}b_{r-2}\cdots b_0)_2,$$

then

$$J(n) = (b_{r-1}b_{r-2}\cdots b_0 1)_2;$$

the leading binary digit of n is moved to the end.

□

Problem 5. (Conway's Soldiers). On an infinite square grid, every square on or below a fixed horizontal line initially contains a soldier, and every square strictly above the line is empty. A legal move jumps one soldier horizontally or vertically over an adjacent soldier into the next square, which must be empty; the jumped soldier is removed. Prove that no soldier can reach the fifth row above the initial line.

Solution. Fix a target square T in the fifth row and put

$$q = \frac{\sqrt{5}-1}{2}, \qquad q + q^2 = 1.$$

Give each square Q the weight

$$w(Q) = q^{d(Q,T)},$$

where d is the Manhattan distance, and let W be the sum of the weights of all occupied squares.

Consider three consecutive squares in the direction of a legal jump. Their distances from T have one of the forms

$$(m+2, m+1, m), \qquad (m, m+1, m+2), \qquad (m+1, m, m+1).$$

In the first case, the jump moves toward T and

$$q^{m+2} + q^{m+1} = q^m$$

because $q+q^2 = 1$. In the second case, $q^m+q^{m+1} > q^{m+2}$, and in the third case, $q^{m+1}+q^m > q^{m+1}$. Thus no legal move increases W.

Take the initial boundary to be $y = 0$ and $T = (0, 5)$. The initial weight is

$$W_0 = \sum_{m=0}^{\infty} \sum_{k \in \mathbb{Z}} q^{|k|+m+5}$$
$$= q^5 \left(1 + 2\sum_{k=1}^{\infty} q^k\right) \left(\sum_{m=0}^{\infty} q^m\right) = 1.$$

Before a soldier could first reach T, only finitely many moves would have occurred. Hence only finitely many initially occupied squares could have been altered, so at least one untouched soldier of positive weight would remain. If T were occupied, it alone would contribute weight 1, and the untouched soldier would contribute an additional positive amount. This would give $W > 1$, contradicting $W \le W_0 = 1$.

Therefore no soldier can reach the fifth row.

□

Problem 6. (Three Utilities Problem). Three houses and three utility companies—water, gas, and electricity—are represented by six distinct points in the plane. Is it possible to join every house to every utility company by curves so that no two curves intersect and no curve passes through any of the six points other than its endpoints? Prove the answer.

Solution. Such a drawing would be a planar embedding of the complete bipartite graph $K_{3,3}$. This graph has

$$V = 6, \qquad E = 9.$$

The graph $K_{3,3}$ is simple, bipartite, and has no bridges. Thus every face boundary in a planar embedding would be a closed walk of length at least 4. If F is the number of faces, double counting edge–face incidences gives

$$2E \ge 4F, \qquad \text{so} \qquad F \le \frac{E}{2}.$$

Euler's formula would then give

$$2 = V - E + F \le V - E + \frac{E}{2},$$

and hence

$$E \le 2V - 4.$$

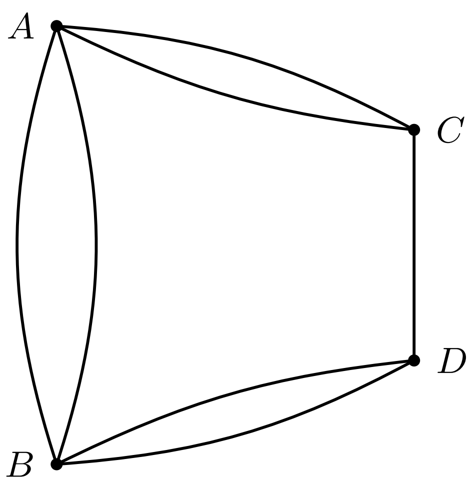
For $K_{3,3}$ this would require

$$9 \le 2 \cdot 6 - 4 = 8,$$

which is impossible. Thus the requested system of nonintersecting connections cannot be drawn in the plane.

□

Problem 7. (Seven Bridges of Königsberg). The city of Königsberg was divided by the Pregel River into four land masses, connected by seven bridges as shown below. Is it possible to start at one land mass, cross each bridge exactly once, and end at some land mass? Prove your answer.



Solution. Replace each land mass by a vertex and each bridge by an edge. Then the problem becomes the following graph-theoretic question: does the resulting connected multigraph have an Euler trail, that is, a trail using every edge exactly once?

In the graph above, the four vertex degrees are

$$3, \quad 3, \quad 3, \quad 5.$$

In particular, all four vertices have odd degree.

A connected graph has an Euler trail if and only if it has exactly zero or two vertices of odd degree. Since this graph has four odd vertices, an Euler trail does not exist.

Therefore it is impossible to cross each of the seven bridges exactly once.

□

Problem 8. (15 Puzzle). In the standard fifteen puzzle, a legal move consists of sliding one tile adjacent to the blank square into the blank square. Starting from the solved position on the left, determine whether one can reach the position on the right, in which only the tiles 14 and 15 have been interchanged.

solved position

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

14–15 challenge

1	2	3	4
5	6	7	8
9	10	11	12
13	15	14	

Solution. Read the tiles row by row from left to right and top to bottom, omitting the blank square. Let I be the number of inversions in this resulting list, and let r be the row number of the blank square counted from the top. We claim that the parity of

$$I + r$$

is invariant under every legal move.

If the blank square moves horizontally, then the order of the numbered tiles in the row-by-row list does not change, so I is unchanged, and r is also unchanged. Hence the parity of $I + r$ is unchanged.

If the blank square moves vertically, then one tile moves up or down by one row. In the row-by-row list, that tile passes across exactly three other tiles, so the inversion number changes by 3, and therefore its parity changes. At the same time, the blank square moves to an adjacent row, so the parity of r also changes. Hence the parity of $I + r$ remains unchanged in this case as well.

In the solved position, the inversion number is

$$I = 0,$$

and the blank square lies in row 4, so

$$I + r \equiv 0 + 4 \equiv 0 \pmod 2.$$

In the target position, the list differs only by the transposition of 14 and 15, so

$$I = 1,$$

while the blank square is still in row 4. Thus

$$I + r \equiv 1 + 4 \equiv 1 \pmod 2.$$

Since the parity of $I + r$ is invariant, the target position cannot be reached from the solved position.

Therefore the 14–15 challenge is impossible.

□

Problem 9. (Chomp). An $m \times n$ rectangular chocolate bar consists of unit-square pieces. The lower-left piece is poisoned.

On each turn, a player chooses one remaining piece and eats that piece together with every remaining piece weakly above it and weakly to its right. The player who eats the poisoned piece loses.

Prove that whenever the initial bar contains more than the poisoned piece alone, the first player has a winning strategy.

Solution. Since the game is finite and has no draws, one of the two players has a winning strategy. Suppose, for contradiction, that the second player has one.

Let the first player eat only the upper-right corner piece. According to the supposed winning strategy, the second player must have a non-poisoned response; otherwise the second player is forced to eat the poisoned piece and loses. Let that response be a remaining piece C. Eating C also removes the upper-right corner, because that corner lies above and to the right of C. Hence the position obtained after these two moves is exactly the position that would have resulted had the first player chosen C on the very first move.

After the original two-move sequence, this position is losing for the player whose turn it is, namely the first player. The first player could instead begin by choosing C and hand the identical losing position to the second player. This contradicts the assumption that the second player has a winning strategy.

Therefore the first player has a winning strategy. The argument proves its existence without identifying the winning first move in general.

□

Problem 10. (Braess's Paradox). A total traffic flow of 4000 units travels from S to T in the directed network with routes

$$S \to A \to T \qquad \text{and} \qquad S \to B \to T.$$

If x denotes the flow on a road, its travel time is given by

Road	$S \to A$	$A \to T$	$S \to B$	$B \to T$
Travel time	$x/100$	45	45	$x/100$

Each driver chooses a route of minimum travel time, taking the total road flows as given.

Determine the equilibrium travel time in the original network. Then a new directed road $A \to B$ with travel time 0 is added. Determine the new equilibrium travel time.

Solution. Let y units use the upper route $S \to A \to T$. Then $4000 - y$ units use the lower route, and their travel times are

$$C_U(y) = \frac{y}{100} + 45, \qquad C_L(y) = 45 + \frac{4000 - y}{100}.$$

At equilibrium both routes are used and have equal travel time. Thus

$$\frac{y}{100} + 45 = 45 + \frac{4000 - y}{100},$$

so $y = 2000$ and

$$C_U = C_L = 65.$$

After the road $A \to B$ is added, let f, g, and h be the flows on the routes

$$S \to A \to T, \qquad S \to B \to T, \qquad S \to A \to B \to T,$$

respectively. Put

$$x = f + h, \qquad z = g + h,$$

for the flows on $S \to A$ and $B \to T$. The three route costs are

$$C_U = \frac{x}{100} + 45, \qquad C_L = 45 + \frac{z}{100}, \qquad C_M = \frac{x + z}{100}.$$

If $f > 0$, equilibrium would require $C_U \le C_M$, which implies $z \ge 4500$, impossible because the total flow is 4000. Hence $f = 0$. Similarly, $g = 0$. Therefore

$$h = 4000, \qquad x = z = 4000,$$

and every driver uses the new middle route with travel time

$$C_M = 40 + 40 = 80.$$

Thus adding the zero-time road increases the equilibrium travel time from

$$\boxed{65} \qquad \text{to} \qquad \boxed{80}.$$

□

Problem 11. (100 Prisoners and 100 Boxes). One hundred prisoners and one hundred boxes are labeled $1, 2, \ldots, 100$. The prisoners' labels are placed in the boxes according to a uniformly random permutation.

The prisoners enter the room one at a time. Each prisoner may open at most 50 boxes and must restore them before leaving. They may agree on a strategy beforehand but may not communicate after the process begins. They all survive if and only if every prisoner finds their own label.

Find a strategy whose probability of saving all the prisoners is greater than 30%.

Solution. Regard the contents of the boxes as a permutation π of $\{1, 2, \ldots, 100\}$, where box i contains the label $\pi(i)$. Prisoner i first opens box i. If it contains label j, the prisoner next opens box j, and continues following the labels in this way for at most 50 boxes.

This procedure follows the cycle of π containing i. Therefore every prisoner finds their own label if and only if every cycle of π has length at most 50.

For $51 \leq k \leq 100$, the probability that a uniformly random permutation contains a k-cycle is $1/k$. Moreover, two cycles of lengths greater than 50 cannot occur simultaneously. Hence the probability that some cycle is too long is

$$\sum_{k=51}^{100} \frac{1}{k}.$$

The probability that all prisoners survive is therefore

$$1 - \sum_{k=51}^{100} \frac{1}{k} = 1 - (H_{100} - H_{50}) \approx 0.3118278207.$$

Thus the cycle-following strategy saves all the prisoners with probability approximately

$$\boxed{31.18\%},$$

which is greater than 30%.

$\square$

Problem 12. (St. Petersburg Paradox). A fair coin is tossed until heads appears for the first time. If the first head appears on the nth toss, the player receives

$$2^n$$

units of money. Compute the expected payoff. If decisions are based only on expected monetary value, determine how large an entry fee it would be rational to pay, and explain why the conclusion is regarded as paradoxical.

Solution. The probability that the first head appears on the nth toss is

$$\left(\frac{1}{2}\right)^n.$$

Therefore the expected payoff is

$$\mathbb{E}[X] = \sum_{n=1}^{\infty} 2^n \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} 1 = \boxed{\infty}.$$

Under the rule of maximizing expected monetary value alone, subtracting any finite entry fee still leaves an infinite expected net payoff. Thus there is no finite upper bound on the fee that this rule recommends.

The conclusion conflicts sharply with ordinary judgment. For example, the payoff is at most 2 with probability 1/2 and at most 4 with probability 3/4, so most outcomes are small. The infinite expectation is driven by extremely rare but extremely large prizes. This mismatch shows that expected monetary value alone is not always an adequate decision criterion; finite resources, risk, and diminishing marginal utility of money can all change a rational valuation.

□

Problem 13. (Bachet's Weights Problem). Using a two-pan balance and weights of positive integer numbers of pounds, determine the minimum number of weights needed to weigh every integer load from 1 to 40 pounds. A weight may be placed on either pan. Determine the weights as well.

Solution. With m weights, each weight has three possible roles: on the load pan, unused, or on the opposite pan. Hence at most 3^m signed sums can be produced. These include 0 and occur in opposite pairs, so at most

$$\frac{3^m - 1}{2}$$

positive loads can be distinguished. To cover $1, 2, \ldots, 40$, we need

$$\frac{3^m - 1}{2} \geq 40,$$

and therefore $m \geq 4$.

Four weights suffice: take

$$\boxed{1, 3, 9, 27}.$$

Every integer from -40 to 40 has a balanced ternary representation

$$N = \varepsilon_0 + 3\varepsilon_1 + 9\varepsilon_2 + 27\varepsilon_3, \qquad \varepsilon_i \in \{-1, 0, 1\}.$$

A coefficient 1 places the corresponding weight opposite the load, a coefficient -1 places it with the load, and a coefficient 0 leaves it unused. Thus all loads from 1 through 40 can be weighed, and four weights are minimal.

□

Problem 14. (Hex). On a finite $n \times n$ Hex board, two players alternately place stones of their own colors in empty cells. Each player wins by forming a connected path joining a designated pair of opposite sides. Prove that the first player has a winning strategy.

Solution. A completely filled Hex board cannot be a draw. Indeed, consider the component of one color attached to one of its target sides. If that component does not reach the opposite target side, its boundary across the hexagonal cells produces a connected chain of the other color between the other pair of opposite sides. Thus exactly one player has a winning connection when the board is full.

Since the game is finite and has no draws, one of the two players has a winning strategy. Suppose, for contradiction, that the second player has one. The first player places a stone in any cell and thereafter follows the supposed second-player strategy, using the symmetry that interchanges the two colors and their target sides. If the strategy ever calls for the already occupied initial cell, the first player makes any legal move instead and continues.

An extra stone of one's own color can never destroy a connection, so this stolen strategy still wins. That contradicts the assumption that the second player had a winning strategy. Hence the first player has one. The argument is nonconstructive and need not identify a winning opening move.

□

Problem 15. (Secretary Problem). There are $n \geq 2$ applicants with distinct ranks, presented in a uniformly random order. After each interview, the current applicant must be accepted or rejected permanently. Only the relative ranking of the applicants seen so far is known. Determine a strategy maximizing the probability of selecting the best applicant, and find its asymptotic success probability as $n \to \infty$.

Solution. Only an applicant who is best among those seen so far can be accepted optimally. The optimal rule has a threshold form: reject the first r applicants, then accept the first subsequent applicant who is better than all of the first r.

For a fixed r, the best applicant must occur at some position $k > r$, and then the best among the first $k-1$ applicants must lie among the first r. Therefore the success probability is

$$P_n(r) = \sum_{k=r+1}^{n} \frac{1}{n} \frac{r}{k-1} = \frac{r}{n} \sum_{j=r}^{n-1} \frac{1}{j}.$$

Moreover,

$$n\big(P_n(r+1) - P_n(r)\big) = \sum_{j=r+1}^{n-1} \frac{1}{j} - 1.$$

Hence $P_n(r)$ increases until the displayed harmonic tail passes below 1 and then decreases. Thus an optimal cutoff is the integer where that crossing occurs.

Since

$$\sum_{j=r}^{n-1} \frac{1}{j} = \ln\left(\frac{n}{r}\right) + o(1),$$

the optimal cutoff satisfies

$$\frac{r}{n} \to \frac{1}{e}.$$

Substitution in $P_n(r)$ gives the limiting success probability

$$\boxed{\frac{1}{e}}.$$

□

Problem 16. (Nim, Bouton's Theorem). Several heaps contain

$$a_1, a_2, \ldots, a_m$$

stones. Two players alternately choose one heap and remove any positive number of stones from it. The player taking the last stone wins. Characterize the positions from which the next player has a winning strategy.

Solution. Let

$$s = a_1 \oplus a_2 \oplus \cdots \oplus a_m$$

be the bitwise XOR, or nim-sum, of the heap sizes.

Suppose first that $s = 0$. Any legal move changes one heap from a_i to a smaller value b_i. At the highest binary digit where a_i and b_i differ, the new nim-sum has a 1, so every move leads to a position with nonzero nim-sum.

Now suppose that $s \neq 0$. Choose the highest binary digit in which s has a 1. Some heap a_i also has a 1 in that position. Set

$$b_i = a_i \oplus s.$$

At the highest chosen digit, b_i has 0 while a_i has 1, so $b_i < a_i$ and this is a legal reduction. The new nim-sum is

$$s \oplus a_i \oplus b_i = 0.$$

Thus every nonzero nim-sum position can move to a zero nim-sum position, while every move from a zero nim-sum position makes it nonzero. Therefore

$$\boxed{a_1 \oplus \cdots \oplus a_m \neq 0 \text{ exactly for the next-player winning positions}}.$$

$\square$

Problem 17. (100 Prisoners and a Light Bulb). One hundred prisoners are isolated. Each day, one prisoner is selected independently and uniformly at random and taken to a room containing a light bulb, initially off. A prisoner in the room may inspect and change the bulb. Before the process begins, the prisoners may agree on a strategy. At any visit, a prisoner may declare that everyone has visited the room; a correct declaration frees all prisoners, while an incorrect one kills them. Find a strategy that eventually permits a correct declaration with probability 1.

Solution. Designate one prisoner as the counter. Each of the other 99 prisoners follows this rule: the first time that prisoner enters and finds the bulb off, turn it on; after having done so once, never turn it on again.

The counter never turns the bulb on. Whenever the counter enters and finds it on, the counter turns it off and increases a private count by 1. When the count reaches 99, the counter declares that everyone has visited.

Every counted signal was produced by a different non-counter, so the declaration is certainly correct. Under independent uniform random selection, every prisoner is selected infinitely often with probability 1. Consequently, each uncounted prisoner eventually encounters an off bulb and signals, and the counter eventually records every signal. Thus the correct declaration occurs with probability 1.

$\square$

Problem 18. (Sicherman Dice). Label the six faces of each of two dice by positive integers, allowing repeated labels. Find a pair of dice, not both standard, whose sum has exactly the same probability distribution as the sum of two standard dice.

Solution. For a die with face labels $d_1, \ldots, d_6$, encode its sums by the polynomial

$$P(x) = x^{d_1} + \cdots + x^{d_6}.$$

The coefficient of x^k in the product of the two dice polynomials is the number of outcomes with sum k.

Take dice with labels

$$\boxed{1, 2, 2, 3, 3, 4}$$

and

$$\boxed{1, 3, 4, 5, 6, 8}.$$

Their polynomials are

$$P(x) = x + 2x^2 + 2x^3 + x^4$$

and

$$Q(x) = x + x^3 + x^4 + x^5 + x^6 + x^8.$$

Direct factorization gives

$$P(x)Q(x) = (x + x^2 + x^3 + x^4 + x^5 + x^6)^2.$$

Hence every possible sum occurs with exactly the same multiplicity as for two standard dice.

$\square$

Problem 19. (Number Erasing Game). The integers

$$1, 2, \dots, 100$$

are written on a board. Repeatedly erase two numbers a, b and replace them by

$$a + b + ab.$$

Determine the final number and prove that it is independent of all choices.

Solution. The operation is designed so that

$$(a + b + ab) + 1 = (a + 1)(b + 1).$$

Thus if every current board entry x is replaced conceptually by $x + 1$, each move simply replaces two factors by their product. The product of all shifted entries is therefore invariant.

Initially this product is

$$\prod_{k=1}^{100} (k + 1) = 101!.$$

If the final entry is N, the final shifted product is $N + 1$. Hence

$$N + 1 = 101!,$$

and the last number is

$$\boxed{101! - 1}.$$

□

Problem 20. (Sprouts). Start with n spots in the plane. A move joins two spots, possibly the same spot, by a non-self-intersecting curve that crosses no existing curve, places one new spot on the new curve, and keeps every spot's degree at most 3. Prove that the game cannot continue indefinitely.

Solution. Call each unused incidence available at a spot a *life*. A spot of degree d has $3 - d$ lives. Initially there are $3n$ lives.

Every move uses two lives at its endpoints. The new spot lies in the interior of the new curve and immediately has degree 2, so it creates one new life. Therefore each move decreases the total number of lives by exactly

$$2 - 1 = 1.$$

Since the number of lives is always nonnegative, there can be at most $3n$ moves. In particular, every game of Sprouts terminates.

□

Problem 21. (Seven Hats Problem). Seven players are independently given red or blue hats, each color having probability 1/2. Every player sees the other six hats but not their own. Simultaneously, each player guesses red, guesses blue, or passes. The group wins if at least one guess is correct and no guess is incorrect. Find a strategy with winning probability 7/8.

Solution. Encode red and blue by 0 and 1, so a hat configuration is a vector in $\mathbb{F}_2^7$. Let $\mathbf{H}$ be the 3×7 matrix whose columns are the seven nonzero vectors of $\mathbb{F}_2^3$, and let

$$C = \{\mathbf{x} \in \mathbb{F}_2^7 \mid \mathbf{H}\mathbf{x} = \mathbf{0}\}$$

be the binary Hamming code. Since $\mathbf{H}$ has rank 3, the code has $2^{7-3} = 16$ words. No nonzero codeword has weight 1 or 2, because the columns of $\mathbf{H}$ are nonzero and distinct; hence the minimum distance is at least 3. The radius-1 Hamming balls around the 16 codewords are disjoint and contain

$$16(1+7) = 128$$

vectors in total. They therefore partition $\mathbb{F}_2^7$, so every non-codeword differs from a unique codeword in exactly one coordinate.

Player i considers the two full configurations consistent with the six visible hats. If exactly one is a codeword, player i guesses the opposite of the ith bit of that codeword. Otherwise, player i passes.

If the actual configuration is a codeword, all seven players guess the opposite of their actual hats, so the group loses. If the actual configuration is not a codeword, let the unique nearest codeword differ in coordinate i. Player i guesses correctly, while every other player passes because changing another coordinate would produce a word at distance 2 from that codeword, not another codeword. Thus the group wins on all $128 - 16 = 112$ non-codewords. The success probability is

$$\boxed{\frac{112}{128} = \frac{7}{8}}.$$

□

Problem 22. (Circular Gas Station Problem). Around a circular road are n gas stations. At station i, a car gains g_i units of fuel, and the trip to the next station costs c_i units. Suppose that

$$\sum_{i=1}^{n} g_i \geq \sum_{i=1}^{n} c_i.$$

Prove that some station permits a car with an initially empty tank to complete one full circuit without ever running out of fuel.

Solution. Put

$$d_i = g_i - c_i, \qquad S_0 = 0, \qquad S_k = \sum_{i=1}^{k} d_i.$$

Choose $j \in \{0, 1, \ldots, n-1\}$ s.t. S_j is minimal, and start at station $j+1$, with indices understood cyclically.

Before wrapping around the circle, the fuel after t legs is

$$S_{j+t} - S_j \geq 0.$$

After wrapping around, the corresponding amount has the form

$$S_n + S_k - S_j.$$

Here $S_n \geq 0$ by hypothesis and $S_k \geq S_j$ by the minimality of S_j, so this quantity is also nonnegative. Hence the car completes the full circuit from station $j+1$.

□

Problem 23. (2013 Oxford MAT). Charlie writes one integer from 1 to 10 on Alice's forehead and one on Bob's forehead. The two integers differ by 1. Alice and Bob can see each other's number but not their own, and all three people always tell the truth. They have the following conversation:

> Alice: "I do not know my number. Will you tell me whether my number is a square?"
>
> Charlie: "If I tell you that, then you will know your number."
>
> Bob: "I still do not know my number."

Determine Alice's number.

Solution. Let Bob's number be b. Alice initially does not know her number, so $b \neq 1, 10$; otherwise only one adjacent integer would be allowed.

Charlie's statement means that among Alice's two candidates

$$b - 1 \qquad \text{and} \qquad b + 1,$$

exactly one is a square. The squares from 1 to 10 are

$$1, 4, 9,$$

so this condition leaves

$$b \in \{2, 3, 5, 8\}.$$

Bob sees Alice's number a and still cannot determine his own number. Under the preceding restriction, this ambiguity occurs only when

$$a = 4,$$

because then Bob's number could be either 3 or 5. For every other possible value of a, exactly one adjacent number belongs to $\{2, 3, 5, 8\}$. Hence Alice's number is

$$\boxed{4}.$$

□

Problem 24. (1998 All-Russian MO). An integer from 1 to 144 is chosen. One may repeatedly choose any subset S and ask whether the chosen integer belongs to S. A "Yes" answer costs 2 rubles, while a "No" answer costs 1 ruble. Determine the minimum cost that is sufficient in the worst case to identify the chosen integer.

Solution. Let $N(c)$ be the greatest number of possibilities that can always be distinguished with remaining worst-case cost at most c. With no useful question available,

$$N(0) = N(1) = 1.$$

For a first question, the "No" branch has remaining budget $c - 1$ and the "Yes" branch has remaining budget $c - 2$. Therefore

$$N(c) = N(c - 1) + N(c - 2).$$

Conversely, this bound is attainable by partitioning the current set of possibilities into subsets of these two maximum sizes and asking whether the chosen number lies in the second subset.

Thus

$$N(c) = F_{c+1},$$

where $F_1 = F_2 = 1$. Since

$$N(10) = F_{11} = 89 < 144$$

but

$$N(11) = F_{12} = 144,$$

the minimum guaranteed cost is

$$\boxed{11 \text{ rubles}}.$$

□

Problem 25. (1976 Waseda Entrance Examination). One card is chosen from a standard deck of 52 cards and placed in a box without being seen. The remaining 51 cards are shuffled, and three cards are drawn. All three drawn cards are diamonds. Determine the conditional probability that the card in the box is a diamond.

Solution. Regard the entire experiment as placing the 52 shuffled cards into labeled positions, with the first position designated as the box and the next three positions designated as the observed draws. After the three observed cards are known to be diamonds, the remaining 49 unobserved positions contain exactly

$$10$$

diamonds. By symmetry, each of these 49 positions is equally likely to be the box position. Therefore the desired conditional probability is

$$\boxed{\frac{10}{49}}.$$

□

Problem 26. A traditional epitaph describes the life of Diophantus as follows. He spent one sixth of his life as a child, one twelfth more before growing a beard, and one seventh more before marrying. Five years later his son was born; the son lived half as long as Diophantus, and Diophantus died four years after his son. Determine Diophantus' age at death.

Solution. Let x be Diophantus' age at death. The epitaph gives

$$x = \frac{x}{6} + \frac{x}{12} + \frac{x}{7} + 5 + \frac{x}{2} + 4.$$

Since

$$\frac{1}{6} + \frac{1}{12} + \frac{1}{7} + \frac{1}{2} = \frac{25}{28},$$

we obtain

$$\frac{3x}{28} = 9.$$

Therefore,

$$\boxed{x = 84}.$$

□

Mathematical Intuition Problems

Problem 1. A geodesic triangle is drawn on a surface of constant Gaussian curvature. Predict whether its angle sum is less than, equal to, or greater than π in each case.

(a) negative curvature;

(b) zero curvature;

(c) positive curvature.

Solution. By the Gauss–Bonnet relation

$$\alpha + \beta + \gamma - \pi = K \operatorname{area}(\triangle ABC),$$

the sign of the angle excess is the sign of K. Therefore,

$$\boxed{\begin{array}{l}\text{(a) less than } \pi, \\ \text{(b) equal to } \pi, \\ \text{(c) greater than } \pi.\end{array}}$$

□

Problem 2. (1996 Putnam). Two circles have radii 1 and 3, and the distance between their centers is 10. A point X moves on the first circle and a point Y moves on the second. Determine the locus of the midpoint M of $\overline{XY}$.

Solution. Let the centers of the two circles be A and B, and let O be the midpoint of $\overline{AB}$. Write

$$\overrightarrow{AX} = \mathbf{u}, \qquad \overrightarrow{BY} = \mathbf{v},$$

where

$$\|\mathbf{u}\| = 1, \qquad \|\mathbf{v}\| = 3.$$

Since M is the midpoint of X and Y,

$$\overrightarrow{OM} = \frac{\mathbf{u} + \mathbf{v}}{2}.$$

The possible lengths of $\mathbf{u} + \mathbf{v}$ range from

$$3 - 1 = 2 \qquad \text{to} \qquad 3 + 1 = 4.$$

Every intermediate length and every direction occur by varying the angle between $\mathbf{u}$ and $\mathbf{v}$ and rotating both vectors. Therefore the locus is the closed annulus

$$\boxed{1 \le OM \le 2}.$$

□

Problem 3. (1996 Putnam). Two squares have total area 1. Their sides, and the sides of a containing rectangle, must remain parallel, and the interiors of the two squares may not overlap. Determine the least number A s.t. every such pair of squares can be packed into a rectangle of area at most A.

Solution. Let the side lengths be $x \ge y > 0$, so

$$x^2 + y^2 = 1.$$

Since two axis-parallel squares with disjoint interiors must be separated horizontally or vertically, any containing rectangle has one side at least $x + y$ and the other at least x. This bound is attained by placing the squares side by side. Thus the least area required for this pair is

$$x(x + y).$$

Put $t = y/x$, where $0 < t \le 1$. Then

$$x^2 = \frac{1}{1 + t^2},$$

and the required area is

$$\frac{1 + t}{1 + t^2}.$$

Differentiation shows that this expression is maximized at

$$t = \sqrt{2} - 1.$$

Its maximum value is

$$\boxed{\frac{1+\sqrt{2}}{2}}.$$

□

Problem 4. (1994 CSAT). A cube is intersected by a plane. Among the following shapes, determine how many can occur as the cross-section:

(1) a triangle;
(2) a nonsquare rectangle;
(3) a nonsquare rhombus;
(4) a pentagon;
(5) a hexagon.

Solution. Work with the cube

$$0 \le x, y, z \le 1.$$

Each of the five shapes occurs.

The plane

$$x + y + z = \frac{1}{2}$$

produces a triangle. The plane

$$z = \frac{x}{2}$$

produces a rectangle whose side lengths are 1 and $\sqrt{5}/2$, so it is not a square.

The plane

$$4z = x + y + 1$$

produces a parallelogram with adjacent side vectors

$$\left(1, 0, \frac{1}{4}\right) \qquad \text{and} \qquad \left(0, 1, \frac{1}{4}\right).$$

These vectors have equal lengths but nonzero dot product, so the cross-section is a nonsquare rhombus.

The plane

$$x + y + 2z = \frac{11}{10}$$

meets five edges of the cube, at

$$\left(\frac{1}{10}, 1, 0\right), \left(1, \frac{1}{10}, 0\right), \left(1, 0, \frac{1}{20}\right), \left(0, 0, \frac{11}{20}\right), \left(0, 1, \frac{1}{20}\right),$$

and therefore produces a pentagon. Finally,

$$x + y + z = \frac{3}{2}$$

produces a hexagon. Thus all five listed shapes are possible, and the required number is

$$\boxed{5}.$$

□